

Developments in Alkali-Metal Atomic Magnetometry

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Abstract

Alkali-metal magnetometers use the coherent precession of polarized atomic spins to detect and measure magnetic fields. Recent advances have enabled magnetometers to become competitive with SQUIDs as the most sensitive magnetic field detectors, and they now find use in a variety of areas ranging from medicine and NMR to explosives detection and fundamental physics research. In this thesis we discuss several developments in alkali-metal atomic magnetometry for both practical and fundamental applications.

We present a new method of polarizing the alkali atoms by modulating the optical pumping rate at both the linear and quadratic Zeeman resonance frequencies. We demonstrate experimentally that this method enhances the sensitivity of a potassium magnetometer operating in the Earth's field by a factor of 4, and we calculate that it can reduce the orientation-dependent heading error to less than 0.1 nT. We discuss a radio-frequency magnetometer for detection of oscillating magnetic fields with sensitivity better than $0.2 \text{ fT} / \sqrt{\text{Hz}}$, which we apply to the observation of nuclear magnetic resonance (NMR) signals from polarized water, as well as nuclear quadrupole resonance (NQR) signals from ammonium nitrate. We demonstrate that a spin-exchange relaxation-free (SERF) magnetometer can measure all three vector components of the magnetic field in an unshielded environment with comparable sensitivity to other devices. We find that octadecyltrichlorosilane (OTS) acts as an anti-relaxation coating for alkali atoms at temperatures below 170°C , allowing them to collide with a glass surface up to 2,000 times before depolarizing, and we present the first demonstration of high-temperature magnetometry with a coated cell. We also

describe a reusable alkali vapor cell intended for the study of interactions between alkali atoms and surface coatings. Finally, we explore the use of a cesium-xenon SERF comagnetometer for a proposed measurement of the permanent electric dipole moments (EDMs) of the electron and the ^{129}Xe atom, with projected sensitivity of $\delta d^e = 9 \times 10^{-30} \text{ e-cm}$ and $\delta d^{\text{Xe}} = 4 \times 10^{-31} \text{ e-cm}$ after 100 days of integration; both bounds are more than two orders of magnitude better than the existing experimental limits on the EDMs of the electron and of any diamagnetic atom.

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Chapter 1

Introduction

DETECTION AND MEASUREMENT of magnetic fields have been of great importance to civilization beginning with the invention of the compass in ancient China for navigational purposes. In 1832, Carl Friedrich Gauss invented the first device for measuring the strength of a field, comprised of a bar magnet suspended in air ([Gauss, 1832](#)). Measurement technology subsequently improved with the development of detectors such as the Hall probe, fluxgate, and proton precession magnetometer. For the past few decades, superconducting quantum interference devices (SQUIDs) have been the most effective detector of magnetic fields, with sensitivity potentially approaching $1 \text{ fT}/\sqrt{\text{Hz}}$ for devices without superconducting shields. Magnetic field characterization is ubiquitous in the modern world and finds application in a wide variety of areas, including medicine, information storage, mineral and oil detection, contraband detection, space exploration, and fundamental physics experiments.

Recent developments in the technology of atomic magnetometers have enabled them to overtake SQUIDs as the most sensitive devices for detecting and measuring magnetic fields. [Dehmelt \(1957a\)](#) originally proposed the observation of precessing alkali spins in

order to determine the strength of a field, and [Bell and Bloom \(1957\)](#) provided the first experimental demonstration. Over the next few decades, much effort was spent on improving the accuracy and precision of atomic magnetometers, which have the advantage over SQUIDs of not requiring cryogenics for operation. In the past several years, the SERF and rf magnetometers have been introduced with demonstrated sensitivity below $1 \text{ fT}/\sqrt{\text{Hz}}$ and the capability to eventually detect attotesla-level fields. A comprehensive review of the current state of atomic magnetometer technology is presented by [Budker and Romalis \(2007\)](#).

The basic principle behind atomic magnetometry, shown in [Figure 1.1](#), is simple: we measure the Larmor precession frequency ω of atomic spins in a magnetic field $\mathbf{B}$, given by

$$\omega = \gamma|\mathbf{B}|, \quad (1.1)$$

where the gyromagnetic ratio γ serves as the conversion factor between the frequency and the field strength. We employ a vapor of alkali-metal atoms for magnetometry because they each have only a single valence electron, so the atomic spin is given by the vector sum of the spins of the nucleus and of the valence electron. We polarize the atoms through optical pumping ([Happer, 1972](#)), which transfers angular momentum to the ensemble of atoms from a beam of circularly polarized light that is tuned to an atomic resonance. There are numerous methods of monitoring the spin polarization, but for the work presented here we use a linearly polarized probe beam propagating along a direction orthogonal to the pump beam. As the probe beam travels through the alkali vapor, its plane of polarization rotates by an angle proportional to the spin component along that direction, and we detect this rotation in order to observe the spin behavior. We contain the alkali metal within a glass cell, which we heat in order to increase the saturated vapor density of the atoms.

Magnetometers are generally characterized by their sensitivity, which determines the precision of the device; we may think of this either as the smallest change in the field level

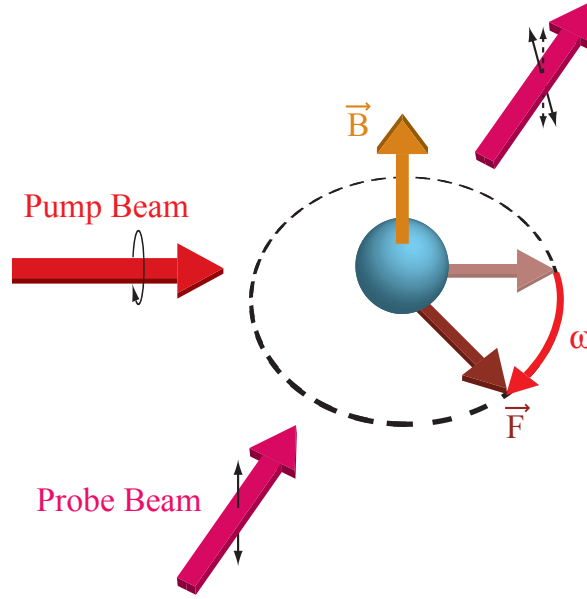


Figure 1.1: Basic principle of atomic magnetometry: we polarize alkali-metal spins by optical pumping and monitor their precession in a magnetic field with a probe beam. The precession frequency is proportional to the amplitude of the magnetic field.

that the sensor can discern, or as the size of the smallest field that it can detect. On a fundamental level, the magnetometer actually measures the energy splitting between the Zeeman sublevels of the atomic ground state due to the magnetic field. The linewidth of such a spectroscopic measurement is given by the coherence lifetime T_2 of the atomic spins:

$$\Delta B = \frac{\Delta\omega}{\gamma} = \frac{1}{\gamma T_2}. \quad (1.2)$$

The construction of a sensitive magnetometer therefore depends on achieving the maximum possible polarization lifetime.

Alkali spins depolarize immediately after colliding with the glass walls of the vapor cell, so it is necessary to prevent these collisions. One method is to fill the cell with a high pressure of an inert buffer gas to inhibit diffusion, which has the advantage of allowing atoms in different parts of the cell to act as independent magnetometers, enabling the measurement of magnetic field gradients. The other method is to coat the surface with a chemical

that prevents depolarization (Robinson *et al.*, 1958; Bouchiat and Brossel, 1966). Paraffin is the most effective known coating, allowing atoms to collide up to 10,000 times off the surface without depolarizing (Graf *et al.*, 2005), but it melts at 60-80°C and so can not be used for higher-temperature applications. We showed that a coating of octadecyltrichlorosilane (OTS) can allow up to 2,000 collisions with the surface at temperatures up to 170 °C (Seltzer *et al.*, 2007). Coated cells have the advantages of providing larger optical rotation signals, reducing the effect of magnetic field gradients on the spin polarization lifetime, and lowering the power requirements of the lasers used for pumping and probing.

From a purely phenomenological point of view, the magnetometer sensitivity depends on the signal-to-noise ratio (S/N) of the Zeeman resonance signal as well as the linewidth,

$$\delta B = \frac{\Delta B}{(S/N)} . \quad (1.3)$$

Thus, magnetic field noise should be attenuated if possible, and care should be taken to ensure that the optical detection system is stable. Diode lasers, especially distributed feedback (DFB) diodes, are easily tunable and can be very stable, allowing for extremely low-noise measurements of optical rotation. We can enhance the resonance signal by increasing the number of atoms N in the spin ensemble, either by increasing the vapor density or by using a larger vapor cell. This has the added benefit of improving the atomic shot noise due to quantum fluctuations in the expectation value of the spin polarization, $\delta\langle S \rangle \propto 1/\sqrt{N}$, which sets a fundamental limit on the magnetometer sensitivity.

However, the rate of depolarizing collisions between alkali atoms scales with the vapor density, and at high density spin-exchange collisions can limit the polarization lifetime of the atoms. Sensitive magnetometers therefore traditionally have used large vapor cells and operated at low density, typically at or near room temperature. First presented in 2002, the spin-exchange relaxation-free (SERF) magnetometer eliminates this effect by operating at

zero field to enable long polarization lifetimes (Allred *et al.*, 2002), with demonstrated sensitivity of $0.5 \text{ fT}/\sqrt{\text{Hz}}$ in a cell with volume less than 1 cm^3 (Kominis *et al.*, 2003) and the potential to achieve sensitivity better than $1 \text{ aT}/\sqrt{\text{Hz}}$. The radio-frequency (rf) magnetometer that we presented in 2005 detects oscillating magnetic fields at frequencies in the kilohertz to megahertz range (Savukov *et al.*, 2005); it partially suppresses spin-exchange relaxation by achieving high spin polarization, and we have attained sensitivity of $0.2 \text{ fT}/\sqrt{\text{Hz}}$ (Lee *et al.*, 2006), with at least an order of magnitude improvement possible. For both the SERF and rf magnetometers, we heat the vapor cell to $100\text{-}200^\circ\text{C}$, depending on the alkali species, in order to operate with density of $10^{12}\text{-}10^{14} \text{ cm}^{-3}$. One of the main engineering challenges in developing these high-sensitivity magnetometer systems is to construct the oven out of completely nonmagnetic materials, so as not to introduce additional magnetic noise into the measurement.

Another important characteristic of a magnetometer is its accuracy. Atomic magnetometers operating in the Earth's magnetic field exhibit heading errors, or shifts of the measured resonance frequency depending on the orientation of the sensor with respect to the field. This effect is due to the quadratic and higher-order Zeeman splitting of the ground-state energy levels and typically limits the accuracy of a magnetometer to $1\text{-}10 \text{ nT}$. While the Earth's magnetic field has an amplitude of approximately $50 \mu\text{T}$, the uncertainty due to the heading error can nevertheless obscure the signal given by a magnetic anomaly. One common method for suppressing the heading error is the use of multiple pump beams (Yabuzaki and Ogawa, 1974). We introduced a different approach, which involves simultaneous excitation of both the linear (Larmor) and quadratic magnetic resonances and can potentially reduce the error below 0.1 nT , as well as improve the magnetometer sensitivity (Seltzer *et al.*, 2007).

In addition to the techniques described in this thesis, there are other varieties of alkali-metal magnetometers currently in use. For example, magnetometers based on nonlinear

magneto-optical rotation (NMOR) feature parallel pump and probe beams and measure the magnetic field along the direction of beam propagation (Budker *et al.*, 2002). NMOR magnetometers have the advantages of operating near room temperature and of being all-optical (*i.e.*, they do not require magnetic field compensation or excitation), and they can achieve sensitivity on the order of $1 \text{ fT}/\sqrt{\text{Hz}}$ (Budker *et al.*, 2000). Unfortunately, they require large vapor cells with volumes on the order of 1000 cm^3 to do so, but they can also be modified to detect rf fields (Ledbetter *et al.*, 2007). Magnetometers based on coherent population trapping (CPT) can reach picotesla-level sensitivity and are also all-optical (Stähler *et al.*, 2001; Affolderbach *et al.*, 2002).

Atomic magnetometers have been developed recently using microfabricated vapor cells with volumes of about 10 mm^3 (Schwindt *et al.*, 2004; Knappe *et al.*, 2006); such devices are eminently portable, with power consumption less than 200 mW and total physics package volume less than 10 cm^3 . Shah *et al.* (2007) demonstrated sensitivity below $70 \text{ fT}/\sqrt{\text{Hz}}$ with a SERF magnetometer using a cell with volume of 6 mm^3 . Although not necessarily portable, Bose-Einstein condensates of alkali atoms can compose a magnetometer with very high spatial resolution on the order of $1\text{-}10 \text{ }\mu\text{m}$ and sensitivity better than $1 \text{ pT}/\sqrt{\text{Hz}}$ (Wildermuth *et al.*, 2006; Vengalattore *et al.*, 2007). An evanescent-wave vapor magnetometer can achieve spatial resolution less than $100 \text{ }\mu\text{m}$ near the surface of the vapor cell with sensitivity of $10 \text{ pT}/\sqrt{\text{Hz}}$ (Zhao and Wu, 2006). Finally, we note that atomic magnetometers have also been demonstrated using metastable ^4He instead of alkali atoms, with sensitivity potentially reaching the femtotesla level and no inherent heading error (McGregor, 1987).

SQUIDs are the main competitors of atomic magnetometers. They measure the magnetic flux through a loop consisting of two Josephson junctions, and thus do not sense the magnetic field directly, although the sensitivity of a low- T_c SQUID can reach the equivalent of $1 \text{ fT}/\sqrt{\text{Hz}}$ in systems that do not use superconducting shields (Clarke and Braginski,

2004). SQUIDs must operate at cryogenic temperatures necessary to reach a superconducting state, so detection systems tend to be bulky and require a steady supply of coolant, making them expensive to operate and unfeasible for many portable applications. Atomic magnetometers have the potential to be significantly cheaper to construct and to maintain while also exhibiting better sensitivity.

Fluxgate magnetometers are often used for field operation because of their portability and high measurement accuracy, although the sensitivity of commercial fluxgates is typically limited to about $1 \text{ pT}/\sqrt{\text{Hz}}$. Portable atomic devices can be much more sensitive, but the heading error makes them less accurate. Inductive pick-up coils are widely used for high-field magnetic resonance applications because of extremely good sensitivity at high frequencies, but their sensitivity scales linearly with frequency, rendering them much less effective below several megahertz. Atomic magnetometers and SQUIDs can therefore outperform inductive coils for low-frequency applications. Figure 1.2 compares the sensitivities of these devices, including both the demonstrated sensitivity and fundamental limits of the SERF and rf magnetometers, as well as the characteristic size of common magnetic signals for reference. Field sensitivity is given in units of $\text{T}/\sqrt{\text{Hz}}$ and represents the precision obtained after 1 second of integration; this improves as the square root of the measurement time, so short-lived signals require greater sensitivity than persistent signals for detection.

As the capabilities of alkali-metal magnetometers have improved, they have found use in applications traditionally dominated by other devices. They have been demonstrated for detection of low-field nuclear magnetic resonance (NMR), both near zero frequency (Yashchuk *et al.*, 2004; Savukov and Romalis, 2005b) and at tens of kilohertz (Savukov *et al.*, 2007), and for magnetic resonance imaging (MRI) (Xu *et al.*, 2006). RF magnetometers can more efficiently detect nuclear quadrupole resonance (NQR) signals at 0.1-10 MHz from explosives and narcotics than pick-up coils because their sensitivity is nearly independent

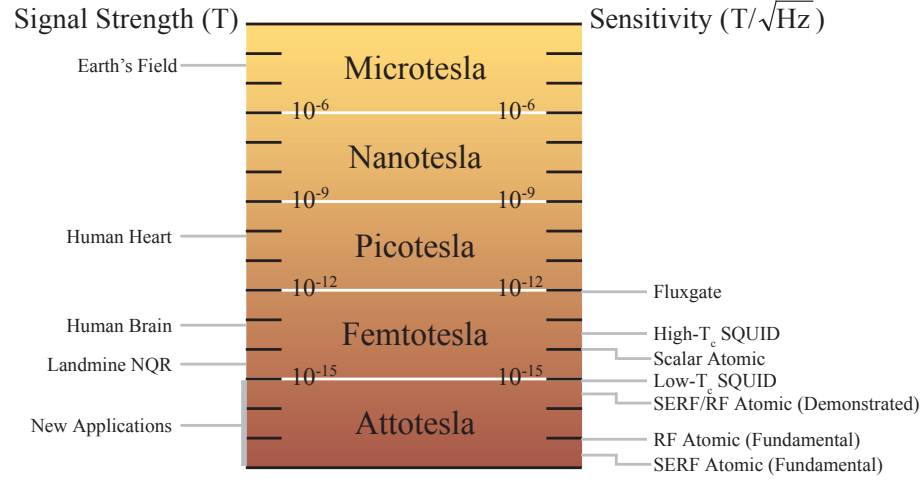


Figure 1.2: Comparison of the demonstrated sensitivity of various magnetic field detectors, as well as the fundamental sensitivity limits of the SERF and rf magnetometers. We also show the amplitudes of several magnetic signals for reference.

of the measurement frequency (Lee *et al.*, 2006). Atomic magnetometers have also been employed for detection of biomagnetic signals from the human heart (Bison *et al.*, 2003; Belfi *et al.*, 2007) and brain (Xia *et al.*, 2006), for geophysical exploration (Nabighian *et al.*, 2005; Mathé *et al.*, 2006), for archaeology (David *et al.*, 2004), and for tests of fundamental physics (Berglund *et al.*, 1995; Groeger *et al.*, 2005; Kornack *et al.*, 2008). There are doubtless many undiscovered applications that will be realized as the sensitivity of atomic magnetometers continues to improve toward the attotesla level.

In this thesis we discuss several recent developments in the technology and application of alkali-metal atomic magnetometers. Chapter 2 is intended as a reference for the chapters that follow and provides an introduction to the basic concepts underlying the operation of an atomic magnetometer, such as optical pumping, optical rotation, and spin relaxation. Chapter 3 discusses the operation of a magnetometer in the Earth's field, in particular a new method of suppressing heading errors and improving sensitivity through resonant excitation of the nonlinear Zeeman splitting. Chapter 4 describes the rf magnetometer and its

use for detection of NMR and NQR signals. Chapter 5 details the detection of all three vector components of the magnetic field with a SERF magnetometer in an unshielded environment. Chapter 6 discusses the advantages of wall coatings for high-temperature operation, the application of OTS-coated cells for SERF magnetometry, and the development of an experiment to identify additional high-temperature coatings. Finally, Chapter 7 presents a proposed experiment to search for the electric dipole moments (EDMs) of the electron and the ^{129}Xe atom using a cesium-xenon SERF comagnetometer.

Chapter 2

General Magnetometry

RECENT DEVELOPMENTS IN ATOMIC MAGNETOMETRY have led to a variety of methods for detecting and measuring magnetic fields, and different types of magnetometers have their own unique characteristics and idiosyncrasies. However, the magnetometers discussed in this thesis all share certain basic features that we describe in general in this chapter; we then move on to discussing the individual magnetometers in subsequent chapters. We begin with a basic overview of the atomic energy structure before describing techniques for polarizing alkali atoms and measuring their spin direction. We consider the atomic response to magnetic fields, and we detail the effects that limit the spin-polarization coherence lifetime. We also analyze the fundamental limit of magnetometer sensitivity due to quantum fluctuations. Finally, we discuss the density matrix formalism and how it can be used to determine the evolution of the atomic spins.

2.1 Atomic Energy Levels

Alkali metal atoms are useful for a variety of applications because they have a single unpaired electron in the outer energy shell that can be easily manipulated. The energy of

the atom can be very well approximated by considering only the valence electron and the nucleus, ignoring the electrons in the filled inner energy shells. Atomic magnetometers operate by exploiting the energy structure of the ground and excited states to polarize the atoms and measure the magnetic field, so it is useful to briefly review the energy levels of the alkali atom.

The valence electron has spin $S=1/2$, and the ground state is an s shell with orbital angular momentum $L=0$, so that total electron angular momentum $J=L+S=1/2$. The first excited state is a p shell with $L=1$; the fine structure splits this state into the $^2P_{1/2}$ ($J=1/2$) and $^2P_{3/2}$ ($J=3/2$) levels. These can be thought of as states with the spin and orbital angular momenta lying anti-parallel and parallel, respectively. Here we use the standard spectroscopic notation, with the superscript denoting the spin multiplicity $2S+1$ and the subscript denoting the total angular momentum J , so that the ground state can be written as $^2S_{1/2}$. The energy transitions between the ground state and the $^2P_{1/2}$ and $^2P_{3/2}$ levels are respectively referred to as the D1 and D2 transitions.

All natural alkali metal isotopes have nonzero nuclear spin I , so the hyperfine interaction between the electron and nuclear spins further splits the atomic energy levels into states with total atomic spin $F=I+J$. According to the Wigner-Eckart theorem, the electron angular momentum vector J must be parallel to the total atomic angular momentum vector F (see for example [Cohen-Tannoudji *et al.* \(1977\)](#)), so a measurement of the direction of the electron spin vector is essentially equivalent to a determination of the direction of the atomic spin vector, and vice versa. The $^2S_{1/2}$ and $^2P_{1/2}$ states are split into levels with $F=I \pm 1/2$ separated by the hyperfine energy splitting E_{hf} . These can be thought of as states with the atomic and nuclear spins lying parallel to one another, with the electron spin either parallel ($F=I+1/2$) or anti-parallel ($F=I-1/2$) to both. The $^2P_{3/2}$ state is split into levels with $F = \{I - 3/2, I - 1/2, I + 1/2, I + 3/2\}$. The fine and hyperfine structure of the ground and first excited states of an alkali atom are shown in Figure 2.1.

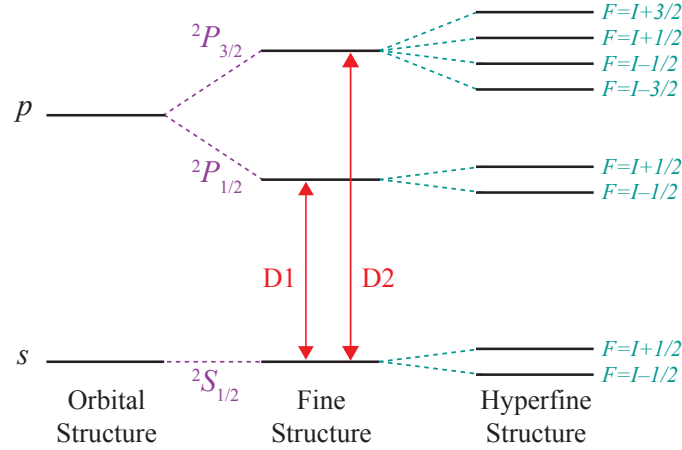


Figure 2.1: Energy level splitting of the ground state and first excited state of an alkali metal atom. The fine structure splits the first excited state into levels with $J=1/2$ and $J=3/2$, and the hyperfine structure further splits the energy levels due to the nonzero nuclear spin. Not drawn to scale.

Finally, interaction with external magnetic fields lifts the degeneracy between different Zeeman sublevels with projection $m_F = \{-F, -F + 1, \dots, F - 1, F\}$ of the atomic angular momentum along some quantization axis. The ground-state Zeeman sublevels for the case of $I=3/2$ are shown in Figure 2.2. The resulting energy splitting ΔE_L depends on the strength of the field and gives rise to Larmor spin precession with frequency $\omega_L = \Delta E_L / \hbar = \gamma |\mathbf{B}|$, where γ is the gyromagnetic ratio of the atomic spin. The valence electron couples much more strongly than the nuclear spin to an external field, so to first order the gyromagnetic ratio is simply that of a bare electron, except reduced because the electron spin must effectively drag the nuclear spin along as it precesses. Then $\gamma \approx \pm 2\pi \times (2.8 \text{ MHz/G}) / (2I + 1)$, where the sign depends on the hyperfine level $F=I \pm 1/2$; we define the gyromagnetic ratio more precisely in Section 3.2. The energy level splittings for the alkali isotopes that are most commonly used for magnetometry are included in Table A.1.

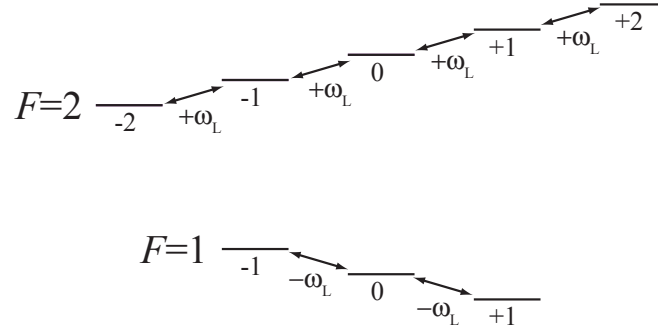


Figure 2.2: Ground-state Zeeman sublevels for the case $I=3/2$. Sublevels are labeled by their projection m_F of the atomic spin along a quantization axis. Note that the energy splitting changes sign depending on the hyperfine level.

2.2 Optical Absorption and the Optical Lineshape

Atomic magnetometers require resonant or near-resonant light to both polarize the alkali atoms and probe their spin orientation. The rate $R_{\text{abs}}(\nu)$ at which an atom absorbs photons of frequency ν is

$$R_{\text{abs}}(\nu) = \sum_{\text{res}} \sigma(\nu) \Phi(\nu), \quad (2.1)$$

where $\Phi(\nu)$ is the total flux of photons of frequency ν incident on the atom in units of number of photons per area per time, and the sum is over all atomic resonances. Most modern magnetometers, including those described in this thesis, use lasers with linewidths that are much narrower than those associated with the atomic D1 and D2 transitions, so the incident light may be treated as monochromatic. The photon absorption cross-section $\sigma(\nu)$ is determined by the atomic frequency response about the resonance frequency ν_0 , which generally depends on three effects: the lifetime of the excited state, pressure broadening due to collisions with other gas species, and Doppler broadening due to thermal motion of the alkali atoms (see for instance [Corney \(1977\)](#)). Regardless of the form of the frequency response, the integral of the absorption cross-section associated with a given resonance is

a constant,

$$\int_0^{+\infty} \sigma(\nu) d\nu = \pi r_e c f_{\text{res}}, \quad (2.2)$$

where $r_e = 2.82 \times 10^{-15}$ m is the classical electron radius, and $c = 3 \times 10^8$ m/s is the speed of light. The oscillator strength f_{res} is the fraction of the total classical integrated cross-section associated with the given resonance. For alkali atoms, the oscillator strengths are approximately given by $f_{\text{D1}} \approx 1/3$ and $f_{\text{D2}} \approx 2/3$; however for heavier elements the actual values deviate slightly due to the spin-orbit interaction and core-valence electron correlation (Migdalek and Kim, 1998). The precise measured values are given in Table A.1.

2.2.1 The Natural Lifetime and Pressure Broadening

The $^2P_{1/2}$ and $^2P_{3/2}$ states have natural lifetimes τ_{nat} of about 25-35 ns, given in Table A.1. The uncertainty principle requires that for a given resonance

$$\Delta E \Delta t \gtrsim \hbar. \quad (2.3)$$

The uncertainty in time is the natural lifetime, $\Delta t = \tau_{\text{nat}}$. The uncertainty in frequency is $\Delta \nu = \Delta E / 2\pi\hbar$, giving the natural linewidth

$$\Gamma_{\text{nat}} = \Delta \nu = 1/2\pi\tau_{\text{nat}}. \quad (2.4)$$

For the D1 and D2 transitions in alkali atoms, the natural linewidth is about 4-6 MHz.

Collisions with buffer gas atoms and quenching gas molecules perturb the excited alkali atoms due to electromagnetic interactions, resulting in both a shift and broadening of the optical resonance line. The magnitudes of these effects are proportional to the number density of perturbing atoms or molecules, and so they are referred to as the pressure shift and pressure broadening; some measured values are included in Table A.2. The amount of pressure broadening is approximately given by the average time between collisions τ_{pr} while in the excited state,

$$\Gamma_{\text{pr}} \approx 1/\pi\tau_{\text{pr}}. \quad (2.5)$$

For typical pressures of buffer and quenching gas used in magnetometer cells, the pressure broadened linewidth is on the order of 1-100 GHz.

The linewidths due to the natural lifetime and pressure broadening add together to give a single linewidth, $\Gamma_L = \Gamma_{\text{nat}} + \Gamma_{\text{pr}}$. The resulting lineshape of the atomic frequency response around the resonance frequency ν_0 has the form of a Lorentzian curve with full width at half maximum (FWHM) Γ_L ,

$$\mathcal{L}(\nu - \nu_0) = \frac{\Gamma_L/2\pi}{(\nu - \nu_0)^2 + (\Gamma_L/2)^2}, \quad (2.6)$$

as shown in Figure 2.3(a). Here the Lorentzian has been written in normalized form, so that the absorption cross-section given by Equation 2.2 becomes

$$\sigma_L(\nu) = \pi r_e c f \mathcal{L}(\nu - \nu_0), \quad (2.7)$$

and the cross-section on resonance is

$$\sigma_L(\nu_0) = \frac{2r_e c f}{\Gamma_L}. \quad (2.8)$$

Therefore, the rate of photon absorption is inversely proportional to the gas pressure in the cell, so that applications using higher gas pressure require the use of more intense lasers.

2.2.2 Doppler Broadening

Atoms with mass M at temperature T move with a root-mean-square thermal velocity $v_{\text{th}} = \sqrt{3k_B T/M}$. If an atom's velocity has some component v_z along the direction of a laser's propagation, then the frequency of the laser as experienced by the atom is shifted due to the Doppler effect,

$$\nu' = \nu \left(1 - \frac{v_z}{c}\right). \quad (2.9)$$

The Doppler effect causes broadening of the atomic resonance lines because light of frequency ν detuned from the resonance frequency ν_0 is experienced as being on resonance

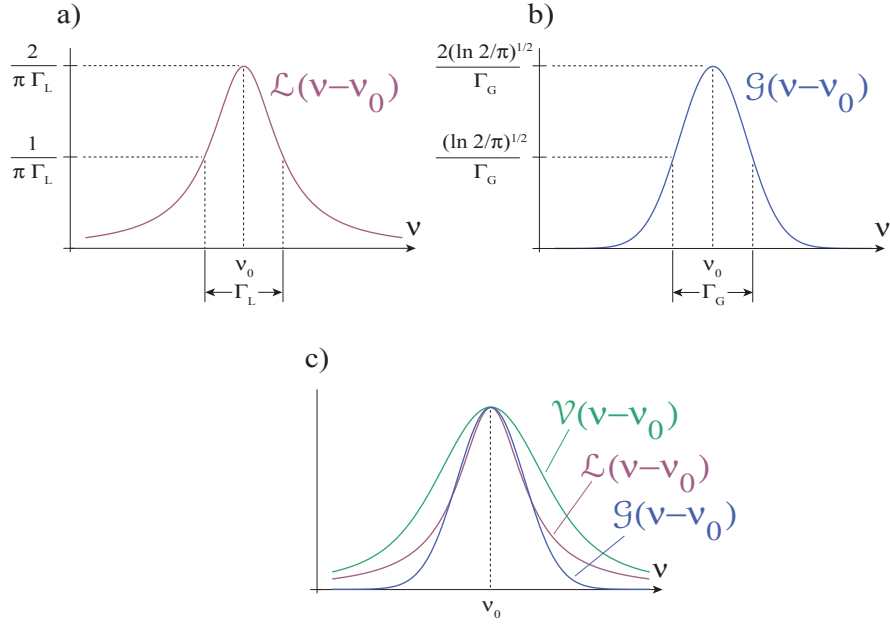


Figure 2.3: Comparison of the Lorentzian, Gaussian, and Voigt lineshapes with $\Gamma_G = \Gamma_L$. The curves have been scaled to have the same value on resonance.

by any atoms moving with the appropriate velocity such that

$$v_z = c \frac{\nu - \nu_0}{\nu}. \quad (2.10)$$

Thus, some subset of the atomic population absorbs the off-resonance light. The probability $P(v_z)dv_z$ of an atom having a velocity in the range from v_z to $v_z + dv_z$ is given by the Maxwellian distribution,

$$P(v_z)dv_z = \sqrt{\frac{M}{2\pi k_B T}} \exp\left(\frac{-Mv_z^2}{2k_B T}\right) dv_z. \quad (2.11)$$

The resulting frequency response has the form of a Gaussian curve,

$$\mathcal{G}(\nu - \nu_0) = \frac{2\sqrt{\ln 2/\pi}}{\Gamma_G} \exp\left(\frac{-4 \ln 2 (\nu - \nu_0)^2}{\Gamma_G^2}\right), \quad (2.12)$$

which has full width at half maximum (FWHM) given by

$$\Gamma_G = 2 \frac{\nu_0}{c} \sqrt{\frac{2k_B T}{M} \ln 2}. \quad (2.13)$$

Alkali Isotopes:	³⁹ K	⁴¹ K	⁸⁵ Rb	⁸⁷ Rb	¹³³ Cs
Γ_{nat}	5.94	5.94	5.75	5.75	4.57
$\Gamma_G, 273 \text{ K}$	737	719	484	478	344
373 K	862	841	566	559	402
473 K	971	947	637	630	452

Table 2.1: Comparison of the natural linewidth and the Doppler broadened linewidth of the D1 transition within the typical range of magnetometer operational temperatures. All linewidths are given in units of MHz.

The Gaussian profile is shown in Figure 2.3(b); note that the wings of the Gaussian approach zero more quickly than the wings of the Lorentzian. The Gaussian has been written in normalized form, so the absorption cross-section given by Equation 2.2 becomes

$$\sigma_G(\nu) = \pi r_e c f \mathcal{G}(\nu - \nu_0), \quad (2.14)$$

and the cross-section on resonance is

$$\sigma_G(\nu_0) = \frac{2r_e c f \sqrt{\pi \ln 2}}{\Gamma_G}. \quad (2.15)$$

At typical magnetometer operating temperatures, the Doppler broadened linewidth is significantly larger than the natural linewidth; see Table 2.1. In the absence of pressure broadening the optical lineshape can therefore be well approximated by the Gaussian lineshape described by Equations 2.12 and 2.13.

2.2.3 The Voigt Profile

In general, the atomic frequency response depends on all three effects described above: the natural lifetime, pressure broadening, and Doppler broadening. The Lorentzian lineshape that results from the first two effects is further broadened by the Maxwellian distribution of thermal velocities, since some fraction of the atomic population experiences incident light of frequency ν to be Doppler shifted onto resonance. The resulting lineshape is the

Voigt profile (Happer and Mathur, 1967),

$$\mathcal{V}(\nu - \nu_0) = \int_0^\infty \mathcal{L}(\nu - \nu') \mathcal{G}(\nu' - \nu_0) d\nu'. \quad (2.16)$$

It is convenient to write the Voigt profile in complex form,

$$\mathcal{V}(\nu - \nu_0) = \frac{2\sqrt{\ln 2/\pi}}{\Gamma_G} w\left(\frac{2\sqrt{\ln 2}[(\nu - \nu_0) + i\Gamma_L/2]}{\Gamma_G}\right), \quad (2.17)$$

where the complex error function $w(x)$ is given by

$$w(x) = e^{-x^2} (1 - \operatorname{erf}(-ix)). \quad (2.18)$$

The Voigt profile for the case that $\Gamma_L = \Gamma_G$ is shown in Figure 2.3(c), along with the Lorentzian and Gaussian profiles for comparison. The absorption cross-section is

$$\sigma_V(\nu) = \pi r_e c f \operatorname{Re}[\mathcal{V}(\nu - \nu_0)]. \quad (2.19)$$

In the case of no pressure broadening such that $\Gamma_G \gg \Gamma_L$ the Voigt profile becomes nearly Gaussian, while in the case of large pressure broadening such that $\Gamma_G \ll \Gamma_L$ the Voigt profile becomes nearly Lorentzian. It is therefore best to describe the optical lineshape with the general Voigt profile, using appropriate values of the linewidths Γ_G and Γ_L , rather than the more specialized Lorentzian and Gaussian curves.

2.2.4 Hyperfine Splitting of the Optical Resonance

In cases where the ground and/or excited state hyperfine splittings are comparable to or larger than the optical linewidth, it is necessary to separately consider the individual resonances $F \rightarrow F'$, as shown in Figure 2.4. The allowed transitions are $F - F' = \{0, \pm 1\}$. Using the Wigner-Eckart theorem, the matrix element for the dipole transition between the ground state $|F, m_F\rangle$ and the excited state $|F', m'_F\rangle$ is given by

$$\langle F, m_F | \hat{\mathbf{e}} \cdot \mathbf{r} | F', m'_F \rangle^2 = \langle F || \hat{\mathbf{e}} \cdot \mathbf{r} || F' \rangle^2 (2F + 1) \begin{pmatrix} F' & 1 & F \\ m'_F & m_F - m'_F & -m_F \end{pmatrix}^2, \quad (2.20)$$

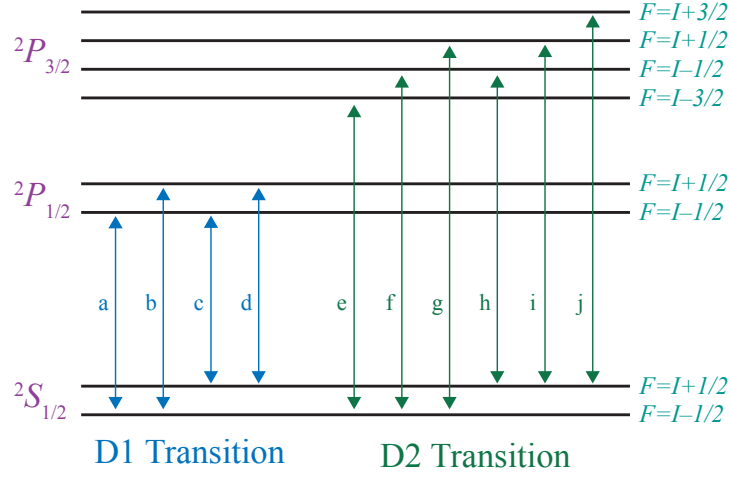


Figure 2.4: Allowed transitions between hyperfine levels of the ground and excited states of the D1 and D2 transitions.

where $\langle F || \hat{\mathbf{e}} \cdot \mathbf{r} || F' \rangle$ is a reduced matrix element, $\hat{\mathbf{e}}$ is the polarization of the incident light, $\mathbf{r}$ is the dipole moment of the atom, and the parentheses denote the Wigner 3-j symbol. If the vapor is unpolarized, then all m_F states are weighted equally. For a given transition $F \rightarrow F'$ and a given value $m_F - m'_F = \{0, \pm 1\}$, there is then a sum rule

$$\sum_{m_F, m'_F} \begin{pmatrix} F' & 1 & F \\ m'_F & m_F - m'_F & -m_F \end{pmatrix}^2 = \frac{1}{3}, \quad (2.21)$$

so that we may write

$$\begin{aligned} \langle F || \hat{\mathbf{e}} \cdot \mathbf{r} || F' \rangle^2 &= \sum_{m_F, m'_F} \langle F, m_F | \hat{\mathbf{e}} \cdot \mathbf{r} | F', m'_F \rangle^2 \\ &= \frac{2F+1}{3} \langle F || \hat{\mathbf{e}} \cdot \mathbf{r} || F' \rangle^2. \end{aligned} \quad (2.22)$$

Again applying the Wigner-Eckart theorem,

$$\langle F || \hat{\mathbf{e}} \cdot \mathbf{r} || F' \rangle^2 = \langle J || \hat{\mathbf{e}} \cdot \mathbf{r} || J' \rangle^2 \frac{(2F+1)(2F'+1)(2J+1)}{3} \left\{ \begin{matrix} J & J' & 1 \\ F' & F & I \end{matrix} \right\}^2, \quad (2.23)$$

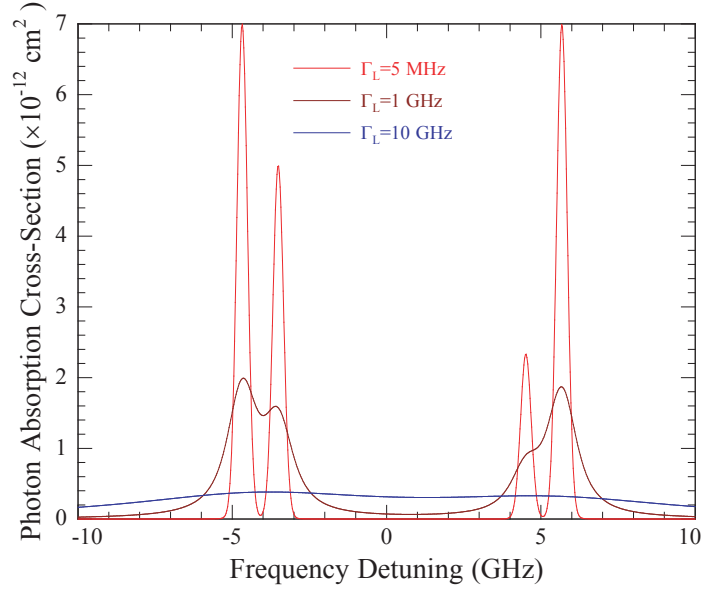


Figure 2.5: Optical lineshape of the cesium D1 transition, taking into account the hyperfine splitting of the ground and excited states. The frequency detuning is taken from the resonance frequency without hyperfine splitting. Photon absorption cross-sections have been calculated using Equation 2.26, with $\Gamma_G=402$ MHz (Doppler broadening at 373 K) and $\Gamma_L=5$ MHz (the natural linewidth), 1 GHz, and 10 GHz.

where the curly brackets denote the Wigner 6-j symbol. For $J=1/2$ and a particular value of $J' = \{1/2, 3/2\}$, corresponding to the D1 or D2 transitions, there is the sum rule,

$$\sum_{F, F'} (2F+1)(2F'+1) \left\{ \begin{matrix} J & J' & 1 \\ F' & F & I \end{matrix} \right\}^2 = 2I+1. \quad (2.24)$$

If we consider only the individual hyperfine transitions $F \rightarrow F'$ within either the D1 or D2 resonances, we may therefore write the relative strength $A_{F, F'}$ of the transition in the normalized form

$$A_{F, F'} = \frac{(2F+1)(2F'+1)}{2I+1} \left\{ \begin{matrix} J & J' & 1 \\ F' & F & I \end{matrix} \right\}^2, \quad (2.25)$$

where the sum of the strengths $\sum_{F, F'} A_{F, F'} = 1$.

The transition strengths are given in Table 2.2, where the individual resonances are labeled according to Figure 2.4. The total photon absorption cross-section at frequency ν is

Transition	$I = 3/2$	$I = 5/2$	$I = 7/2$
a	1/16	5/54	7/64
b	5/16	35/108	21/64
c	5/16	35/108	21/64
d	5/16	7/27	15/64
e	1/16	1/8	5/32
f	5/32	35/216	21/128
g	5/32	7/54	15/128
h	1/32	5/108	7/128
i	5/32	35/216	21/128
j	7/16	3/8	11/32

Table 2.2: Relative strengths $A_{F,F'}$ of the individual D1 (a-d) and D2 (e-j) hyperfine resonances for photon absorption, labeled according to Figure 2.4 and calculated according to Equation 2.25.

given by

$$\sigma_{\text{total}}(\nu) = \pi r_e c f \sum_{F,F'} A_{F,F'} \text{Re}[\mathcal{V}(\nu - \nu_{F,F'})], \quad (2.26)$$

where $\nu_{F,F'}$ is the resonance frequency of the transition $F \rightarrow F'$. For illustration, the frequency response of cesium near the D1 transition is shown in Figure 2.5. The Doppler linewidth is taken at 373 K ($\Gamma_G=402$ MHz), and the lineshape is compared at three values of the total Lorentzian linewidth: $\Gamma_L=5$ MHz (the natural linewidth), 1 GHz, and 10 GHz. If the optical linewidth is small compared to the hyperfine splitting, then the individual hyperfine transitions can be resolved. However, when the linewidth is large compared to the hyperfine splitting the individual transitions are unresolved, and only a single transition is observed.

2.3 Optical Pumping

For the types of magnetometers discussed in this thesis, the magnetometer signal scales with the polarization of the alkali metal vapor (see the discussion on optical rotation in Section 2.4). Sensitive magnetometers therefore require large atomic spin polarization to

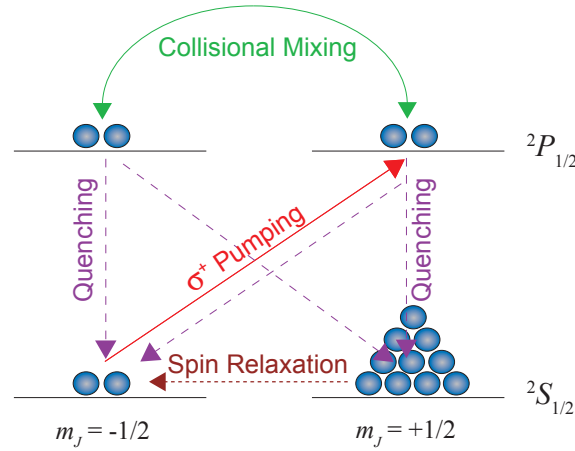


Figure 2.6: Optical pumping of the electron spin of an alkali atom with D1 σ^+ polarized light. Only atoms in the $m_J = -1/2$ sublevel may absorb a photon and become excited to the $^2P_{1/2}$ state. Atoms in the excited state mix between Zeeman levels due to collisions with buffer gas atoms and then decay to the ground state via radiation quenching, with equal probability of ending up in either Zeeman level. Atoms in the $m_J = +1/2$ sublevel remain there unless they undergo spin relaxation, while atoms in the $m_J = -1/2$ sublevel may absorb another photon and undergo the same process again. Over time most or all of the atoms are transferred to the $m_J = +1/2$ sublevel, polarizing the alkali vapor.

provide useful measurement signals. The thermal polarization of an ensemble of alkali atoms, given by

$$P_{\text{ther}} = \tanh\left(\frac{\frac{1}{2}g_s\mu_B B}{k_B T}\right) \quad (2.27)$$

where $g_s \approx 2$ is the electron g-factor and $\mu_B = 9.274 \times 10^{-24}$ J/T is the Bohr magneton, is typically too small to allow for magnetometry measurements. For example, at room temperature the thermal polarization in the Earth's magnetic field ($B \sim 0.5$ G) is only 1×10^{-7} , while in a very large field of 10 T it is only 0.02. Large nonthermal spin polarization, with $P \sim 1$, can be obtained by optical pumping, which transfers angular momentum from resonant light to the atoms. The optimal degree of polarization depends on the specific type of magnetometer but is generally on the order of unity. An introductory primer on optical pumping is given by [Happer and van Wijngaarden \(1987\)](#), and a more comprehensive review is given by [Happer \(1972\)](#).

For simplicity, we ignore the nuclear spin and consider only optical pumping of the electron spin; the processes involved are shown in Figure 2.6. The optical pumping technique used in this thesis is depopulation pumping with circularly polarized light; other kinds of magnetometers (as well as other applications such as atomic clocks) may use different techniques. A pump beam, resonant with the D1 transition, is circularly polarized so that all photons in the beam have the same spin projection along the direction of the beam's propagation. We define this direction as the $\hat{z}$ axis. For σ^+ polarized light, all photons have angular momentum of $+1$ along this axis, in units of the electron-spin angular momentum $\hbar$. An atom in the $m_J=-1/2$ sublevel of the ground state may absorb a photon, in which case conservation of angular momentum requires it to absorb the photon's angular momentum and thus be excited to the $m_J=+1/2$ sublevel of the $^2P_{1/2}$ state. However, an atom in the $m_J=+1/2$ sublevel of the ground state is forbidden from absorbing a photon because there is no level in the excited state with an additional $+1$ angular momentum. Thus, atoms with $m_J=+1/2$ in the ground state remain in that level unless they experience some relaxation mechanism (examples of which are discussed in Section 2.7).

Magnetometer cells typically contain other species of gas besides the alkali vapor, and the presence of this gas affects the efficiency of optical pumping. A chemically inert buffer gas (usually a noble gas such as He or Xe) is often used to prevent wall collisions (see Section 2.7.3). Collisions with buffer gas atoms depolarize the alkali atoms; the scattering cross-section in the excited state is significantly larger than in the ground state, due to coupling of the orbital angular momentum L of the p -shell electron with the rotation of the combined molecule that temporarily forms during the collision. Thus, there is very rapid collisional mixing between the Zeeman levels of the excited state that equalizes the populations of the levels.

Atoms that spontaneously decay back to the ground state do so by emitting a randomly polarized, resonant photon that can depolarize another atom if reabsorbed. In very dense

alkali vapor the probability of absorption becomes large, and a phenomenon known as radiation trapping can occur in which reabsorption of spontaneously emitted photons limits the polarization of the alkali vapor (see Section 2.3.4). To prevent spontaneous decay a quenching gas, typically a diatomic molecule such as N_2 , is added to the cell. During collisions of excited alkali atoms with the quenching gas, atoms transfer their excess energy to the rotational and vibrational modes of the quenching gas molecules and decay back to the ground state without radiating a resonant photon. In the presence of both buffer and quenching gases, there is an equal probability of decaying to the two Zeeman levels of the ground state. Atoms that decay to the $m_J=+1/2$ sublevel must remain there because they can not absorb another photon from the pump beam, while atoms that decay to the $m_J=-1/2$ sublevel may absorb another photon and get excited to the $^2P_{1/2}$ state again. In the absence of relaxation mechanisms, eventually all atoms are placed into the $m_J=+1/2$ sublevel and the alkali vapor is fully polarized with angular momentum $+1/2$ along the $\hat{z}$ axis. Similarly, pumping with σ^- light results in polarization with angular momentum $-1/2$.

The optical pumping rate R_{OP} is defined as the average rate at which an unpolarized atom absorbs a photon from the pump beam, as given by Equations 2.1 and 2.26. The rate with which an atom in the $m_J=-1/2$ sublevel of the ground state absorbs a σ^+ photon is then $2R_{OP}$, since atoms in the $m_J=+1/2$ are unable to absorb photons. The amplitude A of the decay channel from the excited state $|J' = 1/2, m_J'\rangle$ to the ground state $|J = 1/2, m_J\rangle$ is given by the matrix element

$$A \propto \langle J, m_J | \hat{\epsilon} \cdot \mathbf{r} | J', m_J' \rangle, \quad (2.28)$$

where $\hat{\epsilon}$ is the polarization of the emitted light. The branching ratios BR of the decay channels are then given by the Clebsch-Gordan coefficients,

$$BR = \langle J, m_J, 1, \Delta m_J | J', m_J' \rangle^2. \quad (2.29)$$

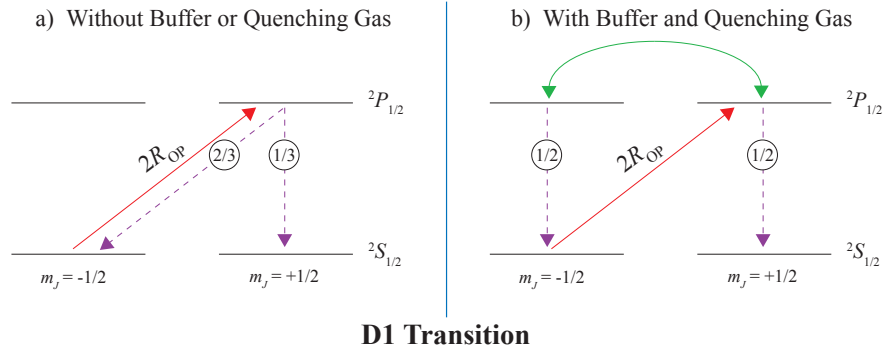


Figure 2.7: Branching ratios for decay of excited atoms in D1 pumping. **(a)** In the absence of buffer gas there is no collisional mixing, so excited atoms remain in the $m_J=+1/2$ sublevel. In the absence of quenching gas the atoms decay by radiating a photon, and the branching ratios are determined by the Clebsch-Gordan coefficients given in Equation 2.29. **(b)** In the presence of buffer and quenching gases, collisional mixing equalizes the populations of the excited state Zeeman levels, and there is an equal probability of decaying to each of the Zeeman levels of the ground state.

In the absence of buffer and quenching gases, all excited atoms remain in the $m_J=+1/2$ sublevel of the $^2P_{1/2}$ state and decay to the $m_J=-1/2$ and $m_J=+1/2$ sublevels of the ground state with branching ratios of 2/3 and 1/3, respectively, as shown in Figure 2.7(a). On average each absorbed photon adds +1/3 angular momentum to the atom. However, in the presence of sufficient buffer gas pressure there is rapid collisional mixing in the excited state, resulting in equal number densities of the Zeeman levels. The atoms then decay to the ground state with equal probability of decaying to the $m_J=-1/2$ and $m_J=+1/2$ sublevels, as shown in Figure 2.7(b), and on average each absorbed photon adds +1/2 angular momentum to the atom.

The optical pumping efficiency parameter a is the probability that an atom excited from the $m_J=-1/2$ sublevel decays to the $m_J=+1/2$ sublevel and runs from 1/3 in the case of no collisional mixing to 1/2 in the case of complete collisional mixing. Alternatively, a is the average angular momentum added by each absorbed photon. We define the number densities $\rho(-1/2)$ and $\rho(+1/2)$ of atoms with $m_J=-1/2$ and $m_J=+1/2$ in the ground state,

respectively, and the rates of change of these number densities are

$$\frac{d}{dt}\rho(-1/2) = -2R_{\text{OP}}\rho(-1/2) + 2(1-a)R_{\text{OP}}\rho(-1/2), \quad (2.30)$$

$$\frac{d}{dt}\rho(+1/2) = +2aR_{\text{OP}}\rho(-1/2). \quad (2.31)$$

We assume that the total density is constant, *i.e.*, $\rho(-1/2)+\rho(+1/2)=1$, since the atoms spend significantly more time in the ground state than in the excited state. Therefore $\rho(-1/2)$ and $\rho(+1/2)$ are the occupational probabilities of the $m_J=-1/2$ and $m_J=+1/2$ sub-levels of the ground state. The spin polarization of the atoms $\langle S_z \rangle$ is given by

$$\langle S_z \rangle = \frac{1}{2} [\rho(+1/2) - \rho(-1/2)], \quad (2.32)$$

and its rate of change is

$$\frac{d}{dt}\langle S_z \rangle = 2aR_{\text{OP}}\rho(-1/2) = aR_{\text{OP}}(1 - 2\langle S_z \rangle). \quad (2.33)$$

The average photon absorption rate per atom is

$$\langle \Gamma_{\text{OP}} \rangle = 2R_{\text{OP}}\rho(-1/2) = R_{\text{OP}}(1 - 2\langle S_z \rangle). \quad (2.34)$$

The solution to Equation 2.33 for the starting condition of no polarization, $\langle S_z \rangle=0$ at time $t=0$, is

$$\langle S_z \rangle = \frac{1}{2}(1 - e^{-2aR_{\text{OP}}t}). \quad (2.35)$$

The total number of photons absorbed by an unpolarized atom is therefore

$$N = \int_0^\infty \frac{d\langle \Gamma_{\text{OP}} \rangle}{dt} dt = \frac{1}{2a}. \quad (2.36)$$

On average an atom must absorb 3/2 photons to become fully polarized without collisional mixing, compared to only one photon with complete mixing. Collisional mixing makes the optical pumping process more efficient since the atoms have a greater probability of decaying to the $m_J=+1/2$ sublevel.

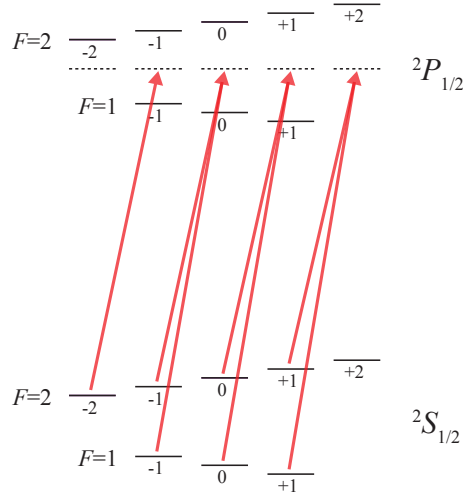


Figure 2.8: Optical pumping of the total atomic spin of an alkali atom with $I=3/2$ using σ^+ polarized D1 light. The atom is pumped into the end state $|F = 2, m_F = 2\rangle$, which is transparent to the pumping light.

If there are no relaxation mechanisms, then a polarized atom will remain in the $m_J=+1/2$ sublevel of the ground state indefinitely. However, if there is some nonzero relaxation rate R_{rel} , then Equation 2.33 must be modified:

$$\frac{d}{dt}\langle S_z \rangle = aR_{\text{OP}}(1 - 2\langle S_z \rangle) - R_{\text{rel}}\langle S_z \rangle, \quad (2.37)$$

and the solution for an initially unpolarized atom is

$$\langle S_z \rangle = \frac{aR_{\text{OP}}}{2aR_{\text{OP}} + R_{\text{rel}}}(1 - e^{-(2aR_{\text{OP}} + R_{\text{rel}})t}). \quad (2.38)$$

We define the electron polarization $P = 2\langle S_z \rangle$, which tends toward an equilibrium value,

$$P = \frac{2aR_{\text{OP}}}{2aR_{\text{OP}} + R_{\text{rel}}}. \quad (2.39)$$

In order to achieve large polarization it is necessary for the pumping rate to be much larger than the relaxation rate. In general we assume that there is sufficient collisional mixing that $a=1/2$, so that $P = R_{\text{OP}}/(R_{\text{OP}} + R_{\text{rel}})$.

The electron and nuclear spins of the atom are strongly coupled, so optical pumping of the electron spin results in polarization of the total atomic spin $\langle F_z \rangle$ (Franzen and Emslie, 1957). The details of this process depend on several factors, including the rate of spin-exchange collisions and the resolution of the hyperfine splitting in the ground and excited states. Pumping σ^+ D1 photons add angular momentum to the atom, which is eventually placed in the $m_F = +F$ end state of the $F = I + 1/2$ hyperfine level. As shown in Figure 2.8, atoms in this state may not absorb photons because of the unavailability of a level with an additional +1 angular momentum in the $^2P_{1/2}$ excited state. Thus the atom can be polarized with $\langle F_z \rangle = +F$ through the optical pumping process, although the actual atomic spin polarization achieved depends on the spin-relaxation and spin-exchange rates.

2.3.1 Optical Pumping on the D2 Transition

In order to demonstrate why the magnetometers discussed in this thesis use optical pumping on the D1 transition, we consider pumping on the D2 transition, as shown in Figure 2.9. In this case, atoms in either ground-state Zeeman level may absorb a σ^+ photon because the excited $^2P_{3/2}$ state includes an $m_J = +3/2$ sublevel. The relative absorption rates and decay branching ratios are determined by the Clebsch-Gordan coefficients given in Equation 2.29 and are shown in Figure 2.9(a). The optical pumping rate equations are

$$\frac{d}{dt}\rho(-1/2) = -\frac{1}{4}R_{\text{OP}}\rho(-1/2) + (1/3)\frac{1}{4}R_{\text{OP}}\rho(-1/2) \quad (2.40)$$

$$\frac{d}{dt}\rho(+1/2) = +(2/3)\frac{1}{4}R_{\text{OP}}\rho(-1/2), \quad (2.41)$$

giving the solution for spin polarization

$$\langle S_z \rangle = \frac{1}{2}(1 - e^{-R_{\text{OP}}t/6}). \quad (2.42)$$

Although atoms in the $m_J = +1/2$ sublevel of the ground state may absorb σ^+ photons, full polarization is nevertheless possible because atoms in the $m_J = +3/2$ sublevel of the excited state must decay back to the $m_J = +1/2$ ground-state sublevel.

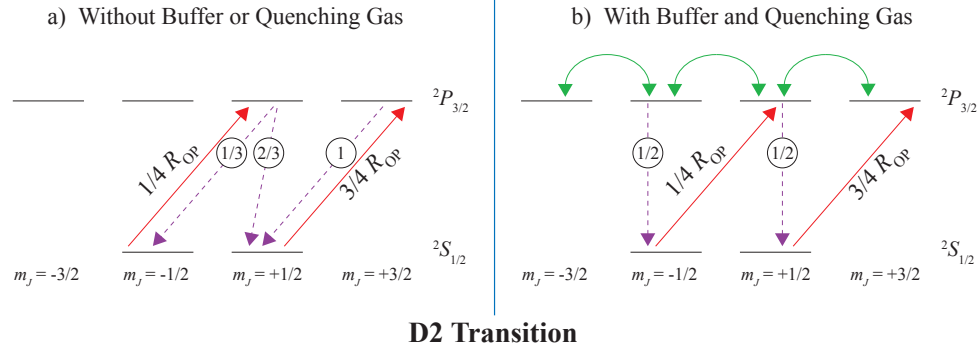


Figure 2.9: Branching ratios for decay of excited atoms in D2 pumping. **(a)** In the absence of buffer gas there is no collisional mixing, so excited atoms remain in the $m_j = +1/2$ and $m_j = +3/2$ sublevels. In the absence of quenching gas the atoms decay by radiating a photon, and the branching ratios are determined by the Clebsch-Gordan coefficients given in Equation 2.29. **(b)** In the presence of buffer and quenching gases, collisional mixing equalizes the populations of the excited state Zeeman levels, and there is an equal probability of decaying to each of the Zeeman levels of the ground state.

However, in the presence of buffer and quenching gases, as shown in Figure 2.9(b), collisional mixing results in excited atoms having equal probability of decaying to either of the ground-state Zeeman levels, giving the optical pumping rate equations

$$\begin{aligned} \frac{d}{dt}\rho(-1/2) &= -\frac{1}{4}R_{OP}\rho(-1/2) + (1/2)\frac{1}{4}R_{OP}\rho(-1/2) \\ &\quad + (1/2)\frac{3}{4}R_{OP}\rho(+1/2) \end{aligned} \quad (2.43)$$

$$\begin{aligned} \frac{d}{dt}\rho(+1/2) &= -\frac{3}{4}R_{OP}\rho(+1/2) + (1/2)\frac{3}{4}R_{OP}\rho(+1/2) \\ &\quad + (1/2)\frac{1}{4}R_{OP}\rho(-1/2). \end{aligned} \quad (2.44)$$

The evolution of the spin polarization is then given by

$$\frac{d}{dt}\langle S_z \rangle = -\frac{1}{8}R_{OP}(1 + 4\langle S_z \rangle), \quad (2.45)$$

and the solution for an initially unpolarized atom is

$$\langle S_z \rangle = \frac{1}{4}(e^{-R_{OP}t/2} - 1). \quad (2.46)$$

The spin polarization in this case is only half of its value in the case of D1 pumping, limiting the maximum polarization to $P=1/2$. The inclusion of buffer and quenching gases in

the magnetometry cell prevents full polarization of the alkali spins, so it is therefore preferable to optically pump the alkali atoms using light at the D1 transition. Note also that the spin is polarized with negative angular momentum, compared to the positive angular momentum achieved with D1 pumping, or with D2 pumping without collisional mixing.

2.3.2 Optical Pumping with Light of Arbitrary Polarization

We briefly consider the case of D1 optical pumping with light of arbitrary polarization $\hat{\epsilon}$, where the average photon spin s is given by

$$s = i\hat{\epsilon} \times \hat{\epsilon}^*. \quad (2.47)$$

It is convenient to characterize the light polarization by the photon spin component along the pumping direction, $s = s \cdot \hat{z}$; s ranges from -1 to +1, where $s=-1$ corresponds to σ^- light, $s=0$ corresponds to linearly polarized π light, and $s=+1$ corresponds to σ^+ light. The absorption rate for an unpolarized atom is $R=\sigma(\nu)\Phi$. The optical pumping rate equations are

$$\begin{aligned} \frac{d}{dt}\rho(-1/2) = & -(1+s)R\rho(-1/2) + (1-a)(1+s)R\rho(-1/2) \\ & + a(1-s)R\rho(+1/2) \end{aligned} \quad (2.48)$$

$$\begin{aligned} \frac{d}{dt}\rho(+1/2) = & -(1-s)R\rho(+1/2) + (1-a)(1-s)R\rho(+1/2) \\ & + a(1+s)R\rho(-1/2), \end{aligned} \quad (2.49)$$

where a is the optical pumping efficiency. The equation for the evolution of the spin is

$$\frac{d}{dt}\langle S_z \rangle = aR(s - 2\langle S_z \rangle) - R_{\text{rel}}\langle S_z \rangle, \quad (2.50)$$

and the solution for an initially unpolarized atom is

$$\langle S_z \rangle = s \frac{aR}{2aR + R_{\text{rel}}} (1 - e^{-(2aR + R_{\text{rel}})t}). \quad (2.51)$$

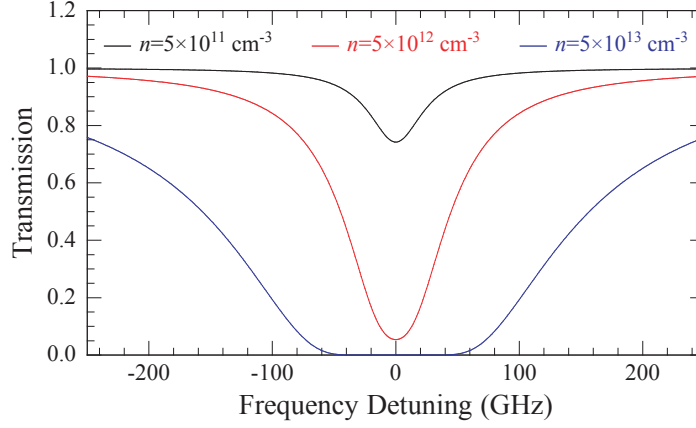


Figure 2.10: The transmission of linearly polarized D1 light through a cell of length $l=5$ cm with $\Gamma_L=50$ GHz, for alkali vapor density $n=5 \times 10^{11} \text{ cm}^{-3}$ (OD=0.28 on resonance), $5 \times 10^{12} \text{ cm}^{-3}$ (OD=2.8), and $5 \times 10^{13} \text{ cm}^{-3}$ (OD=28), as calculated from Equation 2.55.

The average photon absorption rate per atom is

$$\begin{aligned} \langle \Gamma \rangle &= R[(1+s)\rho(-1/2) + (1-s)\rho(+1/2)] \\ &= R(1 - 2s\langle S_z \rangle). \end{aligned} \quad (2.52)$$

A similar analysis reveals that the average absorption rate per atom for D2 light is

$$\begin{aligned} \langle \Gamma \rangle_{D2} &= R[(1 - \frac{1}{2}s)\rho(-1/2) + (1 + \frac{1}{2}s)\rho(+1/2)] \\ &= R(1 + s\langle S_z \rangle). \end{aligned} \quad (2.53)$$

2.3.3 Light Propagation

As on- or near-resonant light propagates through the vapor cell, it becomes partially or completely absorbed by the alkali vapor. The attenuation of the light can result in nonuniform polarization throughout the cell and thus reduce the sensitivity of the magnetometer. The reduction in laser intensity I near the D1 transition is given by Equation 2.52,

$$\frac{d}{dz}I = -n\sigma(\nu)I(1 - 2s\langle S_z \rangle), \quad (2.54)$$

where n is the density of the alkali vapor (see Section A.1). For linearly polarized light ($s=0$), such as that used for optical probing (see Section 2.4), the solution is exponential attenuation,

$$I(z) = I(0) \exp(-n\sigma(\nu)z), \quad (2.55)$$

where z is the position in the cell and $I(0)$ is the intensity entering the front of the cell. The optical depth OD describes the total attenuation by a cell of length l ,

$$\text{OD} = n\sigma(\nu)l, \quad (2.56)$$

such that the intensity of light transmitted through the cell is $I(0)e^{-\text{OD}}$. At a given light frequency, the alkali vapor is referred to as “optically thin” if $\text{OD} \lesssim 1$, so that most or all of the light is transmitted. The vapor is referred to as “optically thick” if $\text{OD} \gg 1$ and the light is completely absorbed. If the optical lineshape is known, then measuring the transmission of incident linearly polarized light is a useful method for determining the alkali vapor density. Alternatively, if the density is known, then the transmission as a function of laser frequency provides a measurement of the atomic frequency response and thus the buffer gas pressure in the cell. Figure 2.10 shows the light transmission for several values of the alkali vapor density as determined by Equation 2.55. At small optical depth the light is almost completely transmitted even on resonance, while at large optical depth the light is completely absorbed even if the frequency is detuned far from resonance.

In general, the absorption of incident light depends on both the alkali and photon polarization. For example, if the atoms are fully polarized with $\langle S_z \rangle = +1/2$, then there is no absorption of σ^+ light, and the vapor becomes transparent to such light. While there is no general solution to Equation 2.54, for the case of circularly polarized light ($s=\pm 1$) the solution is the transcendental equation

$$I(z) \exp\left(\frac{\sigma(\nu)I(z)}{R_{\text{rel}}}\right) = I(0) \exp\left(\frac{\sigma(\nu)I(0)}{R_{\text{rel}}} - n\sigma(\nu)z\right), \quad (2.57)$$

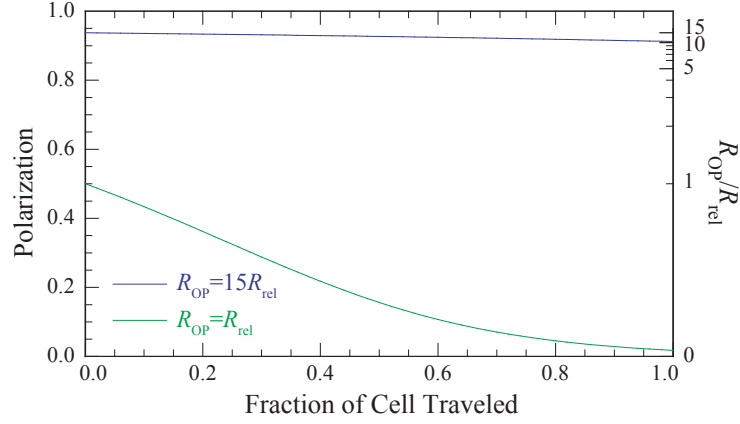


Figure 2.11: Propagation of circularly polarized light through an optically thick cell ($OD=5$) for low ($R_{OP}=R_{rel}$) and high ($R_{OP}=15R_{rel}$) pumping rates at the front of the cell. Polarization and pumping rate throughout the cell are calculated from Equation 2.58. Low pumping rate results in a large polarization gradient, while high pumping rate results in nearly uniform polarization.

which can be solved using the Lambert W -function:¹

$$I(z) = \frac{R_{rel}}{\sigma(\nu)} W \left[\frac{\sigma(\nu)I(0)}{R_{rel}} \exp \left(\frac{\sigma(\nu)I(0)}{R_{rel}} - n\sigma(\nu)z \right) \right]. \quad (2.58)$$

The polarization and light attenuation through the cell are shown in Figure 2.11 for a cell with nominal $OD=5$ and low ($R_{OP}=R_{rel}$) and high ($R_{OP}=15R_{rel}$) pumping rates at the front of the cell. At low pumping rate the beam is almost completely absorbed because of the low polarization induced in the alkali vapor, resulting in a large polarization gradient throughout the cell. However, at high pumping rate the vapor becomes nearly fully polarized throughout the cell, and the beam is barely attenuated. Some types of magnetometers operate with less than full polarization, so polarization gradients can cause problems in optically thick cells.

¹ The Lambert W -function is the inverse function of $f(W) = We^W$ and is also referred to as the product log.

2.3.4 Radiation Trapping and Quenching

As discussed previously, the emission of resonant light by spontaneously decaying atoms can limit the atomic polarization in an optically thick cell where the emitted light is likely to get reabsorbed by other atoms before leaving the cell. Radiation trapping is a complicated process, and [Molisch and Oehry \(1998\)](#) give an extensive review of various methods for the treating this problem. Each photon is emitted in a random direction and is unpolarized, and its probability of escaping the cell without being reabsorbed depends on the optical lineshape, the size and shape of the cell, and the vapor density. The escape probability becomes very small at large optical depth, and so an emitted photon is highly likely to be reabsorbed by another atom within the cell, depolarizing that atom. In turn, if the second atom decays to the ground state by emitting a second photon, then that photon as well is likely to be reabsorbed and depolarize another atom. The emitted radiation is trapped within the vapor cell for several absorption and emission cycles before an emitted photon finally escapes the cell. In this way a single pumping photon can actually cause the depolarization of several atoms within an optically thick vapor, thus limiting the attainable polarization within the vapor.

The addition of a molecular gas to the vapor cell, typically nitrogen, can suppress or eliminate the problem of radiation trapping ([Franz, 1968](#)). The molecules have a large number of rotational and vibrational energy states that are coincident with the excess energy of an excited alkali atom, and during a collision the alkali atom can give up this energy to the molecule and decay back to the ground state without radiating a resonant photon, a process known as “quenching.” The rate R_Q at which an excited alkali atom undergoes a quenching collision is given by

$$R_Q = n_Q \sigma_Q \bar{v}, \quad (2.59)$$

where n_Q is the density of quenching gas molecules, σ_Q is the quenching cross-section, and $\bar{v}$ is the relative velocity between an alkali atom and a quenching molecule (see Section 2.7).

Alkali Metals:	Potassium	Rubidium	Cesium
$\sigma_Q^{N_2}, {}^2P_{1/2}$	3.5×10^{-15}	5.8×10^{-15}	5.5×10^{-15}
${}^2P_{3/2}$	3.9×10^{-15}	4.3×10^{-15}	6.4×10^{-15}
$p_Q', D1$ at 100°C	5.9	3.9	3.5
200°C	6.7	4.4	3.9
$p_Q', D2$ at 100°C	5.4	5.6	3.4
200°C	6.1	6.3	3.8

Table 2.3: Quenching cross-sections between alkali atoms and nitrogen molecules in units of cm^2 and corresponding characteristic pressures in units of Torr, including the slight temperature variation of $p_Q' \propto \bar{v}$. Potassium cross-sections are from [McGillis and Krause \(1968\)](#), rubidium cross-sections are from [Hrycshyn and Krause \(1970\)](#), and cesium cross-sections are from [McGillis and Krause \(1967\)](#).

The quenching factor Q is the probability that an excited atom decays via spontaneous emission rather than quenching and is given by the ratio of the quenching rate and the spontaneous emission rate, which is the inverse of the natural lifetime,

$$Q = \frac{1}{1 + R_Q \tau_{\text{nat}}} = \frac{1}{1 + p_Q/p_Q'} \quad (2.60)$$

where p_Q is the pressure of quenching gas, and p_Q' is the characteristic pressure necessary to achieve $Q=1/2$. The quenching cross-sections with nitrogen gas for the D1 and D2 transitions and the corresponding characteristic pressures are given in [Table 2.3](#).

[Rosenberry *et al.* \(2007\)](#) present a simple model for the attainable polarization in an alkali vapor cell in the regime of large optical density, which we modify slightly to account for the fact that polarized atoms do not absorb pump photons. The spin-relaxation rate R_{RT} due to radiation trapping is

$$R_{\text{RT}} = K(M - 1)QR_{\text{OP}}(1 - P), \quad (2.61)$$

where we add the factor of $(1-P)$ to the original model. Here M is the average number of times that a photon is emitted before it leaves the vapor cell, so that $(M - 1)$ is the average number of times that a photon is reabsorbed; M grows with increasing vapor

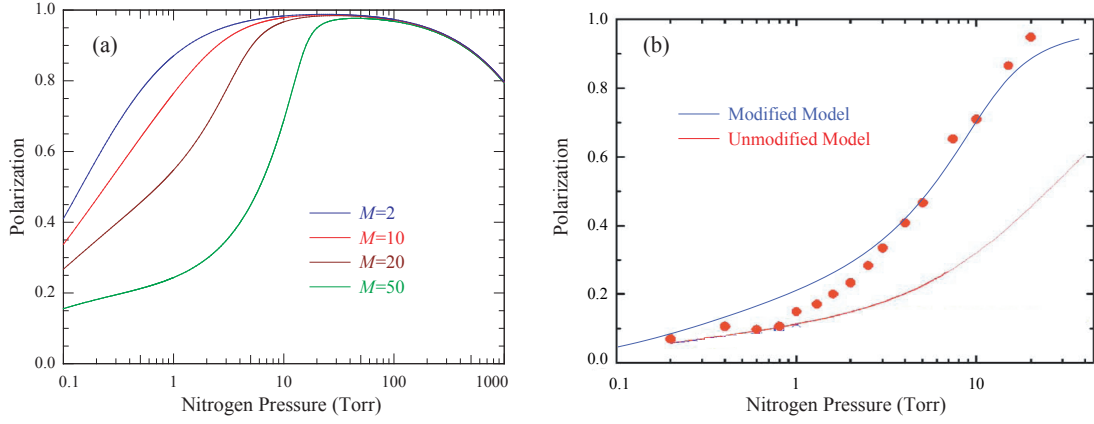


Figure 2.12: (a) Polarization attainable in an optically thick rubidium vapor limited by radiation trapping, calculated according to Equation 2.62 using $K=0.1$ and $R_{OP}=2000 \text{ s}^{-1}$, and including the effects of spin-destruction collisions with nitrogen molecules and the cell wall. (b) Comparison of experimental data and the polarization predicted by the original and modified versions of Equation 2.62 at a density of 10^{13} cm^{-3} . Both models use $M=63$, the unmodified model uses $K=0.12$ and $R_{OP}=150 \text{ s}^{-1}$, and the modified model uses $K=0.06$ and $R_{OP}=80 \text{ s}^{-1}$. Adapted from [Rosenberry et al. \(2007\)](#).

density n . The coefficient K describes the degree of depolarization caused by a reabsorbed photon. In general $K < 1$ because reabsorbed photons are not perfectly depolarizing, and the polarized nuclear spin slows down the depolarization of the total atomic spin (see Section 2.7). The maximum attainable polarization is approximately given by

$$P = \frac{1}{1 + R_{\text{rel}}/R_{OP} + K(M-1)Q(1-P)} , \quad (2.62)$$

where R_{rel} is the rate of spin relaxation due to effects other than radiation trapping.

Figure 2.12(a) displays the polarization predicted by this model as a function of nitrogen pressure for rubidium atoms at 100°C in a spherical cell of radius 2.5 cm; we set $K=0.1$ and $R_{OP}=2000 \text{ s}^{-1}$, and we include the effects of spin-destruction collisions with nitrogen molecules (see Section 2.7.2) as well as diffusion to the cell wall (see Section 2.7.3). We see that additional quenching gas pressure is necessary to maintain high polarization as the vapor density increases, but an excessive amount leads to spin relaxation and limits

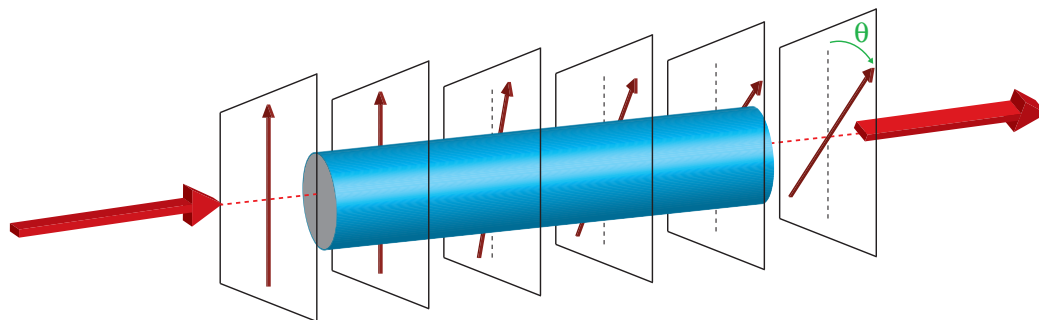


Figure 2.13: The principle of optical rotation: propagation through a vapor of polarized atoms causes rotation of the plane of polarization of linearly polarized light by an angle proportional to $\langle S_x \rangle$, the projection of atomic spin along the propagation direction.

the polarization that can be achieved. Using $K = 0.12$, [Rosenberry *et al.*](#) show qualitative agreement between their model and experimental measurements of rubidium polarization at high density in the presence of nitrogen and hydrogen quenching gases, although their model performs poorly at high optical density. Figure 2.12(b) compares the polarization predicted by their original model and our modified version at a density of 10^{13} cm^{-3} , showing that the additional factor of $(1-P)$ in Equations 2.61-2.62 is necessary to fit the experimental data. This also agrees with our own observations; for example, we are able to achieve nearly 100% potassium polarization at a density of about $6 \times 10^{13} \text{ cm}^{-3}$ using only 70 Torr of nitrogen (see Sections 4.1.1 and 4.3). A typical high-density magnetometer cell with several amagat² of buffer gas contains about 50-100 Torr of nitrogen gas for quenching.

2.4 Measuring Spin Polarization: Optical Rotation

Detection of a magnetic field with an atomic magnetometer requires monitoring the spin precession due to the field, and there are numerous techniques for measuring the atomic spin. The magnetometry techniques described in this thesis all use the optical rotation

² One amagat is defined as the number density of an ideal gas at standard temperature and pressure. This unit is convenient because number density does not vary with temperature and so it can be used to unambiguously describe the amount of gas in a cell. It is abbreviated amg, and $1 \text{ amg} = 2.69 \times 10^{19} \text{ cm}^{-3}$.

of an off-resonant, linearly polarized probe beam. The probe beam propagates along the $\hat{x}$ direction, orthogonal to the pump beam. Its plane of polarization rotates by an angle $\theta \propto \langle S_x \rangle$ due to a difference in the indices of refraction³ $n_+(v)$ and $n_-(v)$ experienced by σ^+ and σ^- light, respectively. The polarization of the light is compared before and after traveling through the cell, yielding a measurement of the projection of the atomic spin along the propagation direction. This effect is illustrated in Figure 2.13. Although this method is sensitive specifically to the electron spin $\langle S_x \rangle$, the electron and total atomic spins are aligned either parallel or anti-parallel, so that measuring the direction of one serves to determine the direction of the other.

The analysis in this section follows the semi-classical derivation by Erickson (2000); the quantum-mechanical theory was first presented by Opechowski (1953) and is summarized by Wu *et al.* (1986). The electric field vector at position $x=0$ of light linearly polarized in the $\hat{y}$ direction can be decomposed into two circularly polarized components of opposite helicity:

$$\begin{aligned} E(0) &= \frac{E_0}{2} e^{i\omega t} \hat{y} + \text{c.c.} \\ &= \frac{E_0}{4} e^{i\omega t} (\hat{y} + i\hat{z}) + \frac{E_0}{4} e^{i\omega t} (\hat{y} - i\hat{z}) + \text{c.c.}, \end{aligned} \quad (2.63)$$

where c.c. denotes the complex conjugate. After traveling a distance $l = tc/n(v)$, the electric field becomes

$$E(l) = \frac{E_0}{4} e^{i\omega n_+(v)l/c} (\hat{y} + i\hat{z}) + \frac{E_0}{4} e^{i\omega n_-(v)l/c} (\hat{y} - i\hat{z}) + \text{c.c.} \quad (2.64)$$

We define the quantities

$$\bar{n}(v) = [n_+(v) + n_-(v)] / 2 \quad (2.65)$$

$$\Delta n(v) = [n_+(v) - n_-(v)] / 2 \quad (2.66)$$

³ The general index of refraction $n(v)$, and the indices of refraction of circularly polarized light $n_+(v)$ and $n_-(v)$, should not be confused with the alkali vapor density n .

and rewrite Equation 2.64,

$$E(l) = \frac{E_0}{4} e^{i\omega\bar{n}(v)l/c} e^{i\omega\Delta n(v)l/c} (\hat{\mathbf{y}} + i\hat{\mathbf{z}}) + \frac{E_0}{4} e^{i\omega\bar{n}(v)l/c} e^{-i\omega\Delta n(v)l/c} (\hat{\mathbf{y}} - i\hat{\mathbf{z}}) + \text{c.c.} \quad (2.67)$$

If we ignore the common phase factor $\exp(i\omega\bar{n}(v)l/c)$ and define the rotation angle

$$\theta = \frac{\pi\nu l}{c} [n_+(v) - n_-(v)], \quad (2.68)$$

then we may rewrite Equation 2.67,

$$E(l) = E_0(\cos\theta \hat{\mathbf{y}} - \sin\theta \hat{\mathbf{z}}), \quad (2.69)$$

which indicates that the plane of polarization of the light rotates by an angle θ after traveling a distance l through a birefringent medium with $n_+(v) \neq n_-(v)$.

We now show that a polarized atomic vapor constitutes such a medium. The propagation of light through the vapor can be written in terms of the complex wave number $\tilde{k}(v)$ of the light and the complex dielectric constant $\tilde{\epsilon}(v)$ of the vapor,⁴

$$\tilde{k}(v) = k + i\kappa, \quad (2.70)$$

$$\tilde{\epsilon}(v) = \left(\frac{\tilde{k}c}{\omega}\right)^2, \quad (2.71)$$

giving the standard plane-wave solution

$$\begin{aligned} E(x, t) &= E_0 e^{i(\tilde{k}x - \omega t)} \\ &= E_0 e^{-\kappa x} e^{i(kx - \omega t)}. \end{aligned} \quad (2.72)$$

The intensity of the light is proportional to $|E|^2$, so comparing to Equation 2.55 reveals that the imaginary component of the complex wave number is in fact the absorption coefficient,

$$\kappa = \frac{1}{2} n\sigma(v). \quad (2.73)$$

⁴ Here we assume that the permeability of the vapor $\mu \approx 1$, so its deviation from unity can be ignored.

The real component of the complex wave number is related to the index of refraction in the standard manner,

$$k = \frac{n(\nu)\omega}{c}. \quad (2.74)$$

We operate in the regime that $(kc/\omega) - 1 \ll 1$ and $(\kappa c/\omega) \ll 1$, so we may approximate the dielectric constant,

$$\tilde{\epsilon}(\nu) \approx 1 + 2 \left(\frac{kc}{\omega} - 1 \right) + \frac{2i\kappa c}{\omega}. \quad (2.75)$$

The real and imaginary parts of the dielectric constant are related by the Kramers-Kronig relations (see for instance [Jackson \(1999\)](#)) given by

$$\text{Re}[\epsilon(\nu)] = 1 + \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Im}[\epsilon(\nu')]}{\nu' - \nu} d\nu' \quad (2.76)$$

$$\text{Im}[\epsilon(\nu)] = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Re}[\epsilon(\nu')] - 1}{\nu' - \nu} d\nu', \quad (2.77)$$

where the $\mathcal{P}$ implies the Cauchy principal value of the integral, from which we find the relationship between the index of refraction and absorption cross-section:

$$\begin{aligned} n(\nu) &= 1 + \left(\frac{nc}{4\pi\nu} \right) \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\sigma(\nu')}{\nu' - \nu} d\nu' \\ &= 1 + \left(\frac{nr_e c^2 f}{4\nu} \right) \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Re}[\mathcal{V}(\nu' - \nu_0)]}{\nu' - \nu} d\nu'. \end{aligned} \quad (2.78)$$

Therefore it is possible to obtain the index of refraction of light for a medium from the absorption profile of the light.

We have defined the complex Voigt profile such that its real and imaginary components obey the relations

$$\text{Re}[\mathcal{V}(\nu - \nu_0)] = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Im}[\mathcal{V}(\nu' - \nu_0)]}{\nu' - \nu} d\nu' \quad (2.79)$$

$$\text{Im}[\mathcal{V}(\nu - \nu_0)] = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Re}[\mathcal{V}(\nu' - \nu_0)]}{\nu' - \nu} d\nu', \quad (2.80)$$

so that the index of refraction is given by

$$n(\nu) = 1 + \left(\frac{nr_e c^2 f}{4\nu} \right) \text{Im}[\mathcal{V}(\nu - \nu_0)]. \quad (2.81)$$

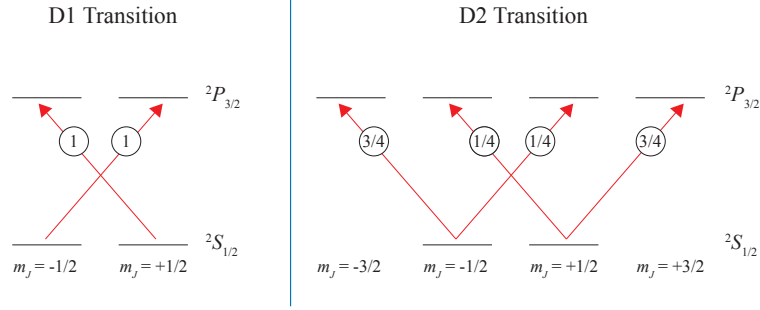


Figure 2.14: Branching ratios for the allowed optical transitions for the D1 and D2 resonances.

Note that it is often convenient to redefine the Lorentzian lineshape in complex form so that its real and imaginary components satisfy the same relations given in Equations 2.79 and 2.80 for the Voigt profile:

$$\mathcal{L}(\nu - \nu_0) = \frac{\Gamma_L/2\pi + i(\nu - \nu_0)/\pi}{(\nu - \nu_0)^2 + (\Gamma_L/2)^2}. \quad (2.82)$$

Replacing the Voigt profile with the complex Lorentzian lineshape in any equation provides a valid approximation in the limit that $\Gamma_L \gg \Gamma_G$.

The absorption coefficients for σ^- and σ^+ light, and thus the indices of refraction $n_-(\nu)$ and $n_+(\nu)$, depend on the ground state populations $\rho(-1/2)$ and $\rho(+1/2)$,⁵ as well as the branching ratios of the optical excitation transitions given by Equation 2.29 and shown in Figure 2.14. For the D1 transition the indices of refraction are

$$n_-(\nu) = 1 + 2\rho(+1/2) \left(\frac{nr_e c^2 f_{D1}}{4\nu} \right) \text{Im}[\mathcal{V}(\nu - \nu_{D1})] \quad (2.83)$$

$$n_+(\nu) = 1 + 2\rho(-1/2) \left(\frac{nr_e c^2 f_{D1}}{4\nu} \right) \text{Im}[\mathcal{V}(\nu - \nu_{D1})], \quad (2.84)$$

while for the D2 transition they are

$$n_-(\nu) = 1 + 2 \left(\frac{3}{4}\rho(-1/2) + \frac{1}{4}\rho(+1/2) \right) \left(\frac{nr_e c^2 f_{D2}}{4\nu} \right) \text{Im}[\mathcal{V}(\nu - \nu_{D2})] \quad (2.85)$$

$$n_+(\nu) = 1 + 2 \left(\frac{1}{4}\rho(-1/2) + \frac{3}{4}\rho(+1/2) \right) \left(\frac{nr_e c^2 f_{D2}}{4\nu} \right) \text{Im}[\mathcal{V}(\nu - \nu_{D2})]. \quad (2.86)$$

⁵ Recall that we are probing the atomic polarization along the $\hat{x}$ direction, so $\rho(-1/2)$ and $\rho(+1/2)$ are measured with respect to the $\hat{x}$ axis.

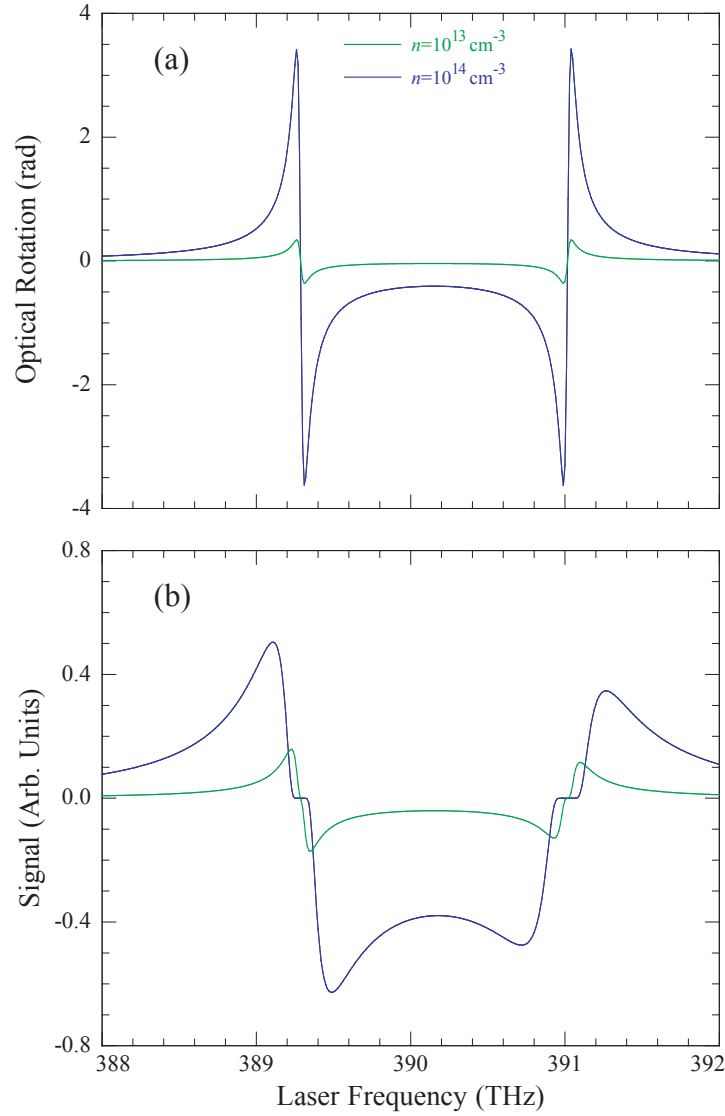


Figure 2.15: (a) Optical rotation angle θ calculated for potassium vapor due to the D1 and D2 transitions (389.3 and 391.0 THz, respectively) in a cell with length 2.5 cm, polarization $P_x=0.5$, $\Gamma_L=50$ GHz, and density $n=10^{13} \text{ cm}^{-3}$ and $n=10^{14} \text{ cm}^{-3}$. (b) Transmitted optical rotation signal θe^{-OD} , taking into account absorption of the probe beam. The optimal frequency detuning for maximum signal depends on experimental parameters such as optical depth. Note that for potassium the D1 and D2 transitions are closely spaced, and their rotation spectra overlap.

Thus, the atomic vapor is birefringent when $\rho(-1/2) \neq \rho(+1/2)$, such that there is some nonzero polarization $P_x = 2\langle S_x \rangle = \rho(+1/2) - \rho(-1/2)$. The total optical rotation is therefore given by Equation 2.68,

$$\theta = \frac{\pi}{2} \ln r_e c P_x \left(-f_{D1} \text{Im}[\mathcal{V}(\nu - \nu_{D1})] + \frac{1}{2} f_{D2} \text{Im}[\mathcal{V}(\nu - \nu_{D2})] \right), \quad (2.87)$$

where ν_{D1} and ν_{D2} are the resonance frequencies of the D1 and D2 transitions, respectively. Recall that $f_{D1} \approx \frac{1}{2} f_{D2}$. The optical rotation spectrum has a dispersive shape, as shown in Figure 2.15(a). For rubidium and cesium the D1 and D2 transitions are well-separated, so we need only consider the resonance on which we are probing; however, for potassium the transitions are separated by only 3.4 nm, and the resonance lineshapes may overlap so that it is necessary to consider their sum as in Equation 2.87. The overall signal is determined by both optical rotation and beam absorption, so the optimal probe beam detuning for maximum signal depends on the experimental parameters such as optical depth, as shown in Figure 2.15(b).

2.4.1 The Effect of Hyperfine Splitting

When the ground- and excited-state hyperfine splittings are optically resolved, we must calculate the optical rotation signal resulting from each of the transitions $F \rightarrow F'$, as we did in Section 2.2.4 for the optical absorption cross-section. In this case we must also consider individually the contribution from each Zeeman sublevel $|F, m_F\rangle$, with appropriate weighting determined by the atomic spin polarization; for simplicity we only consider the D1 transition. We assume that the ensemble is in the spin-temperature distribution, which is valid under conditions of rapid spin exchange and is discussed in Section 2.7.1. If we measure m_F along the direction of spin polarization, then the number density $\rho(F, m_F)$ of

a particular state does not depend on F and is given by

$$\rho(F, m_F) = C e^{\beta m_F} \quad (2.88)$$

$$\beta = \ln\left(\frac{1+P}{1-P}\right), \quad (2.89)$$

where C is a normalization factor,⁶ and β is the spin-temperature parameter.

The optical rotation angle is proportional to the difference between the indices of refraction for the σ^+ and σ^- components of the probe beam. We may decompose the light polarization $\hat{\epsilon}$ into spherical components,

$$\epsilon_{+1} = -\frac{1}{\sqrt{2}}(\epsilon_x + i\epsilon_y) \quad (2.90)$$

$$\epsilon_0 = \epsilon_z \quad (2.91)$$

$$\epsilon_{-1} = \frac{1}{\sqrt{2}}(\epsilon_x - i\epsilon_y). \quad (2.92)$$

If the light polarization is measured along the direction of propagation, then the σ^+ and σ^- components of the polarization have amplitudes ϵ_{+1} and ϵ_{-1} , respectively. If the direction of light propagation is at an angle φ with respect to the atomic spin polarization vector, then we must consider the light polarization measured along the direction of atomic polarization:

$$\epsilon'_{+1} = \frac{1}{2}\epsilon_{+1}(1 + \cos\varphi) + \frac{1}{2}\epsilon_{-1}(-1 + \cos\varphi) \quad (2.93)$$

$$\epsilon'_0 = \frac{1}{\sqrt{2}}(\epsilon_{+1} + \epsilon_{-1})\sin\varphi \quad (2.94)$$

$$\epsilon'_{-1} = \frac{1}{2}\epsilon_{+1}(-1 + \cos\varphi) + \frac{1}{2}\epsilon_{-1}(1 + \cos\varphi). \quad (2.95)$$

From Equations 2.20 and 2.25, we therefore see that the relative weight $\Theta_{F,F'}$ of the transition $F \rightarrow F'$ is

$$\Theta_{F,F'} \propto A_{F,F'} \sum_{m_F} \rho(F, m_F) \left[|\epsilon'_{+1}|^2 \begin{pmatrix} F' & 1 & F \\ m_F + 1 & -1 & -m_F \end{pmatrix}^2 - |\epsilon'_{-1}|^2 \begin{pmatrix} F' & 1 & F \\ m_F - 1 & +1 & -m_F \end{pmatrix}^2 \right], \quad (2.96)$$

⁶ We assume that the excited-state population is insignificant since the atoms spend nearly all of their time in the ground state, and we normalize the total ground-state population including both hyperfine levels $F = I \pm 1/2$.

Transition	$I = 3/2$	$I = 5/2$	$I = 7/2$
a	$\frac{1-p^2}{16(1+p^2)}$	$\frac{5-2p^2-3p^4}{18(3+10p^2+3p^4)}$	$\frac{7+7p^2-11p^4-3p^6}{64(1+7p^2+7p^4+p^6)}$
b	$\frac{-5(1-p^2)}{16(1+p^2)}$	$\frac{-7(5-2p^2-3p^4)}{18(3+10p^2+3p^4)}$	$\frac{-9(7+7p^2-11p^4-3p^6)}{64(1+7p^2+7p^4+p^6)}$
c	$\frac{3(5+3p^2)}{16(1+p^2)}$	$\frac{10(7+14p^2+3p^4)}{18(3+10p^2+3p^4)}$	$\frac{7(15+63p^2+45p^4+5p^6)}{64(1+7p^2+7p^4+p^6)}$
d	$\frac{5+3p^2}{16(1+p^2)}$	$\frac{2(7+14p^2+3p^4)}{18(3+10p^2+3p^4)}$	$\frac{15+63p^2+45p^4+5p^6}{64(1+7p^2+7p^4+p^6)}$

Table 2.4: Relative strengths $A_{F,F'}^{\text{Rot}}$ of the individual D1 hyperfine resonances for optical rotation as a function of polarization P , labeled according to Figure 2.4 and calculated according to Equations 2.96-2.98.

where $A_{F,F'}$ is the absorption amplitude listed in Table 2.2. We sum over all possible transitions $|F, m_F\rangle \rightarrow |F', m'_F\rangle$, and we find that the total rotation angle is proportional to

$$\sum_{F,F'} \Theta_{F,F'} = \frac{P \cos \varphi}{3(2I+1)}, \quad (2.97)$$

so that we may write $\Theta_{F,F'}$ in normalized form,

$$\Theta_{F,F'} = A_{F,F'}^{\text{Rot}} P \cos \varphi, \quad (2.98)$$

where $\sum_{F,F'} A_{F,F'}^{\text{Rot}} = 1$. We see that the optical rotation angle is proportional to the polarization component $P_x = P \cos \varphi$ along the direction of probe beam propagation, regardless of the precise optical lineshape.

Combining Equations 2.87 and 2.98, we see that the optical rotation angle measured at probe beam frequency ν is

$$\theta = -\frac{\pi}{2} \ln r_e c f_{D1} P_x \sum_{F,F'} A_{F,F'}^{\text{Rot}} \text{Im}[\mathcal{V}(\nu - \nu_{F,F'})], \quad (2.99)$$

where $\nu_{F,F'}$ is the resonance frequency of the transition $F \rightarrow F'$. In the limit that the individual transitions are unresolved, this simply gives Equation 2.87. The relative transition

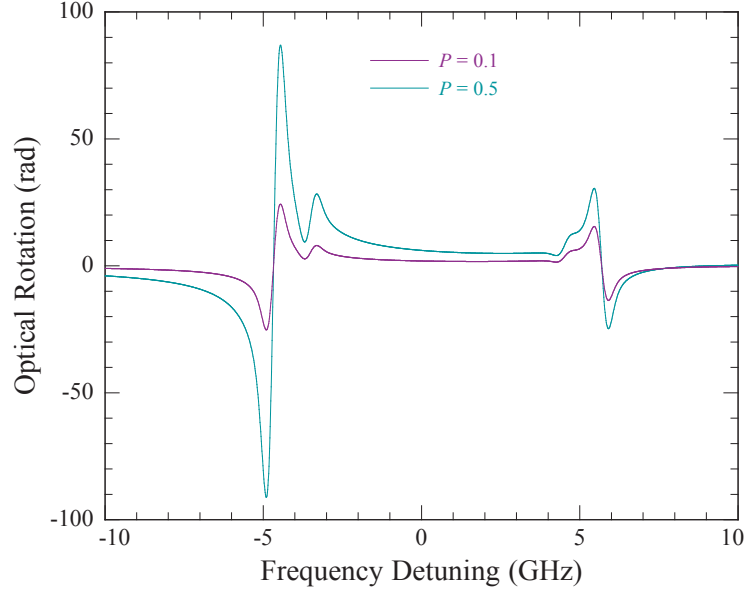


Figure 2.16: Optical rotation angles in cesium vapor calculated from Equation 2.99 for the case of $\varphi=0$ and resolved hyperfine structure with $\Gamma_G=400$ MHz and $\Gamma_L=5$ MHz, in a cell of length $l=2.5$ cm and density $n=10^{13}$ cm $^{-3}$.

amplitudes $A_{F,F'}^{\text{Rot}}$, are listed in Table 2.4; note that they are themselves functions of the polarization. We show the optical rotation spectrum calculated for cesium vapor in Figure 2.16 for the case that $\Gamma_G=400$ MHz and $\Gamma_L=5$ MHz (the natural linewidth), so that the separate resonances are well resolved. As the polarization increases, there is an increasingly larger population in the upper ($F = I + 1/2$) hyperfine level compared to the lower ($F = I - 1/2$) hyperfine level, so we see that the relative weights of the transitions from the upper hyperfine level become larger. We also see that the transition with the largest frequency, transition b with $I - 1/2 \rightarrow I + 1/2$, gives optical rotation angles that have the opposite sign compared to the other three transitions.

2.4.2 Optical Polarimetry

Sensitive magnetometry requires detection of extremely small optical rotation angles. There are numerous methods for detecting the optical rotation of the probe beam; here we briefly

discuss the three optical polarimetry methods most commonly used for the magnetometry techniques discussed in this thesis. The first method makes use of a Faraday modulator, as shown in Figure 2.17(a), placed before the alkali vapor cell. A magnetic field oscillating at a frequency ω_{mod} of a few kilohertz modulates the direction of the probe beam polarization by a small angle α due to the Faraday effect. The beam then travels through the cell and experiences an optical rotation of angle θ . The beam finally passes through a linear polarizer set at 90° to the initial beam polarization direction, so that the detected light intensity is

$$\begin{aligned} I &= I_0 \sin^2 [\theta + \alpha \sin(\omega_{\text{mod}} t)] \\ &\approx I_0 [\theta^2 + 2\theta\alpha \sin(\omega_{\text{mod}} t) + \alpha^2 \sin^2(\omega_{\text{mod}} t)], \end{aligned} \quad (2.100)$$

where I_0 is the light intensity transmitted through the cell. The Fourier component of the detected light intensity at frequency ω_{mod} is proportional to the optical rotation angle:

$$I_{\omega_{\text{mod}}} \approx 2I_0\theta\alpha. \quad (2.101)$$

In the absence of optical rotation, the detected intensity is primarily modulated at the second harmonic $2\omega_{\text{mod}}$. Note that we have assumed that the angles $\theta, \alpha \ll 1$ are very small; large optical rotation angles result in a nonlinear signal.

In the second optical polarimetry method, shown in Figure 2.17(b), the beam passes through the vapor cell, a quarter waveplate, and then a photoelastic modulator, which causes it to experience an oscillating birefringence. The modulation frequency ω_{mod} is typically on the order of 10-100 kHz. The optical axis of the quarter waveplate is set parallel to the initial beam polarization, and the optical axis of the modulator is set at 45° to the initial polarization. If there is no optical rotation in the cell, then the beam polarization is modulated symmetrically between σ^- and σ^+ circular polarizations with an amplitude $\Delta s = 2\beta$; if there is optical rotation, then the modulation is asymmetric. After passing through a linear polarizer set at 90° to the initial beam polarization direction, the transmitted intensity

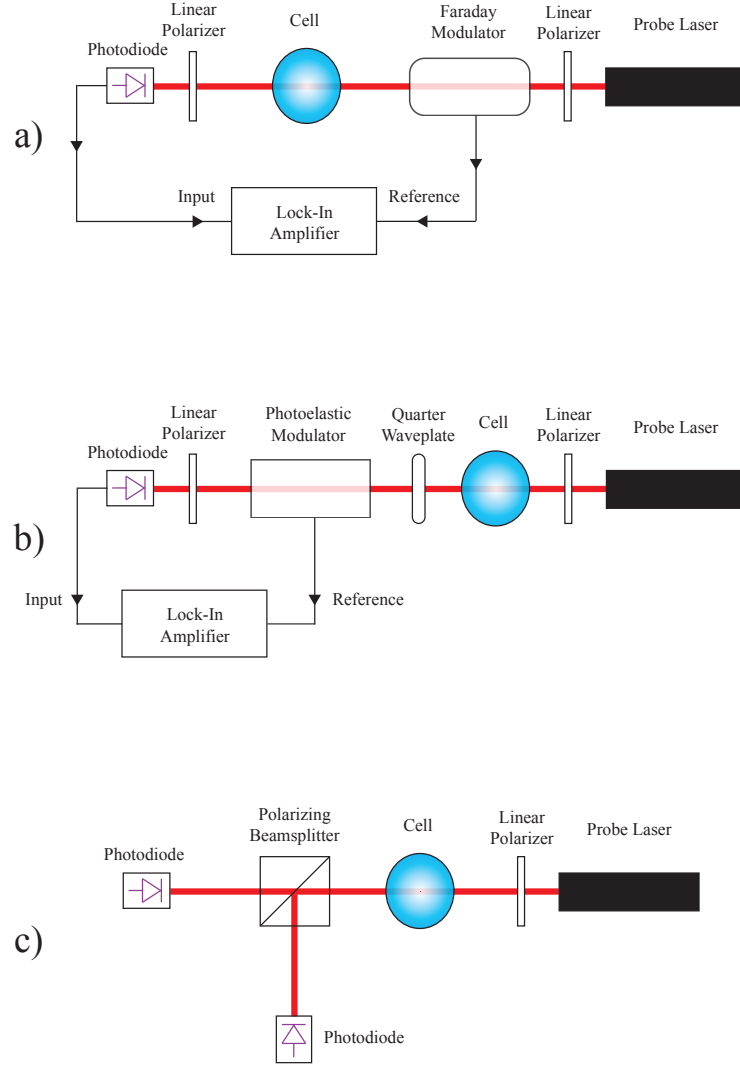


Figure 2.17: Methods for detecting optical rotation of the probe beam: **(a)** The probe beam passes through a Faraday modulator before the cell, followed by a cross-polarizer after the cell. **(b)** The probe beam passes through a photoelastic modulator after the cell followed immediately by a cross-polarizer. **(c)** After the cell the probe beam passes through a polarizing beamsplitter set at 45° to the initial polarization.

is

$$I = I_0 \sin^2 [\theta + \beta \sin (\omega_{\text{mod}} t)], \quad (2.102)$$

so that for $\theta, \beta \ll 1$ the first-harmonic intensity signal is again proportional to the optical rotation angle,

$$I_{\omega_{\text{mod}}} \approx 2I_0\theta\beta. \quad (2.103)$$

The detection methods described above involve modulation of the detected beam intensity, which has the benefit that the rotation signal is separated from $1/f$ and other low-frequency noise resulting from mechanical and technical sources, such as vibration of optical components or drift of the laser frequency and intensity. These methods are appropriate when the alkali spin orientation changes slowly compared to the modulation frequency, such as in the SERF magnetometer described in Chapter 5, but they can not be used for high-frequency applications. For such applications we instead use the balanced polarimetry technique shown in Figure 2.17(c): After the cell, the beam is analyzed by a polarizing beamsplitter set at 45° to the initial polarization direction, resulting in two separate beams with individual intensities given by

$$I_1 = I_0 \sin^2 \left(\theta - \frac{\pi}{4} \right) \quad (2.104)$$

$$I_2 = I_0 \cos^2 \left(\theta - \frac{\pi}{4} \right), \quad (2.105)$$

such that $I_1 + I_2 = I_0$, and the intensities are balanced when $\theta=0$ and unbalanced otherwise. The rotation angle is given directly by the difference between the two intensities for small rotations $\theta \ll 1$,

$$\theta \approx \frac{I_1 - I_2}{2(I_1 + I_2)}. \quad (2.106)$$

While the optical rotation angle can always be extracted from the intensity of one individual channel, the noise of the measurement is generally lower if both channels are used.

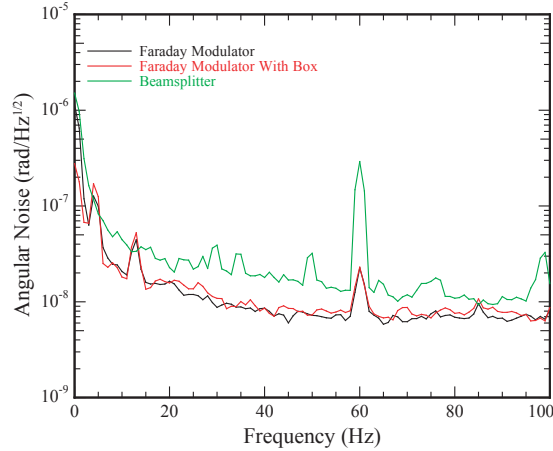


Figure 2.18: Typical optical polarimeter angular sensitivity spectra, taken using the Faraday modulation and balanced beamsplitter detection methods. Placing a box around the optical components reduces beam convection and lowers the detection noise at low frequencies $f \lesssim 1$ Hz.

Using these polarimetry techniques, angular sensitivity to probe beam polarization of better than 10^{-8} rad/ $\sqrt{\text{Hz}}$ can be achieved. However, it is difficult to attain high sensitivity at frequencies below about 1 Hz⁷ due to convection effects slowly changing the alignment of the probe beam with respect to the various components of the detection system. Low-frequency performance can be improved by enclosing the optical system (Kornack, 2005), and it can be improved even further by evacuating the enclosure or filling it with a gas such as helium with a low index of refraction (Smullin *et al.*, 2006), in order to suppress or eliminate convection effects. Typical angular sensitivity spectra are shown in Figure 2.18, though very little effort has been taken to improve the low-frequency performance for these particular measurements. Modulated detection methods typically provide better sensitivity than the beamsplitter method because the $1/f$ noise is suppressed at the modulation frequency.

⁷ For the modulated detection techniques, the frequency components of the signal are in fact sidebands of the first-harmonic ω_{mod} component of the detected intensity.

2.5 Light Shifts

On- or near-resonant light slightly shifts the atomic Zeeman energy levels as a result of two effects, the light shifts due to virtual transitions and real transitions. Our description of these effects summarizes the analysis by [Appelt *et al.* \(1998\)](#). The first effect is the result of the Stark shift due to the oscillating electric field of the light wave and is therefore also known as the ac Stark shift. The atomic interaction with the light can be described by the effective Hamiltonian term

$$\delta\mathcal{H} = \Delta E_v - \frac{i\hbar}{2}\langle\Gamma\rangle = -\mathbf{E}^* \cdot \boldsymbol{\alpha}(\nu)\mathbf{E}, \quad (2.107)$$

where ΔE_v is the energy level shift, and $\boldsymbol{\alpha}(\nu)$ is the atomic polarizability. In analogy to the complex dielectric constant (Equation 2.71), we define the polarizability to be complex, such that its real and imaginary components are Kramers-Kronig transforms of each other (Equations 2.76 and 2.77, where $\varepsilon(\nu)$ is replaced by $\boldsymbol{\alpha}(\nu)$). $\langle\Gamma\rangle$ is the average photon absorption rate given by Equation 2.52, which can be generalized for arbitrary direction of light propagation:

$$\langle\Gamma\rangle = \sigma(\nu)\Phi(1 - 2\mathbf{s} \cdot \mathbf{S}), \quad (2.108)$$

where $\mathbf{s}$ is the photon spin vector defined by Equation 2.47, $\mathbf{S}$ is the atomic spin vector, Φ is the photon flux, and $\sigma(\nu)$ is the absorption cross-section defined by Equation 2.19. Applying the Kramers-Kronig relations to Equation 2.107, we see that the energy shift is given by

$$\begin{aligned} \Delta E_v &= \frac{\hbar}{2}\Phi(1 - 2\mathbf{s} \cdot \mathbf{S}) \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\pi r_e c f \text{Re}[\mathcal{V}(\nu' - \nu_0)]}{\nu' - \nu} d\nu' \\ &= \frac{\hbar}{2} \pi r_e c f \Phi(1 - 2\mathbf{s} \cdot \mathbf{S}) \text{Im}[\mathcal{V}(\nu - \nu_0)]. \end{aligned} \quad (2.109)$$

This has a similar form to the $\mathbf{B} \cdot \mathbf{S}$ Zeeman interaction, so we may treat the ac Stark shift as if it were due to a fictitious magnetic field $\mathbf{B}_{LS}$ in the direction of the photon spin. The

energy shift is therefore

$$\Delta E_v = \hbar \gamma^e \mathbf{B}_{LS} \cdot \mathbf{S}, \quad (2.110)$$

where $\gamma^e = g_s \mu_B / \hbar$ is the gyromagnetic ratio of the electron, and the fictitious magnetic field is

$$\mathbf{B}_{LS} = \frac{-\pi r_e c f \Phi}{\gamma^e} \text{Im}[\mathcal{V}(\nu - \nu_0)] \mathbf{s}, \quad (2.111)$$

where we ignore the constant term in Equation 2.109 that does not depend on the photon and atomic spins and is thus common to all sublevels of the ground state. Comparing Equations 2.52 and 2.53, we see that the contribution to the ac Stark shift due to the D2 resonance can be found by replacing $\mathbf{S}$ with $-\mathbf{S}/2$ in the preceding analysis, giving

$$\Delta E_{v,D2} = \frac{\hbar}{2} \pi r_e c f_{D2} \Phi (1 + \mathbf{s} \cdot \mathbf{S}) \text{Im}[\mathcal{V}(\nu - \nu_{D2})] \quad (2.112)$$

$$\mathbf{B}_{LS,D2} = \frac{\pi r_e c f_{D2} \Phi}{2\gamma^e} \text{Im}[\mathcal{V}(\nu - \nu_{D2})] \mathbf{s}. \quad (2.113)$$

If we define the frequency shift parameter

$$\Delta \omega_v = \frac{E_v}{\hbar} = \pi r_e c f \Phi \text{Im}[\mathcal{V}(\nu - \nu_0)], \quad (2.114)$$

which describes the shift in the Zeeman frequency for light with $|\mathbf{s}|=1$, then we can write the pumping rate for unpolarized atoms and the frequency shift parameter as the real and imaginary components of a single complex optical pumping rate:

$$R + i\Delta \omega_v = \pi r_e c f \Phi \mathcal{V}(\nu - \nu_0), \quad (2.115)$$

hence the phrase “virtual transition” to refer to the ac Stark shift. The alkali atoms respond to the fictitious field $\mathbf{B}_{LS}$ as if it were a real magnetic field, and they precess about the total effective field $\mathbf{B} = \mathbf{B}_0 + \mathbf{B}_{LS}$, where $\mathbf{B}_0$ is the actual ambient field. This effect can be very important when operating in low ambient field such that $\mathbf{B}_{LS} \gtrsim \mathbf{B}_0$, in which case care should be taken to eliminate the ac Stark shift. Note that under nominal operating conditions, with the pump beam tuned directly on resonance (so that $\text{Im}[\mathcal{V}(\nu - \nu_0)] = 0$) and the

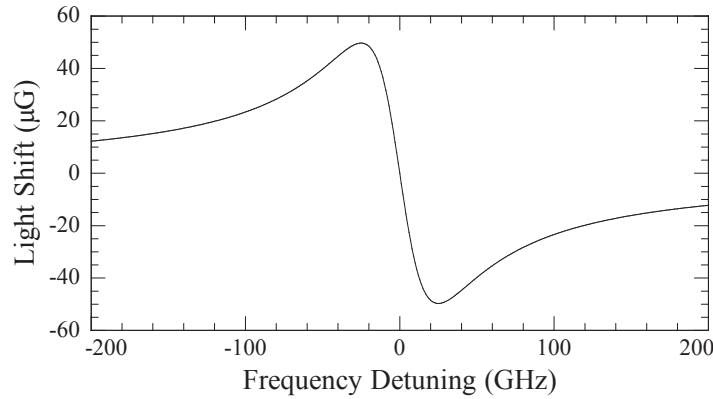


Figure 2.19: Values for the fictitious magnetic field given by Equation 2.111 due to the ac Stark shift caused by light with complete circular polarization $s = 1$, for the potassium D1 transition with $\Gamma_L = 50$ GHz and a laser flux of 1 mW/cm^2 .

probe beam linearly polarized (so that $s=0$), the ac Stark shift is indeed eliminated. However, imperfections in the optical conditions can lead to the atoms experiencing the light shift due to virtual transitions. Figure 2.19 shows the magnitude of the fictitious field B_{LS} due to the ac Stark effect for the potassium D1 transition due to circularly polarized light with $\Gamma_L = 50$ GHz and a modest laser power flux of 1 mW/cm^2 . Under normal operating conditions the light shift field is generally on the order of $1 \mu\text{G}$ or less and should be taken into account when the ambient magnetic field is of comparable size.

The second type of light shift is referred to as the shift due to real transitions. While an atom is in an excited state after absorbing a photon, the electron spin becomes randomized due to rapid collisional mixing. However, the hyperfine interaction is too weak for the nuclear spin to become depolarized in the short period of time before the atom decays back to the ground state, and the nuclear spin remains coherent before and after the excitation. The gyromagnetic ratio is generally different in the ground and excited states, so the spin precession gains some phase difference with respect to the atoms that were not excited during that time period. The result is a shift in the Zeeman transition frequency ν_{Fm_F}

between the ground-state levels $|F, m_F\rangle$ and $|F, m_F - 1\rangle$ given by

$$\Delta\nu_{Fm_F} = -\nu_{Fm_F} \left(\frac{R_{OP}}{R_Q} \right) \frac{[I]^2 + 4\bar{m}_F^2 - 1 - 4(-1)^{I+1/2-F} [I] s \bar{m}_F}{4[I]^2}, \quad (2.116)$$

where R_Q is the quenching rate of excited atoms (or $1/\tau_{\text{nat}}$ in the absence of quenching gas), $\bar{m}_F = m_F - 1/2$, and $[I] = 2I + 1$ (Appelt *et al.*, 1998). The shift of the Zeeman resonance frequency due to real transitions is in general significantly smaller than the frequency itself since $R/R_Q \ll 1$ and so is not typically a concern in magnetometry. We therefore use the term “light shift” to refer exclusively to the ac Stark shift.

2.6 The Magnetometer Response

Atomic magnetometers characterize a magnetic field by observing the response of the alkali spins to the field. The coupling of the electron spin to a magnetic field is approximately given by the Hamiltonian

$$\mathcal{H} = \gamma \hbar \mathbf{B} \cdot \mathbf{S}, \quad (2.117)$$

and the response of the spin to the magnetic field is

$$\frac{d}{dt} \mathbf{S} = \frac{i}{\hbar} [\mathcal{H}, \mathbf{S}]. \quad (2.118)$$

The components of $\mathbf{S}$ commute according to the normal angular momentum property $[S_x, S_y] = iS_z$, so that

$$\frac{d}{dt} S_z = \frac{i}{\hbar} \gamma \hbar (B_x [S_x, S_z] + B_y [S_y, S_z]) = i\gamma (-iB_x S_y + iB_y S_x), \quad (2.119)$$

with similar equations for S_x and S_y . We therefore see that the spin precesses in the magnetic field according to

$$\frac{d}{dt} \mathbf{S} = \gamma \mathbf{B} \times \mathbf{S}, \quad (2.120)$$

which is the classical equation for a dipole in a magnetic field. The electron and nuclear spins of the atom act as coupled oscillators; the electron spin is forced to drag the nuclear

spin along with it as it precesses, resulting in slower precession than would be experienced by a bare electron:

$$\gamma = \frac{\gamma^e}{2I + 1}, \quad (2.121)$$

where $\gamma^e = g_s \mu_B / \hbar = 2\pi \times 2.8 \text{ MHz/G}$ is the gyromagnetic ratio of the bare electron.

The overall evolution of the atomic spin is described phenomenologically by the Bloch equation (Bloch, 1946),

$$\frac{d}{dt} \mathbf{S} = \gamma \mathbf{B} \times \mathbf{S} + \frac{1}{q} \left[R_{\text{OP}} \left(\frac{1}{2} s \hat{\mathbf{z}} - \mathbf{S} \right) - R_{\text{rel}} \mathbf{S} \right], \quad (2.122)$$

where q is the nuclear slowing-down factor (see Section 2.7). The second term in this equation describes the effect of optical pumping to polarize the spin along the $\hat{\mathbf{z}}$ axis, while the third term describes the effect of spin relaxation to depolarize the spin. In the absence of any magnetic field, the equilibrium spin polarization S_0 is given by

$$S_0 = \frac{s R_{\text{OP}}}{2(R_{\text{OP}} + R_{\text{rel}})}. \quad (2.123)$$

We consider the case of an ambient magnetic field aligned parallel to the pumping axis, $\mathbf{B}_0 = (\omega_0/\gamma) \hat{\mathbf{z}}$. The transverse components of the spin, and any applied transverse magnetic field, can be written in the complex form

$$\tilde{S} = S_x + iS_y \quad (2.124)$$

$$\tilde{B} = B_x + iB_y, \quad (2.125)$$

giving the transverse Bloch equation

$$\frac{d}{dt} \tilde{S} = i\gamma \left(\frac{\omega_0}{\gamma} \tilde{S} - S_0 \tilde{B} \right) - \frac{\tilde{S}}{T_2}, \quad (2.126)$$

where $T_2 = q/(R_{\text{OP}} + R_{\text{rel}})$ is the transverse polarization lifetime, which is discussed in Section 2.7.

An applied oscillating field $\mathbf{B}' = B' \cos(\omega t) \hat{\mathbf{y}}$ can be decomposed into two counter-propagating rotating fields, such that

$$\tilde{B}' = \frac{iB'}{2} \left(e^{i\omega t} + e^{-i\omega t} \right). \quad (2.127)$$

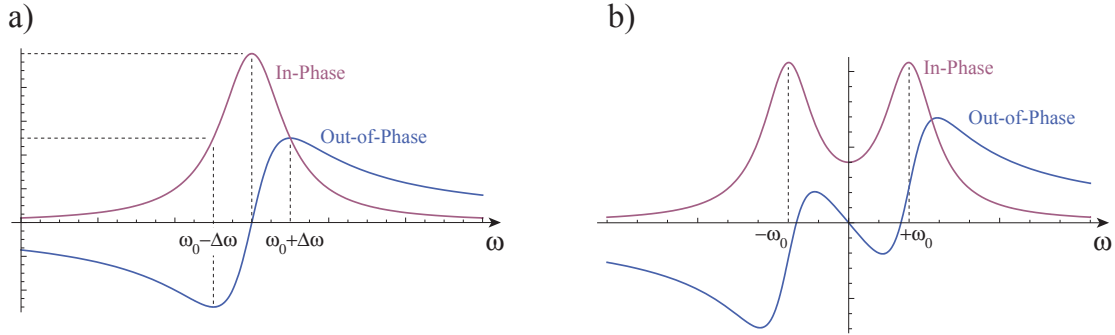


Figure 2.20: Magnetometer frequency response around the resonance frequency ω_0 to an oscillating field of frequency ω , consisting of an absorptive in-phase component and a dispersive out-of-phase component. **(a)** If the linewidth $\Delta\omega$ is much smaller than ω_0 , then the spectrum is given by a single Lorentzian lineshape centered at ω_0 . **(b)** If the linewidth is large compared to the resonance frequency, then the spectrum is given by overlapping Lorentzian curves centered at ω_0 and $-\omega_0$.

The response to such a field is given by the solution to Equation 2.126:

$$\tilde{S} = \frac{1}{2} S_0 \gamma B' \left[\frac{e^{i\omega t}}{\Delta\omega + i(\omega - \omega_0)} + \frac{e^{-i\omega t}}{\Delta\omega - i(\omega + \omega_0)} \right], \quad (2.128)$$

where $\Delta\omega = 1/T_2$ is the magnetic linewidth. Thus, an oscillating magnetic field induces coherent precession of the component of atomic spin in the plane perpendicular to the ambient magnetic field. Note that a rotating field $\tilde{B} = B' e^{i\omega t}$ induces rotating precession with twice the amplitude of that induced by the component of the oscillating field rotating in the same direction. The probe beam measures the spin component $S_x = \text{Re}[\tilde{S}]$, given by

$$S_x = \frac{1}{2} S_0 \gamma B' \left[\frac{\Delta\omega \cos(\omega t) + (\omega - \omega_0) \sin(\omega t)}{(\Delta\omega)^2 + (\omega - \omega_0)^2} + \frac{\Delta\omega \cos(-\omega t) + (\omega + \omega_0) \sin(-\omega t)}{(\Delta\omega)^2 + (\omega + \omega_0)^2} \right]. \quad (2.129)$$

The atomic response described by Equation 2.129 is the sum of two Lorentzian curves centered at frequencies $\pm\omega_0$ with half width at half maximum (HWHM) $\Delta\omega = 1/T_2$.⁸ Each

⁸ We adopt the somewhat confusing convention that magnetic resonance lineshapes are described by their half width at half maximum (HWHM), but optical lineshapes are described by their full width at half maximum (FWHM). Also note that we sometimes refer to the magnetic linewidth as $\Delta\omega$ and sometimes as $\Delta\nu = \Delta\omega/2\pi$, and it should always be clear from the units given which we are using.

curve has a component with an absorptive character that is in-phase with the oscillating field and a component with a dispersive character that is 90° out-of-phase with the oscillating field, as shown in Figure 2.20. The magnetometer response can thus be written using the complex form of the Lorentzian lineshape given in Equation 2.82, with the in- and out-of-phase components represented by the real and imaginary axes of the complex plane. If the linewidth $\Delta\omega \ll \omega_0$, then the counter-propagating rotational response centered at $-\omega_0$ can be ignored, as its contribution becomes insignificant at frequencies near $+\omega_0$; otherwise the spectrum consists of two overlapping Lorentzian curves. A qualitatively identical spectrum with linewidth $\Delta B = \Delta\omega/\gamma$ is obtained if the oscillation frequency is kept fixed and the ambient magnetic field is varied. Although here we consider the specific case of the ambient field parallel to the pumping axis and an oscillating field perpendicular to it, the Lorentzian lineshape is in fact typical of the atomic response to excitation near the Zeeman resonance frequency. A useful figure-of-merit for magnetometer performance is the slope of the out-of-phase dispersion curve at its zero-crossing at the resonance frequency ω_0 , as this slope characterizes the size of the magnetometer response to a change in the magnetic field experienced by the atoms.

2.7 Spin Relaxation

As discussed in Chapter 1 and in Section 2.6, perhaps the most important method for optimizing the magnetometer sensitivity is to maximize the spin polarization lifetime, thus minimizing the magnetic linewidth. If the total magnetic field $\mathbf{B}$ lies along the $\hat{\mathbf{z}}$ axis,⁹ then the polarization lifetime can be characterized by the lifetime T_1 of the longitudinal component $\langle F_z \rangle$ and the lifetime T_2 of the transverse components $\langle F_x \rangle$ and $\langle F_y \rangle$. Many spin-relaxation mechanisms are due to collisions with buffer gas atoms, quenching gas

⁹ We have chosen the direction $\hat{\mathbf{z}}$ randomly here, and in general the magnetic field may be directed along some arbitrary orientation with respect to the pumping and probing axes.

molecules, and other alkali atoms. The general rate for a collision is given by

$$R = n\sigma\bar{v}, \quad (2.130)$$

where n is the density of the other gas species, σ is the effective collisional cross-section, and $\bar{v}$ is the relative thermal velocity. These three quantities will normally be identified by the type of collision and the perturbing atomic or molecular species. The relative thermal velocity is

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi M}}, \quad (2.131)$$

where the reduced mass M given by the masses m of the alkali atom and m' of the perturber is

$$\frac{1}{M} = \frac{1}{m} + \frac{1}{m'}. \quad (2.132)$$

In general we are interested in measuring the spin coherence of precessing atoms, so mechanisms that destroy that coherence are considered to cause spin relaxation.

Most spin-relaxation mechanisms cause depolarization of the electron spin while leaving the nuclear spin unaffected, so the total atomic spin maintains some amount of coherence with the ensemble after the electron spin is destroyed. The degree to which spin coherence is maintained is characterized by the nuclear slowing-down factor q , which depends on the polarization of the ensemble as listed in Table 2.5. Note that at high polarization ($P \approx 1$), the slowing-down factor is simply given by $q = 2I + 1$.

The longitudinal lifetime is given by the rates of the various mechanisms that affect the expectation value of the spin component along the quantization axis, which is defined by the direction of the magnetic field:

$$\frac{1}{T_1} = \frac{1}{q} (R_{SD} + R_{OP} + R_{pr}) + R_{wall}. \quad (2.133)$$

Here R_{SD} is the rate of relaxation due to spin-destruction collisions and is given by

$$R_{SD} = R_{SD}^{\text{Self}} + R_{SD}^{\text{B}} + R_{SD}^{\text{Q}}, \quad (2.134)$$

I	$q(P)$	q_{lp}	q_{hp}
3/2	$\frac{6+2P^2}{1+P^2}$	6	4
5/2	$\frac{38+52P^2+6P^4}{3+10P^2+3P^4}$	38/3	6
7/2	$\frac{22+70P^2+34P^4+2P^6}{1+7P^2+7P^4+P^6}$	22	8

Table 2.5: Nuclear slowing-down factors as a function of spin polarization, as calculated by [Ap-pelt *et al.* \(1998\)](#). q_{lp} and q_{hp} represent the low-polarization and high-polarization limits, respectively, with $q_{hp}=2I+1$.

where the first term is due to collisions with other alkali atoms, the second term is due to collisions with buffer gas atoms, and the third term is due to collisions with quenching gas molecules. The second term in Equation 2.133 is the optical pumping rate, since absorption of a pump beam photon changes an atom's angular momentum so that it lies along the pumping direction. The relaxation rate R_{rel} used in Sections 2.3 and 2.6 is effectively the relaxation rate described here without the contribution from the optical pumping rate. The third term R_{pr} in Equation 2.133 is the absorption rate of photons from the probe beam. Probe photons are polarized along a direction orthogonal to the pump beam, and individual photons have arbitrary helicity, so the polarization of an atom is destroyed upon absorption. The contribution of the probe beam to spin relaxation can be minimized by detuning the frequency sufficiently far from resonance that absorption becomes negligible, although this reduces the optical rotation signal. The fourth term in Equation 2.133 is the rate of depolarization due to collisions with the wall of the vapor cell, which destroy both the electron and nuclear spins. The total effect of these four terms is to drive the longitudinal polarization toward the equilibrium value described by Equation 2.39.

The transverse components of the atomic spin precess in the magnetic field, so any mechanisms that cause dephasing between precessing atoms contribute to relaxation of

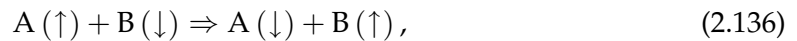
the average transverse polarization of the ensemble without affecting the longitudinal component. In addition, the mechanisms that relax the longitudinal component also relax the transverse components because they randomize the spin direction, so the transverse polarization lifetime is

$$\frac{1}{T_2} = \frac{1}{T_1} + \frac{1}{q_{\text{SE}}} R_{\text{SE}} + R_{\text{gr}}, \quad (2.135)$$

where R_{SE} is the rate of spin-exchange collisions between alkali atoms, and R_{gr} is the broadening due to the magnetic field gradient across the vapor cell.¹⁰ The spin-exchange broadening factor q_{SE} is different from the slowing-down factor q and depends on the size of the ambient magnetic field and the alkali vapor density. In order to attain maximal polarization lifetime, a sensitive magnetometer is designed to minimize the various spin-relaxation mechanisms, each of which is discussed in the following sections.

2.7.1 Spin-Exchange Collisions

At high alkali densities, spin-exchange collisions between alkali atoms are typically the dominant cause of spin relaxation (Purcell and Field, 1956). In such a collision, the direction of the electron spins of the two atoms can be reversed while the total spin is conserved. This process may be represented symbolically by



where the arrows refer to spin-up and spin-down. This effect is due to the alkali-alkali interaction having both attractive singlet and repulsive triplet potential components, with the alkali-alkali dimer existing in a superposition of the singlet and triplet states. There is a large splitting between the singlet and triplet potentials, causing an overall phase shift of the dimer wavefunction during the collision and potentially resulting in the exchange of electron spin states.

¹⁰ Usually a distinction is made between T_2 and T_2^* , where the latter includes the effects of magnetic field gradients and the former does not, but we do not make this distinction here.

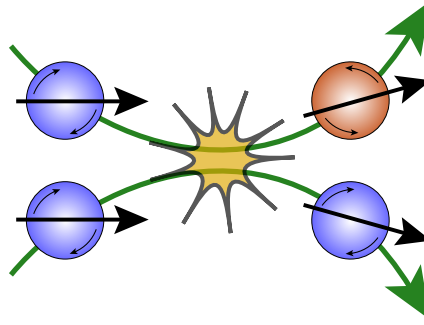


Figure 2.21: During spin-exchange collisions the total angular momentum $F_1 + F_2$ is conserved, but one or both atoms may switch between hyperfine levels. Atoms in different hyperfine levels precess in opposite directions and decohere. The hyperfine levels $F = I \pm 1/2$ are represented here by the colors red and blue.

Spin-exchange collisions are sudden with respect to the hyperfine interaction and so do not affect the nuclear spin of the colliding atoms. However, one or both atoms may change hyperfine states as a result of the collision, as shown in Figure 2.21; the overall effect of spin-exchange collisions is to redistribute the atoms among the atomic m_F Zeeman sublevels while preserving the total atomic spin. Atoms in the two ground-state hyperfine levels precess with approximately the same frequency but in opposite directions (see Section 3.2):

$$\omega_+ = \gamma |\mathbf{B}| = -\omega_-, \quad (2.137)$$

where the subscripts denote the hyperfine levels $F = I \pm 1/2$. Therefore, the populations of the two hyperfine levels decohere as they precess, causing relaxation of the transverse spin components. The contribution of spin-exchange collisions to the polarization lifetime T_2 is characterized by the broadening factor q_{SE} , as shown in Equation 2.135. At high magnetic field such that the precession frequency $|\omega_{\pm}| \gg R_{SE}$, the broadening factor is given by Happer and Tam (1977) as

$$\frac{1}{q_{SE}} = \frac{2I(2I-1)}{3(2I+1)^2}. \quad (2.138)$$

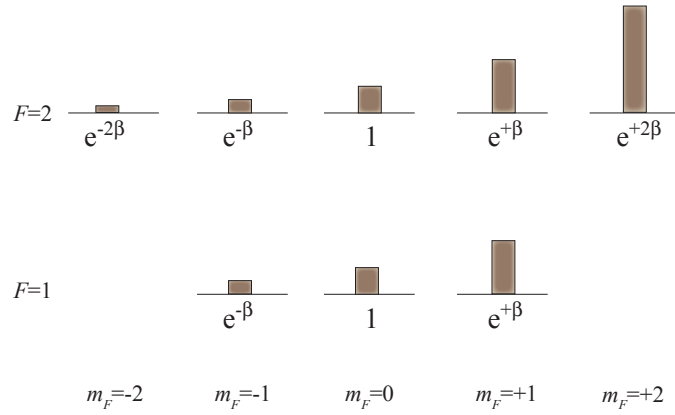


Figure 2.22: In the regime of rapid spin-exchange collisions, the Zeeman sublevel populations are given by the spin-temperature distribution with relative amplitudes $e^{\beta m_F}$. The case of $I=3/2$ is shown.

As the magnetic field approaches zero, $1/q_{\text{SE}} \rightarrow 0$, and spin-exchange no longer affects the polarization lifetime. Operation in this spin-exchange relaxation-free regime is crucial to achieve the ultra-high sensitivity of the SERF magnetometer. The dependence of q_{SE} on the spin-exchange rate and precession frequency is discussed in detail in Section 5.1.

When the alkali vapor density is high enough that the spin-exchange rate is much larger than the optical pumping rate or any other spin-relaxation mechanism, the atomic spins arrive at an equilibrium state described by the spin-temperature distribution (Anderson *et al.*, 1959, 1960). This state is analogous to thermal equilibrium, and the relative populations $\rho(F, m_F)$ of the $|F, m_F\rangle$ ground-state sublevels are given by a Boltzmann distribution with effective spin temperature β :

$$\rho(F, m_F) = C e^{\beta m_F}, \quad (2.139)$$

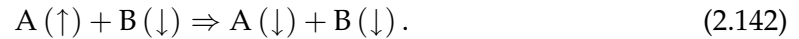
where C is a normalization factor, as shown in Figure 2.22. The spin temperature is given by the atomic polarization P ,

$$\beta = \ln \left(\frac{1+P}{1-P} \right) \quad (2.140)$$

$$P = \tanh(\beta/2). \quad (2.141)$$

2.7.2 Spin-Destruction Collisions

After spin-exchange collisions, the next most-important mechanism for spin relaxation is spin-destruction collisions that do not preserve the total spin of the alkali ensemble. Spin-destruction collisions may occur between alkali atoms or with the buffer and quenching gases. Such collisions between alkali atoms transfer spin angular momentum to the rotational angular momentum of the colliding pair of atoms; in analogy to Equation 2.136, we may represent an alkali-alkali spin-destruction collision symbolically as



The transfer of angular momentum is believed to be the result of a tensor spin-axis interaction of the form

$$V_{SA} = \frac{2}{3} \lambda \mathbf{S} \cdot (3\hat{\mathbf{R}}\hat{\mathbf{R}} - \mathbf{1}) \cdot \mathbf{S}, \quad (2.143)$$

where $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2$ is the total electron spin of the colliding alkali atoms, and $\hat{\mathbf{R}}$ is the direction of the internuclear axis of the colliding pair (Bhaskar *et al.*, 1980). The coupling coefficient λ depends on the spin-spin magnetic dipole interaction, as well as to second order on the spin-orbit interaction (Mies *et al.*, 1996). The spin-destruction cross-sections are included in Table A.2 and are significantly smaller than the spin-exchange cross-sections, with the difference ranging from two orders of magnitude in cesium to four orders of magnitude in potassium. Alkali-alkali spin-destruction collisions therefore do not play an important role in determining the polarization lifetime unless spin-exchange relaxation is

suppressed, such as through light narrowing (see Section 4.1.1) or operation in the SERF regime (see Section 5.1).

Collisions with noble gas atoms can also destroy the alkali polarization by transferring spin angular momentum to rotational angular momentum of the alkali-noble gas pair. A theory for the spin-rotational coupling is given by [Wu *et al.* \(1985\)](#). Representative spin-destruction cross-sections with helium and xenon are given in Table A.2; note that the cross-sections are much smaller than the physical size of the noble gas atoms. In the excited state, the p -shell valence electron is easily depolarized by collisions with noble gas atoms, causing rapid collisional mixing as discussed in Section 2.3. However, in the ground state the alkali atom can collide many times with noble gas atoms before losing its polarization. In cells containing a noble gas species for the purpose of slowing alkali diffusion to the walls, the buffer gas pressure is typically chosen so that the relaxation due to spin-destruction collisions with noble gas atoms is comparable to the other dominant spin-relaxation mechanisms. Collisions with quenching gas molecules can also destroy the alkali polarization, and the quenching gas pressure is chosen to ensure sufficient quenching of excited atoms without greatly affecting the polarization lifetime.

2.7.3 Wall Collisions

When an alkali atom encounters the bare glass surface of the cell wall, it becomes adsorbed into the surface for a finite period of time before being ejected back into the cell volume. During this time the atom experiences the large local electric and magnetic fields produced by ions and molecules within the glass, and its motion on the surface causes these fields to change over time. Although the adsorption period is short, measured to be in the range of 10^{-5} - 10^{-7} s ([de Freitas *et al.*, 2002](#)), by the time the atom leaves the surface its spin direction becomes completely randomized. Wall collisions are therefore completely depolarizing and can dominate all other spin-relaxation mechanisms unless suppressed. When the cell

contains only alkali vapor, with no buffer gas to inhibit motion, the atoms move in straight lines in between collisions with the cell walls. The average thermal velocity is

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}}, \quad (2.144)$$

where m is the mass of the atom. The average time between collisions with the cell walls is

$$T_{\text{wall}} = \frac{4V}{\bar{v}A}, \quad (2.145)$$

where V and A are the cell volume and surface area, respectively. For example, a ^{39}K atom at 373 K has a thermal velocity $\bar{v}=449$ m/s, and in a spherical cell of radius 2.5 cm the time between collisions is $74 \mu\text{s}$, leading to a magnetic linewidth of at least $\Delta\omega = 2\pi \times 2100$ Hz. Such a broad linewidth can prevent high-sensitivity magnetometer performance, and the effect of wall depolarization becomes more pronounced in smaller cells, as the lifetime scales linearly with the characteristic length of the cell.

The two common methods for suppressing wall depolarization are the use of anti-relaxation surface coatings and buffer gas. Surface coatings are molecules that cover the surface and prevent the alkali atoms from approaching the glass walls. These coatings must be chemically inert and can ideally allow the atoms to bounce off the surface thousands of times without depolarizing. We discuss in detail the use of surface coatings and their advantages in Chapter 6. The most effective known coating is paraffin, which melts at temperatures of about 60-80°C; above these temperatures effective coatings were previously unavailable before our discovery of the effectiveness of OTS up to 170°C, as described in Section 6.2, necessitating the use of buffer gas.

At high buffer gas pressures the atoms experience diffusive motion, increasing the time necessary to reach the cell wall. In order to determine the average wall collision rate, we consider the atomic polarization in the absence of other relaxation mechanisms or further optical pumping. The decay of polarization is given by the diffusion equation (Franzen,

1959),

$$\frac{\partial}{\partial t}P = D\nabla^2P, \quad (2.146)$$

where $D = \frac{1}{3}\lambda\bar{v}$ is the diffusion constant of the alkali atom within the buffer gas, and λ is the mean free path length between collisions with buffer gas atoms. The diffusion constant scales inversely with buffer gas pressure and so is given relative to its value D_0 at a pressure p_0 by $D = D_0p_0/p$. As an example, we consider a distribution of polarization with $P=1$ in the center of a spherical cell of radius R . If we assume spherical symmetry, then we can separate the polarization $P(r, t)$ into independent functions

$$P(r, t) = \sum_{m=1}^{\infty} P_m(0, 0)R_m(r)T_m(t), \quad (2.147)$$

where the index m is the order of a particular diffusion mode within the cell, and $P_m(0, 0)$ is the amplitude of the given mode. The time dependence is simply a decaying exponential with time constant τ_m ,

$$T_m(t) = e^{-t/\tau_m}, \quad (2.148)$$

and the radial dependence is a spherical Bessel function,

$$R_m(r) = \frac{\sin(k_m r)}{k_m r}. \quad (2.149)$$

If we approximate the polarization distribution as being given by the fundamental diffusion mode with $m=1$, such that $\tau_1 = T_{\text{wall}}$, then the polarization as a function of position and time is given by

$$P(r, t) = \frac{\sin(kr)}{kr} e^{-t/T_{\text{wall}}}. \quad (2.150)$$

Applying the boundary condition that $P(R)=0$, denoting fully depolarizing walls, gives $k = \pi/R$. Inserting the solution for polarization into the diffusion equation, we find the wall collision time,

$$\frac{1}{T_{\text{wall}}} = D \left(\frac{\pi}{R} \right)^2. \quad (2.151)$$

Following this procedure, similar solutions may be obtained for arbitrary cell geometry.

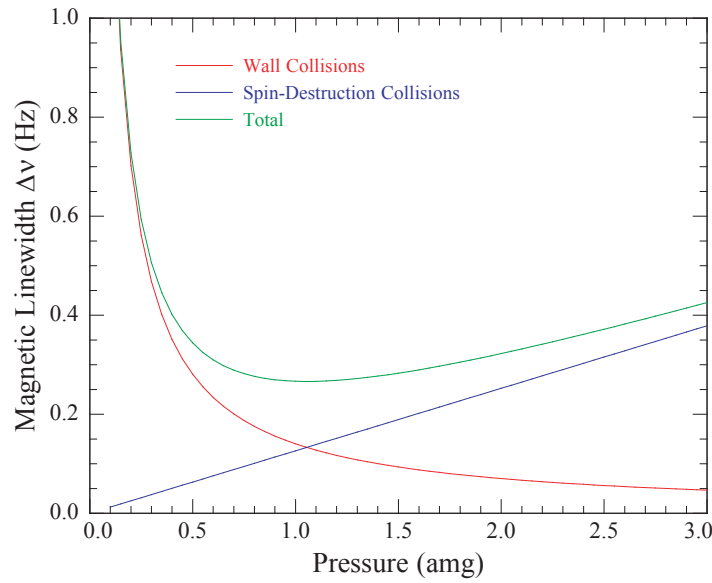


Figure 2.23: Comparison of the broadening of the ^{39}K magnetic linewidth due to wall collisions and spin-destruction collisions, as a function of helium buffer gas pressure in a spherical cell of radius 2.5 cm.

Adding buffer gas to the cell suppresses the wall collision rate but increases the spin-destruction rate. We may compare these two rates for a given cell geometry to determine the optimal buffer gas pressure, as shown in Figure 2.23 for the case of ^{39}K in a helium buffer gas at 373 K within a spherical cell of radius 2.5 cm. In this case the optimal pressure is about 1 amg, although the actual pressure used often depends on the application. For example, a cell used for medical imaging should be at a pressure below 1 atmosphere at operational temperature, in order to avoid the possibility of the cell exploding near a human body due to high internal pressure. On the other hand, much higher pressures are sometimes used to broaden the optical linewidth and ensure full pump beam propagation through the cell.

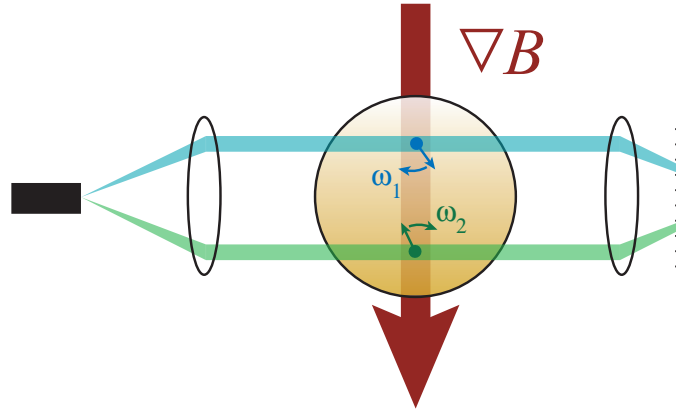


Figure 2.24: Detection of magnetic field gradients. In the presence of high buffer gas pressure, the alkali atoms diffuse slowly and are localized within a small region of the cell. A magnetic field gradient causes atoms in different regions to precess with different frequencies. Imaging the cell with separate parts of the probe beam enables measurement of the field gradient.

2.7.4 Magnetic Field Gradients

In the presence of a magnetic field gradient, atoms in different parts of the cell experience different local fields and therefore precess at different frequencies. In a coated cell without buffer gas, the alkali atoms freely sample the entire cell volume and average the magnetic field throughout, suppressing the effect of the field gradients in an example of motional narrowing. However, in cells where the atoms diffuse very slowly, they remain in only a small region of the cell during a single coherence lifetime and experience only a single value for the magnetic field strength. In this regime the gradient broadening is given by the spread of precession frequencies throughout the cell, $R_{\text{gr}} \sim \gamma \nabla B$, where ∇B is the size of the field gradient. In general, the broadening due to a magnetic field gradient depends on the extent to which atoms diffuse through the gradient and has a complicated dependence on the diffusion constant, cell geometry, and direction of the gradient; see for example [Cates *et al.* \(1988a,b\)](#), [Stoller *et al.* \(1991\)](#), and [Pustelny *et al.* \(2006\)](#).

Operation in the regime of slow diffusion can allow for direct measurement of the magnetic field gradient across the cell. The alkali atoms are localized within a small region of

the cell during a single polarization lifetime and precess according to the local magnetic field. If the probe beam is imaged in multiple channels, then different parts of the probe beam interact with atoms in separate regions of the cell, measuring the local values of the field and enabling the variation in the field to be imaged, as illustrated in Figure 2.24. The spatial resolution is limited by the diffusion length of an atom over a polarization lifetime. [Kominis *et al.* \(2003\)](#) demonstrated detection of the field gradient with 2 mm resolution using a SERF magnetometer, and [Xia *et al.* \(2006\)](#) used this method to spatially map the magnetic fields produced by a human brain; [Affolderbach *et al.* \(2002\)](#) demonstrated gradient imaging using a magnetometer based on coherent population trapping. The main advantage of measuring the gradient of the magnetic field rather than the field itself is that noise common to all measurement channels, such as stray fields produced by distant sources or fluctuations in the probe laser frequency and amplitude, are largely canceled by subtracting the signals from individual channels. An improvement in noise level by an order of magnitude can typically be achieved using this technique.

2.8 Fundamental Magnetometer Sensitivity

The performance of the magnetometer is ultimately limited by quantum fluctuations associated with the two ensembles used to make the measurement, namely the alkali atoms and the probe beam photons. Here we present in more detail the analysis published in [Savukov *et al.* \(2005\)](#), which considered three sources of quantum noise: the spin-projection noise δB_{spn} due to the finite number of alkali atoms used in the measurement, the photon shot noise δB_{psn} due to the finite number of probe photons used, and the light-shift noise δB_{lsn} due to fluctuations in the polarization of the probe beam. In order to write the first two noise sources in terms of uncertainty in the magnetic field measurement, it is necessary to establish the relationship between the alkali spin polarization P_x and the magnetic field to be measured; this is done individually for the radio-frequency magnetometer in

Section 4.1.2 and the SERF magnetometer in Section 5.2, with the latter discussion also based in part on the analysis by Ledbetter *et al.* (2008). The total noise in the magnetic field measurement is given by the quadrature sum of the three individual sources of noise:

$$\delta B = \sqrt{\delta B_{\text{spn}}^2 + \delta B_{\text{psn}}^2 + \delta B_{\text{lsn}}^2}. \quad (2.152)$$

Auzinsh *et al.* (2004) also consider the fundamental sensitivity of an atomic magnetometer and describe the conditions under which spin squeezing improves the measurement noise.

2.8.1 Spin-Projection Noise

The transverse components of the atomic spin do not commute,

$$[F_x, F_y] = iF_z, \quad (2.153)$$

so the product of their uncertainty is given by¹¹

$$\delta F_x \delta F_y \geq \frac{|F_z|}{2}. \quad (2.154)$$

The uncertainty is minimized at full polarization, which we assume for the remainder of this section; a more general form is given in Section 5.2. If we do not employ a spin-squeezing technique, then $\delta F_x = \delta F_y$, and the uncertainty in F_x resulting from N uncorrelated measurements is

$$\delta F_x = \sqrt{\frac{\langle F_z \rangle}{2N}}. \quad (2.155)$$

In an atomic magnetometer we continuously probe the ensemble of atoms, and correlated measurements on an individual atom do not improve the sensitivity of the measurement. The correlation $K(\tau)$ between a measurement at time $t=0$ and at $t=\tau$ is given by the degree to which the spin has lost its coherence with the ensemble over that time frame:

$$K(\tau) = e^{-\tau/T_2}. \quad (2.156)$$

¹¹ Note that we write the atomic spin parameters in dimensionless form throughout this thesis. If we write them with units of angular momentum (*i.e.*, $|S|=\hbar/2$, etc.), then the uncertainty becomes $\delta F_x \delta F_y \geq \hbar|F_z|/2$.

The total uncertainty in a continuous measurement is given by [Gardner \(1990\)](#) as

$$\begin{aligned}\delta\langle F_x \rangle &= \delta F_x \left[\frac{2}{t} \int_0^t \left(1 - \frac{\tau}{t} \right) K(\tau) d\tau \right]^{1/2} \\ &= \delta F_x \left[\frac{2T_2}{t} + \frac{2T_2^2 (e^{-t/T_2} - 1)}{t^2} \right]^{1/2}.\end{aligned}\quad (2.157)$$

We typically measure for time scales $t \gg T_2$, so we may combine Equations 2.155 and 2.157 to give the total uncertainty

$$\delta\langle F_x \rangle = \sqrt{\frac{2F_z T_2 BW}{N}}, \quad (2.158)$$

where $BW=1/2t$ is the bandwidth of the measurement. We interpret $N = nV$ to be the total number of independent spins interrogated by the probe beam, with the active measurement volume V defined by the intersection of the pump and probe beams. It is customary to give the noise in units of root-mean-square uncertainty per $\text{Hz}^{1/2}$,¹² which is given simply by dividing Equation 2.158 by $\sqrt{BW}$:

$$\delta\langle F_x \rangle_{\text{rms}} = \sqrt{\frac{2F_z T_2}{N}}. \quad (2.159)$$

2.8.2 Photon Shot Noise

We consider the photon shot noise for the case of detection using the balanced polarimeter described in Section 2.4.2. The total flux of photons Φ' is given by the integral of the flux per unit area Φ taken over the entire area of the probe beam:

$$\Phi' = \int_A \Phi dA. \quad (2.160)$$

The optical rotation angle is given by Equation 2.106 in terms of the fluxes Φ'_1 and Φ'_2 in the two arms of the polarimeter as

$$\theta = \frac{\Phi'_1 - \Phi'_2}{2(\Phi'_1 + \Phi'_2)}. \quad (2.161)$$

¹² This gives the uncertainty after one second of measurement time, with the uncertainty improving as the square-root of the total measurement time.

We assume that the rotation angle is small, so that $\Phi'_1 \approx \Phi'_2$, and the quantum fluctuation in the number of photons in each channel is given by¹³

$$\delta\Phi'_1 = \delta\Phi'_2 = \sqrt{\frac{\Phi'}{2}}. \quad (2.162)$$

Then the fluctuation in the measured optical rotation angle for a given bandwidth $BW = 1/2t$ is

$$\begin{aligned} \delta\langle\theta\rangle &= \sqrt{2BW \left[\left(\frac{\partial\theta}{\partial\Phi'_1} \delta\Phi'_1 \right)^2 + \left(\frac{\partial\theta}{\partial\Phi'_2} \delta\Phi'_2 \right)^2 \right]} \\ &= \sqrt{\frac{BW}{2\Phi'}}, \end{aligned} \quad (2.163)$$

which we can write as root-mean-square uncertainty per $\text{Hz}^{1/2}$ as

$$\delta\langle\theta\rangle_{\text{rms}} = \sqrt{\frac{1}{2\Phi'}}. \quad (2.164)$$

We combine Equations 2.87 and 2.164 to give the fluctuation in the measurement of the atomic polarization,

$$\delta\langle P_x \rangle_{\text{rms}} = \frac{2}{\pi \ln r_e c f \sqrt{2\Phi'} \text{Im}[\mathcal{V}(\nu - \nu_0)]}, \quad (2.165)$$

where for simplicity we only consider optical rotation due to the D1 transition.

2.8.3 Light-Shift Noise

The fictitious magnetic field B_{LS} produced by the ac Stark light shift, as described by Equation 2.111, is zero for a linearly polarized probe beam. However, quantum fluctuations in the ellipticity of the beam result in fluctuations in the light shift field experienced by the alkali atoms. The polarization s of the probe beam is given by the fluxes Φ'_- of σ^- photons and Φ'_+ of σ^+ photons in the beam:

$$s = \frac{\Phi'_+ - \Phi'_-}{\Phi'_+ + \Phi'_-}. \quad (2.166)$$

¹³ Recall that we have defined the flux as the number of photons per unit time.

For nominally linearly polarized light, $\Phi'_- \approx \Phi'_+$, and the fluctuation in the number of photons of each helicity is

$$\delta\Phi'_- = \delta\Phi'_+ = \sqrt{\frac{\Phi'}{2}}, \quad (2.167)$$

where $\Phi' = \Phi'_- + \Phi'_+$ is the total photon flux, so the fluctuation in the probe beam polarization is

$$\begin{aligned} \delta\langle s \rangle &= \sqrt{2BW \left[\left(\frac{\partial s}{\partial \Phi'_-} \delta\Phi'_- \right)^2 + \left(\frac{\partial s}{\partial \Phi'_+} \delta\Phi'_+ \right)^2 \right]} \\ &= \sqrt{\frac{2BW}{\Phi'}}. \end{aligned} \quad (2.168)$$

Combining Equations 2.111 and 2.168, we get the light-shift noise δB_{lsn} given by the fluctuating fictitious field per unit bandwidth along the probing axis due to the light shift,¹⁴

$$\delta B_{\text{lsn}} = \frac{\pi r_e c f \sqrt{2\Phi'}}{(2I+1)\gamma A} \text{Im}[\mathcal{V}(\nu - \nu_0)], \quad (2.169)$$

where $2I+1=\gamma^e/\gamma$ is the reduction of the alkali atom gyromagnetic ratio compared to that of the bare electron, A is the area of the probe beam, and again we consider only the D1 transition for simplicity. Quantum noise due to light shift fluctuations is considered in much greater detail by Fleischhauer *et al.* (2000).

2.9 The Density Matrix

In an atomic magnetometer, we typically monitor the spins of 10^{12} - 10^{14} or more alkali atoms, each of which can be described by a wavefunction. However, if we wish to model the time evolution of the ensemble of atomic spins, it is impossible to model the evolution of so many independent wavefunctions. Instead, we require another method of determining the statistical average of the spin operator. In addition, the ensemble of atoms

¹⁴ Note that in Equation 2.111 we use the flux per unit area $\Phi \sim \Phi'/A$ rather than the total flux Φ' , where A is the area of the probe beam and we assume an approximately constant beam intensity profile.

is generally not in a pure state, such as when all atoms are in the identical state $|F, m_F\rangle$, but is instead in a mixed state and as such can not be described using an overall wavefunction. We therefore describe the state of the ensemble of spins using the density matrix, which characterizes both the populations of the ground-state levels and the coherences between the levels. General introductions to the density matrix are given by [Fano \(1957\)](#) and [Auzinsh *et al.* \(2009\)](#).

For an ensemble of N atoms, each can be described by an individual wavefunction $|\psi_n\rangle$, where $n = \{1, 2, \dots, N\}$. The density operator is defined as

$$\rho = \frac{1}{N} \sum_n |\psi_n\rangle \langle \psi_n|. \quad (2.170)$$

We describe the atoms in the basis of states $|j\rangle$, where j refers to some complete set of quantum numbers. Then $\langle j|j\rangle$ is the population of a particular state, and the total population is normalized, *i.e.*,

$$\sum_j \langle j|j\rangle = \sum_j |j\rangle \langle j| = 1. \quad (2.171)$$

The average value over the ensemble for an observable X is given by

$$\begin{aligned} \langle X \rangle &= \frac{1}{N} \sum_n \langle \psi_n | X | \psi_n \rangle \\ &= \frac{1}{N} \sum_n \sum_j \langle \psi_n | X | j \rangle \langle j | \psi_n \rangle \\ &= \text{Tr}[X\rho] = \text{Tr}[\rho X]. \end{aligned} \quad (2.172)$$

The evolution of the atomic wavefunction is given by the Schrödinger equation and its Hermite conjugate,

$$i\hbar \frac{d}{dt} |\psi_n\rangle = \mathcal{H} |\psi_n\rangle \quad (2.173)$$

$$-i\hbar \frac{d}{dt} \langle \psi_n | = \langle \psi_n | \mathcal{H}, \quad (2.174)$$

where $\mathcal{H}$ is the atomic Hamiltonian. The time evolution of the density matrix is therefore given by the Liouville equation,

$$\begin{aligned}
 \frac{d}{dt}\rho &= \frac{1}{N} \sum_n \frac{d}{dt} |\psi_n\rangle \langle \psi_n| \\
 &= \frac{1}{N} \sum_n \left[\left(\frac{d}{dt} |\psi_n\rangle \right) \langle \psi_n| + |\psi_n\rangle \left(\frac{d}{dt} \langle \psi_n| \right) \right] \\
 &= \frac{1}{N} \sum_n \left[\frac{1}{i\hbar} \mathcal{H} |\psi_n\rangle \langle \psi_n| - \frac{1}{i\hbar} |\psi_n\rangle \langle \psi_n| \mathcal{H} \right] \\
 &= \frac{1}{i\hbar} [\mathcal{H}, \rho],
 \end{aligned} \tag{2.175}$$

where we denote the commutator $[\mathcal{H}, \rho] = \mathcal{H}\rho - \rho\mathcal{H}$.

The density operator can be written in the form of a matrix with elements given by

$$\rho_{jk} = \langle k | \rho | j \rangle = \frac{1}{N} \sum_n \langle k | \psi_n \rangle \langle \psi_n | j \rangle. \tag{2.176}$$

To give a sense of the physical meaning of the density matrix elements, we consider the wavefunction of a spin-1/2 particle written in the form of a spinor,

$$|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}; \quad \langle \psi| = (c_1^* \ c_2^*), \tag{2.177}$$

so that the density matrix is

$$\rho = \begin{pmatrix} |c_1|^2 & c_1 c_2^* \\ c_1^* c_2 & |c_2|^2 \end{pmatrix}. \tag{2.178}$$

We identify the diagonal element $\rho_{jj} = \langle j | \rho | j \rangle$ as the population of the state $|j\rangle$; we see from Equation 2.171 that $\text{Tr}[\rho]=1$, so that the total population of all states is normalized. The ground-state sublevel number densities $\rho(\pm 1/2)$ described in Section 2.3 are thus given by the matrix elements $\rho_{\pm 1/2, \pm 1/2} = \langle \pm 1/2 | \rho | \pm 1/2 \rangle$. The off-diagonal elements ρ_{jk} characterize the coherence between states $|j\rangle$ and $|k\rangle$. As an example, we solve for the spin

component $\langle S_z \rangle$ using Equation 2.172 and the appropriate Pauli spin matrix:

$$\begin{aligned}\langle S_z \rangle &= \text{Tr}[\rho S_z] \\ &= \text{Tr} \left[\rho \cdot \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix} \right] \\ &= \frac{1}{2} (\rho(+1/2) - \rho(-1/2)).\end{aligned}\tag{2.179}$$

We can similarly solve for the average value of any observable over the ensemble of alkali atoms, provided that we know the complete density matrix at a given time.

In the normal operating regime with the Zeeman splitting much smaller than the hyperfine splitting ($\hbar\omega_L \ll E_{\text{hf}}$), the electron and nuclear spins are strongly coupled and the atomic state can be characterized as $|F, m_F\rangle$, although this is only an approximation. In actuality, the atom is not in an eigenstate of the $|F, m_F\rangle$ basis in a finite magnetic field, since the Zeeman interaction mixes the $|I + 1/2, m_F\rangle$ and $|I - 1/2, m_F\rangle$ states.¹⁵ Appendix B describes the calculation of the physical eigenstates of the alkali atoms in a finite magnetic field, as well as the action of electron and nuclear spin operators on $|F, m_F\rangle$ states. In general, we can use the atomic density matrix defined by

$$\rho_{\bar{F}, m_{\bar{F}}, \bar{F}', m_{\bar{F}}'} = \langle \bar{F}', m_{\bar{F}}' | \rho | \bar{F}, m_{\bar{F}} \rangle,\tag{2.180}$$

where we use the physical eigenstates $|\bar{F}, m_{\bar{F}}\rangle$ described in Appendix B. If there are no coherences between the two ground-state hyperfine levels,¹⁶ then we may write the atomic density matrix in this basis as

$$\rho = \begin{bmatrix} \rho_a & 0 \\ 0 & \rho_b \end{bmatrix},\tag{2.181}$$

¹⁵ Similarly the atom is not in an eigenstate of the $|m_I, m_s\rangle$ basis, since the hyperfine coupling mixes the $|m_I, m_s\rangle$ and $|m_I \pm 1, m_s \mp 1\rangle$ states. Since the hyperfine interaction is much larger than the Zeeman interaction, $|F, m_F\rangle$ is a much better approximation of the physical state than $|m_I, m_s\rangle$.

¹⁶ In other words, there are no external excitations at the hyperfine frequency $\omega_{\text{hf}} = E_{\text{hf}}/\hbar$. In an atomic clock, microwaves at this frequency are applied to the alkali vapor in order to measure the hyperfine splitting, so this approximation can not be made.

where ρ_a is a matrix that spans only the states in the upper hyperfine level with $\bar{F} = I + 1/2$ and $m_{\bar{F}} = \{-I - 1/2, -I + 1/2, \dots, I - 1/2, I + 1/2\}$, and ρ_b spans only the states in the lower hyperfine level with $\bar{F} = I - 1/2$ and $m_{\bar{F}} = \{-I + 1/2, -I + 3/2, \dots, I - 3/2, I - 1/2\}$.

Grossetête (1964) showed that the density matrix of an alkali atom may be written in terms of a purely nuclear part,

$$\varphi = \frac{1}{4}\rho + \mathbf{S} \cdot \rho \mathbf{S}, \quad (2.182)$$

and an electron-polarized part,

$$\Theta \cdot \mathbf{S} = \frac{3}{4}\rho - \mathbf{S} \cdot \rho \mathbf{S} = (\mathbf{S}\rho + \rho\mathbf{S} - 2i\mathbf{S} \times \rho\mathbf{S}) \cdot \mathbf{S}, \quad (2.183)$$

where Θ is a nuclear operator vector, such that the total density matrix is

$$\rho = \varphi + \Theta \cdot \mathbf{S}. \quad (2.184)$$

Using this notation, the overall evolution of the density matrix is given by Appelt *et al.* (1998) as

$$\begin{aligned} \frac{d}{dt}\rho = & \frac{1}{i\hbar} [\mathcal{H}, \rho] + R_{SE} [\varphi (1 + 4\langle \mathbf{S} \rangle \cdot \mathbf{S}) - \rho] \\ & + R_{rel} [\varphi - \rho] + R_{OP} [\varphi (1 + 2\mathbf{s} \cdot \mathbf{S}) - \rho] + D\nabla^2\rho. \end{aligned} \quad (2.185)$$

The first term in this equation is the evolution due to the free-atom Hamiltonian as described by Equation 2.175 and includes the effects of the hyperfine coupling and the Zeeman interaction with external magnetic fields. The second term characterizes the effect of spin-exchange collisions between alkali atoms; the distinction between $\langle \mathbf{S} \rangle$ and $\mathbf{S}$ is that the former is an expectation value, while the latter is an operator that can be applied to a particular state $|\bar{F}, m_{\bar{F}}\rangle$. The third term describes the spin relaxation processes that destroy the electron polarization while leaving the nuclear polarization unaffected; the effect is to drive the electron-polarization part of the density matrix to zero so that the density

matrix is given only by its nuclear part, $\rho \rightarrow \varphi$. The fourth term characterizes the effect of optical pumping. The final term describes the diffusion of spins as given by the diffusion equation, Equation 2.146, and should be used when the pumping and/or relaxation rates are functions of position within the cell. Neither the nuclear slowing-down factor q nor the spin-exchange broadening factor q_{SE} need to be included in the various rates in Equation 2.185, as their effects arise naturally in the solution. In principle the evolution of the density matrix can be solved numerically for any system of atomic spins, in order to simulate the behavior of a magnetometer under any operating conditions. In particular, we use the density matrix to model a scalar magnetometer in Section 3.4 and a SERF magnetometer in Section 7.3.3.

Chapter 3

Scalar Magnetometry: Quantum Revival Beats

SCALAR ATOMIC MAGNETOMETERS measure the precession frequency of alkali atoms in order to determine the amplitude of the ambient magnetic field. However, the small nonlinear dependence of the ground-state energy levels on the field amplitude splits the Zeeman resonance into multiple transitions. Beating between the closely-spaced transitions results in periodic collapses and revivals of the spin polarization expectation value, causing loss of maximum polarization, as well as shifts of the magnetometer measurement that depend on the magnetic field orientation. We describe synchronous optical pumping of the alkali quantum revival beats by double modulation of the optical pumping light at both the Larmor and revival frequencies. We show that synchronous pumping increases spin polarization, and thus magnetometer sensitivity, and greatly reduces orientation-dependent measurement errors. We expect that synchronous pumping of revivals can also be used to improve the precision and accuracy of spectroscopic measurements on other multilevel systems. The work described in this chapter was previously published in [Seltzer *et al.* \(2007\)](#).

3.1 Scalar Measurement of the Magnetic Field

Scalar magnetometers determine the amplitude of the ambient magnetic field by measuring the Larmor precession frequency of the atomic spins, given by

$$|B| = \frac{\omega_L}{\gamma} . \quad (3.1)$$

They are therefore highly insensitive to the orientation within the magnetic field, except for two minor effects: heading errors and dead zones. Heading errors are small, orientation-dependent shifts of the measured resonance frequency, which we discuss in detail in Section 3.2.2. Dead zones are particular orientations of the magnetometer within the field that suffer from reduced (or zero) signal, resulting in a decrease of sensitivity. The dead zone orientations depend on the details of the magnetometer operation, and typically a rotation of 90° from the nominal operating condition is necessary for the magnetometer to completely lose sensitivity. Small changes from the nominal orientation have relatively little effect on the performance of a scalar magnetometer.

In comparison, vector sensors such as the SERF magnetometer are inherently sensitive to one or more individual components of the field and are therefore highly dependent on orientation (see Section 5.3). Small changes in orientation can highly degrade the performance of a vector sensor, and vector magnetometers operating as part of a feedback system may be unable to adjust to fast rotations. Scalar magnetometers are thus preferable for operation on mobile platforms such as vehicles, where the field orientation can be highly unstable. However, a vector measurement provides more information about the field and allows for easier localization of a magnetic source. There are several techniques for retrieving the vector components of the field from scalar measurements, including the application of a bias field and measuring the tilt from the preferred axis (Alexandrov *et al.*, 2004), and the application of a rotating field about the atoms (Vershovskii, 2006).

A scalar magnetometer requires coherent precession of the spin ensemble, so a resonant excitation must be applied in order to force some large fraction of the atoms to precess together with a common phase. Otherwise the phase of individual atoms is random, and the total transverse spin of the ensemble averages to zero. There are numerous techniques for creating this excitation, but in general each involves a feedback system to keep the excitation frequency matched to the Larmor precession frequency. This can be accomplished by monitoring the out-of-phase resonance signal and keeping the response locked to the central zero-crossing. In particular, we now consider the two techniques used in this thesis to detect quantum revival beats, namely radio-frequency excitation and optical excitation. We also briefly consider the fundamental quantum limit on the sensitivity of a scalar magnetometer.

3.1.1 Radio-Frequency Excitation

As shown in Section 2.6, an oscillating radio-frequency (rf) magnetic field applied transverse to the static ambient field will induce resonant, coherent precession of the polarized atomic spins if the oscillation frequency is approximately equal to the Larmor frequency (Bloch, 1946; Hahn, 1950). The use of radio-frequency excitation to measure the Larmor frequency by monitoring the precessing transverse spin components was first proposed by Dehmelt (1957a,b), and the theory and first experimental demonstration were presented by Bell and Bloom (1957). The analysis in Section 2.6 only holds for small rf field amplitude, such that the magnetometer response is linear in the amplitude. In general we must apply the rotating wave approximation and consider the spin response in the co-rotating reference frame; see for example Abragam (1961).

As in Section 2.6, we consider an oscillating field $\mathbf{B}' = B' \cos(\omega t) \hat{y}$, which has two counter-propagating rotational components at frequencies $\pm\omega$ with amplitudes $B'/2$. The

ambient magnetic field is given by

$$\mathbf{B}_0 = \frac{\omega_0}{\gamma} \hat{\mathbf{z}}. \quad (3.2)$$

We assume that the magnetic linewidth is significantly smaller than the resonance frequency, $\Delta\omega \ll \omega_0$, and that the oscillation frequency ω is much closer to $+\omega_0$ than to $-\omega_0$. Shifting to the reference frame that is co-rotating with the precessing spins has three effects: First, we may make the rotating wave approximation and neglect all oscillations that are at frequencies far from the rotation frequency, including particularly the counter-rotating component of the oscillating field at $-\omega$. Second, the static field $\mathbf{B}_0$ is replaced by an effective field

$$\tilde{\mathbf{B}}_0 = \left(B_0 - \frac{\omega}{\gamma} \right) \hat{\mathbf{z}}. \quad (3.3)$$

Third, the co-rotating component of the oscillating field is static in this reference frame. The total effective magnetic field is therefore given by

$$\tilde{\mathbf{B}} = \frac{(\omega_0 - \omega)}{\gamma} \hat{\mathbf{z}} + \frac{B'}{2} \hat{\mathbf{y}}. \quad (3.4)$$

The Bloch equation in this reference frame is

$$\frac{d}{dt} \tilde{\mathbf{S}} = \gamma \tilde{\mathbf{B}} \times \tilde{\mathbf{S}} - \frac{\tilde{S}_x \hat{\mathbf{x}} + \tilde{S}_y \hat{\mathbf{y}}}{T_2} - \frac{(S_z - S_0) \hat{\mathbf{z}}}{T_1}, \quad (3.5)$$

where T_1 and T_2 are the longitudinal and transverse spin coherence lifetimes, and $S_0 = S_0 \hat{\mathbf{z}}$ is the equilibrium atomic spin polarization in the absence of the oscillating excitation field, given by Equation 2.123. We find the steady-state solution to Equation 3.5 by setting $\frac{d}{dt} \tilde{\mathbf{S}} = 0$ and solving for the three individual components of $\tilde{\mathbf{S}}$:

$$\tilde{S}_x = \frac{S_0(\gamma B'/2)T_2}{1 + (\gamma B'/2)^2 T_1 T_2 + (\omega - \omega_0)^2 T_2^2} \quad (3.6)$$

$$\tilde{S}_y = \frac{-S_0(\omega - \omega_0)(\gamma B'/2)T_2^2}{1 + (\gamma B'/2)^2 T_1 T_2 + (\omega - \omega_0)^2 T_2^2} \quad (3.7)$$

$$S_z = \frac{S_0(1 + (\omega - \omega_0)^2 T_2^2)}{1 + (\gamma B'/2)^2 T_1 T_2 + (\omega - \omega_0)^2 T_2^2}. \quad (3.8)$$

The probe beam signal is proportional to S_x as measured in the lab frame, so we shift back from the rotating frame:

$$\begin{aligned} S_x &= \tilde{S}_x \cos(\omega t) - \tilde{S}_y \sin(\omega t) \\ &= \frac{1}{2} S_0 \gamma B' \frac{T_2 \cos(\omega t) + (\omega - \omega_0) T_2^2 \sin(\omega t)}{1 + (\gamma B'/2)^2 T_1 T_2 + (\omega - \omega_0)^2 T_2^2}. \end{aligned} \quad (3.9)$$

As in Section 2.6, the measured signal has a Lorentzian shape with an absorptive in-phase component and a dispersive out-of-phase component. The amplitude of the transverse polarization is proportional to the strength of the applied rf field, and large fields are necessary to measure the coherence frequency precisely. However, comparing Equations 2.129 and 3.9, we see that the magnetic linewidth is actually given by

$$\Delta\omega = \frac{1}{T_2} \sqrt{1 + (\gamma B'/2)^2 T_1 T_2}. \quad (3.10)$$

Large rf excitation amplitudes lead to broadening of the resonance line and decreased magnetometer sensitivity. The amplitude should therefore be chosen to enable a large coherence signal without greatly affecting the linewidth. The magnetometer response given by Equation 3.9 is proportional to the polarization S_0 along the magnetic field direction; the signal is therefore largest when the pumping axis and magnetic field are parallel, and it drops to zero as the axis and field become orthogonal.

3.1.2 Optical Excitation

If the magnetic field lies along an axis orthogonal to the pumping direction, then each atom begins to precess away from the pumping axis immediately after being polarized. Individual atoms are polarized at different instances in time and so precess incoherently with random phases. If the precession period is much smaller than the spin lifetime, then the ensemble has zero average polarization, such as in the case of a scalar magnetometer operating in the Earth's magnetic field. However, if the pumping rate is modulated at nearly the Larmor frequency, then the atoms are pumped at approximately the same time.

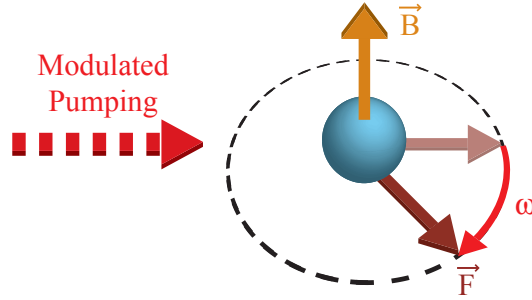


Figure 3.1: Principle of operation of a Bell-Bloom magnetometer: The pumping rate is modulated at the Larmor frequency so that the atoms are pumped for a short fraction of each period, causing the spins to precess coherently in a magnetic field.

They precess coherently with the same phase, and pumping does not occur again until the atomic spins complete one complete precession and are again aligned with the pumping direction. This results in optically driven spin precession and was first demonstrated by [Bell and Bloom \(1961\)](#); magnetometers operating on this principle are sometimes referred to as being of the Bell-Bloom type. This effect can be achieved by modulating either the frequency of the pump beam, so that the light is tuned off-resonance during the time periods when pumping is undesirable, or the amplitude of the beam. The principle of operation of a Bell-Bloom magnetometer is illustrated in [Figure 3.1](#).

We consider a modulated pumping rate given by

$$R(t) = \frac{1}{2}R_0 [1 + \cos(\omega t)] = \frac{1}{4}R_0 [2 + e^{i\omega t} + e^{-i\omega t}], \quad (3.11)$$

varying between 0 and R_0 at a frequency ω near the Larmor frequency ω_0 . We assume that the magnetic field is orthogonal to both the pump and probe directions, so that $\mathbf{B} = (\omega_0/\gamma) \hat{\mathbf{y}}$. Following [Section 3.1.1](#) we consider the spins in the co-rotating reference frame, so that we may ignore both the constant and counter-rotating terms in the pumping rate given by [Equation 3.11](#). We write the transverse components of the spin in complex form, as in [Section 2.6](#):

$$\tilde{S} = \tilde{S}_z + i\tilde{S}_x, \quad (3.12)$$

with time evolution given by the Bloch equation,

$$\frac{d}{dt}\tilde{S} = i(\omega_0 - \omega)\tilde{S} - \frac{\tilde{S}}{T_2} + \frac{1}{4}R_0S_0, \quad (3.13)$$

where S_0 is the polarization in zero magnetic field. If the polarization lifetime is much longer than the precession period, then T_2 is given by the sum of the relaxation time R_{rel} and the time-averaged pumping rate $R_0/2$.

For excitation frequencies near the Larmor frequency, the spins tend to an equilibrium polarization in the rotating frame given by the steady-state solution to Equation 3.13,

$$\tilde{S} = \frac{(R_0/4)S_0}{\Delta\omega - i(\omega - \omega_0)}, \quad (3.14)$$

where $\Delta\omega = 1/T_2$ is the magnetic linewidth. The probe beam measures S_x in the lab frame, given by

$$\begin{aligned} S_x &= \tilde{S}_x \cos(\omega t) + \tilde{S}_z \sin(\omega t) \\ &= \frac{1}{4}R_0S_0 \frac{\Delta\omega \sin(\omega t) + (\omega - \omega_0) \cos(\omega t)}{(\Delta\omega)^2 + (\omega - \omega_0)^2}. \end{aligned} \quad (3.15)$$

The magnetometer response is again characterized by a Lorentzian lineshape, although in this case the dispersive signal is in phase with the driving excitation. The absorptive signal is 90° out of phase, since the atoms take one-quarter of a precession period after being pumped to align with the probing direction. The precession signal is largest when the magnetic field is orthogonal to both the pump and probe directions, and it drops to zero as the field becomes parallel to either beam. The modulation scheme described by Equation 3.11 is not ideal because there is a finite pumping rate throughout the Larmor period; it is preferable to modulate the pump beam so that optical pumping is active with a constant rate for a small fraction of the period and turned off otherwise, so that all atoms truly have about the same phase. The duty cycle should be chosen such that the time-averaged pumping rate gives the optimal equilibrium polarization S_0 .

3.1.3 Fundamental Sensitivity

Smullin *et al.* (2006) consider the fundamental sensitivity limit of a scalar magnetometer, as determined by the three sources of quantum fluctuations described in Section 2.8. They conclude that under optimal conditions $T_2 \sim (R_{\text{SE}} R_{\text{SD}})^{-1/2}$, and the smallest magnetic field that can be detected per unit bandwidth is

$$\delta B_{\min} = \frac{0.94}{\gamma} \sqrt{\frac{\sigma_{\text{SE}} \bar{v}}{V} \left(1 + \frac{1}{\sqrt{2\eta}} \right)}, \quad (3.16)$$

where σ_{SE} is the alkali-alkali spin-exchange cross-section, $\bar{v}$ is the thermal velocity, V is the active measurement volume, and $\eta \sim 0.5$ is the quantum efficiency of the photodiodes used to detect the probe beam. The fundamental sensitivity of a scalar magnetometer thus scales as $\sigma_{\text{SE}}^{1/2}$, compared to $(\sigma_{\text{SE}} \sigma_{\text{SD}})^{1/4}$ for a radio-frequency magnetometer (see Section 4.1.2) and $\sigma_{\text{SD}}^{1/2}$ for a SERF magnetometer (see Section 5.2). The alkali-alkali spin-destruction cross-section σ_{SD} is generally several orders of magnitude smaller than the spin-exchange cross-section, as shown in Table A.2. An ideal scalar magnetometer is therefore less sensitive than an ideal radio-frequency or SERF magnetometer, although as noted earlier the scalar magnetometer has certain practical advantages.

Scalar magnetometer performance can be improved using light narrowing (see Section 4.1.1) to partially suppress spin-exchange broadening if the sensitivity is limited by technical noise, but the sensitivity is ultimately limited by the spin-exchange rate regardless of how narrow the magnetic linewidth becomes. A precise measurement of the atomic precession frequency requires the detection of a large precession signal; it is therefore necessary to generate a large spin coherence between at least two Zeeman sublevels. A fully stretched state with complete polarization provides very narrow linewidth but zero precession signal, while the maximum dispersion curve slope¹ is achieved with lower polarization than would be necessary for light narrowing. Ideally, the alkali density should

¹ Recall from Section 2.6 that the sensitivity is optimized by maximizing the slope of the resonant dispersion curve.

be high enough to achieve a large coherence signal without unnecessarily broadening the magnetic resonance linewidth due to spin-exchange collisions. A fundamental sensitivity of about $1 \text{ fT}/\sqrt{\text{Hz}}$ can be attained using an active measurement volume of 1 cm^3 .

3.2 The Nonlinear Zeeman Splitting

The general assertion that the precession frequency is directly proportional to the magnetic field amplitude is not strictly true. The energy of the $|F, m_F\rangle$ level depends on higher-order powers of Bm_F , leading to splitting of the Zeeman resonance into multiple lines. The ground-state Hamiltonian is given by

$$\mathcal{H} = A_J \mathbf{I} \cdot \mathbf{J} + g_s \mu_B \mathbf{S} \cdot \mathbf{B} - g_I \mu_N \mathbf{I} \cdot \mathbf{B}, \quad (3.17)$$

where $A_J = 2\hbar\omega_{\text{hf}}/(2I + 1)$ is the hyperfine coupling constant, g_I is the nuclear g-factor of the atom, and μ_N is the nuclear magneton. Values for the nuclear g-factor for the alkali isotopes commonly used for magnetometry are given in Table A.1. The first term in Equation 3.17 describes the hyperfine interaction between the electron and nuclear spins, while the second and third terms describe the Zeeman interactions of the electron and nuclear spins, respectively, with the magnetic field. Recall that in the ground state, $\mathbf{J} = \mathbf{S}$.

We define the raising and lowering operators,

$$J_{\pm} = J_x \pm iJ_y \quad (3.18)$$

$$J_{\pm}|J, m_J\rangle = \sqrt{J(J+1) - m_J(m_J \pm 1)} |J, m_J \pm 1\rangle, \quad (3.19)$$

with a similar definition for the operator $I_{\pm}$. If we select the $\hat{\mathbf{z}}$ axis to lie parallel to the magnetic field,² then we may rewrite Equation 3.17:

$$\mathcal{H} = A_J \left[I_z J_z + \frac{1}{2} (I_+ J_- + I_- J_+) \right] + g_s \mu_B B_z J_z - g_I \mu_N B_z I_z. \quad (3.20)$$

² This choice has nothing to do with the direction of the pump beam. From symmetry arguments, the energy levels can not depend on the polarization axis.

The Hamiltonian includes electron and nuclear spin operators, so it is necessary to decompose the state $|F, m_F\rangle$ into the states $|m_I, m_J\rangle = |m_F \pm 1/2, \mp 1/2\rangle$ (see Appendix B). The energy of the state $|F, m_F\rangle$ is therefore given by the eigenvalues of the matrix with elements given by

$$\begin{aligned} \mathcal{H}_{m_I, m_J, m'_I, m'_J} &= \langle m'_I, m'_J | \mathcal{H} | m_I, m_J \rangle \\ &= (A_J m_I m_J + g_s \mu_B B m_J - g_I \mu_N B m_I) \delta_{m_I, m'_I} \delta_{m_J, m'_J} \\ &\quad + \frac{1}{2} A_J \sqrt{[I(I+1) - m'_I(m'_I+1)][J(J+1) - m'_J(m'_J-1)]} \delta_{m_I, m'_I+1} \delta_{m_J, m'_J-1} \\ &\quad + \frac{1}{2} A_J \sqrt{[I(I+1) - m'_I(m'_I-1)][J(J+1) - m'_J(m'_J+1)]} \delta_{m_I, m'_I-1} \delta_{m_J, m'_J+1} \end{aligned} \quad (3.21)$$

Setting $J = 1/2$, we note that both of the square-roots in Equation 3.21 can be written as $\sqrt{(I - m_F + 1/2)(I + m_F + 1/2)}$. The Hamiltonian can therefore be written as

$$\begin{aligned} \mathcal{H}_{m_I, m_J, m'_I, m'_J} &= \begin{bmatrix} \frac{1}{2} A_J (m_F - \frac{1}{2}) + \frac{1}{2} g_s \mu_B B - (m_F - \frac{1}{2}) g_I \mu_N B & \frac{1}{2} A_J \sqrt{(I - m_F + \frac{1}{2})(I + m_F + \frac{1}{2})} \\ \frac{1}{2} A_J \sqrt{(I - m_F + \frac{1}{2})(I + m_F + \frac{1}{2})} & -\frac{1}{2} A_J (m_F + \frac{1}{2}) - \frac{1}{2} g_s \mu_B B - (m_F + \frac{1}{2}) g_I \mu_N B \end{bmatrix} \\ &= -\frac{1}{4} A_J - g_I \mu_N B m_F \\ &\quad + \frac{2I+1}{4} A_J \begin{bmatrix} \frac{2}{(2I+1)A_J} (g_s \mu_B + g_I \mu_N) B + \frac{2m_F}{2I+1} & \frac{2}{2I+1} \sqrt{(I - m_F + \frac{1}{2})(I + m_F + \frac{1}{2})} \\ \frac{2}{2I+1} \sqrt{(I - m_F + \frac{1}{2})(I + m_F + \frac{1}{2})} & -\frac{2}{(2I+1)A_J} (g_s \mu_B + g_I \mu_N) B - \frac{2m_F}{2I+1} \end{bmatrix}. \end{aligned} \quad (3.22)$$

We define

$$x \equiv \frac{2(g_s \mu_B + g_I \mu_N) B}{(2I+1) A_J} = \frac{(g_s \mu_B + g_I \mu_N) B}{\hbar \omega_{\text{hf}}}, \quad (3.23)$$

so that the energy of the state $|F, m_F\rangle$ is given by the eigenvalue of the Hamiltonian described in Equation 3.22:

$$E(F = I \pm 1/2, m_F) = -\frac{\hbar \omega_{\text{hf}}}{2(2I+1)} - g_I \mu_N B m_F \pm \frac{\hbar \omega_{\text{hf}}}{2} \sqrt{x^2 + \frac{4x m_F}{2I+1} + 1}. \quad (3.24)$$

This formula is known as the Breit-Rabi Equation, after its discoverers (Breit and Rabi, 1931).

We readily see from this formula that the energy levels are nonlinear in the magnetic field strength. Figure 3.2 shows the energies of the ground-state $|F, m_F\rangle$ levels of ^{39}K

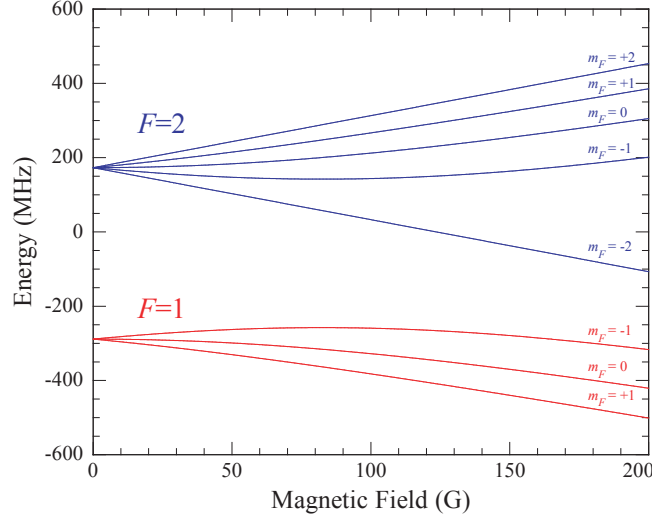


Figure 3.2: Energy levels of the $|F, m_F\rangle$ ground state levels of ^{39}K , as given by Equation 3.24. The Zeeman transitions are between adjacent sublevels with $\Delta m_F = \pm 1$, and it can be clearly seen that in a finite magnetic field the transition frequencies within each hyperfine state become unequal due to nonlinear dependence on the field amplitude. Plots showing this type of data are referred to as Breit-Rabi diagrams.

given by the Breit-Rabi Equation. The nonlinear dependence on the field amplitude is clearly evident for large fields $B \sim 100 - 200$ G, but even at the level of the Earth's field the nonlinearity is large enough to affect the performance of the magnetometer. We note that $x \ll 1$ for geomagnetic fields ($B \sim 0.5$ G), so that we may expand Equation 3.24 about $x = 0$:

$$\begin{aligned}
 E(F = I \pm 1/2, m_F) = & \hbar\omega_{\text{hf}} \left(-\frac{1}{2(2I+1)} \pm \frac{1}{2} \right) \\
 & - g_I \mu_N B m_F \pm \frac{(g_s \mu_B + g_I \mu_N) B m_F}{2I+1} \\
 & \pm \frac{(g_s \mu_B + g_I \mu_N)^2 B^2}{4\hbar\omega_{\text{hf}}} \mp \frac{(g_s \mu_B + g_I \mu_N)^2 B^2 m_F^2}{(2I+1)^2 \hbar\omega_{\text{hf}}} \\
 & \mp \frac{(g_s \mu_B + g_I \mu_N)^3 B^3 m_F}{2(2I+1) \hbar^2 \omega_{\text{hf}}^2} \pm \frac{2(g_s \mu_B + g_I \mu_N)^3 B^3 m_F^3}{(2I+1)^3 \hbar^2 \omega_{\text{hf}}^2}. \quad (3.25)
 \end{aligned}$$

The constant term gives the energy shift of the level F due to the hyperfine splitting.

Following the general description of quantum revivals in multilevel systems (Robinett, 2004), we write the energy levels given by Equation 3.25 in powers of m_F , keeping only the leading B dependence in each term and ignoring the constant term:

$$E(F = I \pm 1/2, m_F) = (-g_I \mu_N \pm \mu_{\text{eff}}) B m_F \mp \frac{\mu_{\text{eff}}^2 B^2 m_F^2}{\hbar \omega_{\text{hf}}} \pm 2 \frac{\mu_{\text{eff}}^3 B^3 m_F^3}{\hbar^2 \omega_{\text{hf}}^2} \quad (3.26)$$

$$\mu_{\text{eff}} \equiv \frac{g_s \mu_B + g_I \mu_N}{2I + 1}. \quad (3.27)$$

We may precisely characterize the Larmor frequency by considering the linear dependence of the energy on the magnetic field,

$$\omega_L = \frac{(-g_I \mu_N \pm \mu_{\text{eff}}) B}{\hbar}. \quad (3.28)$$

This agrees with our earlier assertions that $\gamma = \pm g_s \mu_B / (2I + 1) \hbar$, although we see that there is a small correction due to the nuclear magnetic moment. We also see that the sign of the Larmor frequency depends on the hyperfine level, causing spins in the two levels to precess in opposite directions and leading to the spin-exchange relaxation described in Section 2.7.1. The quadratic and cubic terms in Equation 3.26 lead us to define the quantum beats revival frequency ω_{rev} and super-revival frequency ω_{suprev} as

$$\omega_{\text{rev}} = \frac{\mu_{\text{eff}}^2 B^2}{\hbar^2 \omega_{\text{hf}}} \quad (3.29)$$

$$\omega_{\text{suprev}} = \frac{2 \mu_{\text{eff}}^3 B^3}{\hbar^3 \omega_{\text{hf}}^2}. \quad (3.30)$$

The meaning of these names is made clear in Section 3.2.1. General expressions for the Larmor, revival, and super-revival frequencies are given in Table A.1 for arbitrary magnetic field, and specific values for $B=0.5$ G are given in Table 3.1.³

The effect of the nonlinear dependence of the energy on $B m_F$ is to lift the degeneracy between the different Zeeman transition frequencies in each hyperfine level. The transition frequency between the levels $|F, m_F\rangle$ and $|F, m_F - 1\rangle$ is

$$\omega_{F, m_F} = \frac{E(F, m_F) - E(F, m_F - 1)}{\hbar}. \quad (3.31)$$

³ For notational simplicity we give frequencies ν rather than angular frequencies $\omega = 2\pi \times \nu$.

	I	ν_L (kHz)	ν_{rev} (Hz)	ν_{suprev} (Hz)
^{39}K	3/2	350	266	0.4
^{41}K	3/2	350	483	1.3
^{85}Rb	5/2	233	18.0	0.003
^{87}Rb	3/2	350	17.9	0.002
^{133}Cs	7/2	175	3.3	0.0001

Table 3.1: Larmor, revival, and super-revival frequencies in the Earth's field $B = 0.5$ G for the commonly used alkali isotopes. General expressions for arbitrary field are given in Table A.1.

Adjacent Zeeman transitions are therefore separated by an amount given by Equation 3.26,

$$\omega_{F=I\pm 1/2, m_F} - \omega_{F=I\pm 1/2, m_F-1} = \mp 2\omega_{\text{rev}}, \quad (3.32)$$

where for simplicity we consider only the quadratic dependence; the cubic and higher-order terms lead to small corrections. A resonance spectrum taken at $B=0.5$ G with a Bell-Bloom scalar magnetometer using potassium at natural abundance is shown in Figure 3.3, displaying the individual Zeeman resonances.⁴ The signal is given by the out-of-phase output of a lock-in amplifier referenced to the modulation frequency, so that the $F = I \pm 1/2$ resonance signals have opposite sign. The isotope ^{41}K has a much smaller ground-state hyperfine splitting than the isotope ^{39}K , so the nonlinear Zeeman splitting is larger. Specific Zeeman transition frequencies for ^{39}K and ^{41}K at $B=0.5$ G are given in Table 3.2.

Traditionally, scalar magnetometers operate by exciting only one of the split Zeeman resonances. If the resonances are well-resolved, then this has the effect of exciting a single coherence, forcing the atoms to precess at the chosen frequency. However, the spins are inclined to undergo quantum revival beating due to precession at multiple frequencies, as described in Section 3.2.1. Single-frequency excitation works by fighting this beating effect

⁴ Sidebands due to 60 Hz noise can also be seen and are common in the experimental spectra shown in this chapter.

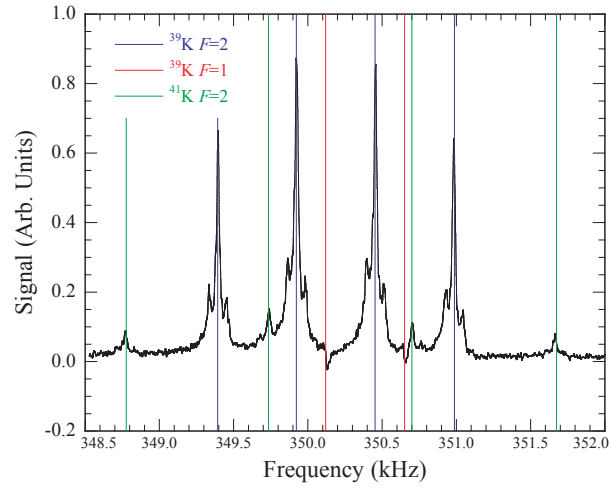


Figure 3.3: Magnetic resonance spectrum for potassium in natural abundance at $B = 0.5$ G, taken with a Bell-Bloom scalar magnetometer. The vertical lines represent the transition frequencies given in Table 3.2. Small sidebands are due to 60 Hz magnetic noise.

Zeeman transition:	^{39}K	^{41}K
$ 2, -1\rangle \rightarrow 2, -2\rangle$	350987	351675
$ 2, 0\rangle \rightarrow 2, -1\rangle$	350453	350701
$ 2, +1\rangle \rightarrow 2, 0\rangle$	349922	349735
$ 2, +2\rangle \rightarrow 2, +1\rangle$	349393	348777
$ 1, 0\rangle \rightarrow 1, -1\rangle$	-350652	-350810
$ 1, +1\rangle \rightarrow 1, 0\rangle$	-350120	-349844

Table 3.2: Individual Zeeman transition frequencies for ^{39}K and ^{41}K at $B=0.5$ G, given in units of Hz.

and results in decreased spin polarization, leading to loss of magnetometer sensitivity, as discussed by [Schwindt *et al.* \(2005\)](#) and [Acosta *et al.* \(2006\)](#).

As a digression, we note that at large magnetic fields, such that $\omega_L \gtrsim \omega_{\text{hf}}$, the Zeeman interaction between the electron spin and the magnetic field becomes comparable to or stronger than the hyperfine interaction. The electron and nuclear spins then become decoupled, and the physical eigenstates of the atom are better described by the $|m_I, m_J\rangle$ basis than the $|F, m_F\rangle$ basis (see Appendix B). The energy levels form two closely-spaced multiplets given by $m_J = \pm 1/2$, each containing $2I + 1$ states; the beginning of the transition from $F = I \pm 1/2$ multiplets to $m_J = \pm 1/2$ multiplets can be seen in Figure 3.2. Note that all of the low-field $F = I - 1/2$ levels are given at high field by $m_J = -1/2$, along with the low-field $|F = I + 1/2, m_J = -F\rangle$ level, while the remaining low-field $F = I + 1/2$ levels are given at high field by $m_J = +1/2$. In this regime $x \gg 1$, so the energy levels given by Equation 3.24 can be approximated as

$$E(m_I, m_J = \pm 1/2) \approx \hbar\omega_{\text{hf}} \left(\frac{-1}{2(2I+1)} \pm \frac{1}{2} \right) - g_I\mu_N B(m_I \pm \frac{1}{2}) \pm \frac{1}{2} (g_s\mu_B + g_I\mu_N) B. \quad (3.33)$$

The Zeeman splitting between adjacent nuclear spin states becomes completely linear in B in high field, as does the splitting between the two $m_J = \pm 1/2$ multiplets. Because the nuclear and electron spins are decoupled in high field, optical pumping polarizes the electron spins while leaving the nuclear spins unpolarized ([Liu *et al.*, 1992](#); [Augustine and Zilm, 1996](#)).

3.2.1 Quantum Revival Beats

The separated Zeeman resonances interfere with one another as the atomic spins precess, resulting in beating as shown in Figure 3.4(a). The figure illustrates free precession of ^{39}K spins fully polarized transverse to a 0.5 G magnetic field in the absence of spin relaxation,

spin exchange, or optical pumping, as determined by the time evolution of the density matrix (see Section 3.4). The spins transition in turn between each of the Zeeman sublevels as they precess, so that the ensemble polarization oscillates at all of the individual transition frequencies simultaneously. The beating creates an envelope about the fast precession at the Larmor frequency, and the polarization experiences periodic collapses and revivals with a period of $\tau_{\text{rev}} = 2\pi/\omega_{\text{rev}}$. There are two revivals per period, with the Larmor precession occurring 180° out of phase between them.

The beating envelope is further modified by the higher-order energy splittings. A comparison of Figures 3.4(a) and 3.4(b) reveals that the quantum beats of ^{39}K take on a different shape as the polarization continues to evolve under the influence of the cubic Zeeman splitting, due to nonequal separation between transitions with $\Delta m_F=2$. Figure 3.4(c) shows that the revival behavior varies on a time scale on the order of the super-revival period $\tau_{\text{suprev}} = 2\pi/\omega_{\text{suprev}}$, with the maximum polarization decreasing at certain times. As the magnetic field increases, the relative effects of successively higher-order terms in Equation 3.26 become greater⁵ and can impact the performance of the magnetometer. In general we may ignore all higher-order energy splittings with characteristic time scales (e.g., τ_{rev} for the quadratic splitting or τ_{suprev} for the cubic splitting) that are significantly longer than the spin coherence lifetime T_2 .

Quantum beats have been studied extensively; see for example [Leichtle *et al.* \(1996\)](#) and [Robinett \(2004\)](#) for the general theory of revivals in multilevel quantum systems. They have been observed in a number of diverse systems, including the Zeeman resonances of nuclear spins ([Chupp and Hoare, 1990](#); [Majumder *et al.*, 1990](#)) and atomic spinor condensates ([Kronjäger *et al.*, 2005](#)), one-atom masers ([Rempe *et al.*, 1987](#)), Rydberg states in atoms ([Yeazell *et al.*, 1990](#)), and molecular states ([Bowman *et al.*, 1989](#); [Rosca-Pruna and Vrakking,](#)

⁵ For simplicity we have only included terms up to $(Bm_F)^3$ in Equation 3.26 because the quartic splitting is generally too small to affect alkali spins in the Earth's field. However, terms up to $(Bm_F)^{2F}$ may affect the behavior of the spin polarization if the field is large enough, since there are two individual transitions with $\Delta m_F = 2F - 1$ in the F hyperfine multiplet with potentially unequal resonance frequencies.

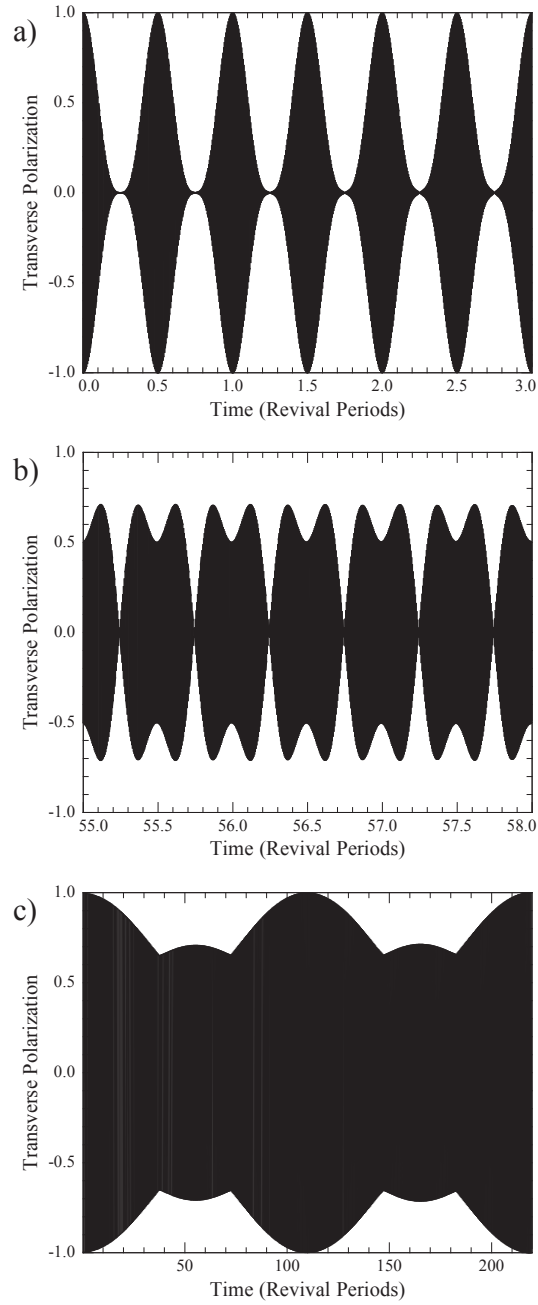


Figure 3.4: Quantum revival beats in polarized ^{39}K at 0.5 G, modeled using the time evolution of the density matrix without spin exchange or spin relaxation. **(a)** Due to interference between the separated Zeeman resonances, the polarization collapses and revives twice per revival period $\tau_{\text{rev}} = 2\pi/\omega_{\text{rev}}$, with the relative phase of the Larmor precession alternating by 180° . **(b)** The shape of the beating envelope changes at different times due to the cubic Zeeman energy splitting. **(c)** The modification of the revival behavior due to the cubic splitting has a period on the order of $\tau_{\text{suprev}} = 2\pi/\omega_{\text{suprev}}$.

2001). Revivals due to quantum beats in the ground-state Zeeman levels of alkali atoms were previously modeled by Alexandrov *et al.* (2005). Quantum beats have generally been observed in freely evolving systems; to the best of our knowledge, this work represents the first demonstration of synchronous pumping of a multilevel system with nonlinear energy splitting in order to generate a coherent superposition state.

3.2.2 Heading Errors

When the Zeeman resonance is separated into several individual, well-resolved transitions, it is necessary to select one particular transition on which to make the frequency measurement to determine the magnetic field amplitude. The choice of which transition to monitor generally depends on the details of the magnetometer operation. The calculated absorptive resonance spectrum of ^{39}K at 0.5 G for a Bell-Bloom magnetometer with the magnetic field along the $\hat{y}$ direction is shown in Figure 3.5(a) (compare to the experimentally measured spectrum shown in Figure 3.3), while the calculated dispersive resonance spectrum is shown in Figure 3.6(a). The relative heights of the resonance lines were calculated assuming equal optical pumping and spin-relaxation rates. Magnetometer operation requires keeping the excitation frequency locked to the dispersion curve zero-crossing of the selected transition, and the frequency offset due to the quadratic splitting must be taken into account in order to determine the field amplitude.

However, changes in the magnetic field orientation alter the resonance spectrum. For example, the nominal operating condition for a Bell-Bloom magnetometer is for the magnetic field to lie orthogonal to the pumping and probing directions. As the spins precess, the populations of the Zeeman sublevels⁶ $\rho(F, m_F)$ and $\rho(F, m_F - 1)$ determine the relative

⁶ The populations of the Zeeman sublevels in this case are measured along the magnetic field direction, as in the analysis of Section 3.2. For nominal operation of a Bell-Bloom magnetometer, there is zero population along the quantization axis, leading to the symmetric spectra shown in Figures 3.5(a,c,e) and 3.6(a,c).

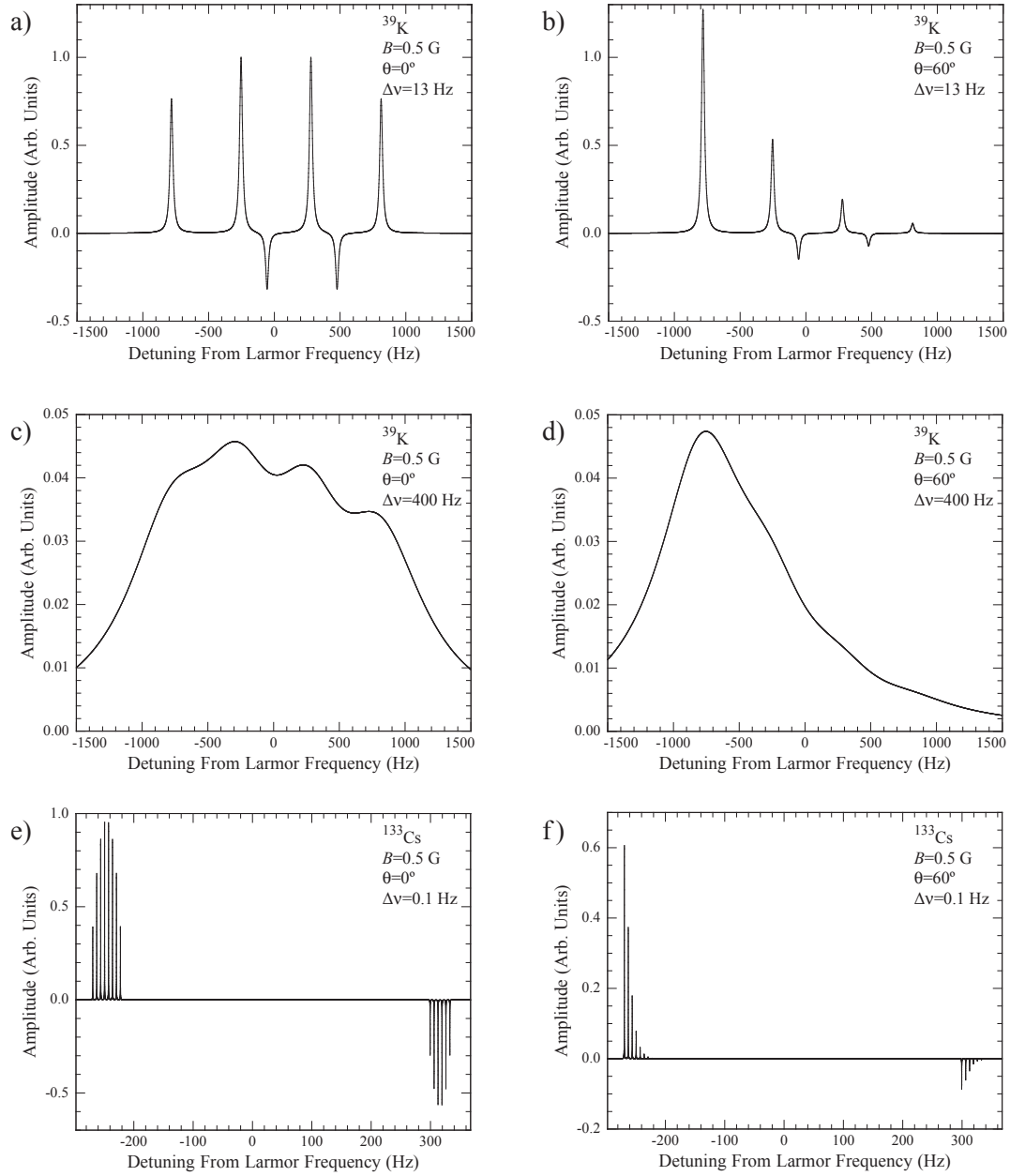


Figure 3.5: Absorptive resonance spectra for a Bell-Bloom magnetometer at 0.5 G, calculated assuming that the optical pumping rate and spin-exchange rate are equal. The field angle θ is relative to the nominal operating condition with the magnetic field orthogonal to both the pump and probe beams. As the field tilts, the spectrum becomes asymmetric, causing heading errors that grow larger as the magnetic linewidth $\Delta\nu = \Delta\omega/2\pi$ increases. The vertical scales are the same for all ^{39}K spectra, as are the scales for the ^{133}Cs spectra.

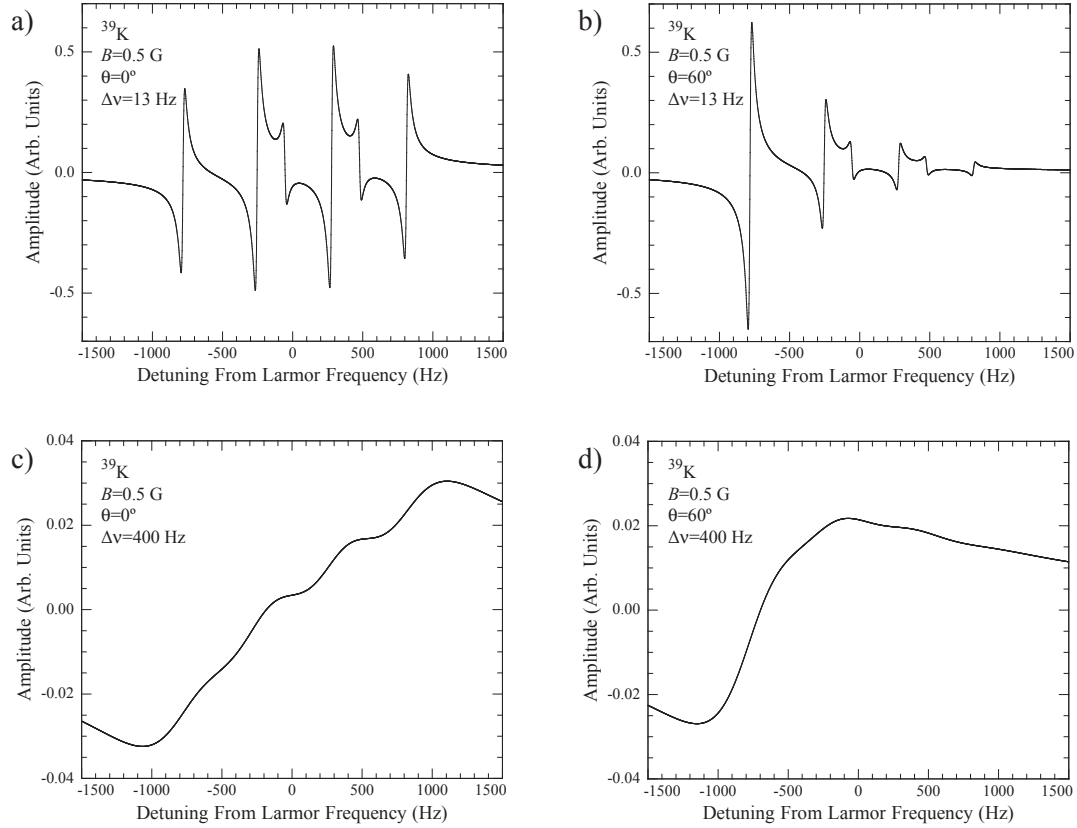


Figure 3.6: Dispersive resonance spectra for a Bell-Bloom magnetometer at 0.5 G, calculated assuming that the optical pumping rate and spin-exchange rate are equal. The field angle θ is relative to the nominal operating condition with the magnetic field orthogonal to both the pump and probe beams. As the field tilts, the spectrum becomes asymmetric, shifting the zero-crossing frequencies and causing heading errors that grow larger as the magnetic linewidth $\Delta\nu = \Delta\omega/2\pi$ increases. If the linewidth is large enough that the individual resonances become unresolved, then the spectrum takes the shape of a single dispersion curve with only one zero-crossing. The vertical scales are the same for all spectra.

strength of the transition between them. If the field tilts into the pumping direction,⁷ then the populations of the sublevels change, and thus so do the relative transition strengths. The resonance spectra that result from the magnetic field tilting by 60° from the nominal operating condition (and thus becoming oriented at 30° from the pumping direction in the $\hat{y}$ - $\hat{z}$ plane) are shown in Figures 3.5(b) and 3.6(b). The spectrum becomes highly asymmetric, with one transition gaining strength at the expense of the others. Although this is an exaggerated example, with a very large orientation change, any tilt into the pumping direction is sufficient to alter the resonance spectrum. A tilt into the direction of the probe beam does not affect the magnetometer response, since the orientation of the magnetic field with respect to the pumping axis remains unchanged, but it does decrease the probe signal. A similar analysis shows that the resonance spectrum of a scalar magnetometer with radio-frequency excitation becomes altered when the magnetic field tilts away from its nominal orientation along the pumping direction.

The change in relative transition strengths with field orientation causes slight shifts in the zero-crossing frequencies of the resonant dispersion curves. The dispersive magnetometer signal due to excitation at a frequency ω is given by the sum of the signals due to each of the individual transitions,

$$S(\omega) = \sum_i A_i \frac{\omega - \omega_i}{(\omega - \omega_i)^2 + \Delta\omega^2}, \quad (3.34)$$

where ω_i and A_i are the resonance frequencies and relative amplitudes of the individual transitions, and the magnetic linewidth $\Delta\omega$ is assumed to be the same for all transitions. We select for monitoring the transition with frequency ω_0 and amplitude A_0 . If the individual resonances are well-resolved, such that $\Delta\omega \ll \omega_{\text{rev}}$, then the zero-crossing of this transition occurs at a frequency ω'_0 that is only slightly shifted from the resonance frequency, *i.e.*, $\omega'_0 - \omega_0 \ll \Delta\omega$. Then the magnetometer signal at the zero-crossing is given by

⁷ This results in finite polarization along the quantization axis, leading to the asymmetric spectra shown in Figures 3.5(b,d,f) and 3.6(b,d).

$$S(\omega'_0) \approx A_0 \frac{\omega'_0 - \omega_0}{\Delta\omega^2} + \sum_i \frac{A_i}{\omega_0 - \omega_i} = 0, \quad (3.35)$$

where the sum is over all other transitions. We therefore see that the zero-crossing is shifted by⁸

$$\omega'_0 - \omega_0 = \frac{\Delta\omega^2}{A_0} \sum_i \frac{A_i}{\omega_i - \omega_0}. \quad (3.36)$$

This shift is due to the small, but finite, tails of the other dispersion curves; each of the transitions essentially pulls the others toward itself. As the field tilts and the amplitudes A_i change, the zero-crossing frequencies tend to shift toward the resonances that grow stronger and away from those that grow weaker. As a result, the magnetometer reading changes with field orientation, an effect known as the heading error.

If the individual transitions are not well-resolved, as in Figures 3.5(c) and 3.6(c), then the spectrum takes the form of a single Lorentzian with only one zero-crossing. This is usually the case for rubidium and cesium magnetometers operating in the Earth's field, as the magnetic linewidth is generally much larger than the quadratic splitting. When the field tilts, as in Figures 3.5(d) and 3.6(d), the shift in the zero-crossing is on the order of the revival frequency ω_{rev} .⁹ Practical scalar magnetometers have therefore often used potassium because of the more easily resolved Zeeman resonances and correspondingly smaller heading errors, despite disadvantages such as lower vapor density compared to rubidium and cesium. For reference we show cesium resonance spectra at 0.5 G in Figures 3.5(e) and 3.5(f), with unrealistically narrow linewidths so that the individual resonance lines are visible; note that the separation between the two hyperfine multiplets is much larger than the separation between adjacent Zeeman transitions.

If the magnetic field orientation is not well known or is unstable, then the heading error places a limit on the accuracy of a scalar magnetometer, just as the sensitivity places a limit

⁸ Note that we have absorbed a minus sign into the denominator on the right-hand side of the equation, switching the order of the two frequencies from Equation 3.35.

⁹ Recall that the individual Zeeman transitions are split by $2\omega_{\text{rev}}$.

on its precision. Most existing scalar alkali-metal magnetometers have heading errors on the order of 1-10 nT, though errors as low as 0.1 nT may be achieved with very narrow resonance lines (Alexandrov, 2003). A common technique to suppress heading errors is to use two pump beams with opposite helicities, in order to pump two separate volumes of alkali atoms with opposite polarization directions. The orientation-dependent effects should cancel when the readings from the two volumes are averaged, but this method adds complexity to the device and requires perfectly matched conditions between the two systems (Yabuzaki and Ogawa, 1974). Another method for reducing the heading error is to excite both of the end transitions within a hyperfine multiplet (for example, the leftmost and rightmost $F=2$ resonance lines shown in Figures 3.5 and 3.6), so that orientation-dependent effects mostly cancel (Balabas *et al.*, 1989). It has also recently been proposed that heading errors can be suppressed by addressing high-rank polarization moments with $\Delta m_F > 1$ (Yashchuk *et al.*, 2003; Acosta *et al.*, 2008). Here we demonstrate simultaneous excitation of *all* Zeeman resonances within a hyperfine multiplet, greatly suppressing the heading error without requiring a particularly narrow linewidth or multiple pump beams or measurement volumes.

3.3 Synchronous Optical Pumping of Quantum Revival Beats

In order to suppress the effect of the nonlinear Zeeman splitting on spin polarization and magnetometer measurement accuracy, we developed the technique of synchronous optical pumping of quantum revival beats by applying double modulation of the optical excitation. We also demonstrated double modulation of an rf excitation field. These methods resonantly excite the spins at both the Larmor and revival frequencies, placing the atoms into a coherent superposition state that survives the collapse and subsequent revival of the spin expectation value. This is analogous to traditional scalar magnetometry techniques,

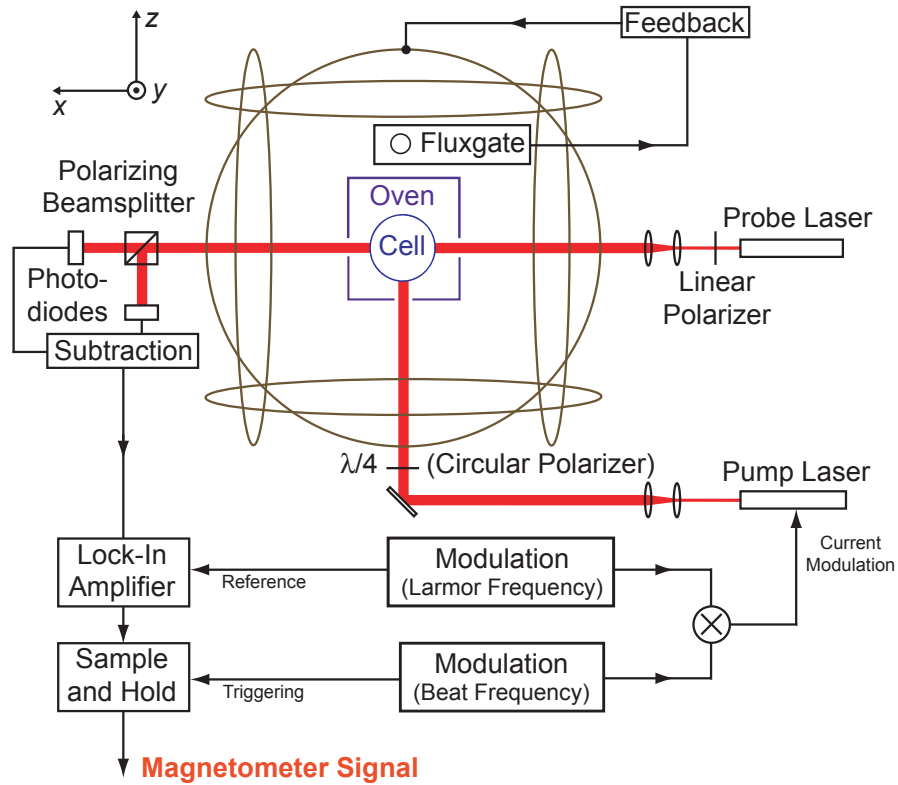


Figure 3.7: Experimental schematic of synchronous optical pumping of quantum revival beats. Double modulation of the pump beam excites the atoms at both the Larmor and revival frequencies. The probe beam signal is analyzed at the Larmor frequency, and a sample-and-hold circuit is used to record the signal only at the instants of maximum spin revival. Three orthogonal Helmholtz coil pairs set the magnetic field amplitude and direction, and feedback using a fluxgate reduces noise along the primary field direction.

which resonantly excite the spins at the Larmor frequency in order to drive coherent precession. However, we allow the spins to experience the periodic quantum revival beats rather than fighting them, as has been the traditional technique.

The experiment is shown schematically in Figure 3.7. The vapor cell contains potassium metal in natural abundance and is heated to 60°C by flowing hot air through a double-walled glass oven. At this temperature, the vapor density is large enough that the polarization can be measured with high signal-to-noise ratio, but the spin-exchange rate is too

small to greatly affect the magnetic linewidth. The cell contains no other gases,¹⁰ and its walls are coated with octadecyltrichlorosilane (see Section 6.2), allowing for suppression of magnetic field gradient broadening through motional narrowing (see Section 6.1.1). The cell is placed within three pairs of orthogonal Helmholtz coils, which are used to control the magnetic field amplitude and direction, as well as five gradient coils used to cancel the ambient magnetic field gradients. No magnetic shielding is used, and magnetic noise along the primary magnetic field direction is actively canceled with feedback from a flux-gate sensor¹¹ located next to the oven. In particular, 60 Hz noise is reduced by about a factor of 30.

The probe beam is detuned about 1 GHz away from the D1 resonance, comparable to the measured optical linewidth of 0.9 GHz (see Figure 3.8), and the probe signal is analyzed by a lock-in amplifier referenced to the fast (near-)Larmor excitation frequency. The output of the lock-in is recorded by a sample-and-hold circuit triggered at the instant of maximum spin revival during each period τ_{rev} . Both output channels of the lock-in are monitored separately, in order to record both the absorptive and dispersive resonance signals. For optimal signal, the probe beam may be turned on only at the instant of maximum spin revival with a short duty factor d , since at all other times the probe signal remains unrecorded due to the sample-and-hold circuit. The instantaneous power of the probe beam can then be increased by a factor of $1/d$ compared to continuous operation, giving the same shot-noise sensitivity and linewidth broadening but increasing the optical polarimeter signal by the same factor of $1/d$ at the instant of detection. We verified that this technique indeed works, but we generally did not employ it as we were neither limited by small signal size nor attempting to achieve shot-noise sensitivity.

¹⁰ At this vapor density, radiation trapping is not a problem, and quenching gas is unnecessary.

¹¹ Bartington Instruments Mag-03MC

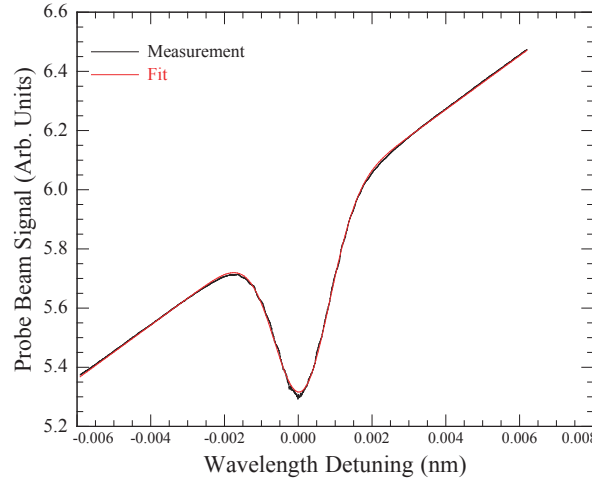


Figure 3.8: Doppler broadened optical spectrum of potassium vapor measured at 60°C in a cell without quenching or buffer gas. The probe beam transmission is monitored as a function of laser frequency, and the spectrum is fitted to a Gaussian lineshape. The measured linewidth of 892 MHz (FWHM) is close to the expected value of 814 MHz given by Equation 2.13. Near the potassium D1 resonance, the conversion from wavelength detuning to frequency detuning is given by $\Delta\nu = \Delta\lambda \times -505.5 \text{ GHz/nm}$. The probe wavelength is tuned by changing the laser current, resulting in the linear increase in signal with wavelength.

3.3.1 Double Modulation

Synchronous optical pumping of quantum revival beats is achieved through double modulation of the pump laser at both the Larmor and revival frequencies, as shown in Figure 3.9. The pump beam is provided by a distributed feedback (DFB) diode laser which can be easily tuned by changing either the laser current or temperature. The laser is detuned by 6.3 GHz from the potassium D1 resonance, much larger than the measured 0.9 GHz optical linewidth (see Figure 3.8). The laser current is then sinusoidally modulated at or near the Larmor frequency, so that the pump beam is tuned on resonance for a short fraction of each precession period. This is a standard implementation of a Bell-Bloom scalar magnetometer, with the atomic spins driven to precess coherently at the Larmor frequency as described in Section 3.1.2.

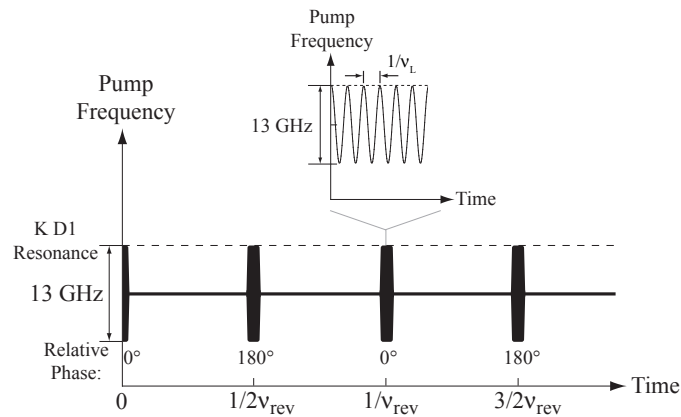


Figure 3.9: The pump laser is modulated at both the Larmor and revival frequencies, in order to synchronously pump quantum revival beats in alkali atoms in the geomagnetic field range. The pump laser is tuned slightly off the D1 resonance, and the laser frequency is modulated at the Larmor frequency so that it is on resonance for a short fraction of each precession period, as in a standard Bell-Bloom type magnetometer. Secondary modulation is applied at twice the revival frequency, turning the fast laser modulation on and off with a duty cycle of 1-10%. The relative phase of the fast modulation can be alternated by 180° between successive modulation pulses.

Secondary modulation is then applied at twice the quantum beats revival frequency, turning the fast Larmor-frequency laser current modulation on and off with a duty cycle in the range of 1-10%. This is achieved by multiplying together a sinusoid at the Larmor frequency and a square wave at the revival frequency, and using the product for analog modulation of the laser current; the function generators that create the two waveforms must have their time bases synchronized. The laser is thus modulated onto resonance for only a short fraction of each revival period. The atoms are polarized and then allowed to precess freely, with quantum beating resulting in collapse and revival of the ensemble spin polarization. At the instant of complete revival, with the atomic polarization $\langle S_z \rangle$ reaching its maximum value along the polarization direction, optical pumping again resumes for a short time. In this way double modulation of optical pumping resonantly drives quantum beating and allows for large spin polarization, as opposed to the standard method of only modulating at the Larmor frequency, which forces continuous precession at a single frequency but limits spin polarization.

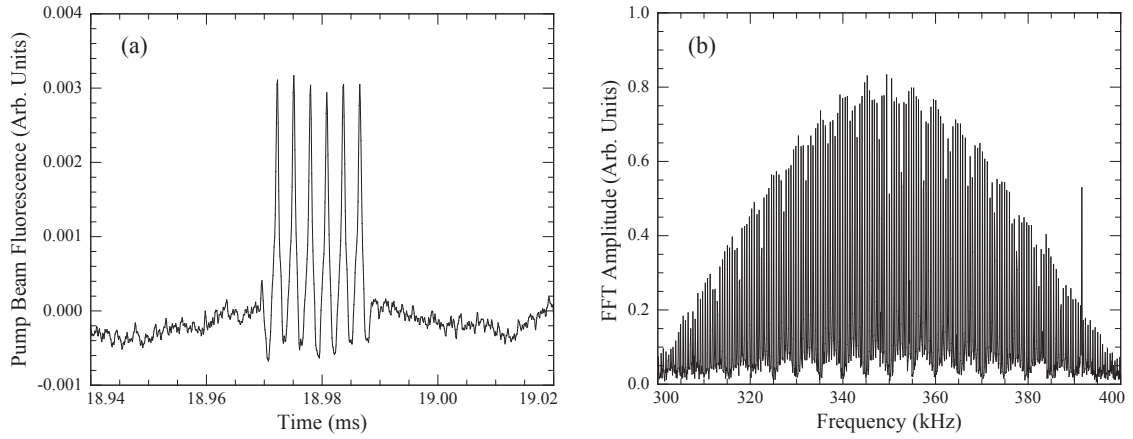


Figure 3.10: (a) The fluorescence signal from potassium atoms at 0.5 G decaying to the ground state after absorbing a pump beam photon. Double modulation with a duty cycle of 1% results in the pump beam being tuned on resonance for a short fraction of each Larmor period, with the fast modulation turned on and off twice each revival period. (b) The Fourier transform of the fluorescence signal. Secondary modulation creates sidebands of the optical pumping rate separated by $2\omega_{\text{rev}}$, allowing for simultaneous excitation of all Zeeman resonances, with a shorter duty cycle resulting in more sidebands.

The ensemble spin polarization revives twice per revival period, with the relative phase of the Larmor precession alternating by 180° between successive revivals. The relative phase of the laser modulation may be changed between successive modulation pulses in order to match the phase of the precessing atoms, as shown in Figure 3.9,¹² or the phase may be kept constant. Figure 3.10(a) shows the fluorescence signal from potassium atoms at 0.5 G experiencing double optical modulation with a duty cycle of 1%; at 60°C the vapor is optically thin, so that in the absence of quenching gas the photons emitted by atoms decaying from the excited state may be easily detected. Modulation of the optical pumping rate is evident, as is the turning off of fast modulation with a short duty cycle. Note that the signal appears at times to drop below the background level during modulation, possibly because of the bandwidth of the detection system.

¹² This may be achieved by alternating the sign of the secondary modulation square wave.

The Fourier transform of the fluorescence signal is shown in Figure 3.10(b). In frequency space, double modulation creates sidebands of the optical pumping rate separated by $2\omega_{\text{rev}}$, with a shorter duty cycle resulting in a broader spectrum with more sidebands. If the fast modulation occurs at frequency ω_{mod} , then for secondary modulation with alternating phase, the sidebands are located at $\omega_{\text{mod}} \pm (2n + 1)\omega_{\text{rev}}$, with $n = 0, 1, 2, \dots$. For secondary modulation with constant phase, the sidebands are located at $\omega_{\text{mod}} \pm 2n\omega_{\text{rev}}$. The sidebands have the same separation as the individual Zeeman resonances, as described by Equation 3.32, so if the fast modulation is tuned properly¹³ then the sidebands simultaneously excite all Zeeman resonances within a hyperfine multiplet. This results in constructive interference between all Zeeman coherences and increased spin polarization compared to excitation of a single Zeeman resonance.¹⁴ For a short duty cycle and the resulting broad optical pumping spectrum, the fast modulation frequency may be tuned very far from the nominal resonance frequency but still result in a large coherent precession signal. Note that for alternating-phase modulation the Zeeman transitions are all excited when the fast modulation frequency ω_{mod} is tuned exactly to the Larmor frequency ω_{L} , resulting in a detectable resonance with frequency directly proportional to the magnetic field strength B without the need to account for any offset due to the quadratic splitting.

The pump beam power is always adjusted to broaden the magnetic linewidth by a factor of 2 for all optical modulation schemes, giving the same average optical pumping rate that approximately maximizes magnetometer sensitivity. The linewidth broadening as a function of total pump beam power is shown in Figure 3.11(a), for the cases of continuous Larmor-frequency modulation (100% duty cycle) and double modulation with a duty cycle of 10%. The linewidth broadening scales inversely with the duty factor d , so that if the laser

¹³ The individual Zeeman resonances within the F hyperfine multiplet are located approximately at $\omega_{\text{L}} \pm (2n + 1)\omega_{\text{rev}}$, with $n = 0, 1, \dots, F - 1$.

¹⁴ This statement is equivalent to our earlier assertion that doubly-modulated optical pumping results in larger polarization than continuous modulation at the Larmor frequency because of excitation only at times of maximum spin revival. One statement considers the effect of double modulation in the time domain, while the other considers the effect in the frequency domain.

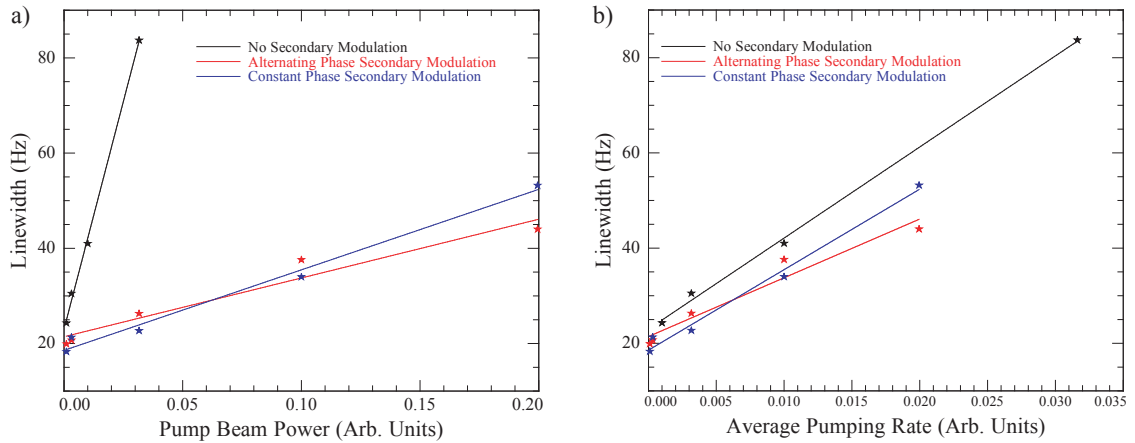


Figure 3.11: (a) Magnetic linewidth broadening due to optical excitation with (10% duty cycle) and without (100% duty cycle) secondary modulation. (b) If the laser power is scaled by the inverse of the duty cycle, then the linewidth broadening and average optical pumping rate become approximately the same for all modulation techniques.

power is scaled by $1/d$ then the average pumping rate and linewidth broadening become nearly the same for all modulation techniques, as shown in Figure 3.11(b).

It has recently been proposed that periodic laser pulses could be used to increase the degree of molecular alignment and maintain it indefinitely (Ortigoso, 2004; Leibscher *et al.*, 2004). In order to demonstrate the general applicability of the double-modulation technique to multilevel quantum systems, we show that it also works for radio-frequency excitation of polarized spins.¹⁵ The rf field induces coherent spin precession and is turned on and off at twice the quantum beats revival frequency, allowing the transverse spin polarization to freely collapse and revive. The relative phase of the rf field can be alternated by 180° or kept constant between successive pulses, and in frequency space the rf excitation has sidebands separated by $2\omega_{\text{rev}}$ and located at the same frequencies as described above for the case of optical excitation. Again, double modulation with alternating phase creates a resonance signal at a frequency directly proportional to the magnetic field amplitude B

¹⁵ All data shown in Section 3.3.2 were taken with a Bell-Bloom type magnetometer with optical excitation, unless otherwise noted. All simulation results discussed in Section 3.4 were modeled with optical excitation.

without an offset due to the quadratic Zeeman splitting. For direct comparison of performance at different duty cycles, the rf-field power should be adjusted to maintain constant magnetic linewidth broadening.

3.3.2 Observations

The experimental observation of quantum revival beats in ground-state potassium atoms at 0.5 G is shown in Figure 3.12(a). Synchronous optical pumping with a 10% duty cycle is started about 150 ms before $t=0$, placing the atoms into a coherent superposition state with simultaneous excitation of all $F=2$ Zeeman transitions. This allows the measured transverse spin expectation value $\langle S_x \rangle$ to freely collapse and revive twice per period $\tau_{\text{rev}}=3.8$ ms. The revivals persist indefinitely, until at $t=60$ ms the pumping light is turned off; the spins then precess freely and remain in the superposition state, showing many quantum revivals with decreasing amplitude as the polarization decays due to spin relaxation. The polarization lifetime $T_2=24$ ms in the absence of pumping light is typical of what was observed in this experiment, so that with pumping light applied the magnetic linewidth is generally about 13-15 Hz. Despite the use of an uncoated cell (allowing motional narrowing) and gradient compensation coils, the polarization lifetime is limited by small residual magnetic field gradients.

The quantum beats revival envelope is shown in close-up in Figure 3.12(b). The envelope is similar to that shown in Figure 3.4(a), except that there are additional instances of partial revival that result from double modulation of the optical pumping rate. The observed envelope is compared to the results of a numerical simulation of the time evolution of the ground-state density matrix (see Section 3.4), shown with red dashed lines that are vertically offset to allow for visibility. The simulation is in excellent agreement with the

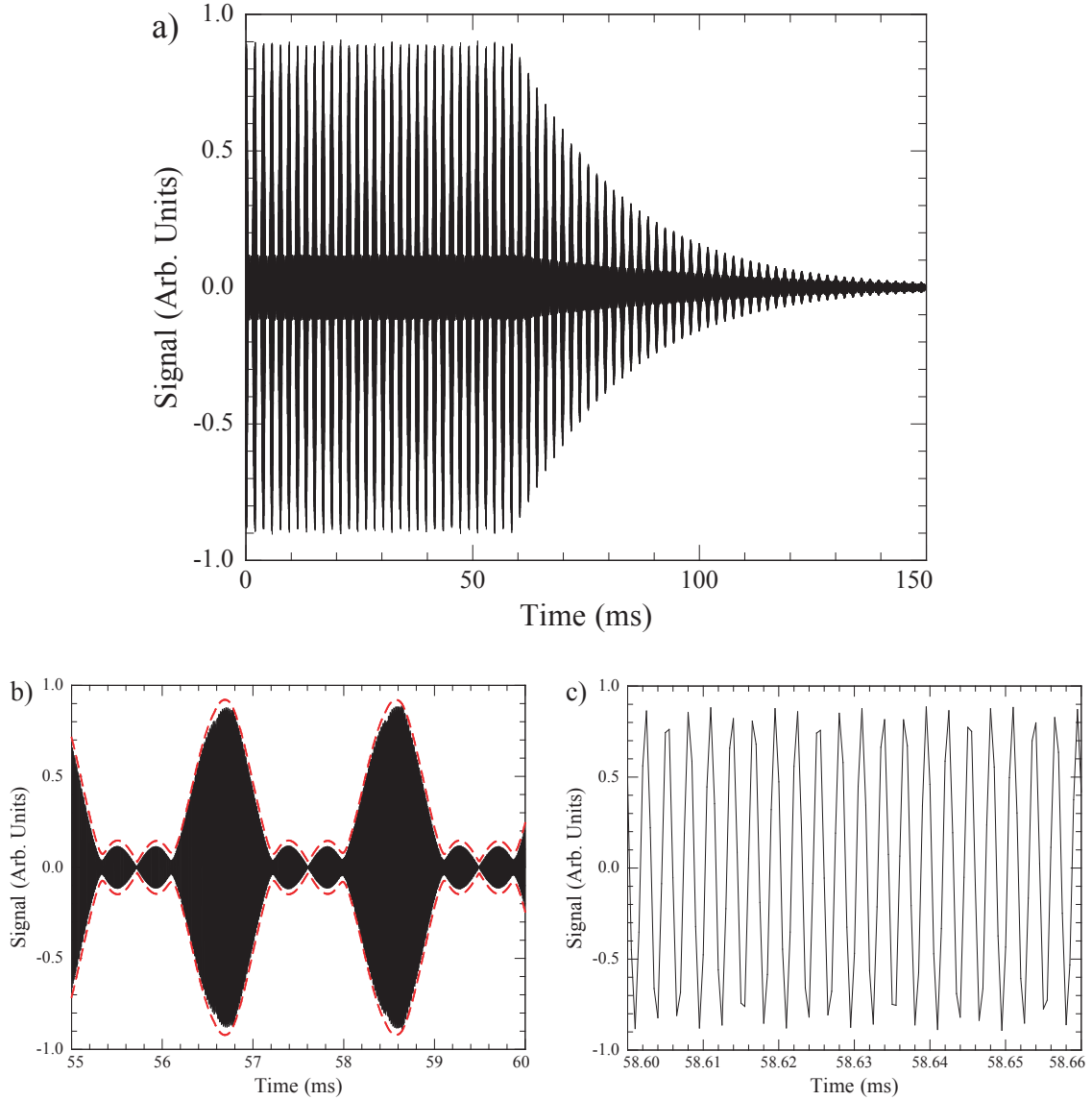


Figure 3.12: Experimental observation of quantum revival beats of ground-state potassium atoms at 0.5 G. **(a)** The probe beam measurement of $\langle S_x \rangle$ shows multiple quantum revivals. Synchronous optical pumping of revivals started about 150 ms before $t=0$ and stopped at $t=60$ ms; the periodic collapses and revivals continued during the free decay of the spin coherence. **(b)** Close-up of the revival envelope. The red dashed lines, which are offset vertically for visibility, represent the results of a density matrix simulation, showing excellent agreement between the simulation and measurement. **(c)** Close-up of the fast Larmor precession.

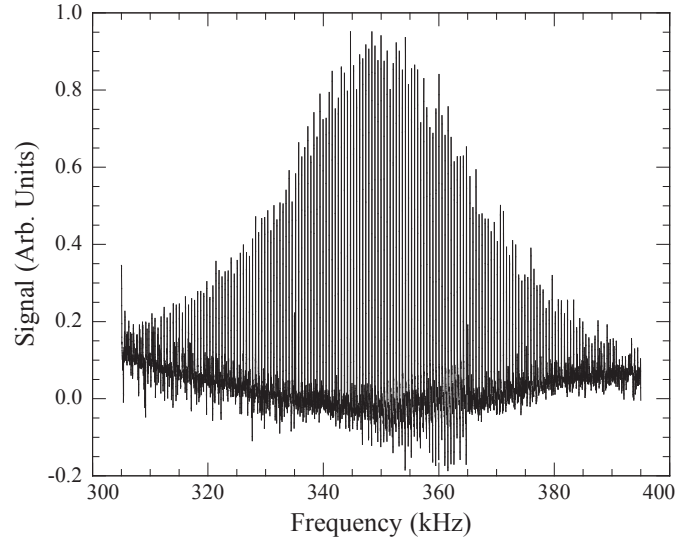


Figure 3.13: The magnetometer spectrum obtained from double modulation with 1% duty cycle. The spectrum contains many resonances, due to the sidebands of the optical pumping rate as shown in Figure 3.10(b). These sidebands allow for magnetometer resonances even when the fast modulation frequency is tuned far from the nominal Larmor frequency, with a smaller duty cycle resulting in a broader spectrum.

measurement, allowing for confidence in the general results of the density matrix simulation. Precession at the Larmor frequency is displayed in Figure 3.12(c), showing that at instants of maximum revival the atoms precess coherently at a single frequency.

The magnetometer response spectrum, shown in Figure 3.13, is generated by keeping the magnetic field at 0.5 G, thus fixing the Larmor frequency at 350 kHz,¹⁶ and sweeping the fast modulation frequency; the probe beam signal is analyzed by a lock-in amplifier referenced to the modulation frequency. The spectrum is very broad with many resonances because the optical pumping rate has numerous sidebands with the same separation as the individual Zeeman transitions (see Figure 3.10(b)). When the fast modulation frequency is tuned properly, several of the sidebands overlap with and excite all $F=2$ Zeeman transitions simultaneously. A shorter duty cycle results in more sidebands, creating a broader

¹⁶ For the following measurements, the magnetic field is tuned by setting the Larmor frequency to precisely 350 kHz, so the field is not exactly 0.5 G.

resonance spectrum and allowing for magnetometer resonance signals even when the fast modulation frequency is tuned very far from the atomic Larmor frequency. In order to observe the full magnetometer spectrum, it is necessary to use a lock-in amplifier with sufficiently high bandwidth; we record all spectra with a time constant of $10\ \mu\text{s}$.

The absorptive resonance spectra measured with the magnetic field tilted by 0° , 60° , and -60° into the pumping direction are shown in Figure 3.14. For each orientation, the spectrum is shown both in broad detail and in close detail near the Larmor frequency. In this figure and those that follow in this section, spectra taken without secondary modulation (100% duty cycle) are shown in black, spectra taken with alternating-phase secondary modulation (1% duty cycle) are shown in red, and spectra taken with constant-phase secondary modulation (1% duty cycle) are shown in blue. For easier visibility, the spectra without secondary modulation are only shown in the close-detail plots. Alternating-phase secondary modulation results in signals with the opposite sign as the other two modulation schemes because the polarization is measured during revivals when the fast modulation has a relative phase of 180° .¹⁷ Although 60 Hz noise is greatly suppressed through feedback, it nevertheless causes random variation in the height of the individual resonance peaks with secondary modulation, as well as noticeable sidebands.

Without secondary modulation, the relative strengths of the individual Zeeman resonances change significantly when the magnetic field direction is tilted. The dispersion curve zero-crossings given by Equation 3.36 therefore experience large shifts, resulting in significant heading errors. In contrast, the spectra obtained with secondary modulation remain symmetrical as the field tilts, and the relative strengths of the resonances do not systematically change. The tails of the various peaks on either side of a particular resonance therefore largely cancel and do not lead to a substantial heading error. In addition, comparing the heights of the resonance peaks at 0° tilt shows that secondary modulation

¹⁷ This allows for better visibility of all three spectra when plotted together, but there is no other practical benefit to triggering the sample-and-hold circuit at either the odd or even revivals.

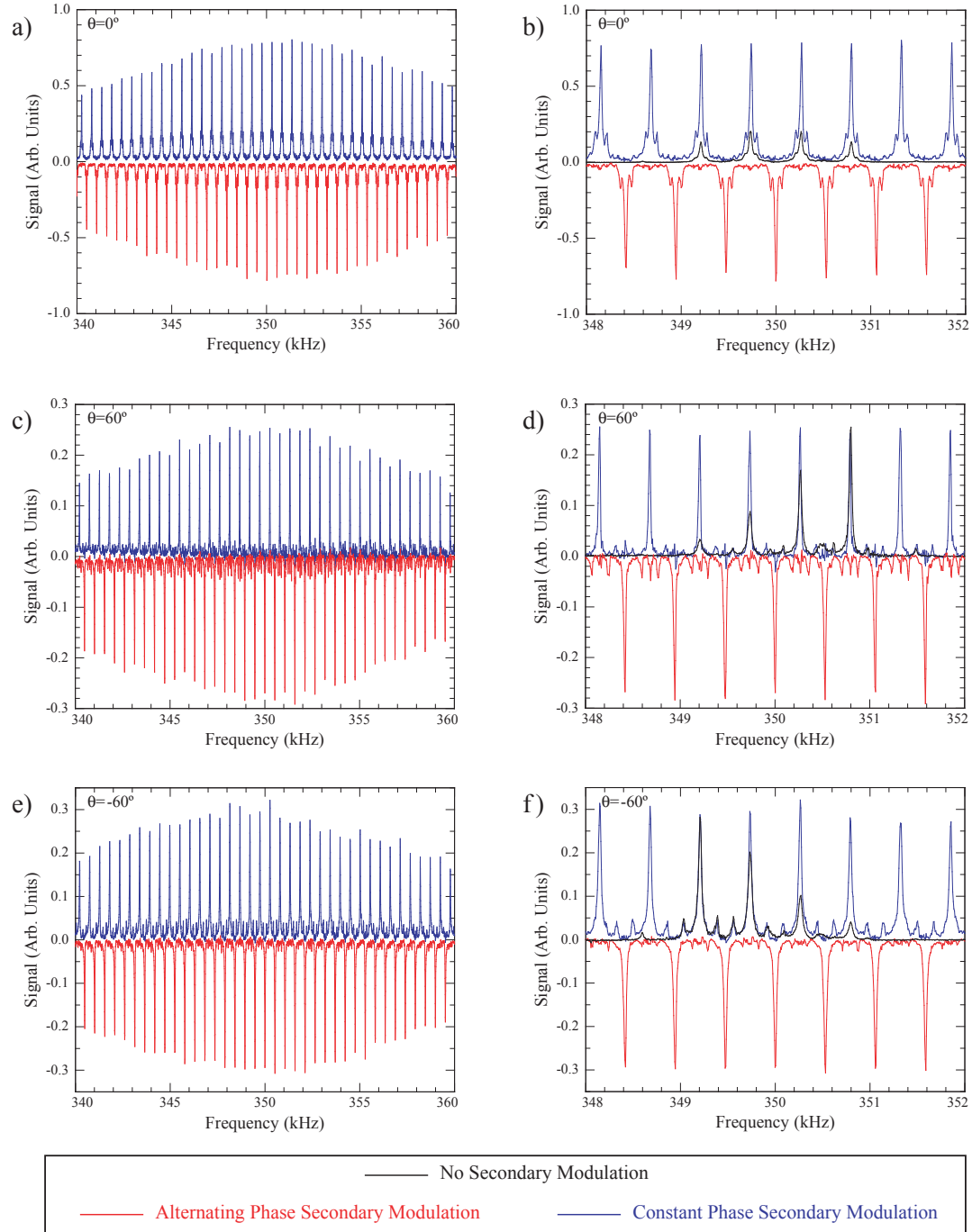


Figure 3.14: Magnetometer resonance spectra measured with the magnetic field tilted by 0° , 60° , and -60° into the pumping direction. The spectrum changes significantly without secondary modulation but remains symmetrical and nearly unchanged with secondary modulation. Random variation in the height of the peaks with secondary modulation and small sidebands are due to 60 Hz noise. The vertical scale is the same for all spectra.

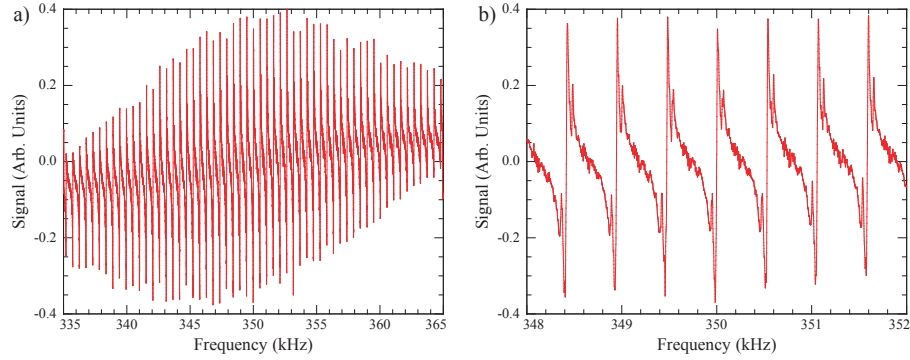


Figure 3.15: Dispersive magnetometer spectrum measured with alternating-phase secondary modulation, showing many individual dispersive resonances and a zero-crossing precisely at the Larmor frequency.

results in an increase in maximum spin polarization by a factor of 3.9; the density matrix simulation predicts an increase by a factor of about 6. The linewidth of the resonances is kept the same, so the slope of the resonant dispersion curve at the zero-crossing increases by the same amount, and thus higher spin polarization translates directly into higher magnetometer sensitivity. The dispersive resonance spectrum taken with alternating secondary modulation is shown in Figure 3.15, showing numerous individual resonances and a central zero-crossing precisely at the Larmor frequency of 350 kHz.

If the magnetic linewidth is comparable to the quadratic splitting, then secondary modulation helps to resolve individual resonances, as shown in Figure 3.16. These spectra were taken at $B=0.21$ G, so that the quadratic splitting given by $\omega_{\text{rev}} = 2\pi \times 47$ Hz was much smaller than at 0.5 G, and the linewidth was purposely broadened by applying a large magnetic field gradient. The heading error, which is usually on the order of ω_{rev} when the resonances are poorly resolved, can be significantly reduced by locking to one of the resolved resonances, which remain symmetrical when the magnetic field is tilted. This can be seen by comparing the spectra with and without secondary modulation in Figures 3.16(a) (0° tilt) and 3.16(b) (60° tilt). The increase in spin polarization by using secondary

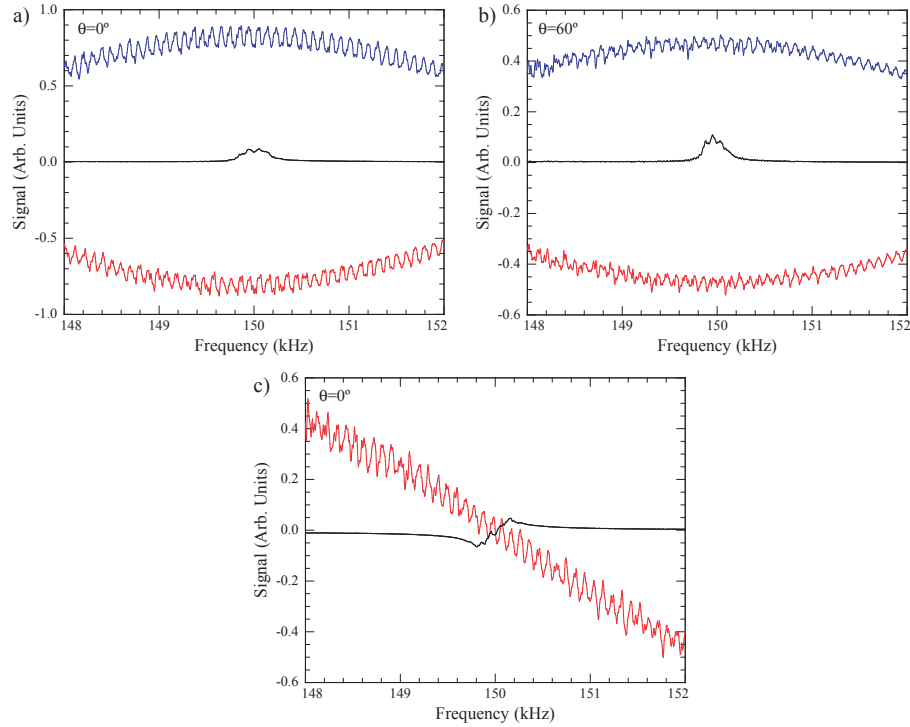


Figure 3.16: Absorptive and dispersive resonance spectra measured at 0.21 G, with the linewidth purposely broadened by applying magnetic field gradients. Secondary modulation provides individual resonances which are poorly resolved without it, and it greatly increases spin polarization and reduces systematic dependence on field orientation. The vertical scale is the same for all spectra.

modulation is much greater in this regime than at narrow linewidth. The dispersive spectra are compared in Figure 3.16(c), showing that double modulation provides resolved dispersion curves (though not all cross zero) and a large increase in slope at the central zero-crossing. These measurements are intended to demonstrate that scalar magnetometers using rubidium or cesium, which have much smaller quadratic Zeeman splitting than potassium (see Table 3.1), have the potential for much greater improvement in sensitivity using double modulation.

The resonance spectra observed with double modulation (1% duty cycle) of a radio-frequency excitation field are shown in Figure 3.17. The tilt angle θ is taken with respect to

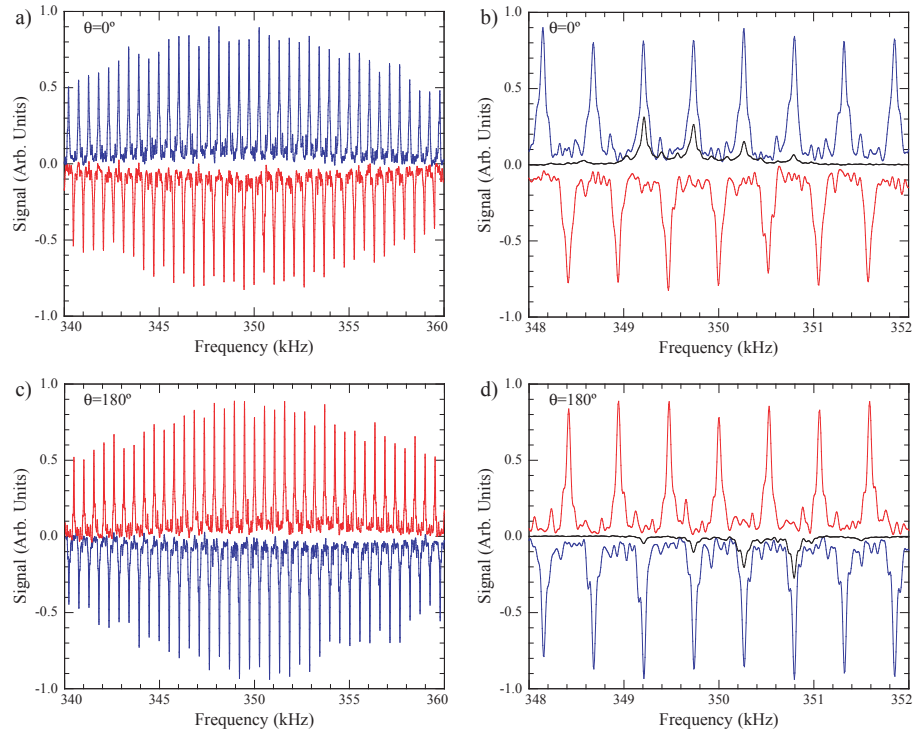


Figure 3.17: Resonance spectra of a scalar magnetometer using radio-frequency excitation with double modulation. As with double optical modulation, there is an increase in sensitivity and suppression of the heading error. The magnetic field is parallel to the pumping direction for $\theta=0^\circ$, and antiparallel for $\theta=180^\circ$. The vertical scale is the same for all spectra.

the nominal magnetic field orientation parallel to the pumping direction. As the magnetic field tilts away from the pumping direction, the magnetometer signal becomes smaller, until it passes through a dead zone with no signal at $\theta=90^\circ$ and then inverts as the field becomes aligned antiparallel to the pumping direction. The height of the largest resonance peak increases by about a factor of 3 using secondary modulation, demonstrating that double modulation increases spin polarization just as for optical excitation; we expect that this is a general result for doubly-modulated excitation of quantum systems with nonlinear energy splitting. Secondary modulation again results in a symmetrical set of resonances that does not cause a significant heading error, compared to the highly asymmetric spectrum produced with only Larmor-frequency modulation.

3.4 Density Matrix Simulation

It is difficult to generate a constant magnetic field amplitude while varying the field orientation, so we use a numerical simulation for a general analysis of heading errors and polarization gains. We calculate the time evolution of the ground-state density matrix of ^{39}K atoms (see Section 2.9) in the presence of doubly modulated optical pumping and evaluate the spin-polarization expectation value $\langle S_x \rangle$. Although the eigenbasis of the Hamiltonian is nearly described by the $|F, m_F\rangle$ states, there are small corrections that result from using the physical eigenstates $|\bar{F}, m_{\bar{F}}\rangle$, as calculated in Appendix B. As shown in Figure 3.12(b), the simulated spin polarization evolution is in excellent agreement with experiment, so we may have confidence in the results of the simulation. The density matrix simulation was also used to generate the quantum revival beat envelopes shown in Figure 3.4, as well as to determine the relative heights of the individual transitions of the resonance spectra shown in Figures 3.5 and 3.6.

The ground-state Hamiltonian is given by Equation 3.17, and the time evolution of the density matrix is calculated according to Equation 2.185. The parameters of the simulation

are set to be identical to the conditions of the experiment described in Section 3.3. For simplicity, we set the spin-exchange rate to zero, which is a good approximation for operation with potassium vapor at 60°C, and we eliminate the spatial diffusion term and assume uniform conditions throughout the vapor cell. We also assume that there are no coherences between the two ground-state hyperfine levels, and we set the average optical pumping rate to be equal to the spin-relaxation rate. Since the eigenbasis of the Hamiltonian depends on the direction of the magnetic field, we fix the field orientation and instead vary the pumping direction as necessary to calculate heading errors.¹⁸

In order to allow for efficient calculation, we make a rotating wave approximation. Each density matrix element $\rho_{\bar{F}, m_{\bar{F}}, \bar{F}', m'_{\bar{F}'}}$ evolves primarily at the frequency $(m'_{\bar{F}'} - m_{\bar{F}})\omega_L$, so we ignore terms in the evolution of that matrix element that evolve at a different frequency. The pumping rate is decomposed into harmonics of the Larmor frequency, with the relative amplitudes of the harmonics taken from the Fourier transform of the optical pumping fluorescence signal.¹⁹ Optical pumping is modulated at the revival frequency, with appropriate duty cycle and change of phase, to create double modulation. Lock-in detection at the Larmor frequency is simulated to generate the absorptive and dispersive components of the resonance signal. Detuning of the magnetic field is represented by a small shift of the atomic resonance frequency from the central frequency at which the rotating wave approximation is made.

Starting from a state of zero polarization, the density matrix evolves under the influence of optical pumping for a long enough time that the atomic polarization reaches its equilibrium behavior (*i.e.*, quantum revivals with constant amplitude). The spin polarization $\langle S_x \rangle$ is measured at the instant of maximum revival. Resonance spectra at 0.5 G generated by the density matrix simulation are shown in Figure 3.18, comparing operation

¹⁸ Even though the pump beam gets tilted in the numerical simulation, in this section we still refer to the magnetic field tilting. Only the relative orientation between the two is relevant, and this matches what was done in the experiment.

¹⁹ The spectrum near the first harmonic is shown in Figure 3.10.

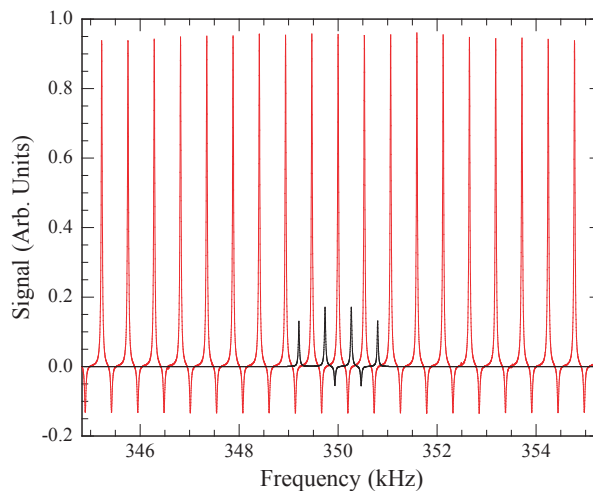


Figure 3.18: Resonance spectra at 0.5 G without secondary modulation (in black) and with alternating-phase modulation (in red), generated by the numerical simulation of the ground-state density matrix.

without secondary modulation (in black) and with alternating-phase secondary modulation (in red). The spectra resemble those observed experimentally, although the increase in polarization with double modulation is approximately a factor of 6, about 1.5 times larger than observed.²⁰

The magnetometer resonance frequency is determined from the zero-crossing of one of the resonant dispersion curves; for the case of alternating modulation the central dispersion curve is used, while for the case of no secondary modulation one of the two transitions nearest to the central frequency is used. The heading error, defined as the shift in the zero-crossing from its value with the magnetic field at 0° tilt, is shown in Figure 3.19(a) for these two modulation schemes and several values of the magnetometer linewidth at 0.5 G. The heading error is seen to be substantially larger without secondary modulation, rising to the level of several nT at 60° tilt for typical operational linewidths. In contrast, using double modulation the heading error is only on the order of 0.1 nT, comparable to the

²⁰ The resonance lines due to the $F=1$ transitions are much less prominent in the experimental observations than in the simulated data; the reason for this is unknown.

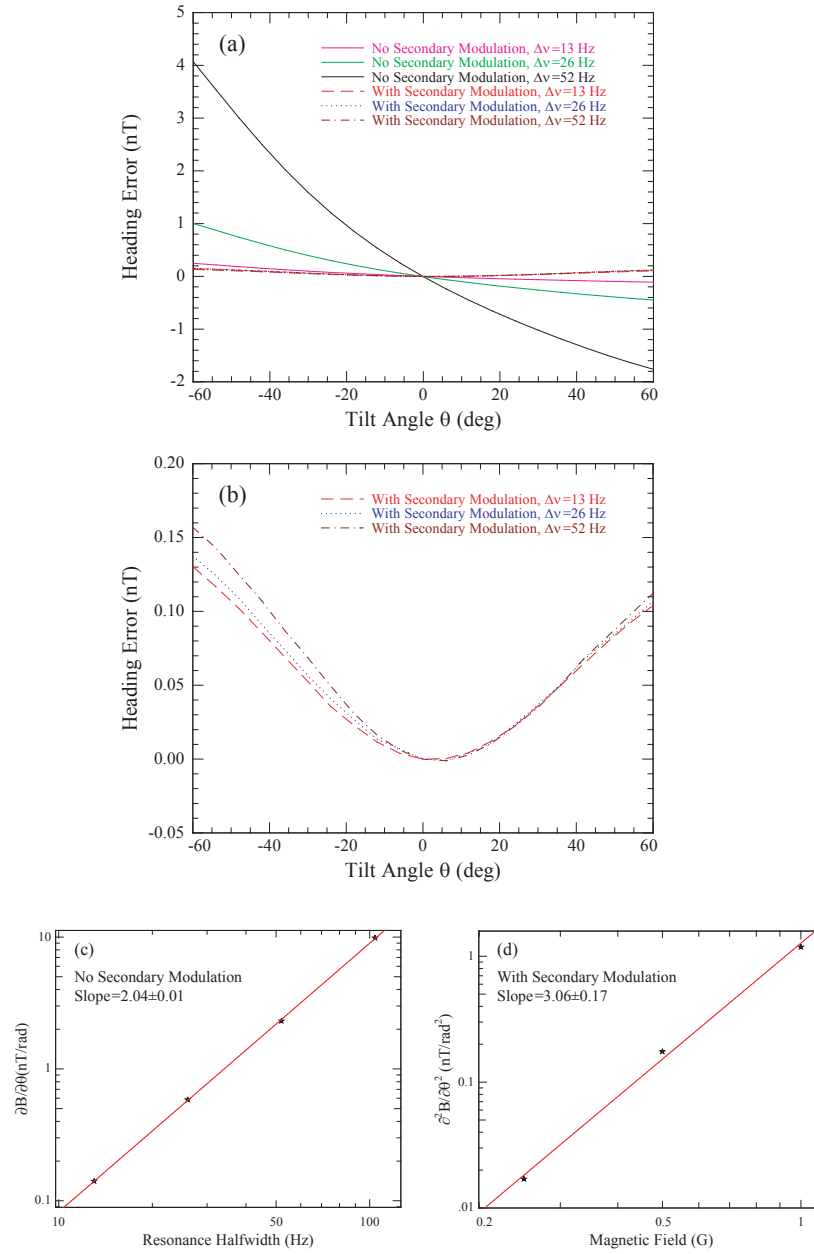


Figure 3.19: (a) Heading error at 0.5 G determined from numerical simulation as a function of the magnetic field tilt angle θ into the direction of the pump beam. Without secondary modulation the heading error is linear near $\theta = 0^\circ$ and depends greatly on the linewidth. (b) Detail of the heading error with double modulation. The heading error is approximately quadratic near $\theta = 0^\circ$ and does not depend significantly on the linewidth. (c) Without secondary modulation, the slope of the heading error near $\theta = 0^\circ$ scales as $\Delta\omega^2$. (d) With secondary modulation, the curvature of the heading error near $\theta = 0^\circ$ scales as B^3 , indicating that it is due to the third-order splitting of the Zeeman levels.

best achieved in practical magnetometers under the condition of very narrow linewidth. The heading error for only the case of double modulation is displayed in detail in Figure 3.19(b), showing that the error has very little dependence on the magnetic linewidth and that a very small heading error can be achieved without requiring particularly narrow resonances. Secondary modulation therefore makes the magnetometer less susceptible to broadening mechanisms such as magnetic field gradients.

The heading error near the nominal operating condition $\theta=0^\circ$ is linear without secondary modulation, so the magnetometer reading varies greatly with even a small change in the field direction. The slope of the heading error as a function of linewidth is shown in Figure 3.19(c); we see that the heading error depends quadratically on the linewidth, as predicted in Equation 3.36, and thus even slight changes in operational conditions can have a large effect on measurement accuracy. On the other hand, the heading error near $\theta=0^\circ$ is quadratic with secondary modulation, so that there is very little change in the magnetometer reading as a result of small field tilts. The curvature of the heading error with secondary modulation, shown in Figure 3.19(d), scales as B^3/ω_{hf}^2 and is due to the cubic energy level splitting. We expect the heading error with secondary modulation to be much smaller for rubidium or cesium, which have larger hyperfine splitting and significantly smaller cubic Zeeman splitting (see Table 3.1).

The magnetometer sensitivity is determined from the slope of the resonant dispersion curve at the zero-crossing and is shown in Figure 3.20(a) for $\theta=0^\circ$ as a function of the ratio of the magnetic linewidth to the revival frequency. The enhancement of sensitivity from applying alternating-phase secondary modulation, thus achieving higher spin polarization, is shown in Figure 3.20(b) for 1% and 30% duty cycles. For small linewidths, the gain is higher for shorter duty cycle and reaches its maximum for $\Delta\omega \approx \omega_{\text{rev}}$ (compare

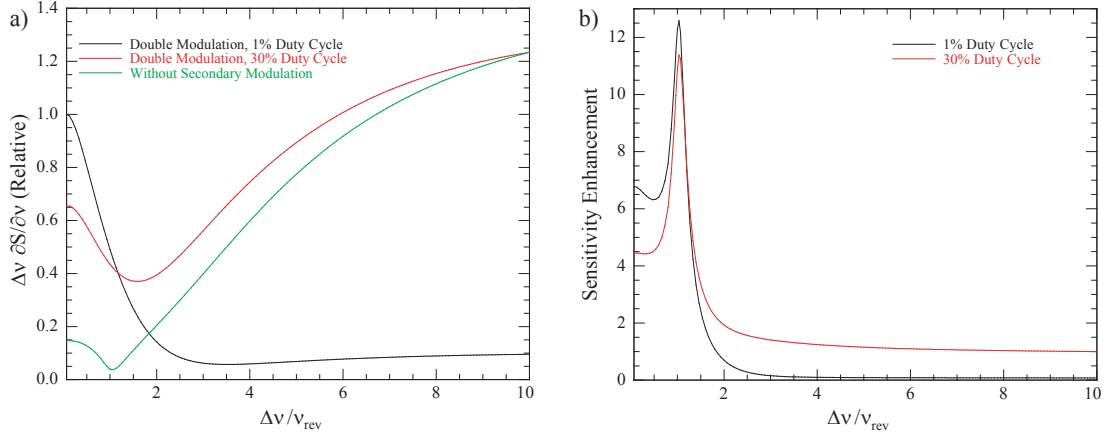


Figure 3.20: (a) Magnetometer sensitivity at 0.5 G and $\theta=0^\circ$ as a function of magnetic linewidth, as determined by the numerical simulation from the slope of the dispersion curve zero-crossing. The average optical pumping rate is set equal to the relaxation rate. **(b)** Calculated gain in magnetometer sensitivity due to higher spin polarization with double modulation compared to the case without secondary modulation.

the simulated enhancement to that observed in Figure 3.16).²¹ However, as the linewidth broadens and becomes much greater than the revival frequency, secondary modulation with short duty cycle becomes detrimental to magnetometer sensitivity. In this case, the polarization lifetime is short compared to the revival period, and the polarization coherence decays to zero in between applications of optical pumping, so that larger duty cycle is necessary to enable build-up of polarization between successive revivals.

In addition to simulating quantum revival beats in potassium as described above, we also model the ground-state density matrix evolution in cesium. Figure 3.21(a) displays the growth of the quantum revival envelope from zero polarization under double modulation; note that there are more instances of partial revival during each period than in potassium, likely due to the larger number of ground-state Zeeman transitions. The gain in spin polarization and magnetometer sensitivity given by double modulation with 1%

²¹ This does not mean that overall magnetometer performance is best when the magnetic linewidth is comparable to the revival frequency, only that the benefit of using double modulation to enhance spin polarization is largest. In general, magnetometer sensitivity is always optimized by decreasing the resonance linewidth.

duty cycle is shown in Figure 3.21(b) and is considerably larger than in potassium, since there are more Zeeman resonances to cause destructive interference without secondary modulation. The resonance spectrum obtained with alternating-phase secondary modulation and $\Delta\omega=2\pi\times 1$ Hz is shown in Figure 3.21(c) for $\theta=0^\circ$ and in Figure 3.21(d) for $\theta=60^\circ$, displaying clear separation between the hyperfine multiplets. The ratio of the amplitudes of the $F=4$ lines compared to the $F=3$ lines increases from 2.5 at 0° to 3.4 at 60° , which may contribute to the heading error; this effect should be more pronounced in the case where the optical hyperfine transitions are well-resolved, leading to hyperfine pumping. Nevertheless, operation with ^{133}Cs and ^{85}Rb should still provide smaller heading errors and greater sensitivity gain than operation with potassium due to smaller cubic splitting and more ground-state energy levels, while operation with ^{87}Rb (which has the same nuclear spin as ^{39}K) should provide smaller heading errors. However, achieving these improvements may be challenging, as the smaller revival frequencies in these elements require much narrower magnetic linewidth.

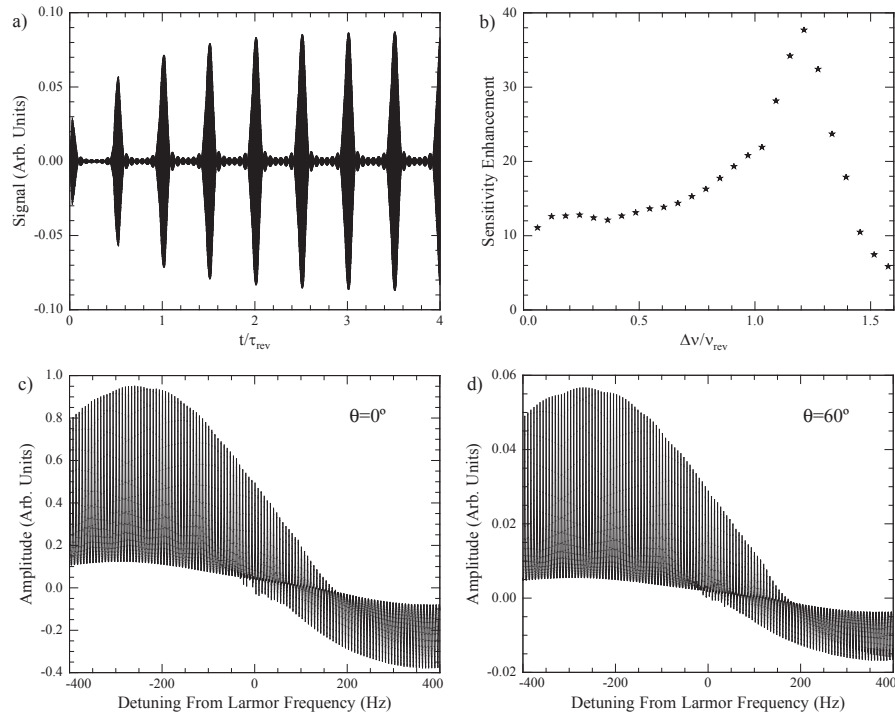


Figure 3.21: (a) Quantum revival beat envelope of cesium spins at 0.5 G, simulated by the evolution of the ground-state density matrix. (b) Enhancement of magnetometer sensitivity with 1% duty cycle compared to operation without secondary modulation. The gain is greater than in potassium due to coherent interference of a larger number of Zeeman levels. (c-d) Absorptive resonance spectra with alternating-phase double modulation at $\theta=0^\circ$ and 60° with 1% duty cycle.

Chapter 4

Radio-Frequency Magnetometry

COILS HAVE TRADITIONALLY BEEN USED to inductively detect oscillating magnetic fields in the radio-frequency (rf) range, such as those produced in magnetic resonance applications. However, the sensitivity of a coil scales linearly with the signal frequency, so detection of weak fields at frequencies below several megahertz requires alternative detection methods. We present a radio-frequency atomic magnetometer that achieves high sensitivity by tuning the Zeeman resonance of the alkali atoms to the rf frequency and partially suppressing spin-exchange relaxation through light narrowing. The magnetometer sensitivity varies minimally with the resonance frequency. We derive an expression for the fundamental limit on the sensitivity of the rf magnetometer and show that it has better intrinsic sensitivity than a pick-up coil of comparable volume at frequencies below about 50 MHz. Using the rf magnetometer, we demonstrate detection of nuclear magnetic resonance (NMR) signals at 62 kHz and nuclear quadrupole resonance (NQR) signals at 423 kHz. The rf magnetometer was first introduced and its fundamental sensitivity was analyzed in [Savukov *et al.* \(2005\)](#), NMR detection was first presented in [Savukov *et al.* \(2007\)](#), and NQR detection was first presented in [Lee *et al.* \(2006\)](#).

4.1 Detection of Radio-Frequency Magnetic Fields

Atomic magnetometers have previously been designed for detection of quasi-static magnetic fields, with sensitivity dropping off as approximately $1/f$ at high frequency. Expanding the range of such a magnetometer to allow for detection of oscillating fields requires an increase in the magnetic linewidth, which decreases overall magnetometer sensitivity; detection at frequencies above several kilohertz is virtually impossible. However, the radio-frequency atomic magnetometer can be made as sensitive to high-frequency fields as traditional magnetometers are to near-dc fields. The alkali atoms are placed in a holding field $B_0\hat{z}$ parallel to the pumping direction, which is set such that the atomic Larmor frequency matches the frequency of the oscillating field to be detected:

$$\omega_0 = \gamma B_0. \quad (4.1)$$

The oscillating field $B_{\text{RF}}\hat{y}\cos(\omega_0 t)$ causes the polarization of each spin to spiral outward from the pumping axis, with the transverse spin component increasing until absorption of a pump photon resets the polarization direction to lie parallel to the pump axis, or until a spin-relaxation mechanism randomizes the polarization direction. The resonant rf field sets the coherent phase of the precessing spins, and the average transverse spin is given by Equation 2.128.¹ The precessing spin component is detected as a small modulation of the polarization of a probe beam. The principle of operation of a radio-frequency magnetometer is illustrated in Figure 4.1.

By adjusting the amplitude of the holding field, the magnetometer may be tuned to detect rf fields anywhere within the range of several kilohertz to many megahertz with minimal variation in the resonance response amplitude.² It can therefore be used to quickly

¹ The rf magnetometer is typically used to detect very faint fields, so rf broadening is negligible, and it is not necessary to use the more general atomic spin response given by Equation 3.9.

² We will see that the signal changes by about a factor of 2 as a function of resonance frequency when using light narrowing to minimize the magnetometer linewidth. However, this variation is insignificant compared to the linear scaling of a pick-up coil signal with the resonance frequency.

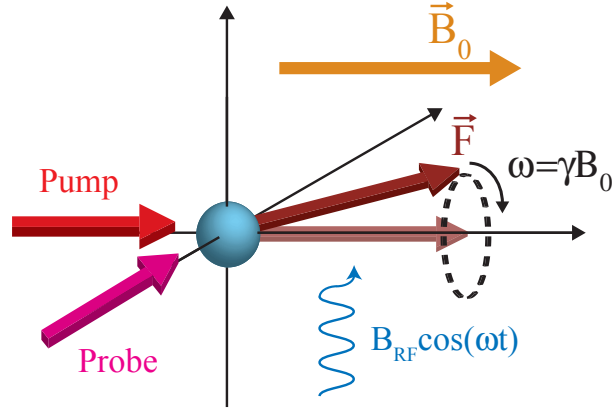


Figure 4.1: Principle of operation of an rf atomic magnetometer: A resonant radio-frequency magnetic field induces a small transverse spin component that precesses coherently at the Larmor frequency and is detected as a small modulation of the probe beam polarization.

scan over a broad frequency range for magnetic resonance signals. The minimum operating frequency is set by the requirement that the resonance frequency be much larger than the magnetic linewidth, while the maximum operating frequency is set by the requirement that the Zeeman splitting remain much smaller than the hyperfine splitting. Under these conditions, if the atoms are nearly completely polarized along $\hat{z}$, then the transverse polarization induced by a resonant rf field is approximately given by

$$P_x = S_x/S_z = \frac{1}{2}\gamma B_{RF}T_2 \sin(\omega_0 t). \quad (4.2)$$

We see that the sensitivity is proportional to the transverse spin-relaxation time T_2 , which should generally be maximized for optimum magnetometer performance. While magnetic resonance broadening due to spin-exchange collisions can be completely suppressed at very low magnetic fields (see Section 5.1), at the higher fields required by rf operation it can be only partially suppressed by light narrowing, which is described in Section 4.1.1. Note that the rf magnetometer is twice as sensitive to rotating magnetic fields as it is to oscillating fields, as shown in Section 2.6.

The fundamental sensitivity limit of the rf magnetometer is on the order of $0.01 \text{ fT}/\sqrt{\text{Hz}}$, as discussed in Section 4.1.2, and we have achieved sensitivity of $0.2 \text{ fT}/\sqrt{\text{Hz}}$. It is easier to attain high sensitivity to fields in the radio-frequency range than to quasi-static fields because at high frequency the magnetometer is relatively insensitive to mechanical and laser noise, as well as to changes in the dc value of the field. [Ledbetter *et al.* \(2007\)](#) have since demonstrated a completely different type of atomic magnetometer for detection of radio-frequency fields based on nonlinear magneto-optical rotation. They have demonstrated sensitivity of $10 \text{ fT}/\sqrt{\text{Hz}}$, with shot-noise limited sensitivity of $2 \text{ fT}/\sqrt{\text{Hz}}$. While not as intrinsically sensitive as the magnetometer described here, the magnetometer invented by [Ledbetter *et al.*](#) has the advantages of operating much closer to room temperature and requiring only one laser, possibly allowing for easier development into a compact, portable device. In addition, [Katsoprinakis *et al.* \(2006\)](#) have proposed a radio-frequency atomic magnetometer based on electromagnetically induced transparency.

4.1.1 Light Narrowing

The rf magnetometer operates at high density, typically on the order of 10^{14} cm^{-3} , in order to improve both the atomic shot-noise limit on sensitivity and the size of the optical rotation signals. In this regime the magnetic resonance linewidth is broadened considerably by spin-exchange collisions. Since the transverse atomic polarization induced by an rf field, and thus the size of the magnetometer signal, is proportional to the coherence lifetime T_2 as shown by Equation 4.2, it is necessary to suppress spin-exchange relaxation in order to achieve high sensitivity. While spin-exchange relaxation can be completely eliminated by operating near zero magnetic field, as in a SERF magnetometer, resonant detection of radio-frequency fields requires operation in a much larger holding field. Instead, spin-exchange relaxation is partially suppressed by light narrowing, which was first discovered

by Appelt *et al.* (1999). Light narrowing can also be used to narrow the microwave end resonances for use by atomic clocks (Jau *et al.*, 2004).

Light narrowing occurs in an ensemble with nearly full atomic polarization, so that almost all alkali atoms are in the $m_F = F$ Zeeman end state with maximum angular momentum. When two atoms in this state undergo a spin-exchange collision, conservation of angular momentum requires that they remain in the end state after the collision. The atoms therefore can not switch hyperfine levels; in a fully-polarized vapor with all atoms in the end state, spin-exchange collisions do not contribute at all to relaxation of the spin polarization. However, in order to detect radio-frequency fields it is necessary to have a finite coherence between at least two Zeeman levels, so there must be a nonzero population in the $m_F = F - 1$ sublevel. Spin-exchange relaxation thus can not be completely eliminated in an rf magnetometer, but it can be highly suppressed.

The transverse spin relaxation time can be calculated using the formalism developed by Appelt *et al.* (1998) for operation in the spin-temperature distribution. The strength of the coherence $|F, m_F\rangle \rightarrow |F, m_F - 1\rangle$ with mean azimuthal quantum number $\bar{m} = m_F - 1/2$ is proportional to the population difference between the two Zeeman sublevels, given by

$$Q_{\bar{m}} = \frac{2P(1+P)^{I+\bar{m}}(1-P)^{I-\bar{m}}}{(1+P)^{2I+1} - (1-P)^{2I+1}}. \quad (4.3)$$

For nearly full polarization with $P \approx 1$, only the end resonance with $F = I + 1/2$, $\bar{m} = I$ has finite amplitude, corresponding to the case where almost all atoms are in the $m_F = F$ end state with a small fraction in the $m_F = F - 1$ states. Note however that for full polarization $P = 1$, this end coherence has zero amplitude, and no transverse spin polarization can be induced.

The relaxation operator Λ has matrix elements given by³

$$\begin{aligned} \text{Re}(F, \bar{m} | \Lambda | F', \bar{m}') &= \delta_{FF'} \left\{ \delta_{\bar{m}\bar{m}'} \left(R' \frac{3[I]^2 + 1 - 4\bar{m}^2}{4[I]^2} - R' s_z' \frac{\bar{m}}{[I]} (-1)^{I+1/2-F} \right) \right. \\ &\quad - \sum_{p=\pm 1} \delta_{\bar{m}, \bar{m}'+p} \frac{R' + p R' s_z'}{2} (F, \bar{m} | S_+) (S_+ | F, \bar{m}') \\ &\quad \left. - R_{\text{SE}} Q_{\bar{m}}(F, \bar{m} | S_+) (S_+ | F, \bar{m}') \right\} \end{aligned} \quad (4.4)$$

$$\text{Im}(F, \bar{m} | \Lambda | F', \bar{m}') = \omega_{F\bar{m}} \delta_{FF'} \delta_{\bar{m}\bar{m}'} \quad (4.5)$$

$$(F, \bar{m} | S_+) = \frac{(-1)^{I+1/2-F}}{2[I]} \sqrt{[F]^2 - 4\bar{m}^2} \quad (4.6)$$

$$R' = R_{\text{SE}} + R_{\text{SD}} + R_{\text{OP}} \quad (4.7)$$

$$R' s_z' = P R_{\text{SE}} + R_{\text{OP}} s_z, \quad (4.8)$$

where R_{SE} is the spin-exchange rate, R_{SD} is the spin-destruction rate, R_{OP} is the optical pumping rate, s_z is the polarization of the pump beam, and $[x] = 2x + 1$. The resonance frequency $\omega_{F\bar{m}}$ of each coherence must include the effects of the nonlinear Zeeman splitting (see Section 3.2), which lifts the degeneracy between the resonance frequencies within each hyperfine multiplet. The eigenvalues of Λ describe the resonance widths of the individual Zeeman transitions, with the smallest eigenvalue giving the width of the $\bar{m} = I$ end resonance in the case of nearly full polarization.

For $P \approx 1$, $R_{\text{SE}} \gg R_{\text{SD}}$, and $s_z=1$, the width of the end resonance can be expanded in powers of R_{SD} to give the transverse polarization lifetime,

$$\frac{1}{T_2} = \Delta\omega = \frac{R_{\text{OP}}}{2I+1} + \frac{R_{\text{SE}} R_{\text{SD}}}{R_{\text{OP}}} G(\omega_0, R_{\text{SE}}). \quad (4.9)$$

The value of the parameter $G(\omega_0, R_{\text{SE}})$ depends on whether or not the individual Zeeman resonances are well-resolved, and it varies as the resonance frequency ω_0 increases and the

³ This analysis is conducted in Liouville space, as opposed to the more customary Schrödinger space, and so matrix elements are denoted as $|\rho\rangle$. A more complete description of Liouville space is provided by Appelt *et al.* (1998).

nonlinear Zeeman splitting becomes larger. Its value depends on the nuclear spin I and is approximately given by⁴

$$G(\omega_0, R_{\text{SE}}) = \text{Re} \left[\frac{R_{\text{SE}} + 8i\omega_0^2/\omega_{\text{hf}}}{5R_{\text{SE}} + 16i\omega_0^2/\omega_{\text{hf}}} \right] \quad (I = 3/2) \quad (4.10)$$

$$G(\omega_0, R_{\text{SE}}) = \text{Re} \left[\frac{3R_{\text{SE}} + 22i\omega_0^2/\omega_{\text{hf}}}{10R_{\text{SE}} + 44i\omega_0^2/\omega_{\text{hf}}} \right] \quad (I = 5/2) \quad (4.11)$$

$$G(\omega_0, R_{\text{SE}}) = \text{Re} \left[\frac{5R_{\text{SE}} + 36i\omega_0^2/\omega_{\text{hf}}}{14R_{\text{SE}} + 72i\omega_0^2/\omega_{\text{hf}}} \right] \quad (I = 7/2), \quad (4.12)$$

where ω_{hf} is the ground-state hyperfine splitting. A comparison of the analytic approximation given by Equations 4.9 and 4.10 to the exact linewidth given by the eigenvalue of Λ is shown in Figure 4.2(a) for the case of ^{39}K with $R_{\text{SE}}=91,000 \text{ s}^{-1}$ and $R_{\text{SD}}=20 \text{ s}^{-1}$, showing good agreement. We see that there is an optimal pumping rate at which the rf magnetometer should be operated that gives the minimum resonance linewidth. Below this point the atomic polarization is too small for complete light narrowing, while above this point the optical pumping rate is too high and the absorption of pump photons destroys the small transverse polarization necessary for detection of rf fields.

From Equation 4.9, we see that the maximum possible transverse polarization lifetime in the light-narrowing regime is given by

$$T_{2,\text{LN}} = \sqrt{\frac{2I + 1}{4G(\omega_0, R_{\text{SE}}) R_{\text{SE}} R_{\text{SD}}}}. \quad (4.13)$$

In comparison, the lifetime in the limit of low polarization at large magnetic field in the absence of other relaxation mechanisms is given by the spin-exchange broadening factor (Equation 2.138) as

$$T_{2,\text{SE}} = \frac{3(2I + 1)^2}{2I(2I - 1)R_{\text{SE}}}. \quad (4.14)$$

The light-narrowing factor $T_{2,\text{LN}}/T_{2,\text{SE}}$ describes the extent to which light narrowing enhances the polarization lifetime due to suppression of spin-exchange relaxation. Values

⁴ The approximation for $I = 3/2$ was previously presented in Savukov *et al.* (2005) and is accurate for large pumping rates, including the point of minimum linewidth. The approximations for $I = 5/2$ and $I = 7/2$ have not been previously published and may not be as accurate, although they are still good to within a few percent and again predict the pumping rate that provides optimal linewidth.

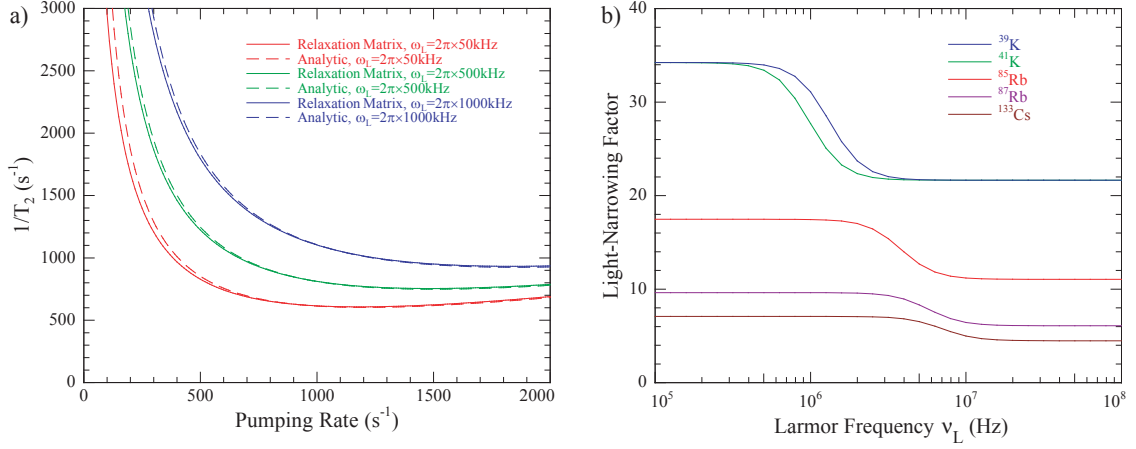


Figure 4.2: Light narrowing of magnetic resonances at high polarization $P \approx 1$. **(a)** Comparison of the analytic expressions for the light-narrowed linewidth given by Equations 4.9-4.12 to the exact linewidth given by the relaxation matrix Λ for ^{39}K with representative $R_{SE}=91,000\text{ s}^{-1}$ and $R_{SD}=20\text{ s}^{-1}$. **(b)** Light-narrowing factors for the alkali isotopes as a function of the resonance frequency, assuming that $n=10^{14}\text{ cm}^{-3}$ and that spin-destruction is solely due to collisions between alkali atoms.

for the light-narrowing factor for the various alkali isotopes are shown in Figure 4.2(b) for density $n=10^{14}\text{ cm}^{-3}$, assuming that only collisions between alkali atoms contribute to the spin-destruction rate R_{SD} and using values for the spin-exchange and spin-destruction cross-sections given by Table A.2. The light-narrowing factor changes near the characteristic frequency at which the quadratic Zeeman splitting becomes comparable to the magnetic linewidth. In the regime of well-resolved Zeeman transitions, the light-narrowing factor ranges from 7 for cesium to 34 for potassium,⁵ allowing for long polarization lifetimes and sensitive detection of rf magnetic fields.

Figure 4.3(a) shows the measured resonance linewidth $\Delta\nu = \Delta\omega/2\pi$ and magnetometer signal response as a function of optical pumping rate in ^{39}K . As predicted by Equation 4.2, the signal is proportional to the coherence lifetime T_2 and reaches a maximum at the ideal optical pumping rate. The linewidth measurements are compared to the analytic

⁵ We use a value for the K-K spin-exchange cross-section of $\sigma_{SE}=1.5 \times 10^{-14}\text{ cm}^2$ for this calculation (Ressler *et al.*, 1969). Using instead a recent measurement of $\sigma_{SE}=1.8 \times 10^{-14}\text{ cm}^2$ (Aleksandrov *et al.*, 1999), as listed in Table A.2, gives a light-narrowing factor of 37, as reported in Savukov *et al.* (2005).

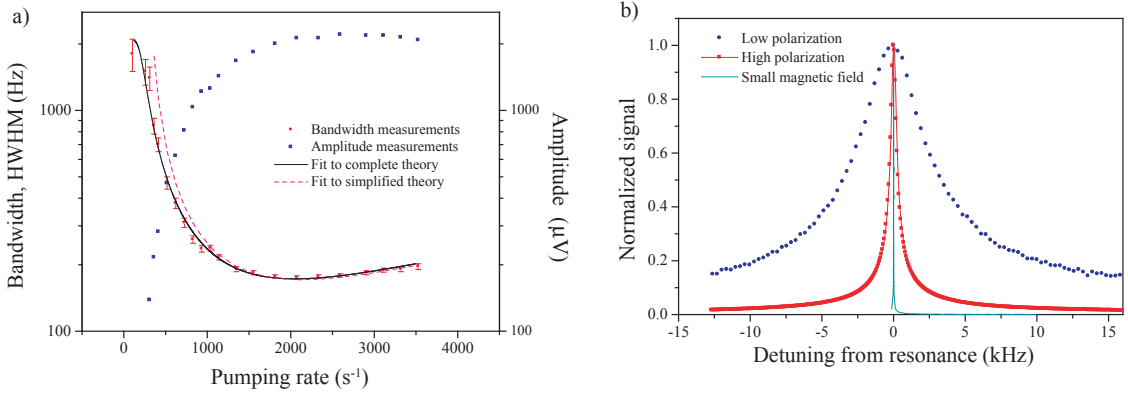


Figure 4.3: (a) Comparison of theory and experiment for the dependence of the Zeeman resonance width $\Delta\nu$ and rf signal response on optical pumping rate. (b) Comparison of Zeeman resonance widths measured in ^{39}K with $R_{\text{SE}}=10^5 \text{ s}^{-1}$ under different modes of operation: in high magnetic field and low polarization, in high magnetic field and high polarization with spin-exchange broadening partially suppressed (the light-narrowing regime), and in very low magnetic field with spin-exchange broadening completely eliminated (the SERF regime).

solution given by Equations 4.9 and 4.10, as well as to a more complete theory including the effect of light attenuation as the pump beam propagates through the cell. Although the complete theory is more accurate at low pumping rate where there is greater absorption of pump light, the simple theory is nevertheless accurate at high pumping rate and accurately predicts the optimal R_{OP} , where there is an observed maximum linewidth narrowing by a factor of about 10. This is smaller than the predicted factor of 34 because collisions with buffer gas atoms and the cell walls increase the spin-relaxation rate; even narrower resonances may be obtainable in coated cells without buffer gas. In Figure 4.3(b) we compare the Zeeman resonance at the optimal pumping rate for light narrowing to that obtained with low polarization and full spin-exchange broadening. At low polarization the linewidth scales as R_{SE} , while at high polarization it scales as $(R_{\text{SE}}R_{\text{SD}})^{1/2}$. For comparison, we also show the magnetic resonance measured at very low magnetic field (the SERF regime, see Section 5.1) with comparable spin-exchange and spin-destruction rates, demonstrating

that spin-exchange broadening is completely eliminated and that the linewidth scales as R_{SD} , which is significantly smaller than R_{SE} .

4.1.2 Fundamental Sensitivity

In Section 2.8 we considered the fundamental limit on magnetometer sensitivity due to quantum fluctuations, given by

$$\delta B = \sqrt{\delta B_{\text{spn}}^2 + \delta B_{\text{psn}}^2 + \delta B_{\text{lsn}}^2}, \quad (4.15)$$

where δB_{spn} is the spin-projection noise due to the finite number of alkali spins used in the measurement, δB_{psn} is the photon shot noise due to the finite number of probe photons detected, and δB_{lsn} is the light-shift noise due to fluctuations in the polarization of the probe beam. We derived the first two effects in terms of the spin polarization; using Equation 4.2, we may give the noise in the measurement of the magnetic field component co-rotating with the atoms at frequency ω_0 in terms of the polarization as

$$\delta \langle P_x \rangle = \frac{\delta \langle F_x \rangle}{F_z} = \frac{\delta \langle S_x \rangle}{S_z} = \delta B_{\text{RF}}(\gamma T_2/2). \quad (4.16)$$

We may then write the spin-projection noise as

$$\delta B_{\text{spn}} = \frac{1}{\gamma} \sqrt{\frac{8}{(I + 1/2)nVT_2}}, \quad (4.17)$$

where $V = N/n$ is the active measurement volume defined by the intersection of the pump and probe beams, and $F_z \approx F = I + 1/2$ for operation with light narrowing. If we take into account the quantum efficiency η of the photodiodes used to detect the probe beam, then the effective number of detected photons is decreased by $1/\eta$, increasing the photon shot noise by $1/\sqrt{\eta}$:

$$\delta B_{\text{psn}} = \frac{4}{\pi \gamma \ln r_e c f T_2 \text{Im}[\mathcal{V}(\nu - \nu_0)] \sqrt{2\Phi'\eta}}. \quad (4.18)$$

The light-shift noise in the co-rotating component of the magnetic field is decreased by a factor of 2 from that given by Equation 2.169,

$$\delta B_{lsn} = \frac{\pi r_e c f \text{Im}[\mathcal{V}(\nu - \nu_0)] \sqrt{2\Phi'}}{2(2I + 1)\gamma A}. \quad (4.19)$$

If there is sufficient buffer gas in the cell that the Voigt profile is approximately described by a Lorentzian, and the probe beam frequency ν is sufficiently detuned from the resonance frequency ν_0 that $(\nu - \nu_0) \gg \Gamma_L$, then we may approximate

$$\text{Im}[\mathcal{V}(\nu - \nu_0)] \simeq \frac{1}{\pi(\nu - \nu_0)}. \quad (4.20)$$

Under these conditions, the probe beam optical pumping rate is given by

$$R_{\text{pr}} = \sigma_L(\nu)\Phi'/A = \frac{\Phi' r_e c f \Gamma_L}{2A(\nu - \nu_0)^2}, \quad (4.21)$$

where A is the probe beam cross-section, and the optical depth of a resonant laser is given by

$$\text{OD} = \sigma_L(\nu_0)nl = \frac{2nlr_e c f}{\Gamma_L}. \quad (4.22)$$

We may then write the total magnetic noise as

$$\delta B = \frac{1}{\gamma\sqrt{nV}} \sqrt{\frac{16}{(2I + 1)T_2} + \frac{8}{R_{\text{pr}}\text{OD}T_2^2\eta} + \frac{R_{\text{pr}}\text{OD}}{2(2I + 1)^2}}. \quad (4.23)$$

In general, the effect of R_{pr} on T_2 can be reduced by sufficient detuning of the probe laser. Also, the sensitivity can be optimized by adjusting the probe beam frequency so that the last two terms in Equation 4.23 are equal, giving

$$R_{\text{pr}}\text{OD} = \frac{4(2I + 1)}{T_2\sqrt{\eta}}. \quad (4.24)$$

For optimal light narrowing such that T_2 is given by Equation 4.13, the fundamental limit on the sensitivity of the rf magnetometer is set by the spin-exchange and spin-destruction cross-sections,

$$\delta B_{\text{min}} = \frac{2}{\gamma} \sqrt{\frac{2\bar{v}\sqrt{G(\omega_0, R_{\text{SE}})}\sigma_{\text{SE}}\sigma_{\text{SD}}}{V(2I + 1)^{3/2}} \left(4 + \frac{1}{\sqrt{\eta}}\right)}, \quad (4.25)$$

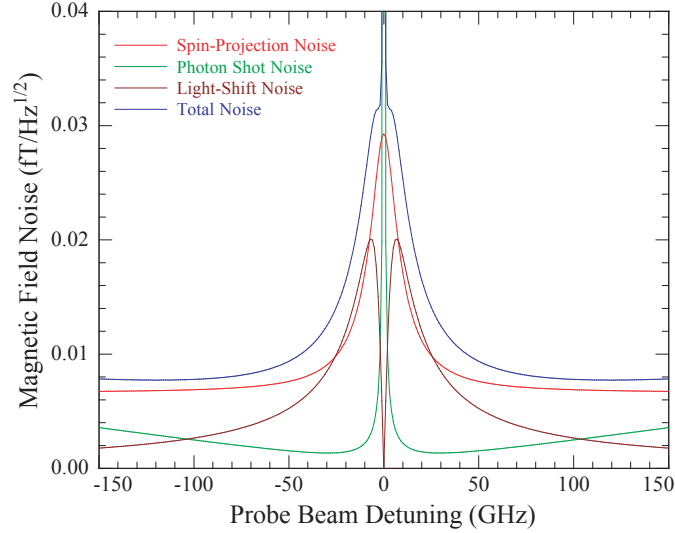


Figure 4.4: Comparison of the size of quantum noise effects in an rf magnetometer operating with potassium under typical conditions as a function of probe beam detuning. The total magnetic noise sets the fundamental limit on magnetometer sensitivity and is generally close to the level of the spin-projection noise. For a cell with volume 100 cm^3 , the fundamental sensitivity is about $0.01 \text{ fT}/\sqrt{\text{Hz}}$.

where we assume that spin destruction is solely due to alkali-alkali collisions, and $\bar{v}$ is the relative thermal velocity of such collisions. Under these conditions, the fundamental sensitivity depends only on the volume of the cell and not the alkali density. The three sources of quantum noise are compared in Figure 4.4 for operation using potassium with $G=1/5$, $n=10^{14} \text{ cm}^{-3}$, $V=100 \text{ cm}^3$, $l=6 \text{ cm}$, $\Gamma_L=13.3 \text{ GHz}$ (given by 1 amg of helium buffer gas), $\eta=0.8$, and probe beam power of 40 mW and area $A=16.7 \text{ cm}^2$. The total noise is generally close to the spin-projection noise, although the photon shot noise blows up at zero detuning because of resonant attenuation of the probe beam. The minimum total magnetic noise occurs when the photon shot noise and light-shift noise are equal, as predicted above, giving fundamental sensitivity of approximately $0.01 \text{ fT}/\sqrt{\text{Hz}}$.

4.1.3 Comparison to an Inductive Pick-Up Coil

The fundamental sensitivity limit of a surface pick-up coil is considered in [Savukov *et al.* \(2007\)](#) and compared to that of an rf magnetometer. Inductive surface coils have traditionally been used for detection of radio-frequency fields in applications such as magnetic resonance imaging (MRI) and NQR, and in such applications the coil can be replaced directly with an atomic magnetometer cell to give approximately the same filling factor. We do not replicate the entire discussion here, but instead we only summarize the results. We consider a coil with mean diameter D and a square-winding cross-section of size $W \times W$ with $W \ll D$, as shown in the inset of Figure 4.5. The coil contains N turns of wire, each with diameter d , that fill the available winding volume $V_w = \pi DW^2$, giving $N = 4W^2/\pi d^2$. The voltage induced in a coil by a uniform magnetic field oscillating at frequency ω with amplitude B is given by

$$V = \frac{1}{4} B \omega \pi D^2 N, \quad (4.26)$$

while the Johnson noise spectral density is given by

$$\delta V = \sqrt{16 k_B T \rho D N / d^2}, \quad (4.27)$$

where ρ is the resistivity of the wire material, and T is the temperature. Combining these relations gives the magnetic field sensitivity limited by Johnson noise, valid at frequencies below a few tens of kilohertz,

$$\delta B_{\text{lf}} = \frac{8}{\omega D} \sqrt{\frac{k_B T \rho}{V_w}}. \quad (4.28)$$

Thus, the ideal sensitivity of an inductive coil improves linearly with the signal frequency.

At higher frequency we must consider the effects of eddy currents that give rise to the skin depth effect, as well as parasitic capacitance between coil turns; a detailed treatment of these effects for circular wires is given by [Butterworth \(1922\)](#). A common technique for improving coil performance is to use Litz wire, which is made of many strands of very thin wire connected in parallel, reducing the skin effect of individual wires shielding

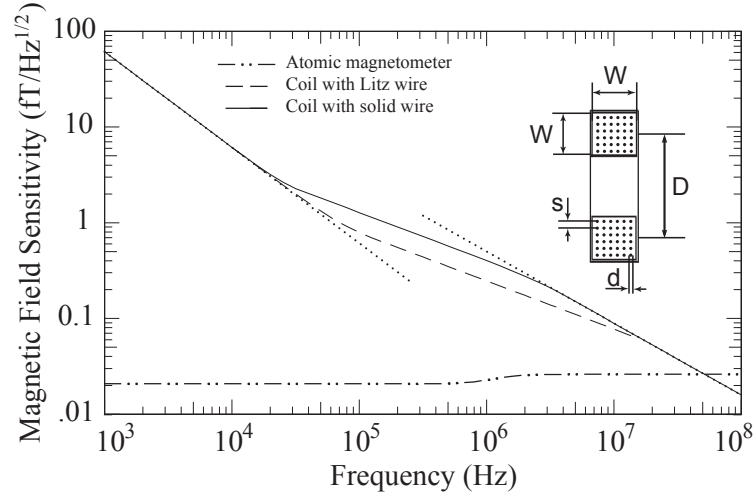


Figure 4.5: Estimated optimal magnetic field sensitivity of a surface pick-up coil with dimensions $D=5$ cm and $W=1$ cm, including both solid wire and Litz wire with 1000 strands, compared to the fundamental sensitivity of a potassium rf magnetometer occupying the same volume. The magnetometer has better sensitivity at frequencies below approximately 50 MHz.

the rf signal. Litz wire is effective for frequencies below a few megahertz, but at higher frequencies it is not practical to make wire with diameter less than the skin depth. At frequencies higher than a few megahertz, the noise is given by a slightly modified form of Equation 4.28,

$$\delta B_{\text{hf}} = \frac{8}{\omega D} \sqrt{\frac{k_B T \rho}{1.8 \pi D W \delta}}, \quad (4.29)$$

where $\delta = \sqrt{2\rho/\omega\mu_0}$ is the skin depth of the rf field in the metal conductor.⁶ In the high-frequency limit, current is only induced in the surface of the coil winding within a skin depth. In the intermediate frequency range, the sensitivity is limited by self-resonance effects or the minimum practical wire diameter; details are given in Savukov *et al.* (2007).

Figure 4.5 shows the expected magnetic field sensitivity for a surface pick-up coil with $D=5$ cm and $W=1$ cm. The diameter of the wires and the spacing between wires are optimized for each frequency, subject to the constraint that the diameter be larger than $30 \mu\text{m}$.

⁶ The skin depth δ should not be confused with the uncertainty in the measurement of some observable, written for instance as δB for the case of the magnetic field.

It can be seen that the sensitivity is well approximated by Equations 4.28 and 4.29 for frequencies below 30 kHz and above 10 MHz, respectively, and that in the intermediate range Litz wire improves sensitivity by about a factor of 2. We also show the fundamental sensitivity of an optimized potassium rf magnetometer occupying the same space as the coil with $V = \pi(D + W)^2 W / 4$, as given by Equation 4.25. While the pick-up coil sensitivity improves linearly with frequency, the magnetometer sensitivity changes only slightly due to the dependence of $G(\omega_0, R_{SE})$ on frequency. As a result, the fundamental atomic magnetometer sensitivity exceeds that of the surface coil at frequencies below about 50 MHz for the given geometry.

Given some characteristic size l of the pick-up coil such that $D, W \propto l$, Equations 4.28 and 4.29 show that the coil sensitivity scales as $l^{-5/2}$ at low frequency and l^{-2} at high frequency. In comparison, the sensitivity of the comparable magnetometer with $V \propto l^3$ scales as $l^{-3/2}$. The crossover frequency below which the magnetometer outperforms the surface coil therefore increases as the sensors become smaller. Thus, miniaturization of the atomic magnetometer (see for example [Schwindt *et al.* \(2004\)](#) and [Shah *et al.* \(2007\)](#)) to replace microcoils for NMR detection is a promising approach, although at small sizes it may be more difficult to achieve close separation between the heated magnetometer cell and the sample.

4.1.4 Counter-Propagating Pump Beams

In the optically thick vapor necessary for sensitive radio-frequency detection, the pump beam is attenuated as it travels through the cell even if it is fully polarized (see Section 2.3.3). As illustrated in Figures 4.2 and 4.3, the resonance linewidth varies rapidly as the pumping rate decreases from its optimal value. Attenuation of the pump beam can therefore cause loss of sensitivity, since the magnetometer signal results from averaging over the entire active volume measured by the probe beam. However, using two counter-propagating

pump beams that enter the cell from opposite sides, it is possible to obtain a nearly uniform pumping rate across the cell. The two beams must have opposite helicity $s=\pm 1$, so that they polarize the alkali spins along the same direction. If the two beams have equal initial pumping rate R_0 when they enter the cell,⁷ with the first beam entering the cell at $z=0$ and the second at $z=l$, then the change in the pumping rates R_1 and R_2 of the two beams as they propagate through the cell is described by Equation 2.54 as

$$\frac{d}{dz}R_1 = -nR_1(z) [1 - P(z)], \quad R_1(0) = R_0 \quad (4.30)$$

$$\frac{d}{dz}R_2 = +nR_2(z) [1 - P(z)], \quad R_2(l) = R_0 \quad (4.31)$$

$$P(z) = \frac{R_1(z) + R_2(z)}{R_1(z) + R_2(z) + R_{\text{rel}}}. \quad (4.32)$$

R_{rel} is the spin-relaxation rate, including the effects of spin-destruction and wall collisions. We can solve this system of differential equations numerically to calculate the polarization throughout the cell.

In order to determine the effectiveness of the counter-propagating beam technique, we compare the magnetometer signals resulting from the use of one and two beams. From Equation 4.2, we see that the signal is proportional to PT_2 . We optimize the initial pumping rate at the front of the cell in order to maximize the total signal, given by the integral of PT_2 across the cell along the pumping direction, under a particular set of experimental conditions. The spin coherence lifetime is calculated from the appropriate eigenvalue of the relaxation matrix Λ described in Section 4.1.1. We find in general that for the optimal cases the initial pumping rate for the single beam is about double the initial pumping rates for each of the counter-propagating beams, so that the total laser power is the same. In Figure 4.6(a) we compare the change in pumping rates throughout a cell containing potassium vapor with density $n=10^{14} \text{ cm}^{-3}$, illustrating that the total pumping rate is significantly more uniform when counter-propagating pump beams are used.

⁷ This can be easily achieved by splitting a single pump beam into two and sending them into the cell from opposite directions, as shown in Figure 4.16.

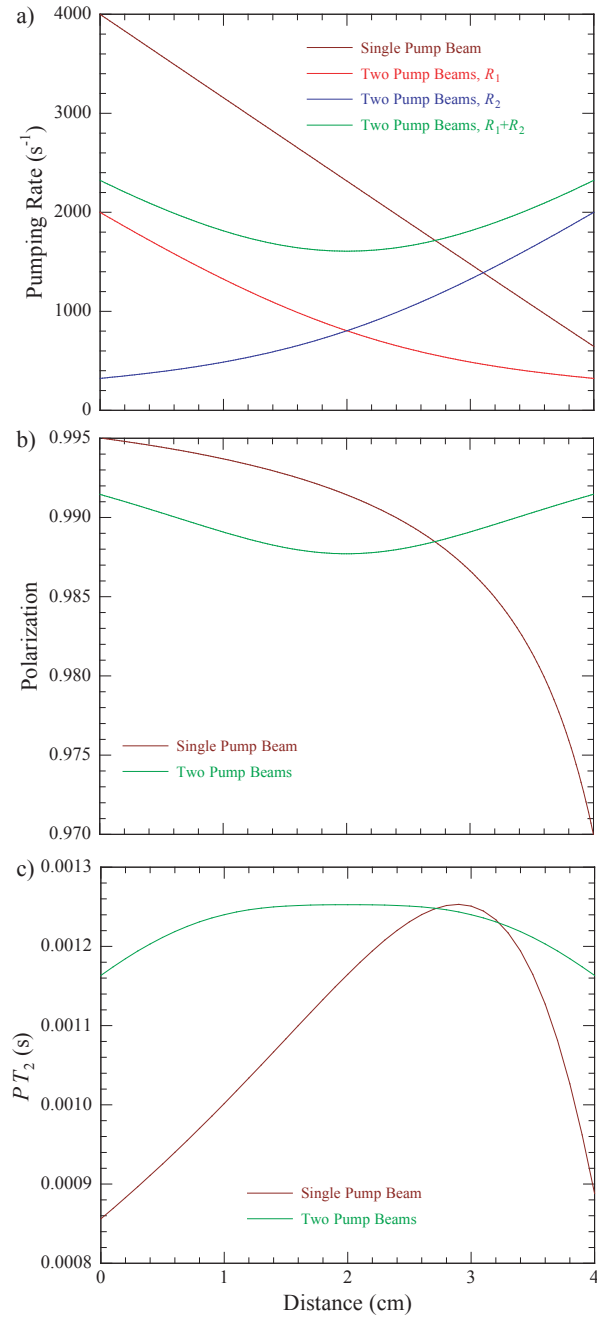


Figure 4.6: Comparison of rf magnetometer performance using a single pump beam versus using two counter-propagating pump beams. Performance is calculated in a 4 cm long cell containing potassium with density $n=10^{14} \text{ cm}^{-3}$ and 1 amg of helium buffer gas, with $R_{\text{rel}}=20 \text{ s}^{-1}$ and $\omega_L=2\pi\times 423 \text{ kHz}$. The total pumping rate is more uniform throughout the cell when it is illuminated by counter-propagating beams and thus remains near the optimal value for light narrowing, resulting in a larger magnetometer signal (which is proportional to PT_2).

Although polarization remains nearly constant throughout the cell for both techniques (as shown in Figure 4.6(b)), the spin coherence lifetime decreases as the pumping rate deviates from its ideal value for light narrowing, causing the magnetometer signal to vary throughout the cell (as shown in Figure 4.6(c)). The magnetometer signal remains near its maximum value when counter-propagating pump beams are used, but it deviates from this value when only a single beam is used. For these data the counter-propagation technique increases the total magnetometer signal by 13%; this improves to 34% when the density increases to $5 \times 10^{14} \text{ cm}^{-3}$. However, these numbers are given assuming that the initial pumping rates are at their optimal values. The total magnetometer signal is more sensitive to variation in the initial pumping rate if only a single pump beam is used, so it is easier to achieve optimal conditions using two beams. In addition, the counter-propagation technique provides more uniform sensitivity across multiple channels in a gradiometric measurement. There is thus practical value in the use of counter-propagating pump beams for an rf magnetometer with optically thick alkali vapor. We demonstrated this technique as part of the NQR detection experiment discussed in Section 4.3.

4.2 Detection of Nuclear Magnetic Resonance

NMR and MRI studies are traditionally performed in magnetic fields on the order of 1-10 Tesla, which give large thermal polarization of samples and allow sensitive detection of high-frequency resonance signals with inductive pick-up coils. However, there are several major advantages to operating in the low magnetic field regime of tens to hundreds of microtesla (on the order of several Gauss) where coil performance is limited. Low-field applications do not require the use of a superconducting magnet, greatly reducing costs and simplifying operation. Resonance linewidths are significantly narrower at low field in both solids (Zax *et al.*, 1985) and liquids (McDermott *et al.*, 2002). In addition, low-field measurement has great potential for biomedical applications, such as simultaneous imaging of

the human brain with both MRI and magnetoencephalography using a single detector (Zotev *et al.*, 2008). SQUIDs have generally been used to detect low-field magnetic resonance signals because their sensitivity does not degrade at low frequencies, unlike coils (Greenberg, 1998). However, SQUIDs require cryogenic cooling, and so their use negates the advantages of operating without a superconducting magnet.

Atomic magnetometers are thus an attractive alternative for low-field imaging, as their sensitivity exceeds that of pick-up coils at frequencies below several tens of megahertz (see Section 4.1.3). Previous NMR and MRI experiments with atomic magnetometers detected either static nuclear magnetization (see for instance Cohen-Tannoudji *et al.* (1969) and Yashchuk *et al.* (2004)) or nuclear precession at very low frequency ~ 20 Hz (Savukov and Romalis, 2005b). Detection with the radio-frequency magnetometer offers several advantages; for example, it can be tuned to detect resonance signals in a wide range of magnetic fields, and it may allow measurements of chemical shifts, as has been demonstrated at low field with SQUIDs (Saxena *et al.*, 2001). The available measurement bandwidth is increased because femtotesla-level sensitivity can be obtained with a broader resonance linewidth compared to a traditional magnetometer, and the effects of transverse magnetic field gradients are suppressed by the application of a large longitudinal bias field. The rf magnetometer also has a number of practical advantages, as discussed in Section 4.1, that make it easier to achieve high sensitivity to radio-frequency fields than to quasi-static fields.

The setup of the experiment to detect nuclear magnetic resonance signals from polarized water protons with an rf magnetometer is shown in Figure 4.7. The potassium cell contains about 2.5 amg of helium buffer gas and is heated to 180°C with hot air flowing through a double-walled G7 oven. The oven is covered with fiberglass insulation so that the room-temperature NMR sample can be placed as close as possible to it. Because vibrational noise is not a concern in the radio-frequency range, the oven and lasers are mounted

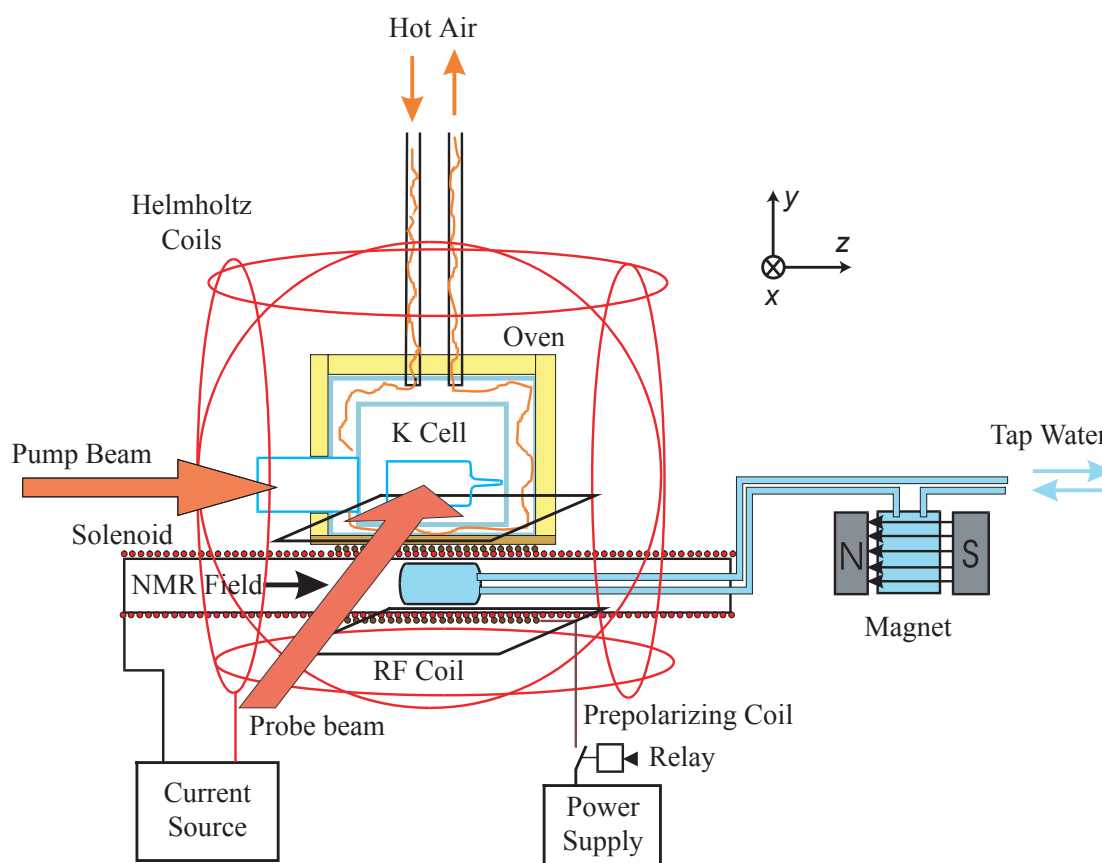


Figure 4.7: Experimental schematic of detection of water NMR with a radio-frequency atomic magnetometer. Helmholtz coils set the atomic resonance frequency, while thin aluminum shields (not shown) partially attenuate high-frequency magnetic noise, and gradient coils (also not shown) cancel field gradients. Water protons are pre-polarized either by flow through a 140 mT permanent magnet or by a 10 mT magnetic field generated *in situ*. A solenoid sets the proton resonance frequency to equal the atomic resonance frequency, and an rf coil applies tipping and spin-echo pulses to the protons. The NMR sample is placed as close as possible to the magnetometer in order to maximize the filling factor.

on an aluminum plate without vibration isolation. The pump beam is an inexpensive multimode diode laser, while the probe beam is a single-mode DFB diode laser,⁸ and the active measurement volume is about 0.5 cm³. Helmholtz coils cancel the ambient magnetic field and generate the bias field along the pump direction, while gradient coils (not shown) cancel the ambient field gradients.

Since the magnetometer is relatively unaffected by near-dc magnetic field noise, we do not use μ -metal shields in this experiment but instead rely only on a 1/16 inch thick aluminum shield to attenuate high-frequency noise, giving sensitivity of about 7 fT/ $\sqrt{\text{Hz}}$. In Figure 4.8 we compare the noise levels of the rf magnetometer operating in an unshielded environment, inside the aluminum shield, and inside a multi-layer μ -metal shield. The broad peak in the magnetometer spectrum is caused by the resonant response of the alkali atoms to ambient magnetic field noise. The height of the peak is equal to the noise level, while the width of the peak gives the magnetometer linewidth and is on the order of 1 kHz. For comparison, with a quasi-static magnetometer operating in an unshielded environment we observed a significantly higher noise level of about 1 pT/ $\sqrt{\text{Hz}}$ (see Section 5.4). The sensitivity can generally be improved by operating as a gradiometer to cancel common-mode noise (see Section 2.7.4), though we make only single-channel measurements in this experiment. Little effort was made to produce an aluminum box that could close tightly, and an aluminum shield with reduced rf leakage could potentially allow for sensitive magnetometer operation without expensive and bulky μ -metal shielding. The magnetometer is shown in Figure 4.9 both with and without the aluminum shield.

The water sample is contained in a cylindrical glass cell with 3 cm diameter and 4 cm length. Tipping and spin-echo pulses are generated by an rf coil wound around the NMR sample. The water protons are either pre-polarized by flowing through a permanent magnet with a field of 140 mT, or they are polarized *in situ* by application of a 10 mT magnetic

⁸ We attempted to use a multimode laser for the probe beam as well, but instability of the laser frequency resulted in excessive measurement noise.

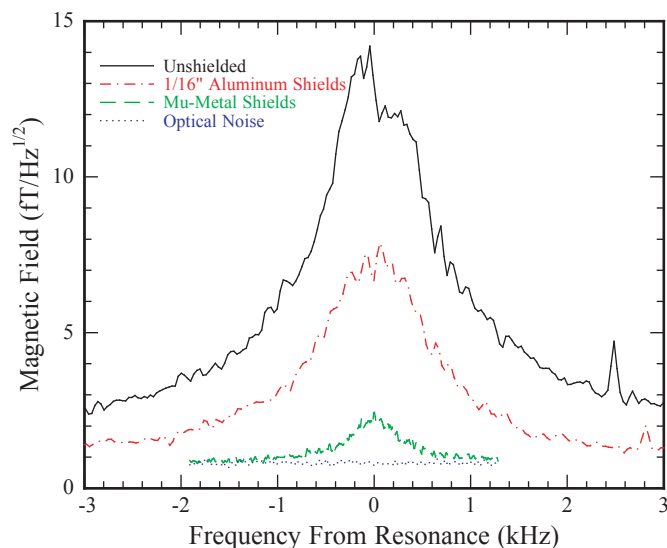


Figure 4.8: Sensitivity of the radio-frequency magnetometer used for NMR detection, limited by ambient field noise. We compare the noise levels without shielding, with thin aluminum shields, and with μ -metal shields. The NMR signals were detected using the aluminum shield, giving sensitivity of about $7 \text{ fT}/\sqrt{\text{Hz}}$.

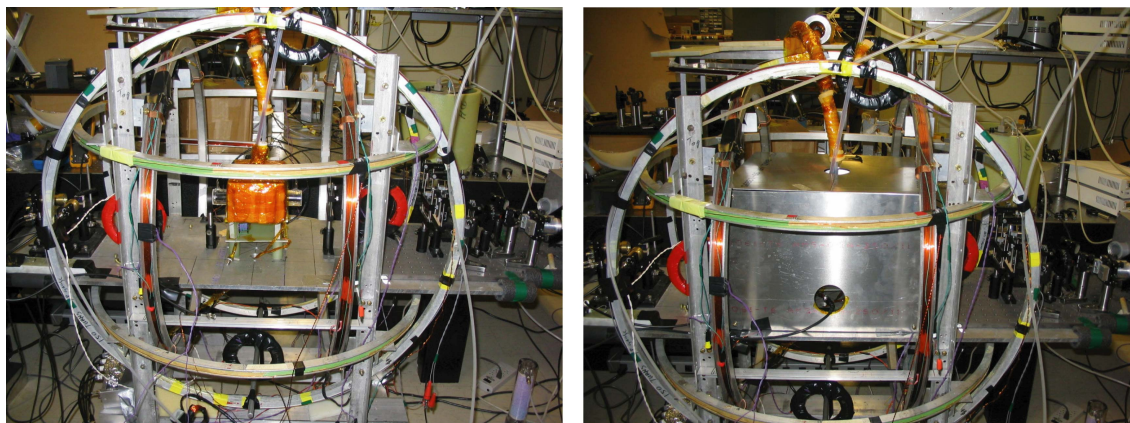


Figure 4.9: Pictures of the rf magnetometer used for NMR detection, both without (left) and with (right) thin aluminum shielding. The oven can be seen covered in amber-colored kapton tape, resting on top of the rf coils used to generate tipping pulses for the NMR sample.

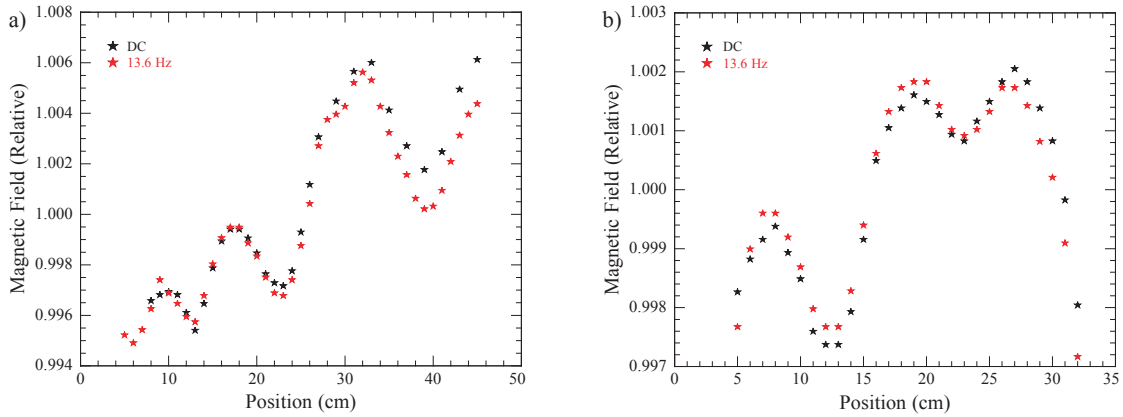


Figure 4.10: Longitudinal magnetic field generated by two different solenoids wound for the NMR sample, measured with a fluxgate sensor and giving homogeneity no better than 10^{-3} over a 4 cm region. The 13.6 Hz measurements were made by applying an oscillating field and measuring with lock-in detection, showing good agreement with measurements taken by applying a static field.

field for several seconds created by a separate coil. The NMR sample is placed inside a solenoid which creates a large field inside for the nuclear spins, in order to match their resonance frequency to that of the alkali spins, while producing relatively little field outside so as not to disturb the magnetometer. The gyromagnetic ratio of the ^1H proton spins in a water molecule is approximately $2\pi \times 4 \text{ kHz/G}$, compared to $2\pi \times 700 \text{ kHz/G}$ for potassium spins, so the water sample must experience a significantly stronger magnetic field than the alkali vapor.

At our NMR frequency of 62 kHz, the solenoid must have a magnetic field homogeneity on the order of 10^{-5} in order for the water free induction decay time T_2^* to be close to the intrinsic spin relaxation time. We attempted to wind several solenoids using different methods, including a lathe with automatic feed control, but we were unable to obtain longitudinal field homogeneity better than 10^{-3} over a 4 cm region within which we could place the NMR cell. Figure 4.10 shows the longitudinal field measured in two of these solenoids; the field appears to vary randomly along the axis, possibly due to slight variation in the size and shape of the wire and its enamel insulation. We therefore use a shimming coil

for the NMR cell that provides longitudinal field homogeneity of 10^{-4} but generates large field gradients away from the axis, giving T_2^* for water NMR signals of 9 ms. The necessity for two separate holding fields for the alkali and nuclear spins is a disadvantage of the rf magnetometer compared to pick-coils and SQUIDs, so it will be necessary to develop magnetic field coils that create more highly uniform fields in order to improve sample spin relaxation times.

The precessing proton spins generate a dipolar field that can be decomposed into two counter-rotating components with unequal amplitude. The alkali and proton spins are separated along the $\hat{y}$ axis by an average distance r , so the field experienced by the atoms due to the precessing proton magnetization $\mathbf{m}(t)$ is

$$\mathbf{B}(t) = \frac{\mu_0}{4\pi r^3} [3(\mathbf{m} \cdot \hat{y}) \hat{y} - \mathbf{m}]. \quad (4.33)$$

The proton spins precess in the $\hat{x}$ - $\hat{y}$ plane, so we may write the magnetization and dipolar field in complex form as in Section 2.6,

$$\tilde{\mathbf{m}} = m_x + im_y \quad (4.34)$$

$$\tilde{\mathbf{B}} = B_x + iB_y. \quad (4.35)$$

If the direction of the solenoid field is such that the magnetization precesses at frequency $-\omega$ as

$$\tilde{\mathbf{m}}(t) = m e^{-i\omega t}, \quad (4.36)$$

then the dipolar field experienced by the alkali atoms is given by

$$\begin{aligned} \tilde{\mathbf{B}}(t) &= \frac{\mu_0}{4\pi r^3} m \left(3i \operatorname{Im} [e^{-i\omega t}] - e^{-i\omega t} \right) \\ &= \frac{\mu_0}{4\pi r^3} m (-2i \sin \omega t - \cos \omega t) \\ &= \frac{\mu_0}{4\pi r^3} m \left(-\frac{3}{2} e^{+i\omega t} + \frac{1}{2} e^{-i\omega t} \right). \end{aligned} \quad (4.37)$$

Since the rf magnetometer is sensitive to the magnetic field component co-rotating with frequency $+\omega$, operation with this orientation of the solenoid field provides a resonance

signal that is three times larger than operation with the opposite orientation. The direction of the magnetic field inside the solenoid should therefore be chosen so that the larger of the two rotating NMR field components is co-rotating with the atomic spins.

The water spins are initially oriented parallel to the solenoid field, so it is necessary to apply a $\pi/2$ tipping pulse in order to induce spin precession that can be detected by the magnetometer (see for instance Fukushima and Roeder (1981)). Applied rf pulses induce ringing in the alkali spins, resulting in a brief dead time as the spins undergo large transverse spin oscillations; the magnetometer is unresponsive to the NMR signal during the dead time, and we must wait until the transverse polarization decays away and the spins are repolarized. In order to reduce the dead time, the bias magnetic field experienced by the alkali atoms is temporarily changed during application of an rf pulse, detuning the atomic Zeeman frequency so that the pulse is no longer on resonance and does not cause significant spin excitation.⁹ It is important that the bias magnetic field is only changed in magnitude and not in direction, so as not to excite spin precession. By increasing the intensity of the pump laser and decreasing the alkali density (allowing faster propagation of the pump beam through the depolarized vapor), the repumping time after the application of an rf pulse can be reduced to about 3 ms, although magnetometer sensitivity is also reduced. In general the dead time should be shorter than the sample relaxation time T_2^* so that the NMR signal does not decay much before the magnetometer is able to measure it.

To further reduce the effect of magnetometer dead time, the NMR signals are acquired using a spin-echo sequence. A π pulse is applied about 12-15 ms after the $\pi/2$ pulse, in order to refocus the transverse proton spin polarization and allow the magnetometer to observe the entire free induction decay signal. The timing of rf pulses and accompanying offsets in the magnetometer bias field are shown in Figure 4.11. The NMR signal following

⁹ It is necessary to reduce eddy currents that prevent quick changes in the bias magnetic field. After finishing this experiment, we later discovered that the aluminum holders of the Helmholtz coils should be cut so that they no longer form complete conductive paths for eddy currents to flow around. This would have aided in the generation of large field offsets that could be quickly turned on and off.

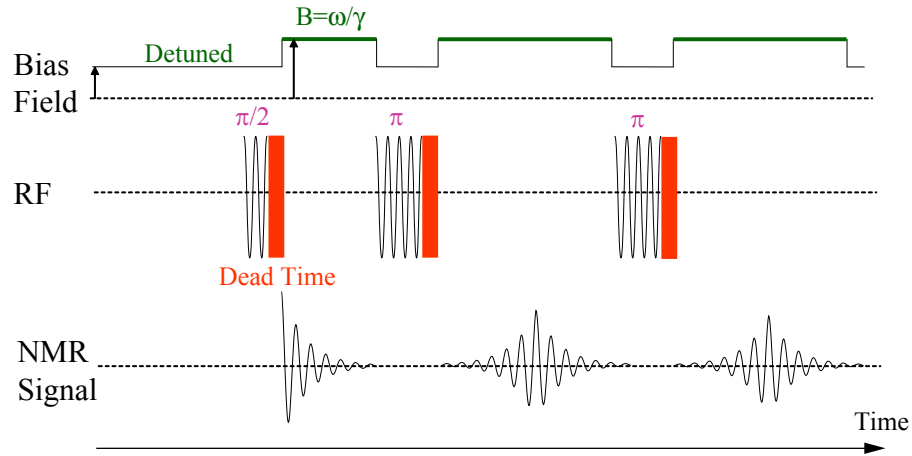


Figure 4.11: Timing of pulses for NMR detection with a radio-frequency atomic magnetometer. During application of pulses to the NMR sample, the atomic resonance frequency is detuned in order to minimize the magnetometer dead time.

a π pulse is shown in Figure 4.12 after 100 averages; the signal is mixed with a reference signal at 60.5 kHz and filtered with a bandwidth of 2 kHz for easier visibility. The $\pi/2$ pulse is applied at $t=0$ ms and is 0.5 ms long, while the π pulse is applied at $t=12$ ms and is 1.0 ms long. Ringing in the alkali spins can be seen during and after application of the π pulse.

The magnetometer output is measured during the entire extent of the spin echo, and its Fourier transform is analyzed to give the NMR signal from the precessing water protons. In Figure 4.13 we show the signal obtained after 10 averages from water pre-polarized by flowing through a permanent magnet, showing the NMR peak near 62 kHz. For comparison, we also show the signal detected with an inductive surface coil with winding volume of 19 cm³; both the magnetometer and coil are located 5 cm away from the water sample. The magnetometer and coil both measure a noise level of $7 \text{ fT}/\sqrt{\text{Hz}}$, so both sensors give the same signal-to-noise ratio limited by the ambient field noise at the detection frequency. However, the sensitivity limit of the coil at this frequency is about $3 \text{ fT}/\sqrt{\text{Hz}}$, so the coil is already near the limit of its performance, while the magnetometer can be further optimized to significantly reduce the measurement noise level.

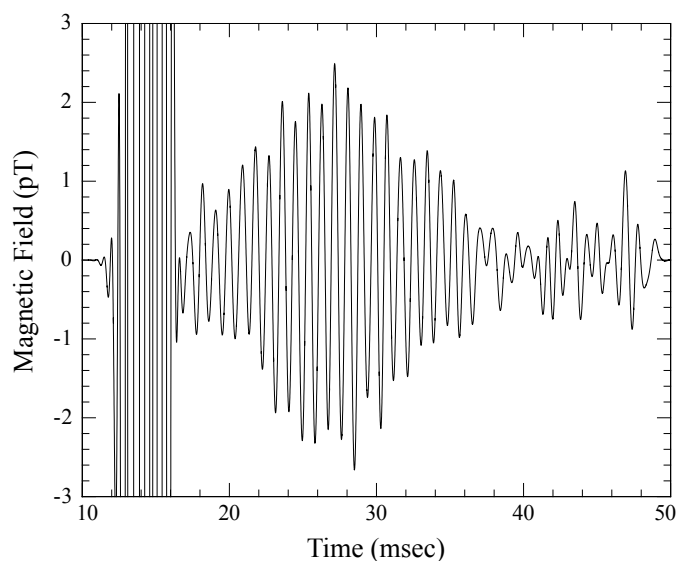


Figure 4.12: NMR signal from polarized water spins after 100 averages. A π spin-echo pulse is applied at $t=12$ ms and lasts 1.0 ms, causing ringing in the alkali atoms. The dead time of the magnetometer is about 3 ms.

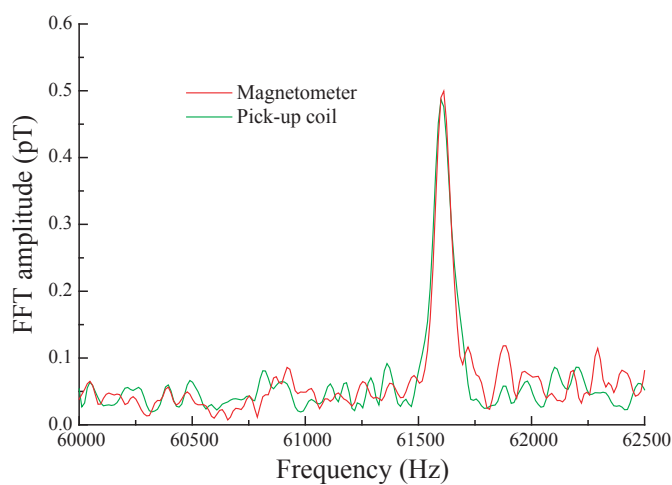


Figure 4.13: Comparison of the water NMR signals obtained after 10 averages with the radio-frequency magnetometer and an inductive pick-up coil. The water is pre-polarized by flowing through a 140 mT permanent magnet before reaching the measurement volume located 5 cm away from both sensors. The signal-to-noise ratio of both detectors is limited by the ambient magnetic field noise.

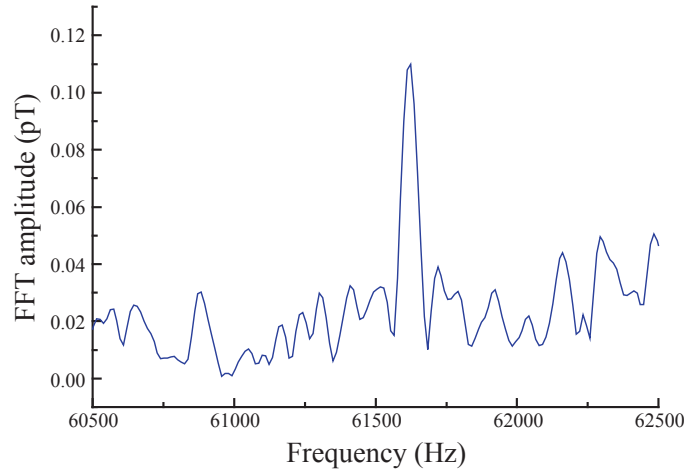


Figure 4.14: NMR signal after 50 averages from water pre-polarized *in situ* by application of a 10 mT magnetic field. The current for the pre-polarizing coil is generated by a power supply connected by a solid-state relay, so that the field can be quickly turned on and off.

We display the signal detected from water pre-polarized *in situ* in Figure 4.14 after 50 averages, showing that the rf magnetometer has the capability of performing MRI without requiring remote encoding, which was used by Xu *et al.* (2006, 2008) for imaging with a quasi-static magnetometer. Operation with better rf shielding and the use of gradient measurements should enable much higher sensitivity to magnetic resonance signals by an rf atomic magnetometer than has been demonstrated so far. Multi-channel gradiometric measurements will also allow for spatial imaging with a single detector. In Section 4.3 we show that sensitivity of $0.2 \text{ fT}/\sqrt{\text{Hz}}$ has been achieved with an rf magnetometer using μ -metal shielding and a larger measurement volume. With such sensitivity, significantly more efficient detection of low-field NMR and MRI will be possible with a magnetometer than with an inductive coil, but without the many complications of operation with SQUIDs. There are thus a wide variety of potential scientific and medical applications to be explored with the radio-frequency magnetometer.

4.3 Detection of Nuclear Quadrupole Resonance

Detection of nuclear quadrupole resonance (NQR) is currently a subject of great interest as a method for detecting explosives (Garroway *et al.*, 2001; Suits *et al.*, 2003) and narcotics (Garroway *et al.*, 1994; Yesinowski *et al.*, 1995). For example, NQR can potentially allow significantly more efficient identification of buried landmines and hidden bombs than other methods currently employed. In particular, ^{14}N has a large nuclear quadrupole moment and is contained in most explosives, and so its presence can be detected with an appropriately tuned NQR spectrometer. The main limitation of NQR is that signals tend to be very small due to low thermal polarization and have very short duration; pick-up coils typically suffer from small signal-to-noise ratio in NQR detection, often less than 1, requiring long measurement times for contraband identification. SQUIDS have also been demonstrated for NQR detection (Hürlimann *et al.*, 1992; He *et al.*, 2006), but the requirement of operating at cryogenic temperatures may render them impractical for field applications.

The theory of nuclear quadrupole resonance is given by Smith (1971) and Lee (2002); we offer only a short summary. The electric quadrupole moment Q of a nucleus is defined by

$$e_c Q = \int \rho (3z^2 - r^2) d\tau, \quad (4.38)$$

where $e_c = 1.60 \times 10^{-19}$ C is the magnitude of the electron charge, and ρ is the nuclear charge density in the volume element $d\tau$. The integral is performed over the volume of the nucleus. The quadrupole moment is zero when the nucleus is spherical, such as when $I < 1$, so only nuclei of spin 1 or greater have $Q \neq 0$. The electric quadrupole moment couples to an electric field gradient, with the Hamiltonian given by

$$\mathcal{H}_Q = \frac{1}{4} e_c^2 q Q \left[2I_z^2 - I_x^2 - I_y^2 + \eta (I_x^2 - I_y^2) \right], \quad (4.39)$$

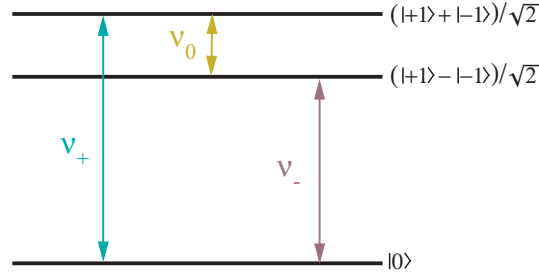


Figure 4.15: Quadrupole energy levels for a spin-1 nucleus such as ^{14}N , characterized by states of m_I . The transition frequencies are given by Equations 4.42-4.44 and become degenerate for $\eta=0$. Adapted from [Suits et al. \(2003\)](#).

where the principal component of the electric field gradient lies along $\hat{z}$. If the local electric potential is denoted as V , then this component of the field gradient is

$$e_c q = \frac{\partial^2 V}{\partial z^2} = V_{zz}. \quad (4.40)$$

The asymmetry parameter η , which ranges from 0 to 1 and characterizes the degree of cylindrical symmetry of the nuclear charge distribution about the $\hat{z}$ axis (with $\eta = 0$ describing perfect symmetry), is defined as

$$\eta = \left(\frac{\partial^2 V}{\partial x^2} - \frac{\partial^2 V}{\partial y^2} \right) / \frac{\partial^2 V}{\partial z^2} = \frac{V_{xx} - V_{yy}}{V_{zz}}. \quad (4.41)$$

For a nucleus with $I=1$, such as ^{14}N , the Hamiltonian has three eigenvalues, giving the energy levels shown in Figure 4.15. The frequencies of the transitions between energy levels are given by

$$\nu_0 = \left(\frac{e_c^2 q Q}{h} \right) \frac{1}{2} \eta \quad (4.42)$$

$$\nu_+ = \frac{3}{4} \left(\frac{e_c^2 q Q}{h} \right) \left(1 + \frac{\eta}{3} \right) \quad (4.43)$$

$$\nu_- = \frac{3}{4} \left(\frac{e_c^2 q Q}{h} \right) \left(1 - \frac{\eta}{3} \right), \quad (4.44)$$

which become degenerate for $\eta=0$. The transition frequencies are extremely sensitive to the characteristics of the local electric field, which is generated by the surrounding atoms in

the molecule. Each compound containing ^{14}N (or another nucleus with a large quadrupole moment) therefore has a unique NQR spectrum that depends on its internal structure, with multiple ^{14}N nuclei within a compound experiencing different resonance frequencies. This unique spectrum acts as a “fingerprint” for the compound and offers a reliable means of identification. However, the precise transition frequencies depend highly on the sample temperature and thus can drift, so a large sensor bandwidth makes it easier to observe the quadrupole resonances without needing to scan the detection frequency over a wide range near the nominal values.

The nuclear magnetic dipole moment is aligned with the principal axis of the electric quadrupole moment, so a resonant rf pulse at one of the NQR transition frequencies ν induces a small net magnetization that oscillates at the frequency $\omega = 2\pi \times \nu$. The magnitude of the induced magnetization in a powder sample can be estimated as

$$m(\omega) = 0.43n \frac{\gamma h}{3} \left(\frac{h\nu}{k_B T} \right), \quad (4.45)$$

where n is the density of quadrupolar nuclei in the sample, and $\gamma h/3$ is the nuclear magnetic moment (Garroway *et al.*, 2001). The numerical factor of 0.43 is due to the fact that crystallites within a powder are randomly oriented, so the rf excitation will only affect the fraction of molecules with principal axes that are properly aligned. Although NQR detection has the advantage of not requiring a large polarizing magnetic field as in NMR, the Boltzmann factor is typically on the order of 10^{-7} for ^{14}N and so the signals tend to be extremely small. For example, the NQR signal from a typical landmine with 100 g of TNT located 10 cm away from a sensor is about 4 fT with a bandwidth of 1 kHz, much too small to be detected with a pick-up coil without extensive signal averaging.

The NQR frequencies for compounds containing ^{14}N are generally in the range of 0.1–10 MHz, so atomic magnetometers are an attractive alternative to pick-up coils for NQR detection, with better intrinsic sensitivity in this frequency range potentially leading to larger signal-to-noise ratios and shorter measurement times. Figure 4.16 shows the setup of the

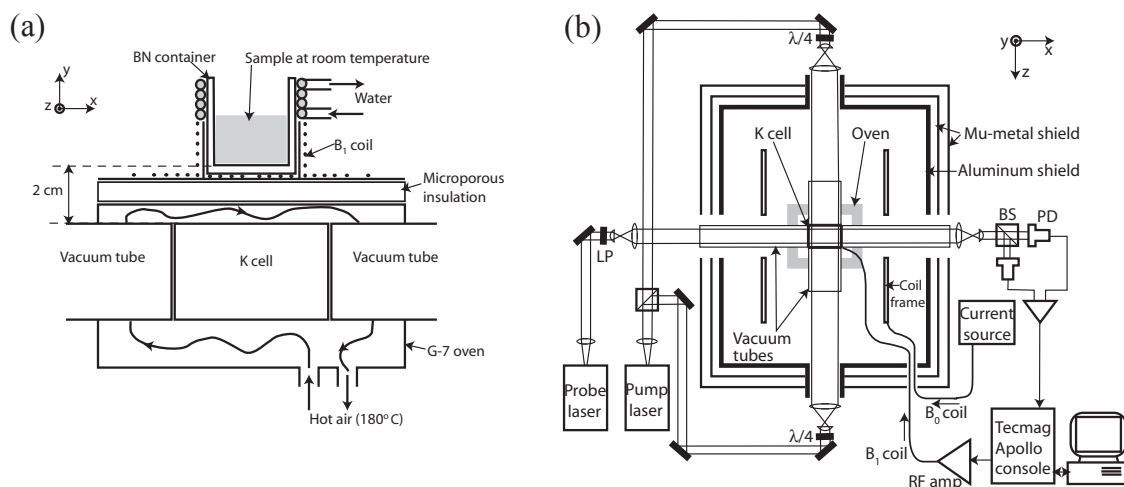


Figure 4.16: Schematic of the NQR experiment. **(a)** The ammonium nitrate sample is kept at a constant temperature by water cooling of its boron nitride container. The alkali cell is located 2 cm away, with thermal insulation between the oven and sample. **(b)** The experiment is located within μ -metal shielding, and counter-propagating pump beams are used to ensure a near-optimal pumping rate throughout the cell.

experiment to detect NQR signals from room-temperature ammonium nitrate [NH_4NO_3], which is often used as a fertilizer and is a key component in many explosive devices.¹⁰ The potassium cell has dimensions of $4 \times 4 \times 6 \text{ cm}^3$, contains 0.8 amg of helium buffer gas, and is heated to 180°C inside a double-walled G7 fiberglass oven. The NQR sample is held in a high-thermal conductivity boron nitride container that is kept at a temperature of $24 \pm 0.5^\circ\text{C}$ by a continuous flow of chilled water so that the resonance frequencies remain nearly constant. The bottom of the sample container and the top of the potassium cell are separated by 2 cm, including 6 mm of microporous thermal insulation. The sample and oven are placed inside a two-layer μ -metal shield, with a single-layer aluminum shield between the sample and the μ -metal, in order to attenuate the applied rf pulses and prevent them from saturating the μ -metal.

¹⁰ This makes ammonium nitrate a good stand-in for an actual explosive, which could potentially be dangerous to use in the laboratory for testing.

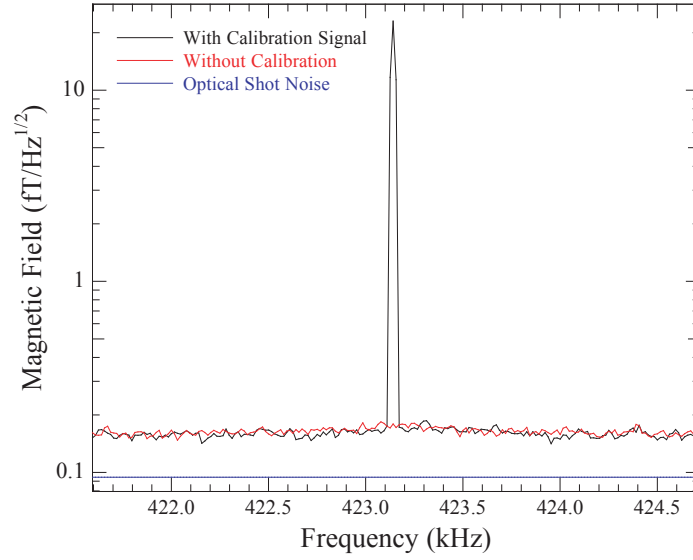


Figure 4.17: Sensitivity of the radio-frequency atomic magnetometer used for detection of NQR signals. The noise level is limited by fluctuations of the probe beam frequency and intensity, giving sensitivity of $0.18 \text{ fT}/\sqrt{\text{Hz}}$. The calibration signal and photon shot-noise level are also shown.

The pump laser is a high-power (400 mW) broadband diode and is split into two beams of equal intensity, which illuminate the cell from both directions in order to maintain a nearly constant pumping rate as described in Section 4.1.4. The probe beam is generated by a DFB diode with a tapered amplifier to increase the laser power, although losses from the optical components limit the power reaching the cell to about 40 mW. We performed a detailed numerical analysis of the magnetometer signal, including the effects of light attenuation and interaction with the D2 resonance. We found that for our experiment a probe beam power of several hundred milliwatts would give the ideal signal-to-noise ratio; in addition, the cell should generally be longer along the probe direction than the other two directions, in order to increase the optical rotation signal. The noise spectrum obtained with the rf magnetometer is presented in Figure 4.17, showing sensitivity of $0.18 \text{ fT}/\sqrt{\text{Hz}}$, about twice the level of the photon shot noise, with a magnetic linewidth of approximately $2\pi \times 200 \text{ Hz}$. The noise level is apparently limited by frequency and intensity noise of the

probe laser, but it is nevertheless significantly lower than that of the NMR magnetometer described in Section 4.2 due to better magnetic shielding, a larger alkali cell, more intense laser beams, and a narrower resonance linewidth given by a uniform pumping rate.

The longitudinal lifetime $T_1=16.6$ s of the nitrogen nuclei is much longer than the transverse lifetime $T_2=4.4$ ms, so we detect the NQR signals using a spin-lock spin-echo sequence that refocuses the nitrogen spins every 2.2 ms, in order to reduce the effect of the long longitudinal repolarization time (Marino and Klainer, 1977). The magnetometer dead time after application of each rf pulse is reduced to 0.8 ms by application of an offset field during the pulse duration (as in Section 4.2), as well as the use of a single-sided excitation coil which produces an rf field that is about 80 times stronger on one side than on the other side, in order to excite the NQR sample without affecting the alkali atoms. The averaged NQR signal from 22 g of ammonium nitrate powder at the resonance frequency 423 kHz after 32 repetitions of a 2048 echo sequence is shown in Figure 4.18(a). The signal is given by the component of the magnetometer output at the resonance frequency, and the slow rise time of about 0.5 ms is determined by the bandwidth of the magnetometer. The repetition rate was limited to about twice per minute due to the need to wait for the spins to repolarize in between echo sequences.

In Figure 4.18(b) we show the NQR signal as a function of the strength of the rf excitation pulse, which varies as a Bessel function (Vega, 1974). For comparison, we also show the signal detected by a tuned surface pick-up coil with $\omega/2\Delta\omega=60$ and comparable volume to the magnetometer cell.¹¹ The coil signal is significantly lower than that of the magnetometer and corresponds to a sensitivity of $3.6 \text{ fT}/\sqrt{\text{Hz}}$. The fundamental sensitivity of the coil is estimated to be about $0.8 \text{ fT}/\sqrt{\text{Hz}}$, although the coil signal improves slowly

¹¹ The ratio $\omega/2\Delta\omega$ of a resonator is usually referred to as its Q , although we avoid this terminology in the main text so as not to cause confusion with the electric quadrupole moment. In general higher Q leads to better sensitivity; the Q of the magnetometer used for this experiment is about 1000.

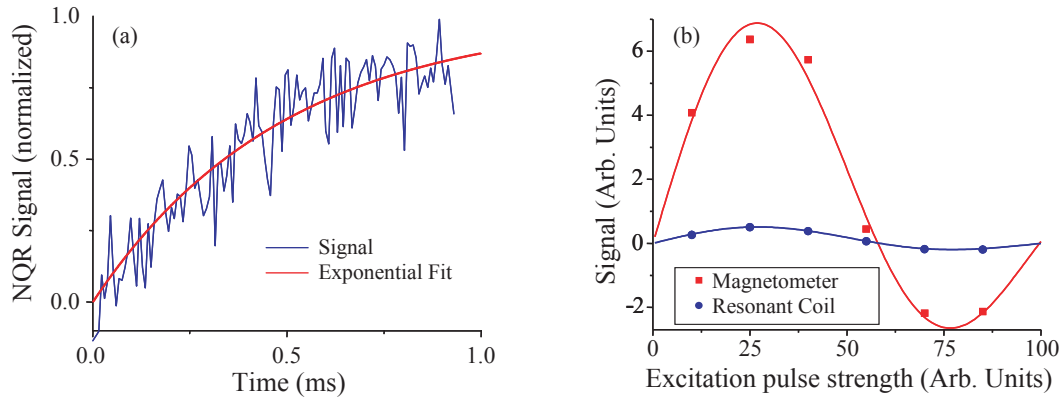


Figure 4.18: (a) NQR signal detected from ammonium nitrate with a radio-frequency magnetometer, averaged after 32 repetitions of a 2048 spin-echo sequence. The exponential fit gives a rise time of about 0.5 ms, limited by the magnetometer bandwidth. (b) Comparison of NQR signals detected by the rf magnetometer and a traditional, tuned pick-up coil of comparable volume. The magnetometer displays significantly better sensitivity. The signal varies as a Bessel function with the rf pulse strength (Vega, 1974).

as $\Delta\omega^{-1/2}$, and a narrow detection bandwidth can prevent the application of a fast spin-echo sequence. We therefore conclude that the rf magnetometer performs better than an inductive coil of similar volume for NQR applications where a surface detection geometry is required, such as the detection of buried landmines.

The demonstrated magnetometer sensitivity is limited by the noise and low power of the probe laser used; with a more stable laser and an increase of probe power entering the cell to 100 mW, a sensitivity of about $0.06 \text{ fT}/\sqrt{\text{Hz}}$ limited by photon shot-noise should be possible. Further improvement in sensitivity should be obtained by performing a gradiometric measurement to cancel common-mode noise. Operation in the field will require the development of techniques to enable high sensitivity without the extensive rf shielding used in this experiment. With these improvements, radio-frequency atomic magnetometers will be able to detect nuclear quadrupole resonance signals with a significantly higher signal-to-noise ratio than is possible with other types of sensors, enabling more efficient and accurate detection of explosives and narcotics.

Chapter 5

Spin-Exchange Relaxation-Free Magnetometry

ONLY BY COMPLETELY ELIMINATING relaxation due to spin-exchange collisions can an atomic magnetometer achieve subfemtotesla sensitivity to slowly changing magnetic fields, such as those produced by the Earth or by magnetic anomalies. The spin-exchange relaxation-free (SERF) magnetometer accomplishes this by operating with high density in very small fields, enabling it to be the most sensitive known detector of static and quasi-static magnetic fields. We present a method for using the SERF magnetometer to measure all three vector components of the magnetic field simultaneously, and we demonstrate operation in an unshielded environment through active cancelation of the ambient field. The work presented in this chapter was originally published in [Seltzer and Romalis \(2004\)](#).

5.1 Suppressing Spin-Exchange Relaxation

The sensitivity of a magnetometer generally improves as the number of atoms involved in the measurement increases. As shown in Section 2.8.1, the spin-projection noise limit, which is typically the dominant source of quantum fluctuations, improves as the square-root of the number of atoms. The optical rotation signal, given by Equation 2.87, is proportional to the density of atoms multiplied by the probe beam path length; although not all magnetometers use optical rotation to measure the atomic spin polarization, this scaling with the number of atoms is typical. For most implementations of an atomic magnetometer, it is impractical to operate with high alkali vapor density because rapid spin-exchange collisions limit the polarization lifetime (see Section 2.7.1), so any improvement in sensitivity from the increased number of atoms is negated by the associated broadening of the magnetic linewidth. In order to achieve femtotesla-level sensitivity to static or slowly changing magnetic fields, traditional low-density magnetometers require measurement volumes of hundreds (Budker *et al.*, 2000) or thousands (Aleksandrov *et al.*, 1995) of cubic centimeters, making them impractical for portable applications as well as for high-resolution field imaging in magnetic resonance and biomedical applications.

However, Happer and Tang (1973) discovered that spin-exchange broadening vanishes at extremely high alkali density. This effect was explained by Happer and Tam (1977) as being similar to motional narrowing. As discussed in Section 2.7.1, spin-exchange collisions redistribute alkali atoms among the ground-state Zeeman sublevels and lead to relaxation because atoms in the two hyperfine levels precess in opposite directions. If the spin-exchange rate is larger than the precession frequency, such as occurs at high density in a sufficiently small magnetic field, then an individual atom precesses by only a small angle in between collisions, as shown in Figure 5.1. Each atom therefore samples all of the ground-state Zeeman sublevels in a short period of time,¹ with statistical weights given

¹ Short relative to the Larmor period in this case.

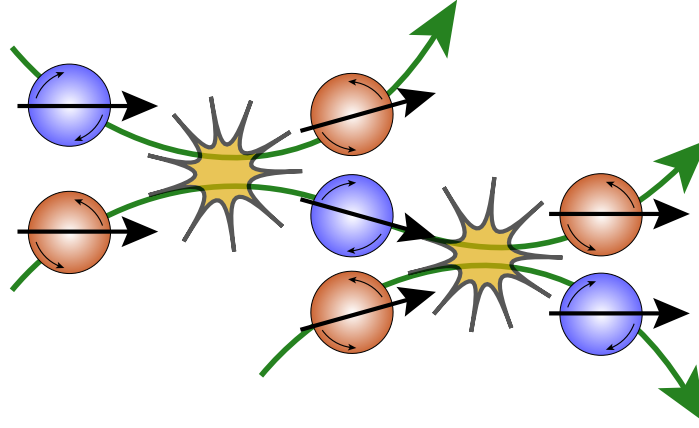


Figure 5.1: In the spin-exchange relaxation-free regime of high alkali density and low magnetic field, alkali atoms precess by an infinitesimal angle in between spin-exchange collisions. Atoms sample all Zeeman sublevels, and the populations of the two hyperfine levels precess together coherently. The $F = I \pm 1/2$ hyperfine levels are represented here by the colors red and blue; compare to Figure 2.21.

by the spin-temperature distribution. There is a slight tendency to spend more time in the upper $F = I + 1/2$ hyperfine level which contains more Zeeman sublevels than the lower hyperfine level, especially in a partially polarized vapor with a large population in the $|F = I + 1/2, m_F = I + 1/2\rangle$ end state, so all atoms experience the same net precession frequency in one direction. In this way, atoms in the two hyperfine levels become locked together as they precess, and spin-exchange collisions no longer cause spin relaxation because the entire alkali ensemble precesses coherently.

This effect is illustrated in Figure 5.2, with the spin-exchange rate increasing by a factor of 10 between each successive plot from top to bottom. If the spin-exchange rate is small compared to the Larmor frequency, then individual atoms spend long periods of time precessing freely in one of the hyperfine states, before possibly switching to the other hyperfine level after a collision with another alkali atom. The relative phases of the individual atoms become randomized, and the coherence time of the ensemble is limited. As the spin-exchange rate increases, individual atoms spend less time in a particular state between collisions, until precession at a single, slowed-down frequency is observed under

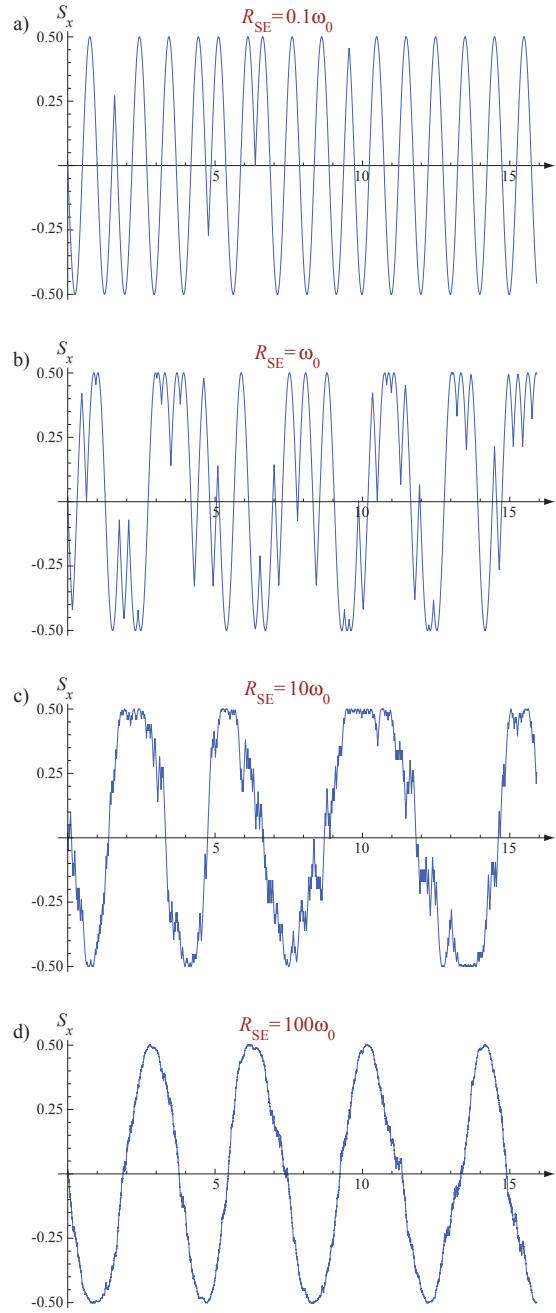


Figure 5.2: Simulated precession of an individual alkali spin, with the spin-exchange rate R_{SE} increasing by a factor of 10 between successive plots from top to bottom. ω_0 is the nominal Larmor frequency, and time is given in units of the Larmor period $2\pi/\omega_0$. If the spin-exchange rate is sufficiently fast, then the atoms precess coherently at a net frequency that is slower than ω_0 . Adapted from [Happer and Tam \(1977\)](#).

sufficiently fast spin exchange. Although the atoms rapidly switch between the two hyperfine levels, over long time periods they undergo precession in the direction determined by the upper hyperfine level.

In the SERF regime, the atomic resonance frequency is reduced from the nominal high-field Larmor frequency by an amount that depends on the density, magnetic field strength, and polarization. In the limit of zero field and low polarization, with the spin-temperature parameter $\beta \ll 1$, the precession frequency becomes

$$\omega_q = \frac{S(S+1)\gamma^e}{S(S+1) + I(I+1)} |\mathbf{B}|, \quad (5.1)$$

where γ^e is the gyromagnetic ratio of the bare electron. As the polarization increases, the precession frequency approaches its high-field value because the atoms spend an increasingly greater fraction of their time in just the upper hyperfine level (Savukov and Romalis, 2005a). We denote the instantaneous precession frequency as ω_0 , given by the high-field Larmor frequency, with atoms in the hyperfine levels $F = I \pm 1/2$ precessing at $\pm\omega_0$. The angular momenta of the atoms in the individual hyperfine levels F_+ and F_- remain nearly parallel and precess together at the net frequency ω_q . If the atoms are initially polarized along the $\hat{z}$ axis and experience a small perpendicular magnetic field, then in a short time period dt the spins of the hyperfine levels change by

$$\omega_0 F_{+,z} dt - \omega_0 F_{-,z} dt = \omega_q (F_{+,z} + F_{-,z}) dt, \quad (5.2)$$

giving the net precession frequency in terms of the Larmor frequency:

$$\omega_q = \omega_0 \frac{F_{+,z} - F_{-,z}}{F_{+,z} + F_{-,z}}. \quad (5.3)$$

Using Equations 2.139-2.141, we may rewrite this equation in terms of the spin polarization P , as shown in Table 5.1. We find that in fact the precession frequency in the SERF regime is given by that of a bare electron reduced by the nuclear slowing-down factor q ,

$$\omega_q = \frac{\gamma^e}{q} |\mathbf{B}|. \quad (5.4)$$

I	ω_q/ω_0	$\Delta\omega/\omega_q^2$ (s)	$n\Delta\nu/\nu_q^2$ (cm ⁻³ /Hz)
3/2	$2 - \frac{4}{3+p^2}$	$5/(3R_{SE})$	8.4×10^9 (³⁹ K at 170°C) 1.2×10^{10} (⁸⁷ Rb at 140°C)
5/2	$3 - \frac{48(1+p^2)}{19+26p^2+3p^4}$	$280/(57R_{SE})$	3.6×10^{10} (⁸⁵ Rb at 140°C)
7/2	$\frac{4(1+7p^2+7p^4+p^6)}{11+35p^2+17p^4+p^6}$	$105/(11R_{SE})$	8.1×10^{10} (¹³³ Cs at 120°C)

Table 5.1: Spin precession frequency of alkali atoms in the SERF regime, compared to the high-field Larmor frequency. At high polarization the the precession frequency approaches the Larmor frequency. Also listed is the ratio $\Delta\omega/\omega_q^2$ at low polarization in the SERF regime, as well as the corresponding factor $n\Delta\nu/\nu_q^2$ for calibration of density measurements.

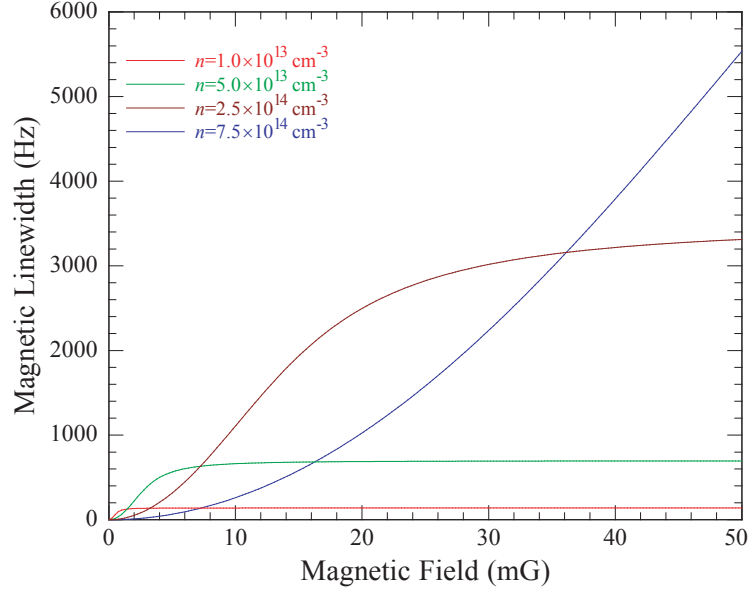


Figure 5.3: Magnetic linewidth $\Delta\omega/2\pi$ calculated from Equation 5.5 for potassium atoms as a function of the magnetic field amplitude. As the vapor density increases, spin-exchange broadening is more effectively suppressed.

The general dependence of the precession frequency ω_q and magnetic linewidth² $\Delta\omega$ on the magnetic field B and spin-exchange rate R_{SE} is given by [Happer and Tam \(1977\)](#) in the limit of low polarization as

$$\Delta\omega + i\omega_q = \frac{([I]^2 + 2)R_{SE}}{3[I]^2} - \sqrt{-\omega_0^2 - \frac{2i\omega_0 R_{SE}}{[I]} + \left(\frac{([I]^2 + 2)R_{SE}}{3[I]^2}\right)^2}, \quad (5.5)$$

where $[I] = 2I + 1$, and the Larmor frequency is given by

$$\omega_0 = \frac{g_s \mu_B B}{\hbar(2I + 1)}. \quad (5.6)$$

The spin-exchange broadening factor $1/q_{SE}$ (see Section 2.7) ranges from 0 at zero field, corresponding to no contribution to spin relaxation from spin-exchange collisions, to the high-field limit given by Equation 2.138. Close to zero field, spin-exchange broadening is approximately quadratic in the magnetic field:

$$\Delta\omega = \frac{R_{SE}}{q_{SE}} = \omega_0^2 \frac{(2I + 1)^2 [q_{lp}^2 - (2I + 1)^2]}{2R_{SE}q_{lp}^3}, \quad (5.7)$$

where q_{lp} is the low-polarization limit of the slowing-down factor given in Table 2.5. The spin-exchange contribution to the magnetic linewidth is plotted in Figure 5.3 for potassium as a function of the magnetic field amplitude. Broadening turns on as the field increases from zero, though suppression improves at higher density, allowing for operation at finite field strength while maintaining a narrow linewidth. For typical operating densities, the magnetic field must be less than about 100 μG to achieve complete elimination of spin-exchange relaxation.

At very high alkali density, the vapor is too optically thick to allow measurement of the density by monitoring absorption of a laser as described in Section 2.3.3. It is instead possible to determine the vapor density by measuring the resonance frequency and spin-exchange linewidth broadening in the SERF regime as a function of magnetic field and

² More specifically, this is the contribution of spin-exchange collisions to the magnetic linewidth. The total linewidth is given by adding the contribution from Equation 5.5 to the contribution from all other processes.

fitting to Equation 5.5. The frequency and linewidth can be fit separately, or they can be directly compared to one another; expanding Equation 5.5 near zero field, we find that

$$\Delta\omega = \omega_q^2 \frac{2I[-3 + I(1 + 4I(I + 2))]}{3[3 + 4I(I + 1)]R_{SE}}. \quad (5.8)$$

This relationship provides a simple means of determining vapor density from the spin-exchange rate. We list the factors $\Delta\omega/\omega_q^2$ for the alkali metals in Table 5.1, as well as the factors $n\Delta\nu/\nu_q^2$ for practical use in calibrating density measurements.

In Figure 5.4(a) we show the magnetic linewidth measured in cesium vapor at low polarization near zero magnetic field. The spin-exchange broadening has the expected quadratic behavior, and at zero field the linewidth is limited to about 50 Hz due to spin-destruction collisions, significantly less than the high-field linewidth dominated by spin-exchange relaxation. Improved suppression of spin-exchange broadening is evident at increased temperature, corresponding to higher alkali density and thus more rapid spin exchange. In Figure 5.4(b) we show the corresponding measurement of the atomic resonance frequency for the same values of the magnetic field amplitude and alkali density. Near zero field this frequency is given by Equation 5.4, while at finite field it approaches the Larmor frequency with a larger effective gyromagnetic ratio. Similar to the linewidth, as the density increases the resonance frequency becomes smaller, remaining at the low-field limit at greater field amplitude. By fitting the measured linewidths and resonance frequencies to the theory given by Equation 5.5, we estimate that the data taken at 90°C correspond to a density of $1 \times 10^{13} \text{ cm}^{-3}$, the data taken at 110°C correspond to a density of $2 \times 10^{13} \text{ cm}^{-3}$, and the data taken at 125°C correspond to a density of $4 \times 10^{13} \text{ cm}^{-3}$; all values are within a factor of two of the expected density given by Equation A.1.

By operating in this regime of high density and low magnetic field, the spin-exchange relaxation-free magnetometer achieves very narrow resonance linewidths limited by spin-destruction and wall collisions. The alkali-alkali spin-destruction cross-sections are typically orders of magnitude smaller than the corresponding spin-exchange cross-sections,

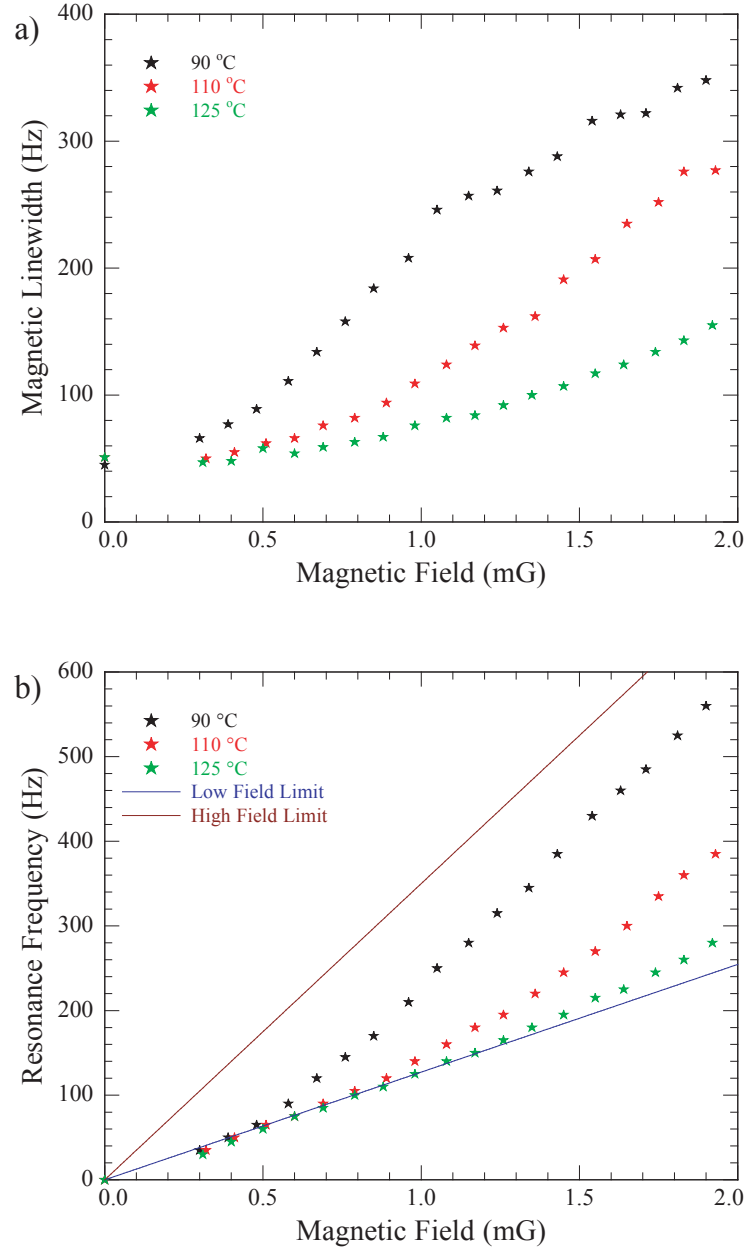


Figure 5.4: (a) Observed suppression of spin-exchange broadening in cesium atoms. As the temperature increases, the higher vapor density results in better suppression at finite magnetic field. The zero-field linewidth is limited by collisions with the buffer and quenching gases. (b) Measured atomic resonance frequencies as a function of magnetic field amplitude, showing slowed alkali spin precession in the SERF regime.

as shown in Table A.2; the relaxation rate due to collisions with the wall or the buffer and quenching gases can be made comparable to the alkali-alkali spin-destruction rate by proper selection of gas pressures, as well as the use of an anti-relaxation surface coating if appropriate. At zero field there is no well-defined quantization axis for the atomic spins, so $T_1=T_2$, enabling the atoms to experience the longest possible coherence lifetime for the given experimental conditions. Ultra-narrow linewidths on the order of 1 Hz can thus be obtained even when the spin-exchange rate is on the order of 10^4 s^{-1} or higher. See Figure 4.3(b) for a comparison of the magnetic resonance spectrum of potassium measured in the SERF regime compared to the high-field regime for $R_{\text{SE}} \sim 10^5 \text{ s}^{-1}$, with narrowing by a factor of several thousand.³ This translates directly to an improvement in the slope of the magnetometer dispersion curve by the same factor, as well as an improvement in the fundamental magnetometer sensitivity by the square root of the factor (see Section 5.2).

The SERF magnetometer was first introduced by Allred *et al.* (2002), who demonstrated a linewidth of 1.2 Hz despite a spin-exchange rate of $1.3 \times 10^5 \text{ s}^{-1}$, resulting in sensitivity of $10 \text{ fT}/\sqrt{\text{Hz}}$ in a single channel, limited by the resistive Johnson noise of the μ -metal shields used in the experiment. By using a magnetic shield made of a nonconductive MnZn ferrite, Kornack *et al.* (2007) demonstrated single-channel sensitivity of $0.75 \text{ fT}/\sqrt{\text{Hz}}$, with the potential for improvement by the typical order of magnitude using a gradiometric configuration; the use of ferrite instead of μ -metal is increasingly important as SERF magnetometers become more compact, with the magnetic shielding located closer to the alkali vapor cell. Kominis *et al.* (2003) demonstrated sensitivity of $0.54 \text{ fT}/\sqrt{\text{Hz}}$ to the magnetic field gradient with a measurement volume of only 0.3 cm^3 , an improvement by a factor of 13 over their single-channel noise level because of cancelation of Johnson noise from the shields. This remains the highest sensitivity obtained by any detector to near-dc magnetic fields, including both atomic magnetometers and SQUIDs.

³ Light narrowing reduces the magnetic linewidth at high field, but it does not help with measurement of static fields, as shown by Smullin *et al.* (2006); see Section 3.1.3.

5.2 Fundamental Sensitivity

We show in Section 5.3 that the SERF magnetometer is primarily sensitive to the magnetic field component along the direction perpendicular to both the pumping and probing axes, which we denote as the $\hat{y}$ axis. Near zero field, the magnetometer signal is approximately (see Equation 5.23)

$$S_x = \frac{P_0 \gamma^e B_y}{2(R_{\text{OP}} + R_{\text{pr}} + R_{\text{SD}})} , \quad (5.9)$$

where $\gamma^e = q\gamma = g_s \mu_B / \hbar$ is the gyromagnetic ratio of a bare electron, P_0 is the equilibrium spin polarization, and $R_{\text{pr}} + R_{\text{SD}}$ is the spin-relaxation rate given by the sum of the probe beam pumping rate and the spin-destruction collision rate. Setting $\partial S_x / \partial R_{\text{OP}} = 0$, we find that the signal is optimized for $R_{\text{OP}} = R_{\text{pr}} + R_{\text{SD}}$, giving spin polarization $P_0 = 0.5$, as shown in Figure 5.5. For larger pumping rates than this the polarization begins to saturate, while the resonance linewidth scales linearly with the pumping rate, so the magnetometer signal is reduced. This pumping rate gives optimal performance when the magnetometer sensitivity is not limited by quantum fluctuations, though we will see that it does not necessarily provide the best quantum noise limit for the magnetometer.

The fundamental noise limit of the SERF magnetometer was originally presented by Ledbetter *et al.* (2008), based in part on the derivation for an rf magnetometer given in Savukov *et al.* (2005) and reproduced in Sections 2.8 and 4.1.2. We may therefore follow our earlier discussions of quantum fluctuations limiting magnetometer sensitivity, but with several modifications. The SERF magnetometer is unaffected by light shifts resulting from fluctuations in the polarization of the probe beam, since the measured spin component S_x is highly insensitive to magnetic fields along the direction of the probe beam. In addition, we must modify the expression for the spin-projection noise given in Section 2.8.1. The uncertainty in N uncorrelated measurements of F_x is given by Ledbetter *et al.* as

$$\delta F_x = \sqrt{\frac{q}{4N}} , \quad (5.10)$$

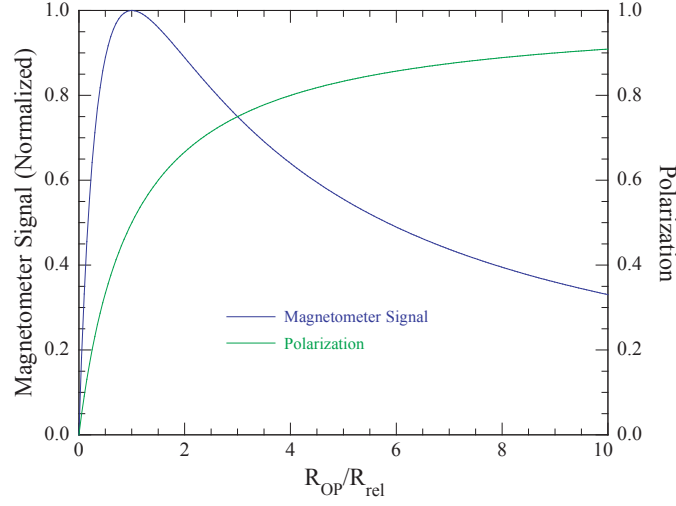


Figure 5.5: The SERF magnetometer signal given by Equation 5.9 is optimized when the pumping rate equals the spin-relaxation rate, giving spin polarization of 50%.

which agrees with the minimum uncertainty given by Equation 2.155 in the case of full polarization with $q = 2F_z = 2I + 1$. Combining this with Equation 2.157, we see that the uncertainty per unit bandwidth in a continuous measurement of the transverse spin component is

$$\delta\langle F_x \rangle = \frac{q}{\sqrt{N(R_{OP} + R_{pr} + R_{SD})}}, \quad (5.11)$$

where $T_2 = q / (R_{OP} + R_{pr} + R_{SD})$. In the spin-temperature distribution, the ratio between the expectation values of the atomic and electron spins is given by the nuclear slowing-down factor (Appelt *et al.*, 1998),

$$S_x = \frac{1}{q} F_x. \quad (5.12)$$

Combining Equations 5.9, 5.11, and 5.12, we see that the spin-projection noise does not depend on the nuclear spin, as it does for the rf magnetometer,

$$\delta B_{spn} = \frac{2}{P_0 \gamma^e} \sqrt{\frac{R_{OP} + R_{pr} + R_{SD}}{nV}}, \quad (5.13)$$

where $V = N/n$ is the active measurement volume defined by the intersection of the pump and probe beams.

To write the photon shot noise in simplified form as in Section 4.1.2, we assume that the vapor cell contains enough buffer gas that the Voigt profile is well-approximated by a Lorentzian, and that the probe beam is sufficiently detuned from the optical resonance that

$$\text{Im}[\mathcal{V}(\nu - \nu_0)] \simeq \frac{1}{\pi(\nu - \nu_0)}. \quad (5.14)$$

We denote the optical depth of a resonant laser as OD, and we take into account the quantum efficiency η of the photodiodes used to detect the probe beam. Combining Equations 2.165 and 5.9, we get the photon shot noise,⁴

$$\delta B_{psn} = \frac{R_{OP} + R_{pr} + R_{SD}}{P_0 \gamma^e} \sqrt{\frac{2}{nV R_{pr} OD \eta}}. \quad (5.15)$$

The fundamental sensitivity limit of the SERF magnetometer is given by the quadrature sum of the spin-projection noise and photon shot noise:

$$\delta B_{\min} = \frac{1}{R_{OP} \gamma^e \sqrt{nV}} \sqrt{4(R_{OP} + R_{pr} + R_{SD})^3 + \frac{2(R_{OP} + R_{pr} + R_{SD})^4}{R_{pr} OD \eta}}. \quad (5.16)$$

We see that if the quantum limit is dominated by the spin-projection noise, then it is minimized by setting $R_{OP} = 2(R_{pr} + R_{SD})$, giving $P_0=0.67$. If the photon shot noise dominates, then the sensitivity limit is optimized by setting $R_{OP} = (R_{pr} + R_{SD})$, giving $P_0=0.5$, which in general results in the largest magnetometer signal as shown earlier in this section. For a potassium SERF magnetometer with characteristic values of $n=1 \times 10^{14} \text{ cm}^{-3}$, $P_0=0.5$, $\frac{1}{2}R_{OP}=R_{pr}=R_{SD}=2\pi \times 1 \text{ Hz}$, $OD=20$, and $\eta=0.8$, the fundamental sensitivity is $.012 \text{ fT}/\sqrt{\text{Hz}}$ in a measurement volume of only 1 cm^3 , or less than $1 \text{ aT}/\sqrt{\text{Hz}}$ for volumes larger than about 150 cm^3 . There is therefore significant potential for improvement in the performance of the SERF magnetometer beyond what has already been demonstrated, and there are likely many new applications that will be discovered as the sensitivity pushes toward the attotesla level.

⁴ Note that [Ledbetter et al.](#) derive the shot noise for probing on the D2 transition, which results in optical rotation angles that are half that given by probing on the D1 transition, accounting for the difference in numerical factors between their expression and the one presented here.

5.3 Three-Axis Vector Detection

The SERF magnetometer is inherently a vector sensor, and its response to a small applied field depends on its orientation with respect to that field. As discussed in Section 3.1, vector magnetometers can be difficult to operate on vehicles and other mobile platforms where the field orientation is unstable. On the other hand, a vector sensor can provide information about the magnetic field's direction in addition to its amplitude, allowing for more complete characterization of the field, as well as better localization of magnetic sources. Although the SERF magnetometer is primarily sensitive to the field component along the direction perpendicular to both the pump and probe beams, we show here that it can be used to measure all three components simultaneously.

Under conditions of rapid spin exchange, the behavior of the atomic spin is most accurately described by the evolution of the density matrix. However, in the SERF regime with sufficiently slow spin precession that $\omega / R_{\text{SD}} \ll 1$, the spin evolution is well-described by the Bloch equation (see Section 2.6):

$$\frac{d}{dt} \mathbf{S} = \frac{1}{q} \left[\gamma^e \mathbf{B} \times \mathbf{S} + R_{\text{OP}} \left(\frac{1}{2} s \hat{\mathbf{z}} - \mathbf{S} \right) - R_{\text{rel}} \mathbf{S} \right], \quad (5.17)$$

where s is the photon polarization of the pump beam, and R_{rel} is the total spin-relaxation rate. For notational simplicity, we introduce the dimensionless magnetic field parameter,

$$\boldsymbol{\beta} = \left(\frac{\gamma^e}{R_{\text{OP}} + R_{\text{rel}}} \right) \mathbf{B}. \quad (5.18)$$

If the magnetic field changes slowly, then we may set $d\mathbf{S}/dt=0$ and find the steady-state solution of Equation 5.17, giving the electronic spin components as

$$S_x = S_0 \frac{\beta_y + \beta_x \beta_z}{1 + (\beta_x^2 + \beta_y^2 + \beta_z^2)} \quad (5.19)$$

$$S_y = S_0 \frac{-\beta_x + \beta_y \beta_z}{1 + (\beta_x^2 + \beta_y^2 + \beta_z^2)} \quad (5.20)$$

$$S_z = S_0 \frac{1 + \beta_z^2}{1 + (\beta_x^2 + \beta_y^2 + \beta_z^2)}, \quad (5.21)$$

where the equilibrium electronic spin polarization S_0 is given by Equation 2.123. The average direction of the spin vector of the ensemble is determined by the competition between three processes: optical pumping tends to align the spin direction along the pumping axis, spin relaxation tends to randomize the spin direction, and the ambient magnetic field induces the spin to precess in the plane perpendicular to the field. For example, a small field B_y results in a steady-state equilibrium with the average atomic spin at a small angle from the pumping direction,

$$\theta = \frac{S_x}{S_0} \simeq \frac{\gamma^e B_y}{R_{\text{OP}} + R_{\text{rel}}} . \quad (5.22)$$

The probe beam signal is proportional to S_x , the spin component along the probing axis. If the ambient magnetic field is small enough that $|\beta| \ll 1$, then the magnetometer response to the field components along the pump and probe axes is suppressed, and the signal is approximately given by

$$S_x \approx S_0 \frac{\gamma^e B_y (R_{\text{OP}} + R_{\text{rel}})}{(R_{\text{OP}} + R_{\text{rel}})^2 + (\gamma^e B_y)^2} . \quad (5.23)$$

The signal has the form of a Lorentzian dispersion curve, with its maximum value occurring when the resonance frequency⁵ $\omega_q = \gamma^e B_y / q$ is equal to the magnetic linewidth $\Delta\omega = (R_{\text{OP}} + R_{\text{rel}}) / q = 1 / T_2$. The slope of the dispersion curve near zero field is proportional to the coherence lifetime; the SERF magnetometer therefore typically achieves the highest sensitivity using potassium because it has a significantly smaller self spin-destruction cross-section than rubidium or cesium.

⁵ We use ω_q to denote the magnetic resonance frequency as determined by the value of q in the SERF regime in order to avoid confusion with ω_0 , which we used to denote the high-field limit given by $q = 2I + 1$ in Section 5.1.

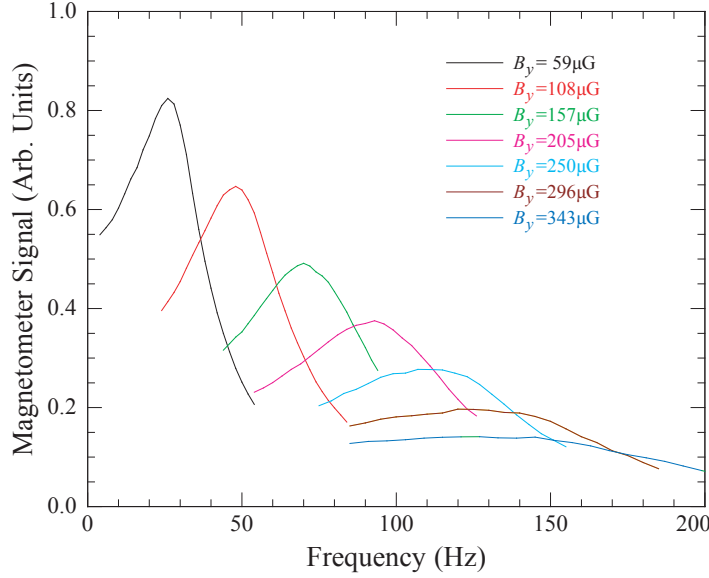


Figure 5.6: Measured frequency response of a rubidium SERF magnetometer operating at 111°C in small but finite magnetic fields. The quadrature sum of the in- and out-of-phase signal is shown. The linewidth at zero field is limited by the quality of the OTS coating on the cell walls, and the measured values of ω_q and $\Delta\omega$ give a density of approximately $9 \times 10^{12} \text{ cm}^{-3}$ when fit to Equation 5.5.

If a small bias field B_z is applied, then the response of the SERF magnetometer to an oscillating field $\mathbf{B}' = B' \cos(\omega t) \hat{\mathbf{y}}$ is given by Equation 2.129 as

$$S_x = \frac{S_0 \gamma^e B'}{2q} \left[\frac{\Delta\omega \cos(\omega t) + (\omega - \omega_q) \sin(\omega t)}{(\Delta\omega)^2 + (\omega - \omega_q)^2} + \frac{\Delta\omega \cos(-\omega t) + (\omega + \omega_q) \sin(-\omega t)}{(\Delta\omega)^2 + (\omega + \omega_q)^2} \right]. \quad (5.24)$$

For low frequencies such that $\omega_q \sim \Delta\omega$, the response takes the form of two overlapping Lorentzian curves centered at $\pm\omega_q$, as shown in Figure 2.20(b). The frequency response curves measured with a rubidium SERF magnetometer for several different values of ω_q are shown in Figure 5.6. The effect of the resonance at negative frequency is most evident for the curve with $\omega_q = 2\pi \times 26 \text{ Hz}$, which has $\Delta\omega = 2\pi \times 11 \text{ Hz}$ and gives a nonzero signal at zero frequency. The signal on resonance decreases as the resonance frequency grows larger, causing the magnetic resonance to broaden due to spin-exchange relaxation turning

on at finite field; operation at larger density allows for improved sensitivity to oscillating fields at higher frequencies because of better suppression of spin-exchange relaxation.

Although the SERF magnetometer is primarily sensitive to B_y , as evidenced by the fact that $\beta_y \gg \beta_x \beta_z$ in Equation 5.19, it can be made sensitive to the other components of the field as well. We apply small field modulations along the x and z axes, such that

$$\beta_x = \beta_x^0 + \beta_x^{\text{mod}} \sin(\omega_x t) \quad (5.25)$$

$$\beta_z = \beta_z^0 + \beta_z^{\text{mod}} \sin(\omega_z t), \quad (5.26)$$

where β_x^0 and β_z^0 are components of the static magnetic field, and β_x^{mod} and β_z^{mod} are the modulation amplitudes. The modulation frequencies ω_x and ω_z should be chosen to be slow enough that the quasi-steady-state approximation $dS/dt \approx 0$ is valid, giving Equations 5.19-5.21 as a solution. If all ambient and applied magnetic fields are small enough that $|\beta| \ll 1$, then we may expand Equation 5.19 to give

$$S_x \approx S_0 \left[\beta_y + \beta_x^0 \beta_z^0 + \beta_x^0 \beta_z^{\text{mod}} \sin(\omega_z t) + \beta_z^0 \beta_x^{\text{mod}} \sin(\omega_x t) + \mathcal{O}(\beta^3) \right]. \quad (5.27)$$

Thus, to first order the dc response of the magnetometer is linear in β_y , while lock-in amplifiers referenced to ω_x and ω_z provide signals that are proportional to β_z^0 and β_x^0 , respectively. The response to these two fields takes the form of a Lorentzian dispersion curve, like for β_y ; for example, if the other two components are zeroed, then the signal given by a finite β_x is

$$S_x \approx S_0 \beta_z^{\text{mod}} \sin(\omega_z t) \frac{\gamma^e B_x (R_{\text{OP}} + R_{\text{rel}})}{(R_{\text{OP}} + R_{\text{rel}})^2 + (\gamma^e B_x)^2}, \quad (5.28)$$

which is smaller than the corresponding signal from a finite β_y due to the necessity to keep $\beta_z^{\text{mod}} \ll 1$.

We can zero the magnetic field experienced by the alkali atoms by taking advantage of this three-axis measurement capability. We adjust B_y so that the dc magnetometer signal is at the zero-crossing of the resonant dispersion curve in order to zero that field component. We zero the other two components using a cross-modulation scheme suggested by

Equation 5.28. We apply a modulation to B_x and adjust B_z until the oscillation of the magnetometer signal at that frequency disappears at the dispersive zero-crossing, and vice versa. This process is repeated iteratively until the total field amplitude becomes as small as possible. The ambient field must already be near zero for this process to be effective; inside magnetic shielding the residual fields are typically small enough, but in an unshielded environment the ambient field should first be approximately canceled using another sensor before using the SERF magnetometer for precise zeroing.

Figure 5.7 shows a schematic of our experiment to measure all three components of the magnetic field simultaneously using a SERF magnetometer. Three pairs of Helmholtz coils adjust the magnetic field experienced by the alkali atoms, and real-time feedback from the magnetometer keeps all field components near zero in order to maintain operation in the SERF regime. We use the dc signal from the magnetometer to null B_y , and we use the cross-modulation scheme for the other two axes: one lock-in amplifier is used for modulation of B_x and feedback on B_z , while another lock-in amplifier is used for modulation of B_z and feedback on B_x . We use simple integral feedback to zero the dispersive magnetometer signal, and the compensating currents in each of the three coils serve as a measure of the corresponding vector component of the magnetic field, as the field produced should exactly cancel out the ambient field. The demonstrated orthogonality of the magnetometer response is shown in Table 5.2 and is probably limited by imperfections in the orthogonality of the coil pairs used for generating both the modulation and feedback fields. For unshielded operation we typically modulate with frequencies $\omega_x, \omega_z \sim 100$ Hz, chosen to be far from noise peaks as well as each other, but also to be small enough to remain comparable to the magnetometer bandwidth.

We find that with three-axis feedback the magnetometer can remain locked to zero field for indefinite periods of time. The lock is robust against external perturbations, such as moving magnetic objects in the vicinity of the magnetometer. The feedback must be able

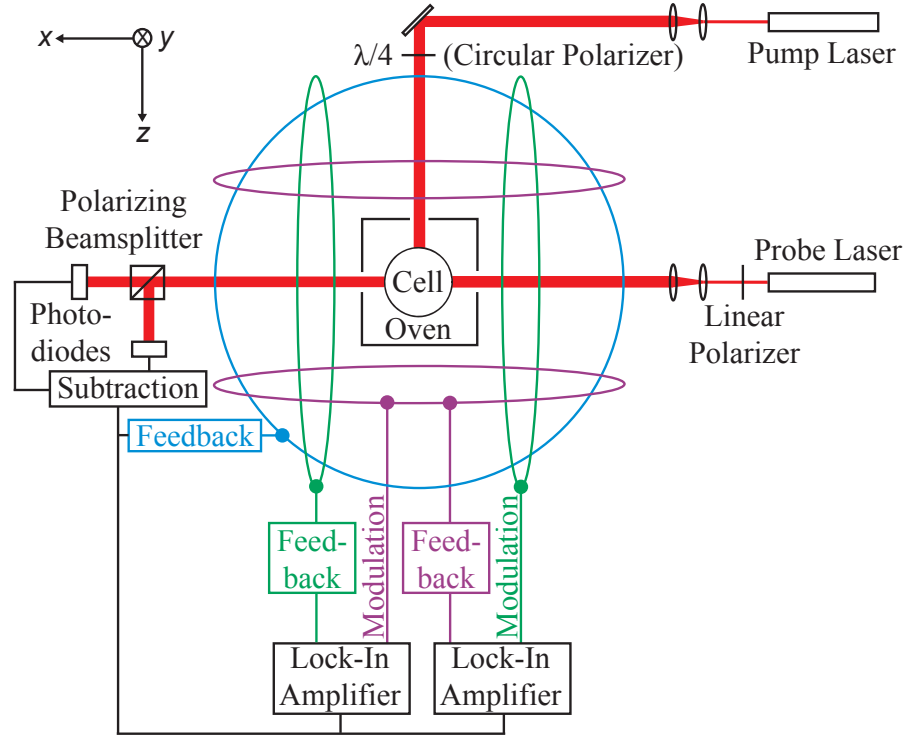


Figure 5.7: Experimental schematic of the three-axis vector SERF magnetometer. Real-time feed-back from the magnetometer signal keeps the magnetic field zeroed so that the alkali atoms remain in the SERF regime. The dc signal is used to cancel B_y , while a cross-modulation scheme is used to cancel the other two components.

		Perturbation		
		B_x	B_y	B_z
Response	B_x	1	0.021	0.059
	B_y	0.033	1	0.028
	B_z	0.068	0.004	1

Table 5.2: Orthogonality of the SERF magnetometer response, determined by measuring the change in compensation current along each of the three directions after application of a small field along one particular direction. It is likely limited by imperfections in the coil construction.

to respond quickly enough to any change in the magnetic field such that the magnetometer signal remains within the central part of the resonant dispersion curve at all times along all three directions; if a field component becomes large enough that $\gamma^e B_i / q > \Delta\omega$, then the slope of the dispersion curve changes sign (see Equations 5.23 and 5.28), and the feedback drives the field away from zero rather than towards it. We show in Section 5.4 that three-axis detection can be used to allow operation of a SERF magnetometer without magnetic shielding, but it can also be used to periodically cancel the magnetic field for applications with partial shielding, as demonstrated by Xia *et al.* (2006) for biomagnetic imaging where large holes were necessary in the μ -metal shields in order to accommodate the presence of a human subject. Sensitivity is lower to the transverse components B_x and B_z than it is to B_y , since the magnetometer response is suppressed along these directions. Li *et al.* (2006) demonstrated a different modulation technique that allows the SERF magnetometer to obtain equal sensitivity to B_x and B_y , but without the ability to measure B_z . It may be possible to detect this field component as well by applying additional modulation and measuring the resulting sidebands in the magnetometer response spectrum, albeit with reduced sensitivity compared to the other two components.

5.4 Unshielded Operation

The SERF magnetometer requires operation in very small magnetic fields, so previous demonstrations used magnetic shielding to attenuate the ambient field, with coils canceling any small residual fields. However, the use of shields is not possible in applications to measure the local magnetic field or detect magnetic anomalies, such as vehicles, unexploded ordinance, or buried mineral deposits or archaeological artifacts. In order to take advantage of the inherent sensitivity of the SERF magnetometer for such applications, it is necessary to instead use feedback to null the ambient field in order to keep the alkali atoms in the low-field regime. We accomplished this using the SERF magnetometer as a

three-axis vector sensor, as described in Section 5.3, canceling all three components of the ambient field through active feedback. This method enables operation of the SERF magnetometer even in environments where the magnetic field is too large for suppression of spin-exchange relaxation.

The unshielded SERF magnetometer is shown in Figure 5.8. The cell containing potassium metal, 2.5 amg of ^4He buffer gas, and 60 Torr of N_2 quenching gas is heated to 170°C by hot air in a double-walled glass oven. The oven and optics are mounted on an aluminum plate in the middle of a set of three orthogonal Helmholtz coils, and the pump and probe beams are generated by single-frequency external cavity diode lasers. We surround the oven with 2-inch-thick aluminum plates in order to attenuate the high level of 60 Hz magnetic noise present in the laboratory without affecting the dc and low-frequency components of the magnetic field. We apply the magnetic field modulations using a second set of small coils located inside the aluminum shield. Although this setup is too bulky to be used for portable measurements, it serves to demonstrate the principle of unshielded SERF magnetometer operation.

We find that the easiest method for zeroing the magnetic field is to use a different sensor, such as a fluxgate, and then turn on feedback from the magnetometer once the field along all three directions is sufficiently small. Zeroing the field using only the magnetometer signal is extremely difficult and inefficient, especially when the field amplitude is much larger than the magnetic linewidth. Once feedback is active, the magnetometer is able to track changes in the local environment, such as motion of distant magnetic objects.⁶ Despite the thick aluminum shields, the sensitivity of the magnetometer is limited by residual magnetic field noise; in particular, the noise level at 60 Hz is still about 0.2 mG inside the shields. We are able to achieve sensitivity to B_y on the order of $1 \text{ pT}/\sqrt{\text{Hz}}$ in the laboratory using

⁶ For example, we are able to detect a magnetic screwdriver turning about 30 feet away.

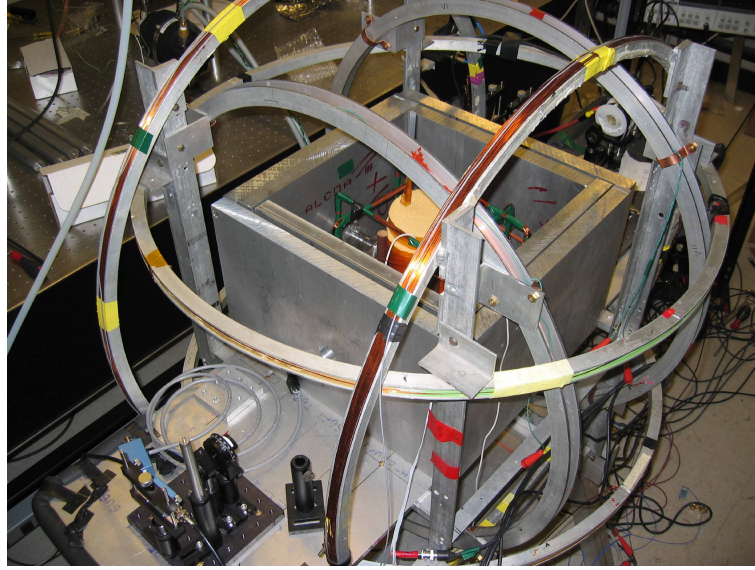


Figure 5.8: Picture of the unshielded SERF magnetometer. The oven can be seen covered in polyimide foam insulation and kapton tape, surrounded by small modulation coils. These are in turn surrounded by thick aluminum shields that attenuate 60 Hz magnetic field noise and a larger set of three orthogonal Helmholtz coils.

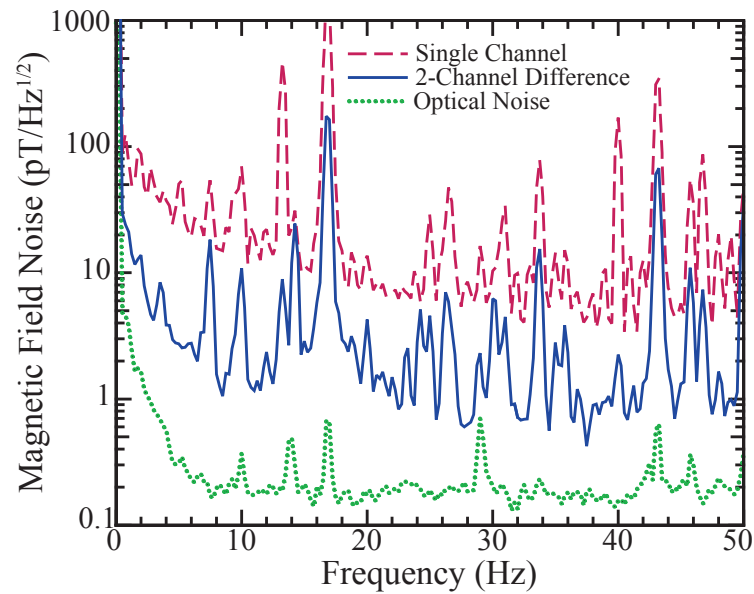


Figure 5.9: Sensitivity of the unshielded SERF magnetometer to the magnetic field gradient is on the order of $1 \text{ pT}/\sqrt{\text{Hz}}$, limited by ambient field noise in the laboratory environment.

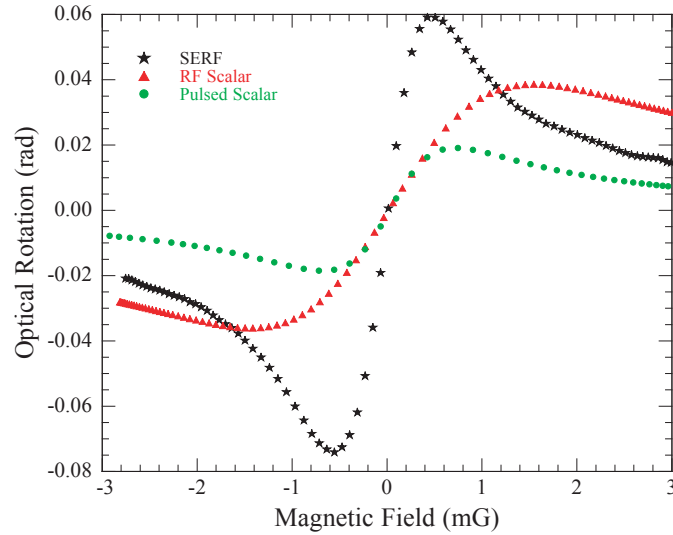


Figure 5.10: Comparison of the dispersion curves obtained near resonance for the magnetometer operating in vector SERF mode and in the two scalar modes. The relative sensitivity is given by the slope near resonance, so the SERF mode exhibits about four times better sensitivity than either scalar mode.

a two-channel gradiometer measurement, as shown in Figure 5.9. In comparison, the optical detection noise obtained with the pump beam blocked is significantly lower, indicating that the magnetometer noise level is dominated by fluctuations of the ambient magnetic field.

In Figure 5.10 we compare the dispersive resonance signal obtained by operating the unshielded magnetometer in the vector SERF mode to the signals obtained near resonance by operating in two different scalar modes, using either radio-frequency (rf) excitation (discussed in Section 3.1.1) or pulsed optical excitation (discussed in Section 3.1.2). For the scalar modes the Larmor resonance frequency is set to 20 kHz, and various operational parameters, such as the pump beam intensity and the rf modulation amplitude, are optimized in order to maximize the slope of the dispersion curve. To allow direct comparison, the density of potassium atoms and the pump beam intensity are kept the same for all

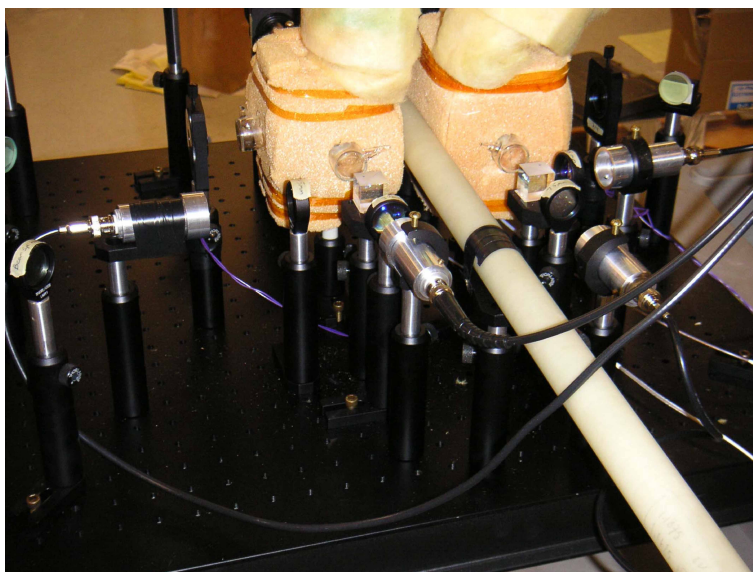


Figure 5.11: Prototype design for an improved unshielded SERF magnetometer: the magnetic field is nulled by feedback from a high-bandwidth fluxgate, while multiple alkali cells measure the field gradient with a large baseline.

operational modes. The magnetometer sensitivity is proportional to the slope of the dispersion curve near resonance, and we observe the slope for the SERF magnetometer to be four times larger than for either of the scalar modes.

The magnetic linewidth $\Delta B = \Delta\omega/\gamma$ is about 0.5 mG for the SERF and optical scalar modes and about 1.4 mG for the rf scalar mode. Although the exact alkali vapor density is not known, from the operating temperature we can estimate that the optimal light-narrowed linewidth for the scalar magnetometers given by Equations 4.9 and 4.10 is 0.1 mG, comparable to the measured linewidths. The sensitivities of the scalar magnetometers are close to the limit set by spin-exchange collisions. In contrast, a resonance linewidth of $2.6 \mu\text{G}$ has been obtained with a SERF magnetometer operating in a magnetically shielded environment (Allred *et al.*, 2002). The sensitivity of the unshielded SERF magnetometer therefore has the potential to improve by over two orders of magnitude if the linewidth can be reduced.

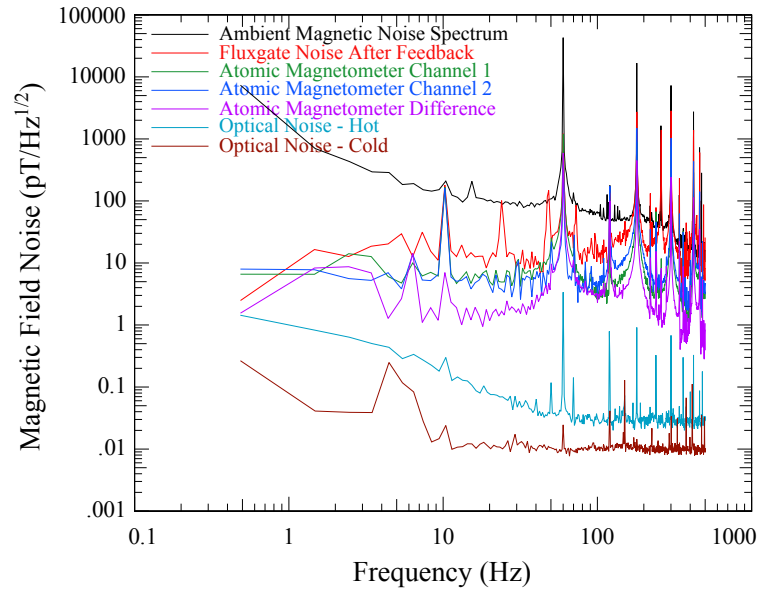


Figure 5.12: Sensitivity of the prototype unshielded SERF magnetometer shown in Figure 5.11, which is comprised of two separate cells that enable a large gradiometer baseline. The sensitivity of $1 \text{ pT}/\sqrt{\text{Hz}}$ is likely limited by the ambient magnetic field noise.

The magnetic linewidth is limited by ambient high-frequency noise, especially at 60 Hz and its harmonics, as well as magnetic field gradients across the cell. The noise can be better suppressed by using a high-bandwidth fluxgate for feedback rather than the magnetometer, since it would be able to respond more quickly to high-frequency oscillations. The SERF magnetometer would measure the field gradient, potentially allowing active feedback to cancel all five independent gradient components and thus reduce the linewidth. Several magnetometer cells could measure the local magnetic field in separate locations in order to give a large gradient baseline, which enables better detection of distant magnetic anomalies; three cells operating as three-axis vector magnetometers are necessary to measure all five gradient components. A prototype for this system is shown in Figure 5.11, with

a fluxgate surrounded by two independent ovens, each containing a potassium cell and illuminated by pump and probe beams.⁷ We achieve sensitivity of approximately $1 \text{ pT}/\sqrt{\text{Hz}}$ with this prototype, as displayed in Figure 5.12, again limited by ambient field noise. In general, substantial improvement in the sensitivity of the unshielded SERF magnetometer should be possible simply by operating in a remote location far from noise sources, enabling its use for portable applications requiring precise characterization of magnetic fields.

⁷ This system was constructed by Tristan Technologies as part of a collaboration with us to develop a SERF magnetometer for magnetic anomaly detection.

Chapter 6

Anti-Relaxation Surface Coatings

TRADITIONALLY, HIGH-TEMPERATURE MAGNETOMETERS and other applications using alkali vapor have required the use of buffer gas to prevent wall relaxation, due to the previous unavailability of effective anti-relaxation coatings that operate above 80°C. However, coatings have numerous advantages over buffer gas, including larger optical rotation signals, lower laser power requirements, and lack of spin-destruction effects. We demonstrate that octadecyltrichlorosilane (OTS) acts as an effective anti-relaxation coating at temperatures up to about 170°C, making it the first known coating for high-temperature applications. Using an OTS-coated potassium cell, we achieve sensitivity of 6 fT/ $\sqrt{\text{Hz}}$ with a single-channel SERF magnetometer. We also describe an experiment to evaluate the anti-relaxation properties of various surface coatings using a reusable alkali vapor cell, in an effort to determine the characteristics of functional coatings and design better high-temperature coatings. The use of OTS as an anti-relaxation coating was first presented in [Seltzer *et al.* \(2007\)](#), the first demonstration of a SERF magnetometer with a coated cell will be presented in [Seltzer and Romalis \(2009\)](#), and initial results from the search for effective high-temperature coatings using the reusable alkali cell will be presented in [Seltzer *et al.* \(2008\)](#).

6.1 Surface Coatings for Alkali Vapor Cells

As discussed in Section 2.7.3, alkali atoms become completely depolarized following collisions with the bare glass walls of the vapor cell due to interaction with the local electromagnetic fields within the glass. The two common methods for suppressing this effect are the use of buffer gas to slow diffusion to the walls and the use of a chemical coating to prevent atoms from reaching the bare surface. All previous demonstrations of high-density magnetometers, including the rf magnetometer described in Chapter 4 and the SERF magnetometer described in Chapter 5, used cells filled with buffer gas because there was no available coating for operation at high temperature. On the other hand, magnetometers operating at or slightly above room temperature often use cells coated with paraffin, which is the most effective known coating. We discuss the advantages of wall coatings over buffer gas in Section 6.1.1.

Paraffin was first discovered to enhance spin polarization lifetimes by Robinson *et al.* (1958). A long, organic hydrocarbon chain, *n*-paraffin [C_nH_{2n+2}] covers the surface and prevents direct interaction between alkali atoms and the cell walls. Bouchiat and Brossel (1966) found that relaxation of alkali spins on paraffin is caused by two independent magnetic interactions experienced by atoms during the brief time they are adsorbed into the surface following a collision: The first effect is due to the dipole-dipole interaction with 1H nuclear spins in the coating, while the second effect is possibly due to the spin-orbit interaction with carbon atoms moving within the coating. Paraffin has very low polarizability and features significantly smaller local fields than bare glass, and the adsorption time is short due to the low adsorption energy ~ 0.1 eV, so these effects are weak, and atoms can bounce off the coated surface up to 10,000 times while maintaining their polarization (see also Graf *et al.* (2005)).

Paraffin melts and loses its effectiveness above 60-80°C, with the precise melting point determined by the length of the molecular chain, but it is commonly employed in low-temperature applications for which this restriction is not problematic.¹ Recent examples of magnetometers operating with paraffin-coated cells include Groeger *et al.* (2005, 2006), Schwindt *et al.* (2005), Balabas *et al.* (2006), and Ledbetter *et al.* (2007), with an observed magnetic linewidth as narrow as 0.35 Hz in a 10 cm diameter spherical cell (Budker *et al.*, 2005). Paraffin coatings have also been studied for use in atomic clocks (Risley *et al.*, 1980; Robinson and Johnson, 1983; Guzman *et al.*, 2006), since buffer gases introduce temperature-dependent frequency shifts, but cells coated with paraffin suffer from unacceptable long-term instability of the clock transition frequency. Alternative coatings that allow more stable measurements of the clock frequency are thus highly desirable. Other applications of paraffin-coated alkali vapor cells include magneto-optical traps (Stephens *et al.*, 1994; Lu *et al.*, 1997; Aubin *et al.*, 2003), quantum memory (Julsgaard *et al.*, 2004), and slow light experiments (Klein *et al.*, 2006).

In addition to paraffin, several silane coatings have also been shown to reduce surface relaxation times for alkali spins (Alley, 1961; Camparo, 1987; Swenson and Anderson, 1988; Fedchak *et al.*, 1997). These are organic molecules that terminate with a silicon end group that becomes chemically bound to the surface, unlike paraffin which simply rests on top of the surface. Silane coatings are generally used for applications at or near room temperature, and they have not been demonstrated previously to preserve spin polarization at temperatures significantly higher than the paraffin melting point. Here we show that octadecyltrichlorosilane (OTS), which resembles paraffin but terminates with a trichlorosilane end group $[\text{CH}_3(\text{CH}_2)_{17}\text{SiCl}_3]$, functions as an anti-relaxation coating at temperatures up

¹ We refer to magnetometers that operate above the paraffin melting point as “high-temperature;” such magnetometers achieve high sensitivity by suppressing spin-exchange relaxation, enabling the use of large alkali density. In contrast, magnetometers that operate below this point are of the “low-temperature” type and achieve high sensitivity through the use of very large vapor cells.

to about 170°C, allowing its use for high-temperature applications. We discuss magnetometry with OTS-coated cells in Section 6.2, but first we consider in general the advantages and evaluation of surface coatings for atomic magnetometry.

6.1.1 Advantages for Magnetometry

Wall coatings offer several practical advantages over buffer gas that allow for easier realization of high magnetic field sensitivity. Although buffer gas slows diffusion of alkali atoms to the cell wall, spin-destruction collisions with buffer gas atoms broaden the magnetic linewidth and can limit magnetometer performance, especially in high-pressure cells. In contrast, a high-efficiency coating causes no spin depolarization. The improvement in linewidth given by coated cells compared to buffer gas cells becomes greater as the cell size decreases, since higher buffer gas pressure is required to maintain the same diffusion time to the walls. Surface coatings are therefore particularly important for obtaining narrow resonance linewidths in miniature cells (Kitching *et al.*, 2002).

Although a small amount of quenching gas is necessary to prevent radiation trapping for high-density applications, in the absence of buffer gas the atoms are free to move between different parts of the cell. The entire cell therefore serves as the active measurement volume even if the pump beam does not fill the cell, since the wall coating preserves some degree of spin polarization in the parts of the cell not actively pumped; we discuss this further in Section 6.1.3. In addition, the probe beam can remain small because the atoms are likely to pass through it at least once during each polarization lifetime. More efficient coatings enable the atoms to sample a larger fraction of the cell volume within a polarization lifetime, so that smaller beams may be used. This allows for easier use of optical fibers to deliver light to and from the cell without precise expansion of the beams.

In high-pressure cells the atoms diffuse slowly and remain within a small part of the cell, so magnetic field gradients cause atoms in different regions to precess with different

frequencies, resulting in broadening of the magnetic linewidth as discussed in Section 2.7.4. On the other hand, in coated low-pressure cells this effect is suppressed because the atoms move throughout the cell and effectively average the field over the entire cell volume, so the spins all precess at the same average frequency. The broadening is quadratic in the size of the gradient and is completely eliminated if all gradients are kept sufficiently small (Cates *et al.*, 1988a; Pustelny *et al.*, 2006). At low field, such that the Larmor period is comparable to or larger than the mean time for an atom to travel across the cell, the spins are equally sensitive to all gradient components. At high field, however, gradients of the field components orthogonal to the main field direction are suppressed. In Figure 6.1 we show the gradient broadening measured in an OTS-coated cell without buffer gas in a magnetic field of 0.93 G along the $\hat{y}$ direction, with all gradients zeroed except the component being adjusted. As predicted, the atoms are most sensitive to the gradient $\partial B_x/\partial y = \partial B_y/\partial x$, less sensitive to $\partial B_x/\partial x$, and significantly less sensitive to $\partial B_x/\partial z$.²

This motional averaging of magnetic field gradients makes gradiometric measurements impossible in a traditional coated cell without buffer gas. Although separate cells may be used to measure the field gradient over a large baseline for detection of distant magnetic anomalies, using a system such as that shown in Figure 5.11, finer spatial resolution is necessary for many imaging applications. We show two cell designs in Figure 6.2 that may potentially solve this problem; on the left we show a cell containing an internal membrane that divides the cell into individual measurement volumes, and on the right we show two separate cells connected by a common stem. In both cases, atoms are mostly restricted to remain within one chamber or the other with minimal exchange of polarized atoms, and a shared alkali metal reservoir in the stem maintains a common alkali vapor density. A single pump beam can illuminate both chambers, and individual probe beams (or individual parts of a single probe beam) are detected separately. Although these designs have yet

² Recall that $\nabla \cdot \mathbf{B} = 0$, so $\partial B_i/\partial x_j = \partial B_j/\partial x_i$ for $i \neq j$, and $\sum_i \partial B_i/\partial x_i = 0$. Therefore, application of a magnetic field gradient $\partial B_x/\partial x$ also results in the application of $\partial B_y/\partial y = \partial B_z/\partial z = -\frac{1}{2}\partial B_x/\partial x$.

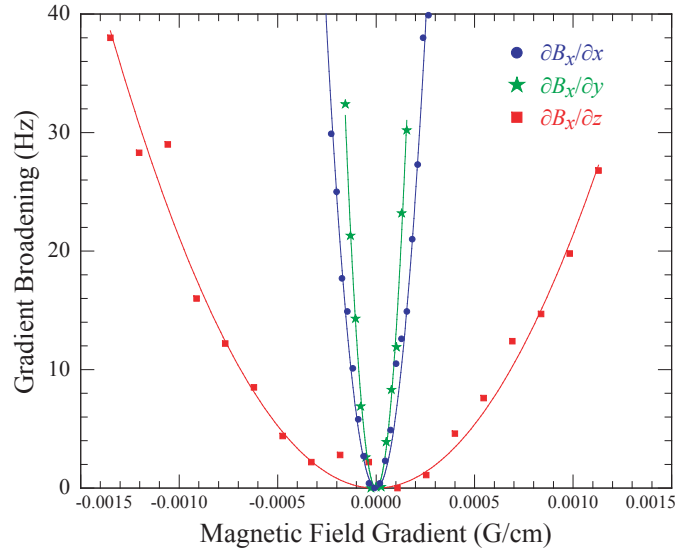


Figure 6.1: Measurements of gradient broadening in an OTS-coated cell without buffer gas in a magnetic field $B=(0.93 \text{ G})\hat{y}$. The lines are quadratic fits to the data and are intended to guide the eye.

to be implemented, they should allow the typical improvement in sensitivity that results from measuring the field gradient rather than the field itself.

Another advantage of using cells with a surface coating rather than buffer gas is that the optical linewidth is significantly narrower. In coated cells with a small quantity of quenching gas, under typical operating conditions both Γ_G and Γ_L are less than 1 GHz, while in a cell with several amg of buffer gas $\Gamma_L \sim 10\text{-}100$ GHz. If the optical lineshape is approximately Lorentzian, then we see from Equation 2.8 that the photon absorption cross-section scales as $1/\Gamma_L$, so we expect the cross-section to be inversely proportional to the buffer gas pressure. This is not entirely accurate however, since at low pressure the hyperfine structure splits the resonance into individual, resolved transitions that do not completely add together. We show in Figure 6.3(a) the photon absorption cross-section for the D1 transition in potassium and cesium given by Equation 2.26 as a function of the

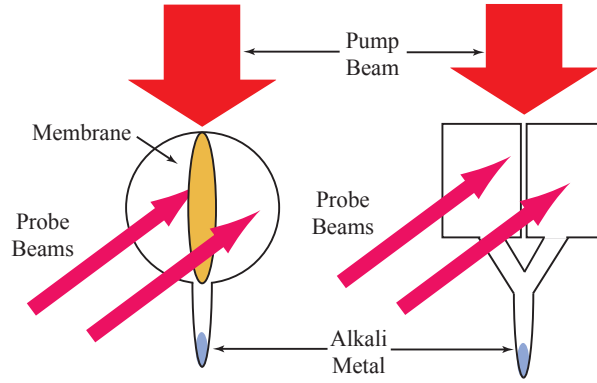


Figure 6.2: Two possible designs for coated cells that would enable measurement of magnetic field gradients, with the cell divided into separate chambers where the atoms independently measure the local field.

pressure-broadened optical linewidth Γ_L , taking the maximum cross-section given by optimizing the light frequency.³ We see that the absorption cross-section has roughly the expected $1/\Gamma_L$ scaling. At low pressure there is stronger interaction between (near-)resonant light and the alkali atoms, allowing for instance the use of lower-intensity pump beams to achieve the same pumping rate as in high-pressure cells. Coated cells are thus particularly advantageous in applications where total power consumed must be minimized, such as portable and miniature magnetometers.

This stronger interaction between alkali atoms and resonant laser light in coated cells compared to buffer gas cells also leads to larger optical rotation signals. We consider the rotation signal given by $\theta e^{-n\sigma l}$, taking into account absorption of the probe beam, which represents the usable signal provided by the magnetometer for measuring the magnetic field. Figure 6.3(b) shows the rotation signal given by optimal detuning of the probe beam as a function of the pressure-broadened linewidth. The rotation signal scales approximately as

³ As an example, looking at Figure 2.5, we take the maximum value of the cross-section for each curve, which occurs at an optical frequency that gets closer to zero detuning as the individual transitions begin to overlap at higher pressure. Similarly looking at Figure 2.15(b), for the optical rotation signals described below we consider the optimal probe beam detuning for each value of the optical linewidth.

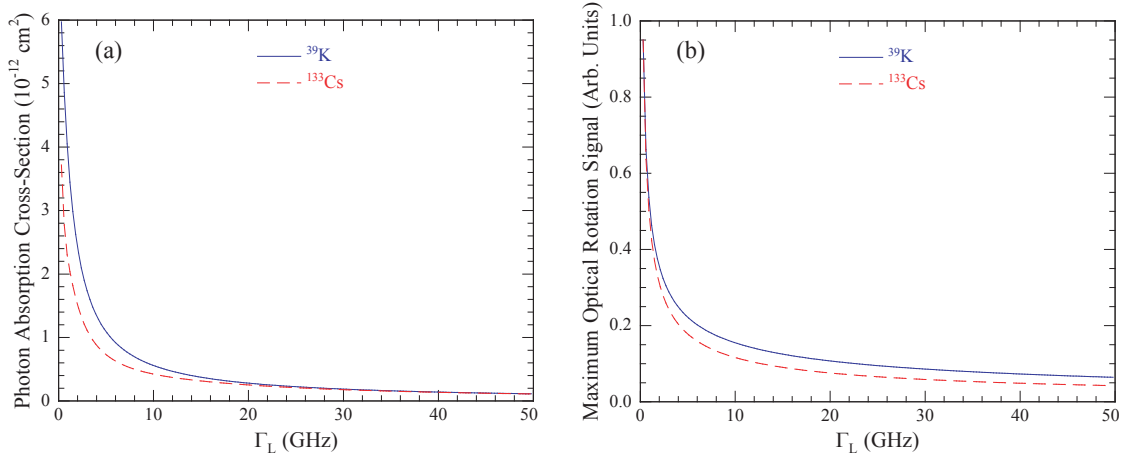


Figure 6.3: (a) Maximum photon absorption cross-section σ calculated from Equation 2.26 as a function of the pressure broadening Γ_L . (b) Maximum optical rotation signal $\theta e^{-n\sigma l}$ calculated from Equation 2.99 and taking into account absorption of the probe beam as a function of the pressure broadening; the potassium and cesium curves are normalized separately and should not be compared. For both plots, we consider only the D1 transition, and if the hyperfine structure is resolved then we select the optimal laser frequency that provides the largest cross-section or rotation signal. We take the Doppler width at 160°C , $P_x=0.5$, $n=5 \times 10^{12} \text{ cm}^{-3}$, and $l=5 \text{ cm}$.

$1/\sqrt{\Gamma_L}$, so coated cells can increase the signal by up to an order of magnitude.⁴ While this does not improve the inherent sensitivity of a magnetometer, it potentially makes it much easier in practice to realize that sensitivity. Magnetometers are often limited by probe beam noise, including $1/f$ noise at near-dc frequencies and optical shot noise at radio frequencies, so a narrower optical linewidth increases the alkali polarization signal amidst the probe noise background. The use of coated cells may therefore be an important part of achieving attotesla-level sensitivity with rf and SERF magnetometers.

6.1.2 Measuring Coating Quality

We characterize the effectiveness of an anti-relaxation coating in terms of the average number of collisions by an alkali atom with the coated surface before the atom depolarizes,

⁴ An analysis considering the rotation angle divided by the shot noise, $\theta e^{-n\sigma l/2}$, rather than signal size provides a qualitatively similar result.

which we refer to as the number of “bounces” permitted by the coating. For a perfect coating, this is given by the fraction of the surface area that is not covered by the coating, so maximum coverage is necessary to achieve long polarization lifetimes. We determine this quantity by polarizing the atomic vapor and then quickly turning off optical pumping;⁵ we subsequently measure T_1 , the time constant with which the longitudinal spin polarization decays. We ensure that absorption of probe photons is slow enough to not affect T_1 by reducing the intensity of the probe beam until further attenuation no longer increases the observed polarization lifetime. We consider the longitudinal rather than transverse lifetime so that spin-exchange collisions and magnetic field gradients have no effect on the measurement. We see from Equation 2.133 that T_1 is then determined only by spin-destruction collisions, which are usually slow in cells with minimal quenching or buffer gas and low alkali density, and interactions with the cell walls. Since the beams used to pump and probe the atoms have no effect on the measured polarization lifetime, this is known as the relaxation-in-the-dark method and was developed by [Franzen \(1959\)](#).

Care must be taken to ensure that there is no alkali condensation on the walls of the cell, since polarized atoms are readily absorbed into volumes of liquid or solid alkali metal, which function as uncoated surface regions that reduce the measured T_1 from the maximum value enabled by the coating. The stem of the cell is often partially insulated by its holder and remains at a slightly lower temperature ($\sim 10\text{--}15^\circ\text{C}$) than the main cell body, so alkali condensation usually occurs in the stem rather than the cell; the temperature differential can be increased by flowing compressed air or cold water through a tube wrapped around the stem. Once we turn off optical pumping, the spins experience multiple decay modes with different time constants (see Section 2.7.3), and we consider only the data

⁵ This is typically accomplished by using an optical chopper to periodically block the pump beam, but in a coated cell with a narrow optical linewidth we may instead detune the pump beam so that its frequency is far from the atomic resonance.

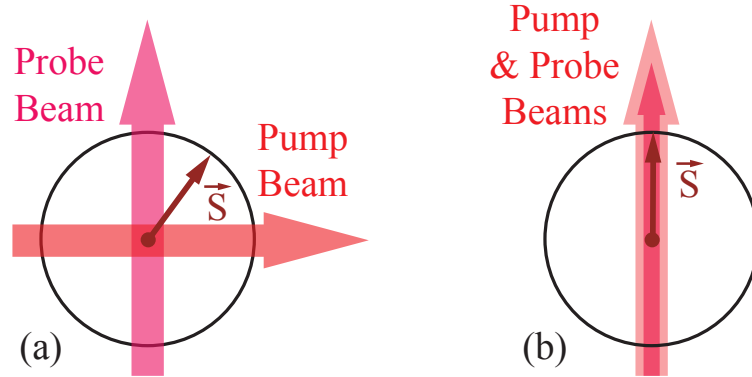


Figure 6.4: Schematics of T_1 measurement techniques. **(a)** Orthogonal pump and probe beams both at 45° to the longitudinal spin direction. **(b)** Pump and probe beams both parallel to the spin direction, with the probe beam polarization oscillated by a photoelastic modulator.

taken after enough time has elapsed that only the lowest-order mode affects the spin relaxation. We estimate the error by starting the data fit at successively later moments in time and noting the variation in the fit value for T_1 .

There are numerous methods for performing this measurement, but we generally use two techniques for the work reported here. The first method, which we employ for determining the effectiveness of surface coatings in actual vapor cells, uses orthogonal pump and probe beams as shown in Figure 6.4(a). The magnetic field is set at 45° to both beams, so that pumped atoms have a large average longitudinal component of spin polarization along the magnetic field direction while their transverse spin components have random phase and average to zero. Once pumping turns off, the atoms have some degree of polarization along the direction of the probe beam, which is linearly polarized and provides an optical rotation signal. We monitor the signal as the spin polarization relaxes and fit it to a decaying exponential in order to determine the time constant T_1 . As an example, we show in Figure 6.5 a measurement performed using this technique of $T_1=137$ ms corresponding to approximately 2000 bounces in Cell #7, an OTS-coated spherical potassium cell with diameter of 1.96 inches and no buffer or quenching gas (see Table 6.1).

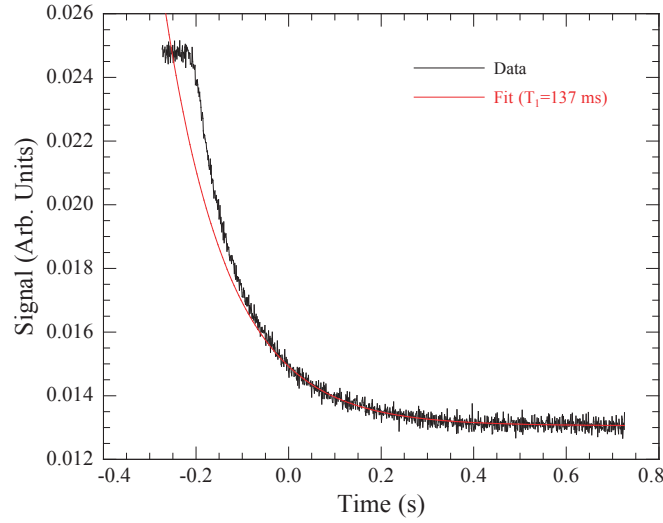


Figure 6.5: Measurement of $T_1=137$ ms in Cell #7, an OTS-coated spherical potassium cell of diameter 1.96 inches without buffer or quenching gas, corresponding to roughly 2000 bounces. The signal is proportional to the longitudinal spin polarization, and the vertical offset is due to imperfect polarization of the probe beam. The fit considers only the lowest-order decay mode.

We use a different technique to measure T_1 for the coated slides used in the reusable alkali vapor cell, as described in Section 6.3, where the geometry of the experiment makes the use of orthogonal pump and probe beams impractical. Instead we employ parallel pump and probe beams, with the magnetic field (and thus the longitudinal spin direction) set parallel to both, as shown in Figure 6.4(b). The probe beam is linearly polarized but passes through a photoelastic modulator (PEM), so that its polarization oscillates between σ^+ and σ^- . When the atoms are fully polarized, they absorb only one helicity of light, leading to an oscillation of the probe beam transmission through the cell at the PEM frequency; as the atomic polarization decays, so does this signal because unpolarized atoms absorb both helicities equally. The PEM is not strictly necessary for this measurement, and we can set the probe beam so that it always has the same sense of circular polarization as the

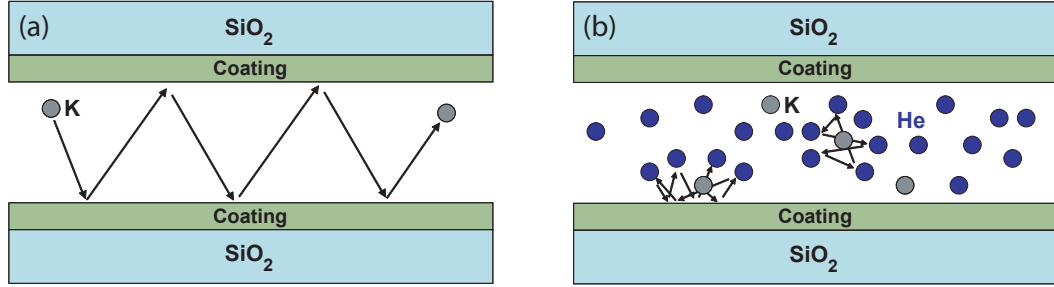


Figure 6.6: (a) In cells without buffer gas, alkali atoms move in straight lines in between collisions with the cell walls. (b) In cells with buffer gas, the atoms require a long period of time to diffuse from the center of the cell to the wall, but once there they collide repeatedly with the surface before they are able to diffuse away.

pump beam, in order to measure an increase in probe beam transmission as the atomic polarization decays, but the use of a PEM at high frequency substantially reduces the noise of the lifetime measurement.

Once we measure the longitudinal spin polarization lifetime T_1 in a cell, we translate it into the number of bounces allowed by the coating. In a cell without any gas species other than the alkali vapor, the atoms experience ballistic motion, following straight paths in between collisions with the cell walls as illustrated in Figure 6.6(a).⁶ The average time between wall collisions T_{wall} is given by Equation 2.145, and the number of bounces N is simply given by

$$N = \frac{T_1}{T_{\text{wall}}} . \quad (6.1)$$

However, in a cell containing buffer and/or quenching gas, the atoms experience diffusive motion, and the conversion between T_1 and N becomes more complicated. In this case, the atoms take a long time to diffuse from the middle of the cell to the wall, but once near the surface they remain there for an extended period of time and collide repeatedly with the surface before diffusing away again, as shown in Figure 6.6(b).

⁶ The alkali vapor density is generally low enough that collisions between alkali atoms occur infrequently and do not affect the trajectories of the individual atoms.

The theory of alkali depolarization at a coated surface in the presence of buffer gas was first developed by [Masnou-Seeuws and Bouchiat \(1967\)](#). The flux of polarized atoms going to the surface (J_+) and returning back from it (J_-) are

$$J_+ = \bar{v} \left[\frac{1}{4} nP - \frac{\lambda}{6} \left(\frac{\partial}{\partial x} nP \right) \right] \quad (6.2)$$

$$J_- = \bar{v} \left[\frac{1}{4} nP + \frac{\lambda}{6} \left(\frac{\partial}{\partial x} nP \right) \right], \quad (6.3)$$

where the surface is orthogonal to the $\hat{x}$ direction, n is the alkali vapor density, P is the spin polarization near the surface, $\bar{v}$ is the thermal velocity of the alkali atoms, and $\lambda=3D/\bar{v}$ is the mean free path. The diffusion constant D of the alkali atoms within the buffer gas depends on the gas pressure. The probability of an atom depolarizing during a particular collision with the surface is $1/N$, so the reflected flux is given by

$$J_- = \left(1 - \frac{1}{N} \right) J_+. \quad (6.4)$$

Combining Equations 6.2-6.4, we find that the gradient of polarization at the coated surface gives the boundary condition

$$\frac{\partial}{\partial x} P = \frac{-\bar{v}P}{2N(2 - 1/N)D}. \quad (6.5)$$

As an example we consider diffusion within a spherical coated cell of radius r_0 , similar to our previous discussion in Section 2.7.3 except with nonzero polarization at the surface. The evolution of spin polarization is described by the standard diffusion equation,

$$\frac{\partial}{\partial t} P = D \nabla^2 P - R_{SD} P, \quad (6.6)$$

where we include the spin-destruction rate R_{SD} , which is a function of the buffer gas pressure. The solution for the fundamental diffusion mode is

$$P(r, t) = \frac{\sin(kr)}{kr} e^{-t/T_1}, \quad (6.7)$$

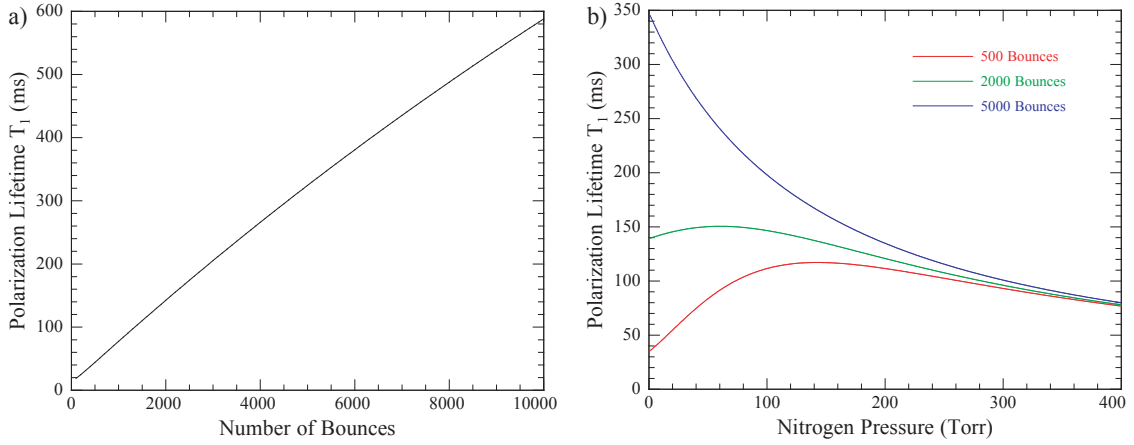


Figure 6.7: (a) Polarization lifetime T_1 as a function of the number of bounces permitted by a surface coating, modeled in a spherical potassium cell of radius 2.5 cm at 150°C containing 10 Torr of nitrogen quenching gas. (b) Polarization lifetimes as a function of quenching gas pressure for coatings permitting 500, 2000, and 5000 bounces. At high enough pressure, the spin-destruction rate dominates the wall relaxation rate.

where k is the radial diffusion wave number. Combining Equations 6.6 and 6.7 gives

$$\frac{1}{T_1} = Dk^2 + R_{SD}, \quad (6.8)$$

and applying the boundary condition (Equation 6.5) gives

$$k \cot(kr_0) = \frac{1}{r_0} - \frac{\bar{v}}{2N(2 - 1/N)D}. \quad (6.9)$$

We combine Equations 6.9 and 6.8 and solve numerically for the number of bounces N characterized by the measured polarization lifetime T_1 in a particular cell. We find the equivalent solution in a cell with rectangular geometry in Section 6.3.

In Figure 6.7(a) we show the polarization lifetime T_1 as a function of the number of bounces in a spherical cell of radius 2.5 cm containing potassium vapor and 10 Torr of nitrogen for quenching. A coating that performs as well as paraffin can allow a linewidth $\Delta\omega < 2\pi \times 1$ Hz. In Figure 6.7(b) we show the lifetime for a particular number of bounces as a function of quenching gas pressure, demonstrating that a small amount of gas improves the lifetime by slowing diffusion to the cell wall. However, the alkali metals all

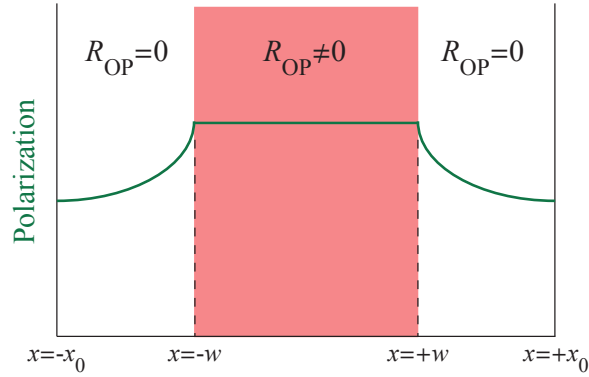


Figure 6.8: Illustration showing partial illumination of a one-dimensional cell by the pump beam; the polarization is described by Equations 6.12-6.23.

have large spin-destruction cross-sections with nitrogen, so excessive quenching gas in the cell can limit the attainable polarization lifetime; better coating effectiveness results in a lower characteristic pressure, above which relaxation due to spin-destruction collisions is greater than relaxation due to wall collisions. It is nevertheless necessary for the cell to contain some amount of quenching gas to eliminate radiation trapping for operation at high alkali density.

6.1.3 Polarization Distribution in Coated Cells

Alkali atoms can remain partially polarized throughout a coated cell with low gas pressure, even in regions not illuminated by the pump beam, because the atoms are free to move around and the coating maintains nonzero polarization at the cell wall. For simplicity, we consider the distribution of polarization within a one-dimensional cell of length $2x_0$, as illustrated in Figure 6.8. The pump beam illuminates a region of width $2w$ and has polarization $s=+1\hat{z}$, and the walls are covered with a coating that allows N bounces. If the atoms experience a magnetic field $\mathbf{B} = B_y\hat{y}$, then the spin polarization components P_x and

P_z are described by a combination of the Bloch and diffusion equations:

$$\begin{cases} D \frac{\partial^2}{\partial x^2} P_x + \gamma B_y P_z - (R_{OP} + R_{SD}) P_x = 0 & ; \quad |x| < w \\ D \frac{\partial^2}{\partial x^2} P_x + \gamma B_y P_z - R_{SD} P_x = 0 & ; \quad |x| \geq w \end{cases} \quad (6.10)$$

$$\begin{cases} D \frac{\partial^2}{\partial x^2} P_z - \gamma B_y P_x - (R_{OP} + R_{SD}) P_z + R_{OP} = 0 & ; \quad |x| < w \\ D \frac{\partial^2}{\partial x^2} P_z - \gamma B_y P_x - R_{SD} P_z = 0 & ; \quad |x| \geq w. \end{cases} \quad (6.11)$$

The solution to these equations has the form

$$P_x = \begin{cases} a + b_1 \cosh(m_1 x) + b_2 \cosh(m_2 x) & ; \quad |x| < w \\ g_1 \cosh(h_1 x + j_1) + g_2 \cosh(h_2 x + j_2) & ; \quad |x| \geq w \end{cases} \quad (6.12)$$

$$P_z = \begin{cases} d - ib_1 \cosh(m_1 x) + ib_2 \cosh(m_2 x) & ; \quad |x| < w \\ -ig_1 \cosh(h_1 x + j_1) + ig_2 \cosh(h_2 x + j_2) & ; \quad |x| \geq w. \end{cases} \quad (6.13)$$

Inspecting the solution, we see that the phases j_1 and j_2 describe the polarization maintained at the surface by the coating, while the offsets a and d describe the polarization at the center of the cell. We do not reproduce here the method for solving these equations, which involves setting the polarization and its derivative to be continuous at the domain boundaries ($x = \pm w$) and applying the boundary condition given by Equation 6.5 at the cell walls ($x = \pm x_0$). We find that

$$a = \frac{R_{OP} \gamma B_y}{(\gamma B_y)^2 + (R_{OP} + R_{SD})^2} \quad (6.14)$$

$$d = \frac{R_{OP}(R_{OP} + R_{SD})}{(\gamma B_y)^2 + (R_{OP} + R_{SD})^2} \quad (6.15)$$

$$m_1 = \sqrt{\frac{R_{OP} + R_{SD} + i\gamma B_y}{D}}; \quad m_2 = \sqrt{\frac{R_{OP} + R_{SD} - i\gamma B_y}{D}} \quad (6.16)$$

$$h_1 = \sqrt{\frac{R_{SD} + i\gamma B_y}{D}}; \quad h_2 = \sqrt{\frac{R_{SD} - i\gamma B_y}{D}} \quad (6.17)$$

$$j_1 = -h_1 x_0 + \tanh^{-1} \left(\frac{-\bar{v}}{2N(2 - 1/N)Dh_1} \right) \quad (6.18)$$

$$j_2 = -h_2 x_0 + \tanh^{-1} \left(\frac{-\bar{v}}{2N(2 - 1/N)Dh_2} \right) \quad (6.19)$$

$$b_1 = \frac{(a + id)h_1 \sinh(h_1 w + j_1)}{2m_1 \cosh(h_1 w + j_1) \sinh(m_1 w) - 2h_1 \sinh(h_1 w + j_1) \cosh(m_1 w)} \quad (6.20)$$

$$b_2 = \frac{(a - id)h_2 \sinh(h_2 w + j_2)}{2m_2 \cosh(h_2 w + j_2) \sinh(m_2 w) - 2h_2 \sinh(h_2 w + j_2) \cosh(m_2 w)} \quad (6.21)$$

$$g_1 = \frac{(a + id)m_1 \sinh(m_1 w)}{2m_1 \cosh(h_1 w + j_1) \sinh(m_1 w) - 2h_1 \sinh(h_1 w + j_1) \cosh(m_1 w)} \quad (6.22)$$

$$g_2 = \frac{(a - id)m_2 \sinh(m_2 w)}{2m_2 \cosh(h_2 w + j_2) \sinh(m_2 w) - 2h_2 \sinh(h_2 w + j_2) \cosh(m_2 w)} . \quad (6.23)$$

We may verify that this solution satisfies the differential equations and boundary condition. The solution simplifies considerably when $B_y=0$, giving $P_x=0$; we provide the general solution for $B_y \neq 0$ because it is useful for modeling the optical rotation signal, which is given by integrating P_x over the path of the probe beam, in a SERF magnetometer with incomplete pump beam illumination of the cell.

In Figure 6.9(a) we model the distribution of spin polarization as a function of the number of bounces in a coated cell that is only partially illuminated by the pump beam. We see that as the number of bounces increases, larger polarization is maintained at the surface and in the regions not actively pumped, although there is some degree of polarization throughout the cell because pumped atoms freely diffuse around. We also see in Figure 6.9(b) that atoms are polarized to some extent everywhere in the cell regardless of the size of the pump beam, though a wider beam gives a broader region of high polarization. In Figure 6.9(c) we show the dependence of polarization distribution on the gas

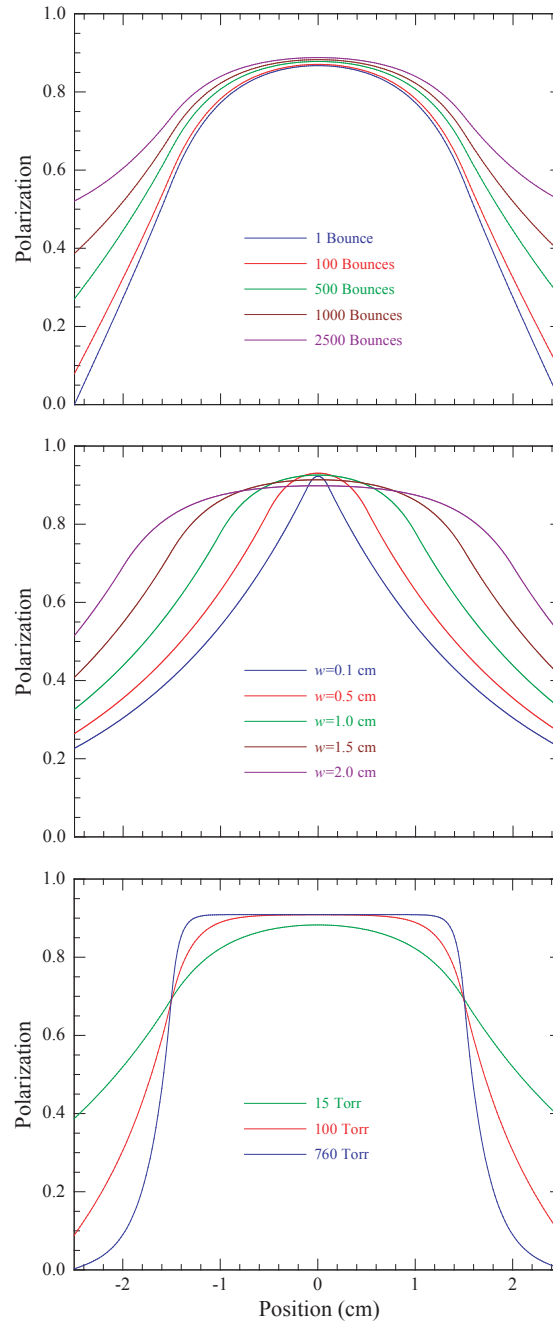


Figure 6.9: Distribution of polarization calculated in a coated cell containing potassium and nitrogen with length $x_0=2.5$ cm, $R_{OP}=2\pi\times 10$ Hz, $R_{SD}=2\pi\times 1$ Hz, and temperature of 150°C . **(a)** Dependence on the number of bounces, with 15 Torr of nitrogen and pump beam width $w=1.5$ cm. **(b)** Dependence on the width of the pump beam, with 1000 bounces and 15 Torr of nitrogen. We set the pumping rate to $2\pi\times 10$ Hz for $w=2.0$ cm and scale it with the beam width in order to maintain constant total power. **(c)** Dependence on the nitrogen pressure, with 1000 bounces and pump beam width $w=1.5$ cm.

pressure, which determines the diffusion constant D . Lower pressure enables the atoms to diffuse freely, while higher pressure inhibits diffusion and leads to low polarization in the regions not actively pumped.

6.2 Octadecyltrichlorosilane

OTS is the first known anti-relaxation coating for high-temperature applications of polarized alkali vapor. It is an organic molecule that resembles paraffin, with eighteen carbon atoms each attached to two or three hydrogen atoms, and a silane end group that reacts with the glass surface and binds to it. The form of an OTS molecule after attaching to the surface is illustrated in the bottom of Figure 6.16. Unlike paraffin, the OTS coating does not melt, and because it is attached to the surface by a chemical bond it remains stable under normal operating conditions. Silane coatings have been used to suppress surface relaxation of polarized noble gas atoms such as ^3He and ^{129}Xe (Zeng *et al.*, 1983; Driehuys *et al.*, 1995), and OTS in particular is also used for magneto-optical traps of radioactive atoms (Guckert *et al.*, 1998) and as an anti-stiction coating for microelectronic devices (Maboudian *et al.*, 2000; Ashurst *et al.*, 2001).

The OTS-coated alkali vapor cells produced in this study are listed in Table 6.1, along with the maximum polarization lifetime T_1 observed in each cell and corresponding number of bounces.⁷ The number of bounces is only approximate for the cells with quenching gas, since the diffusion constant of potassium in nitrogen is not well known (Silver, 1984), so a more precise measurement would greatly benefit this study. The quality of the coating in each cell is highly variable, ranging approximately from 20 to 2100. The cause of this variation is presently unknown, but one possibility is that the coating process is likely to be highly sensitive to even slight variations; we detail the procedure used for these cells in Section 6.2.2, and a careful study in which individual steps are altered could provide a

⁷ The rubidium cells were coated by Thomas Kornack using the procedure described in Section 6.2.2.

Cell	Size and Shape	Alkali	Gas	Coating	Max. T ₁ (ms)	Bounces (Approx.)
1	0.90" Diameter Sphere	K	26.0 Torr N ₂	OTS	23	500
2	0.90" Diameter Sphere	K	26.0 Torr N ₂	OTS	24	550
3	0.90" Diameter Sphere	K	None	ODPA	0.35	15
4	1.90" Diameter Sphere	K	None	OTS	1.6	20
5	1.90" Diameter Sphere	K	None	OTS	33	500
6	1.95" Diameter Sphere	K	None	OTS	61	900
7	1.96" Diameter Sphere	K	None	OTS	145	2100
8	2.00" Diameter Sphere	K	5.0 Torr N ₂	OTS	40	500
9	2.00" Diameter Sphere	K	5.0 Torr N ₂	OTS	33	400
10	2.01" Diameter Sphere	K	5.0 Torr N ₂	OTS	46	600
11	2.00" Diameter Sphere	K	12.6 Torr N ₂	OTS	45	450
-	2.2 mm × 2.2 mm × 3.5 mm	⁸⁷ Rb	5 Torr N ₂	OTS	51	850
-	2.5 mm × 2.5 mm × 4.5 mm	⁸⁷ Rb	10 Torr N ₂	OTS	59	800

Table 6.1: List of coated cells produced in this study. All OTS coatings are multilayers produced by the method of Section 6.2.2, and all cells are made of pyrex (borosilicate) glass. Each cell is etched with its identifying number.

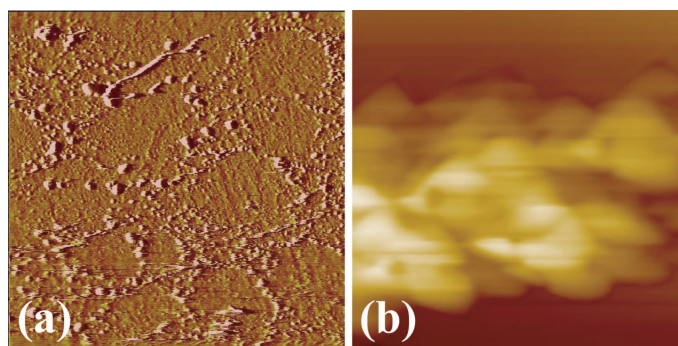


Figure 6.10: $5\text{ }\mu\text{m}\times 5\text{ }\mu\text{m}$ AFM height images of OTS films after exposure to rubidium vapor; lighter colors represent greater surface height and indicate the presence of rubidium clusters. **(a)** AFM image of a monolayer OTS coating, showing small rubidium clusters at domain boundaries. **(b)** AFM image of a large rubidium cluster in a multilayer OTS film. Adapted from [Rampulla *et al.* \(2009\)](#).

better understanding of cell variation.⁸ Nevertheless, nearly all cells demonstrate at least several hundred bounces, so this procedure is highly reliable for the creation of a stable anti-relaxation coating for high-temperature applications.

In parallel to the work described in Section 6.3, [Rampulla *et al.* \(2009\)](#) have examined OTS coatings on silicon slides prepared using this procedure, and they find that the films are 4.9 nm thick by using angle-resolved x-ray photoelectron spectroscopy (XPS). This thickness is approximately equal to twice the length of a single OTS molecule, so we refer to these as “multilayer” coatings. Moisture present in the ambient environment during creation of the coating affects the smoothness and thickness of the film ([Wang and Lieberman, 2003](#); [Wang *et al.*, 2005](#)) and likely results in cross-linking between adjacent OTS molecules. The coating may in fact contain more than two layers of cross-linked OTS molecules tangled together. In contrast, monolayer OTS films with no cross-linking can be produced in a completely dry environment, such as a nitrogen-purged glovebox (see for example [Seltzer *et al.* \(2008\)](#)).

⁸ The coating of vapor cells is sometimes referred to as “black magic” because of the variation and unpredictability of its results, making it as much of an art as a science. Coating procedures tend to be followed religiously in even the smallest details, with little understanding of why they work or how they can be improved.

Rampulla *et al.* have also characterized OTS films on silicon slides after exposure to rubidium vapor. Performing XPS on both monolayer and multilayer coatings, they observe that some rubidium atoms diffuse into the film and bond with the silicon and oxygen atoms at the surface attachment sites. Using atomic force microscopy (AFM), they also find that the monolayer coatings contain numerous small clusters of rubidium atoms, as shown in Figure 6.10(a); the arrangement and separation of the clusters suggest that they are located at domain boundaries formed during the growth of the monolayer. On the other hand, the multilayer coatings contain much larger rubidium islands that are located very far apart, such as the one shown in Figure 6.10(b) (note that the scales are the same for both AFM images), so that rubidium occupies a smaller fraction of the surface compared to the monolayer. We speculate that the presence of rubidium clusters indicates defects in the coating where atoms may depolarize upon contact. Preliminary data discussed in Section 6.3 suggest that multilayer OTS performs better than monolayer OTS as an anti-relaxation coating, which may be a result of lower surface coverage by rubidium clusters. Elimination of defect sites in multilayer OTS may therefore result in a more effective coating.

The alkali density observed in all OTS-coated cells is lower than the saturated vapor pressure predicted by Equation A.1 for the coldest spot in the cell (typically the bottom of the stem), often by a factor of 2 or greater, with a more pronounced difference at higher operating temperatures. Reduced vapor density has been observed previously in silane-coated alkali cells (Zeng *et al.*, 1983, 1985) and may be due to a chemical reaction between the alkali atoms and the coating (Camparo *et al.*, 1987). In Cells #10 and #11 we observe significantly reduced density, by more than an order of magnitude, unless we heat the stem to a higher temperature than the main cell body in order to induce potassium condensation on the walls. Normally alkali condensation is to be avoided because it reduces the polarization lifetime, but in these cells there is no such reduction. The reason for this atypical

behavior is unknown; we speculate that some minimum flux of alkali atoms absorbed at the surface may be necessary to saturate the OTS coating, and the presence of nitrogen gas for quenching may reduce that flux by slowing diffusion to the surface, but this behavior is not observed in the other cells containing nitrogen. At temperatures slightly below the maximum operating point for the OTS coating, we measure densities up to about $9 \times 10^{12} \text{ cm}^{-3}$ in potassium vapor and $7 \times 10^{13} \text{ cm}^{-3}$ in rubidium vapor.⁹

We determine the maximum operating temperature of the OTS coating by increasing the temperature of the cell in small steps and monitoring the change in T_1 when the cell is left at a particular temperature for an extended period of time. We occasionally observe a small reduction of T_1 due to alkali condensation on the cell wall, and this can be reversed simply by cooling the stem in order to increase the temperature differential between it and the main cell body, forcing the alkali vapor to condense in the stem instead. However, at high enough temperature the coating displays permanent degradation, possibly caused by the alkali atoms eating away at the coating by destroying the chemical bonds between individual OTS molecules and the glass, thus exposing bare patches on the surface. We have conducted this experiment on one cell each containing potassium and rubidium, and for both cells we detect a slow reduction of T_1 at 160°C and a rapid reduction at 170°C .

We show our measurements of the polarization lifetime in the rubidium cell in Figure 6.11, taken over several days with the cell at elevated temperature. The cell is at 100°C for the initial measurement each day, heated to 170°C at $t=0$, and cooled back to 100°C after the last measurement. The initial measurements at lower temperature display a clear reduction in the quality of the OTS coating after each day of operation at 170°C . At the higher temperature, T_1 is dominated by spin-destruction collisions between rubidium atoms, so small variations in vapor density can affect the lifetime measurement; nevertheless, we can

⁹ We typically determine the density of an optically thick vapor by measuring the magnetic linewidth and resonance frequency for several small values of the magnetic field in the SERF regime and fitting to Equation 5.5. This provides the spin-exchange rate, which is directly proportional to the vapor density.

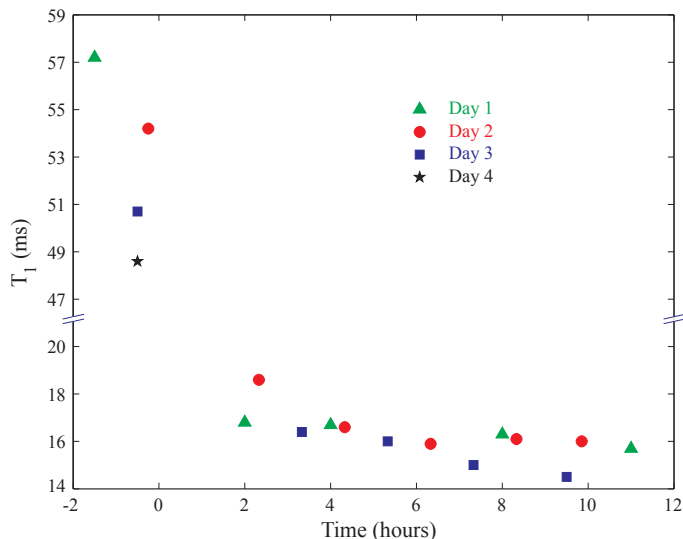


Figure 6.11: Measurements of T_1 in the OTS-coated rubidium cell with 10 Torr of nitrogen. Each day, the cell temperature is 100°C before $t=0$ and 170°C afterward, and the cell is cooled back to 100°C after the final measurement. The data taken at lower temperature evidence permanent degradation of the coating effectiveness. The error bars are smaller than the symbols at each point.

discern a decrease in T_1 with time as the coating is slowly destroyed. Leaving the cell at lower temperature for an extended period of time and cooling the stem with flowing water, which chills the stem to more than 40°C colder than the main cell body, does not cause the observed lifetime to recover, so we conclude that the degradation of the OTS coating after operation at 170°C is permanent.

These observations indicate that operation at 170°C is damaging to the multilayer OTS coating, while operation at slightly lower temperatures may be acceptable for very short periods of time. In comparison, a monolayer OTS coating does not become damaged until heated to 190°C in the presence of rubidium (Yi *et al.*, 2008) and can survive at temperatures up to 400°C when not exposed to alkali vapor (Fedchak *et al.*, 1997). The fact that the maximum operating temperature of multilayer OTS is the same for potassium and rubidium, despite nearly an order of magnitude difference in vapor density, suggests that the destruction of the coating may be caused by a binary collision between individual alkali

atoms and silane groups attached to the surface; some minimum thermal energy may be required for the atoms to break the chemical bonds between the OTS molecules and the glass. However, OTS coatings function at high enough alkali density to enable their use for high-sensitivity SERF and rf magnetometry applications, especially with rubidium and cesium.

6.2.1 Magnetometry With OTS-Coated Cells

In order to demonstrate the application of OTS coatings at high-temperature, we implement a SERF magnetometer using several OTS-coated cells. This is the first demonstration of high-density magnetometry without buffer gas. We use orthogonal pump and probe beams generated by distributed feedback (DFB) diode lasers that are easily tunable over a large frequency range by adjusting the temperature of the diode. The cell is heated by warm air flowing through a double-walled glass oven, which is surrounded by five layers of μ -metal shields that provide a shielding factor of about 10^5 . Magnetic field coils cancel any residual fields, and cancelation of gradients is generally unnecessary.

Figure 6.12 displays the magnetic resonance around zero field recorded in Cell #7 at 150°C with very weak pump and probe beams. The observed magnetic linewidth (HWHM) $\Delta\omega=2\pi\times 2.3$ Hz is about twice that expected from the polarization lifetime of $T_1=121$ ms measured immediately beforehand, but it still demonstrates that extremely narrow linewidths are possible in coated cells at high temperature. This linewidth is comparable to those observed in SERF magnetometers with buffer gas (Allred *et al.*, 2002) and low-temperature magnetometers with paraffin coating (Budker *et al.*, 2005). Although this is the narrowest linewidth observed in an OTS-coated cell, most cells give linewidths below 10 Hz and thus may be employed for high-sensitivity magnetometry.

Radiation trapping due to strong absorption of spontaneously emitted photons is one disadvantage of the narrow optical linewidth that results from the lack of buffer gas (see

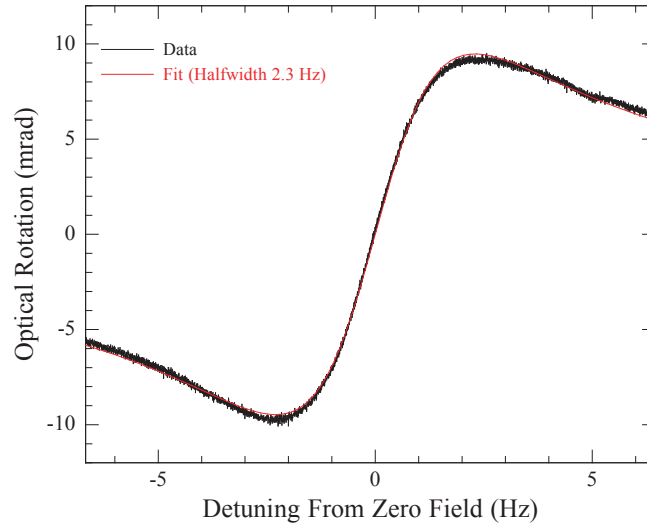


Figure 6.12: Magnetic resonance measured in Cell #7 around zero field in the SERF regime; the fit is to a Lorentzian dispersion curve with linewidth $\Delta\omega=2\pi\times 2.3$ Hz.

Section 2.3.4). In Figure 6.13 we show that maximum polarization of about 2% can be attained at potassium density of about 5×10^{12} cm⁻³ in Cell #5, which contains no quenching gas.¹⁰ The observed polarization is significantly less than predicted by Equation 2.39, and it actually decreases once the pumping rate rises above some critical value because the additional pumping photons lead to an excess of emitted radiation. This behavior is inconsistent with the model described in Section 2.3.4, which predicts that the polarization should saturate at high optical pumping rate, possibly because the pump beam is strongly attenuated and only polarizes atoms near the front of the cell. In Cell #11, which contains 12.6 Torr of nitrogen for quenching, at the same density we measure polarization of 10% when setting the pumping rate to double the magnetic linewidth, as opposed to the predicted 50%. After increasing the pumping rate by about a factor of six, we measure polarization of 45%, roughly half that expected.

¹⁰ We determine polarization by applying a field at some angle to both the pump and probe beams, recording the optical rotation signal as a function of probe beam frequency, and fitting the result to Equation 2.99.

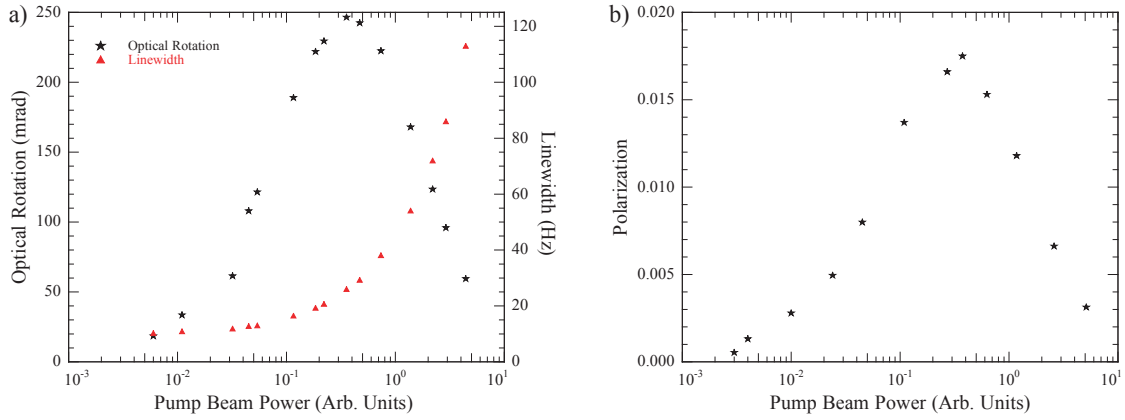


Figure 6.13: Dependence of optical rotation, linewidth, and polarization on pumping rate at potassium density of $5 \times 10^{12} \text{ cm}^{-3}$ in Cell #5, an OTS-coated cell without quenching gas. The polarization is limited to approximately 2% due to radiation trapping.

These measurements underscore the importance of determining the amount of nitrogen necessary for effective quenching in coated cells without buffer gas; ideally the cell should contain just enough quenching gas without unnecessarily broadening the optical resonance. The quenching cross-sections for rubidium and cesium with nitrogen are larger than for potassium with nitrogen, as listed in Table A.2, so cells with those elements should require lower nitrogen pressure. In the rubidium cell listed in Table 6.1 containing 10 Torr of nitrogen, at vapor density of $5 \times 10^{12} \text{ cm}^{-3}$ we measure polarization that agrees with the predicted value for several different pumping rates. The size and nitrogen pressure of this cell are comparable to those of Cell #11, so we may conclude that indeed less nitrogen is necessary for operation with rubidium than potassium. Rubidium and cesium cells may therefore be better suited for high-temperature magnetometry with surface coatings due to their higher alkali density and reduced need for quenching gas.

Even with low polarization due to radiation trapping in the coated potassium cells, we still observe large optical rotation signals. We display in Figure 6.14 a Lorentzian dispersion curve measured around zero field in Cell #10 with an amplitude corresponding to an

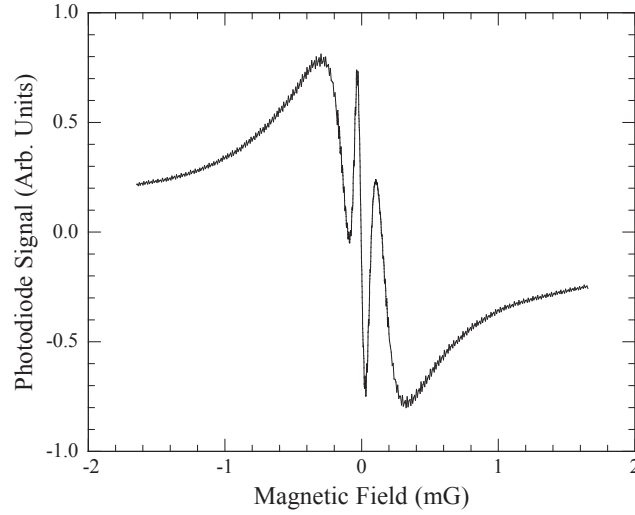


Figure 6.14: Dispersion curve around zero field with amplitude of 1.6 radians, recorded in OTS-coated Cell #10 at 150°C. The signal from the balanced polarimeter (see Section 2.4.2) turns over once the optical rotation angle reaches $\pi/4$ radians.

optical rotation of 1.6 radians; note that the signal saturates and turns over once the rotation angle increases past $\pi/4$ radians, as per Equations 2.104-2.105. This is the largest optical rotation signal yet observed in a SERF magnetometer, and even larger signals should be possible in rubidium and cesium cells with higher vapor density and effective quenching. As noted in Section 6.1.1, large optical rotation signals limit the effect of technical noise on magnetometer sensitivity.

The slope of the resonant dispersion curve acts as a useful figure-of-merit for comparing magnetometer sensitivity under different operating conditions, so we consider the slopes measured in coated cells and buffer gas cells at potassium density of $5 \times 10^{12} \text{ cm}^{-3}$. We tune the probe beam to provide the largest optical rotation signal $\theta e^{-n\sigma l}$, as discussed in Section 6.1.1, and we set the pumping rate to double the magnetic linewidth. In Cell #11 with $\Delta\omega = 2\pi \times 6.5 \text{ Hz}$, an effective probe beam path length of 3.9 cm,¹¹ and polarization

¹¹ We take into account the polarization distribution due to partial illumination of the cell by the pump beam, and we determine the effective path length over which a uniform polarization provides an equivalent optical rotation signal.

of 10%, we measure a slope of 57 rad/mG, comparable to the prediction by Equation 2.99. By tuning the probe laser closer to resonance we achieve a slope of 135 rad/mG. For comparison, in a square cell with length 2.2 cm containing 2.9 amg of helium buffer gas, at 50% polarization with $\Delta\omega = 2\pi \times 4.6$ Hz we measure a slope of 15 rad/mG, close to the prediction of 17 rad/mG at optimal probe beam detuning. Even accounting for the longer optical path length, the slope is larger in the coated cell despite a broader magnetic linewidth and significantly smaller polarization, and the advantage should be even greater in cells with effective quenching.

Large optical rotation signals enable us to easily obtain high magnetic field sensitivity. Figure 6.15 shows a single-channel magnetic noise spectrum taken with Cell #11 at 150°C, exhibiting sensitivity of about 6 fT/ $\sqrt{\text{Hz}}$. This is comparable to previous measurements with SERF magnetometers using buffer gas (Allred *et al.*, 2002; Kominis *et al.*, 2003), and it is possibly limited by thermal Johnson noise produced by the μ -metal shields, which we calculate to be 5.9 fT/ $\sqrt{\text{Hz}}$ according to the formula given by Lee and Romalis (2008).¹² We are able to achieve sensitivity better than 10 fT/ $\sqrt{\text{Hz}}$ using Cell #5 at a temperature of only 120°C, significantly lower than the temperature at which any previous implementation of a potassium SERF magnetometer has operated, although it is difficult to keep the magnetometer in the SERF regime at such low vapor density. Operation in nonconductive ferrite shields should reduce the ambient magnetic noise (Kornack *et al.*, 2007), and we expect that performing a gradiometric measurement using the cell designs illustrated in Figure 6.2 should provide an order of magnitude improvement in sensitivity.

6.2.2 Coating Procedure

We design cells intended for coating to have long stems, so that we can join the cells to a glass manifold and later pull them off without overheating the OTS-coated regions with

¹² There is a layer of Metglas magnetic alloy shielding inside the μ -metal that produces a noise level of 1.2 fT/ $\sqrt{\text{Hz}}$, but there is no indication that it shields the magnetic noise produced by the μ -metal.

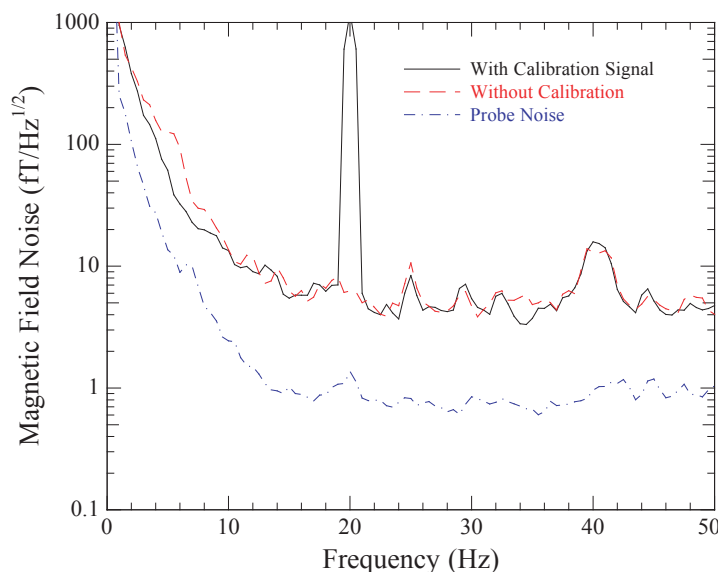


Figure 6.15: Single-channel magnetic noise spectrum taken at zero field with Cell #11 at 150°C, demonstrating sensitivity of about $6 \text{ fT}/\sqrt{\text{Hz}}$, limited by the noise of the magnetic shields.

a torch flame. We also design cells to have as small an aperture as possible between the stem and the main cell body; polarized atoms that enter the stem are unlikely to return to the cell before they depolarize, so the aperture effectively functions as an uncoated region on the cell surface. For example, the opening to the stem makes up approximately 1/2000 of the inner surface area of Cell #7, so the polarization lifetime in that cell may in fact be limited by the stem aperture rather than the quality of the OTS coating. We typically use openings that are slightly larger than 1 mm in diameter, so that we can feed through tubing for filling and emptying the cell of fluids during the coating process. We use a peristaltic pump for the transfer of fluids, and in particular the OTS solution described below should be transferred slowly because otherwise we notice visible artifacts where the OTS squirts onto the glass. When coating flat slides intended for the reusable vapor cell described in Section 6.3, we submerge the slides in the same fluids with which we fill vapor cells during this process. The procedure for coating silicon is the same as for glass.

The procedure for coating multilayer OTS films is adapted from previous work by [Rosen *et al.* \(1999\)](#) and [Bear \(2000\)](#). We operate in a fume hood in an ambient atmosphere that contains moisture, which leads to cross-linking between OTS molecules, as opposed to the dry environment necessary for the formation of monolayer films. Beforehand we clean all beakers by sonicating with detergent (such as a solution of water with 2% Micro 90 cleaner) and then rinsing several times with deionized water. We allow the beakers to dry because OTS reacts with water, and we make sure that there are no traces of organic solvents on the glass in order avoid potentially explosive reactions with piranha solution.

First we clean our cells with piranha solution, which removes all organic material from the surface of the glass so that the reaction with OTS can occur. Piranha solution consists of 3 parts hydrogen peroxide to 7 parts sulfuric acid by volume, with the hydrogen peroxide always added to the acid. The creation of the solution is exothermic, and the solution is initially very hot, so we allow it to cool down for about an hour before use. We place the beaker containing the solution inside a larger beaker for secondary containment in case the glass breaks because of the heat. Once we finish with the piranha solution we allow it to sit and cool down for several hours before neutralizing it slowly with a mild base and disposing of it by pouring down the drain with large amounts of water. Piranha solution must never be stored in a closed container because gaseous byproducts of chemical reactions can build up and cause an explosion.

Piranha solution is highly viscous, so it is faster to fill the cells by submerging their stems in the solution and then pumping out the air from the cells rather than simply pumping in the solution. We expose the glass to piranha solution for one hour, and then we remove the solution by pumping air back in while allowing the solution to dribble out of the stems into a beaker. All other fluids are significantly less viscous and can be directly transferred into and out of the cells. We next rinse the cells by filling them first with deionized water and then with methanol (slides can be rinsed by submerging several times in

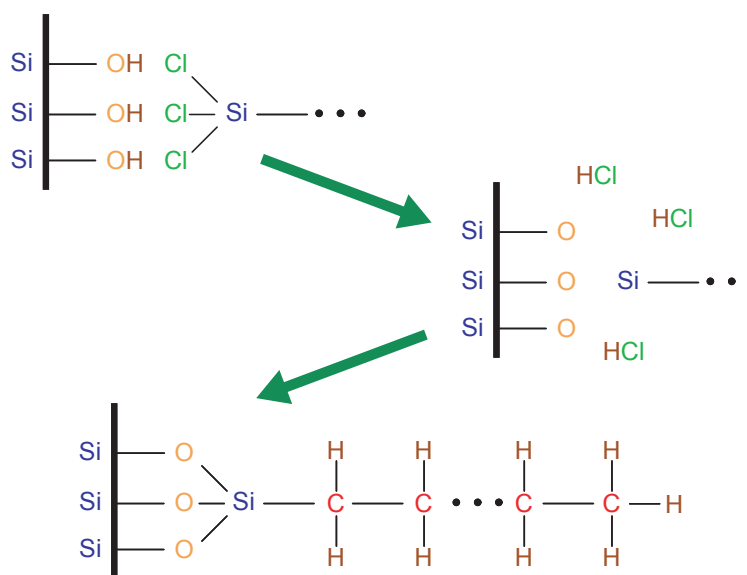


Figure 6.16: Attachment of an OTS molecule to a glass or silicon surface. The chlorine atoms in the end group react with OH groups on the surface to form HCl, and then the silicon atom binds to the remaining oxygen atoms.

each liquid). We then dry out the cells by baking in vacuum at a temperature of at least 100°C for one hour. Afterward the oven can be backfilled with nitrogen before exposing the cells to air, but we do not find this to be necessary. The reaction with the OTS solution is very sensitive to the amount of moisture on the glass surface, which would ideally be covered only by a monolayer of water, but we find it sufficient to simply expose the glass to the ambient environment.

We then return the cells to the fume hood for OTS coating. We store the OTS in a refrigerator in order to slow its reaction with moisture in the air, and this causes it to freeze, so it is necessary to thaw it by submerging its bottle in warm water before use. We form a solution containing 1 part chloroform to 4 parts hexanes by volume, and we add OTS to the solution at a concentration of 0.8 mL OTS per 1 L of solution.¹³ We expose the cells to the OTS solution for five minutes, during which time OTS molecules attach to the glass.

¹³ The precise concentration does not seem particularly important, and we are often off by a factor of 2. The effect of varying OTS content in this solution on the quality of the resultant coating bears future consideration.

The chlorine atoms in the trichlorosilane end group react with OH groups on the surface to form HCl, and then the silicon atom binds to the oxygen atoms remaining on the surface; this process is illustrated in Figure 6.16. The presence of small amounts of water in the solution likely induces cross-linking between OTS molecules so that they bind to each other as well as the surface, and some molecules may in fact link only to other molecules without any attachment to the glass.

After exposure to the OTS solution, we allow the cells to sit in air for five minutes, and we then rinse them with chloroform. We cure the coating by baking the cells in vacuum at 200°C for 24 hours. We then attach the cells to a glass manifold, taking care not to heat the coated areas with the torch flame, and we bake the cells again at 150-200°C before chasing in alkali metal, filling with quenching gas if desired, and pulling the cells off the manifold. Finally, we find that the OTS film must again cure through exposure to alkali vapor for an extended period of time. Initially we detect no vapor in the cells because the coating absorbs alkali atoms until it becomes saturated (Camparo, 1987; Camparo *et al.*, 1987); we therefore heat the cells for several days before use (potassium cells at 120°C and rubidium cells at 100°C). We also observe that the number of bounces improves in some cells for a period of several weeks as the coating continues to cure at high temperature. It is sometimes necessary to chase condensation out of the cell and into the stem after the curing process concludes.

6.2.3 Light-Induced Atomic Desorption

Surface coatings typically absorb a small quantity of alkali atoms before reaching saturation, and they emit these atoms upon exposure to light, a phenomenon known as light-induced atomic desorption (LIAD). Low-intensity white light, such as that produced by a light bulb, is sufficient to cause desorption. LIAD can temporarily produce atomic vapor in a cold cell, enabling control of the vapor density without active heating elements that

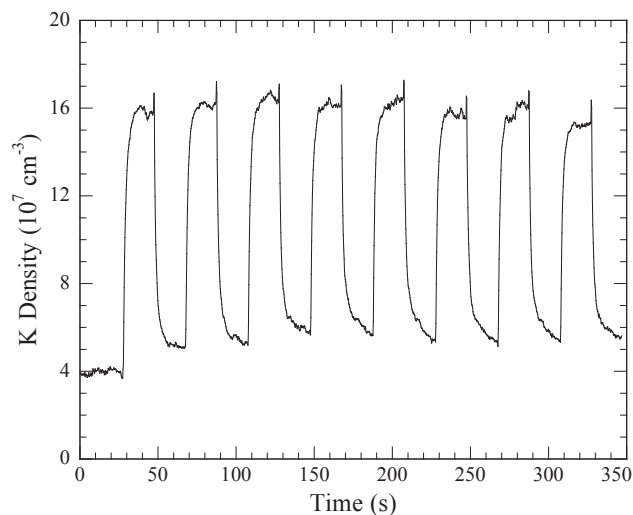


Figure 6.17: Increase in potassium vapor density upon illumination of Cell #7 by white light, demonstrating LIAD in the OTS coating.

can introduce magnetic noise, and it thus may be useful for low-density magnetometry and clock applications (Karaulanov *et al.*, 2008); other demonstrated applications include the loading of magneto-optical traps (Atutov *et al.*, 2003) and production of chip-scale Bose-Einstein condensates (Hänsel *et al.*, 2001). LIAD has been previously observed in organic coatings such as siloxane films (Marinelli *et al.*, 2001) and paraffin (Alexandrov *et al.*, 2002; Gozzini *et al.*, 2008). Here we demonstrate LIAD in OTS with potassium, and it has also recently been demonstrated with rubidium by Cappello *et al.* (2007).

We induce desorption of potassium atoms using a flashlight with an approximate intensity of 200 mW/cm^2 , which we modulate by controlling the flow of current directly across the bulb. The light is unlikely to heat the cell and raise the saturated vapor density because the potassium metal sits at the base of the stem, which is contained in a holder and remains unilluminated. We measure potassium vapor density by monitoring the transmission of a laser tuned precisely to the D1 resonance, and we confirm that there is minimal leakage of light from the flashlight into the photodetector. Figure 6.17 displays the increase

in observed potassium density at room temperature in Cell #7 by a factor of four upon illumination of the coating. Once the light turns off, the vapor density returns to its normal value within about 10 seconds due to both condensation and reabsorption of atoms by the OTS film. We infer that the OTS quickly resaturates with absorbed atoms because the density increases to the same amount upon repeated application of light. Further study may reveal how to achieve higher densities using LIAD in OTS-coated cells.

6.2.4 Alkali Whiskers

As discussed in Section 6.2.1, several of the OTS-coated cells require us to induce condensation of potassium in the main cell body in order to achieve high vapor density. Upon visual inspection of these cells, we notice that the potassium metal does not pool together at the surface, as is typical, but that instead it forms long, thin needle-like whiskers. We speculate that the potassium atoms prefer to condense on alkali-metal deposits rather than on the organic molecules in the coating, so they form thin structures that have minimal contact area with the surface. These whiskers can act as an alkali reservoir within the main cell body, but they are thin enough to not adversely affect the spin polarization lifetime. Alkali-metal whiskers have also been observed in paraffin-coated cells by Balabas *et al.* (2007), who suggest that they might be useful for manufacturing alkali-metal wires.

We can stimulate the growth of whiskers in OTS-coated cells by heating the stems for several weeks while leaving the main cell bodies at a lower temperature. We show images of whiskers in two different cells in Figure 6.18. Individual cells display one of two distinct growth patterns, containing either dense growth with short whiskers (<1 mm long) or sparse growth with long whiskers (~1-2 mm long); lower quenching gas pressure may lead to denser growth. Each cell contains dozens to hundreds of whiskers, some of which are clearly attached to the surface and pointed into the cell (this is difficult to perceive in the two-dimensional photographs), while others appear to rest on the surface, perhaps

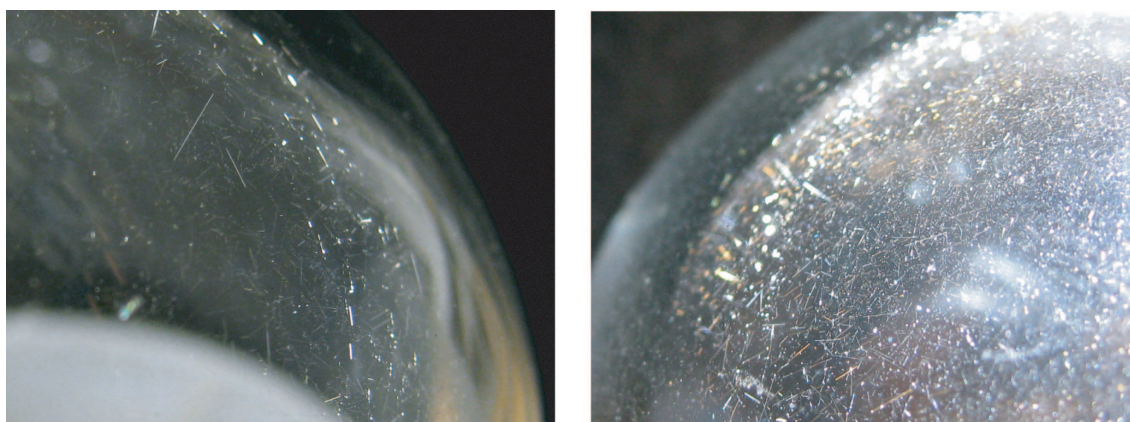


Figure 6.18: Potassium whiskers seen growing on the OTS-coated surfaces of two vapor cells. The spherical curvature makes it difficult to discern the direction in which the whiskers point, or whether they are attached to the near or far surface, but in fact many whiskers are directed away from the surface and into the cell.

after having snapped off the glass due to fragility. Observation of whisker growth may be a method for characterizing the coverage of OTS coating on a glass surface.

6.3 Search for Effective High-Temperature Coatings

Paraffin performs exceptionally well as an anti-relaxation coating at low temperature, and the recent work with OTS described above introduces the prospect of high-temperature magnetometry without buffer gas. However, there is little understanding about why these coatings work while others do not, or about how to improve them. The development of paraffin-quality coatings for high-temperature operation would greatly benefit SERF and rf magnetometers, so we have developed a method for comparing the effectiveness of different coatings under identical experimental conditions. This allows us to test numerous coating types and determine the characteristics of effective coatings, which will hopefully lead to the development of improved surface coatings for high-temperature operation.

Previous experiments investigating the effectiveness of different surface coatings have used individual coated cells, including [Camparo \(1987\)](#), [Swenson and Anderson \(1988\)](#),

Stephens *et al.* (1994), Fedchak *et al.* (1997), and Yi *et al.* (2008). We instead use a single reusable alkali vapor cell that enables us to test multiple coatings without the need for manufacturing many separate cells, which is time-consuming and labor-intensive. We coat glass or silicon slides, which we insert into the reusable cell; we observe the polarization lifetime in the volume between pairs of slides, providing a measurement of the number of bounces allowed by a particular coating. Flat slides can be coated more easily and uniformly than closed-geometry cells, and they allow us to conduct additional surface characterization tests both before and after exposure to alkali vapor, including infrared (IR) spectroscopy, atomic force microscopy (AFM), and x-ray photoelectron spectroscopy (XPS). Other experiments have also studied the interaction between alkali atoms and wall coatings by specifically measuring polarization at the surface (Grafström and Suter, 1996; Zhao and Wu, 2005), and it should be possible to design a modified version of the reusable cell for observation of surface rather than bulk polarization.

6.3.1 The Reusable Alkali Vapor Cell

Here we describe the prototype for the reusable alkali vapor cell, and we discuss improvements to the design in Section 6.3.3. The cell is made of pyrex and consists of two parts, the main body and a lid, that are joined together by a viton o-ring flange held by a high-temperature nylon clamp. We use Apiezon-H vacuum grease to seal the o-ring flange because it has a high working temperature and is chemically compatible with potassium vapor.¹⁴ The coated slides are 2 inches by 1 inch and sit in a grooved pyrex tray that keeps them parallel and separated by approximately 5 mm. The cell includes an ampule of potassium metal in a separate retort; a small amount of potassium is chased into the main body of the cell to act as the source of alkali vapor during the experiment. Figure 6.19 shows

¹⁴ We observe that potassium remains unaffected by vapor produced by the grease at a temperature of at least 150°C. Potassium does react with the solid grease, however, and we have not tested the grease with rubidium or cesium, which may be more reactive.

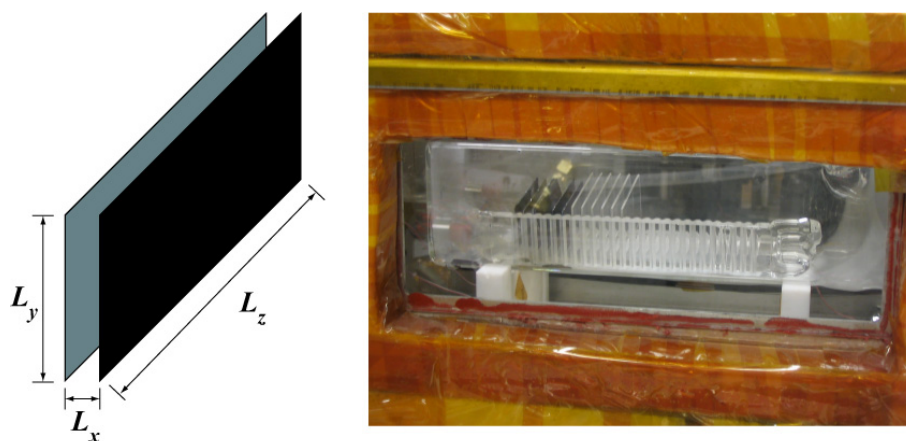


Figure 6.19: Coated glass and silicon slides sitting inside the reusable alkali vapor cell for testing. On the left we illustrate the convention for labeling the dimensions of the slides.

several glass and silicon slides contained inside the cell, as seen through a window of the oven used to heat the cell. A later version of the cell is displayed in Figure 6.24(a), clearly showing the o-ring flange and an advanced design of the nylon clamp.

The main cell body has an input port, normally sealed with an o-ring, used for connecting high-purity nitrogen and helium.¹⁵ When the cell is open during loading and unloading of samples, we maintain a flow of nitrogen through the cell to reduce the flow of air into the cell. A thin layer of oxidation on the surface of the potassium droplets inside the cell typically forms during this procedure. After the cell is sealed, a fresh surface of potassium may be exposed by filling the cell with several hundred Torr of helium to prevent diffusion of alkali vapor to other areas of the cell and then melting the potassium droplets with a torch. While this procedure allows the same droplets of potassium to be used multiple times, the alkali vapor pressure becomes lower with each use due to contamination, until fresh potassium must be chased into the main chamber from the retort. We examine multiple samples of the same coating at one time and then compare to different coatings loaded separately in order to prevent cross-contamination.

¹⁵ We use a filter that purifies the nitrogen and helium to less than 1 ppm of oxygen and water vapor.

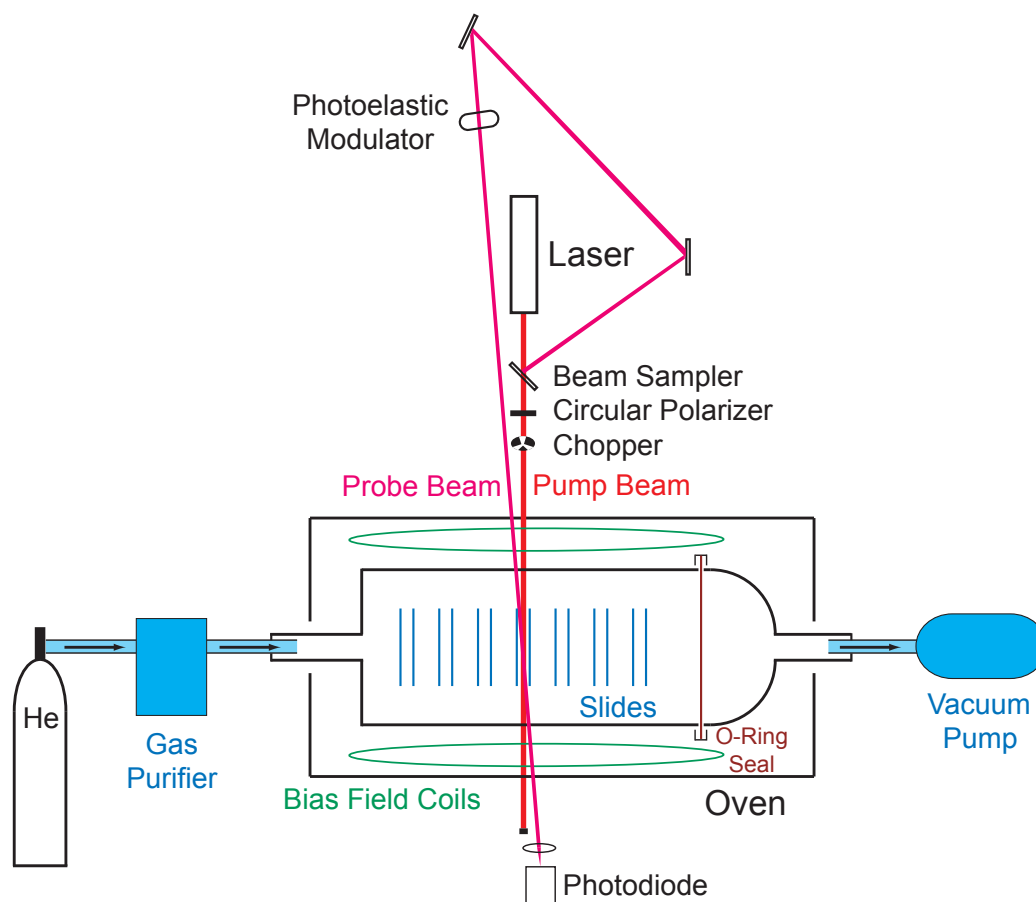


Figure 6.20: Experimental schematic of anti-relaxation coating characterization with the reusable alkali vapor cell. A supply of purified helium and a vacuum pump allow for control of the buffer gas pressure. The pump beam polarizes atoms along the bias field direction in the volume between two adjacent slides, and after an optical chopper blocks the pump beam, we monitor the transmission of the nearly-parallel probe beam through the cell as the polarization decays.

The setup of the experiment to measure the polarization lifetime T_1 in the volume between adjacent coated slides is shown in Figure 6.20. An oven heats the cell as high as 150°C, limited by the operating temperature of the viton o-ring and vacuum grease, though we typically take measurements between 60-100°C. The cell is connected to both high-purity helium and a vacuum pump to allow for *in situ* control of the buffer gas pressure. We use a single DFB diode laser tuned to the D1 resonance. A beam sampler picks off a small amount of laser light to act as the probe beam; the polarization of this beam is modulated by a PEM, and the transmission of light at the modulation frequency is monitored by a photodiode and lock-in amplifier to measure the spin polarization (see Section 6.1.2). The remaining laser light acts as the pump beam and is circularly polarized, expanded to fill the volume between two adjacent slides, and passed through an optical chopper. The paths of the two beams are kept at a small angle so that they intersect and are nearly parallel as they pass through the cell but diverge enough beyond the cell so that a photodiode detects only the probe beam. A pair of magnetic field coils creates a bias field of several Gauss along the direction of the laser beam to allow for efficient optical pumping.

After the chopper blocks the pump beam, we fit the observed decay of spin polarization to the sum of two decaying exponentials. The time constant of the first decaying exponential term is the polarization lifetime T_1 of the atoms due to the influence of the coated surfaces. The time constant of the second term is generally 5-10 times longer than that of the first term and represents the polarization lifetime in the absence of the surfaces; we determine the second time constant by measuring T_1 far from the slides. When fitting the data we vary the first time constant and the relative weights of both exponential terms, although the weight of the second term is typically much smaller than that of the first term.

To prevent the potassium atoms from diffusing out of the region between the sample slides during the measurement, we fill the cell with helium buffer gas. The distance between slides is much smaller than the length of the slides along the other two dimensions,

so the polarization lifetime is dominated by diffusion along the dimension between the slides and interaction with the coated surfaces. We solve Equation 6.6 for rectangular geometry. We label the direction between the slides as $\hat{x}$ according to the convention shown in Figure 6.19; the boundary condition along this direction is given by Equation 6.5, where N is the number of bounces allowed by the coating, and the diffusion constant D is determined by the buffer gas pressure. Along the other two directions the boundaries are defined by the openings out of the volume between the two slides, and the boundary conditions are similar to Equation 6.5, except that $N \sim 1$ since polarized atoms diffusing out of this volume are generally lost to the measurement and replaced by unpolarized atoms diffusing into the volume.

The solution to the diffusion equation for the fundamental diffusion mode of the alkali atoms is

$$P = \cos(k_x x) \cos(k_y y) \cos(k_z z) e^{-t/T_1}, \quad (6.24)$$

where k is the diffusion wave vector. Combining Equations 6.6 and 6.24 gives

$$\frac{1}{T_1} = D (k_x^2 + k_y^2 + k_z^2) + R_{SD}, \quad (6.25)$$

and applying the boundary conditions gives

$$k_x \tan\left(\frac{k_x L_x}{2}\right) = \frac{\bar{v}}{2N(2 - 1/N)D} \quad (6.26)$$

$$k_y \tan\left(\frac{k_y L_y}{2}\right) = \frac{\bar{v}}{2D} \quad (6.27)$$

$$k_z \tan\left(\frac{k_z L_z}{2}\right) = \frac{\bar{v}}{2D}, \quad (6.28)$$

where L_x is the distance between slides, and L_y and L_z are the length and height of the slides. We measure the polarization lifetime at several different pressures of buffer gas, and we determine the value of N that provides the best fit to the data by the numerical solution of Equations 6.25-6.28.

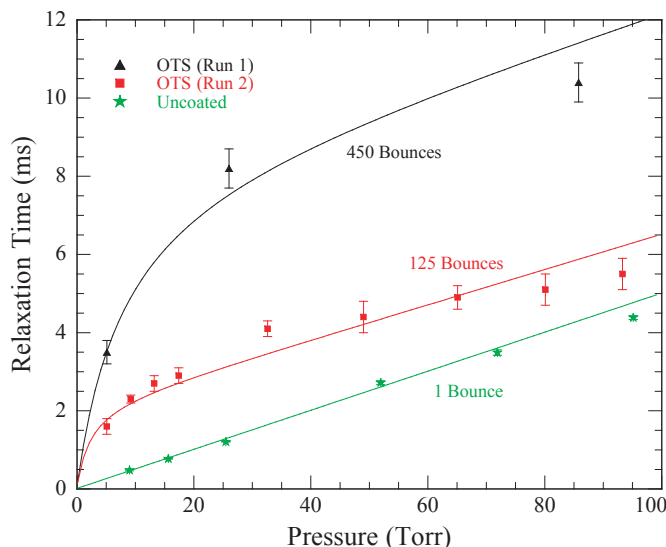


Figure 6.21: Measurements of alkali polarization lifetime as a function of helium buffer gas pressure taken at 70°C. The measurements taken two days apart with multilayer OTS coating show a decrease in bounces, possibly due to condensation of potassium on the slides. The lifetime allowed by the uncoated slides agrees with the prediction for a completely depolarizing surface.

6.3.2 Observations

We have made preliminary measurements on several organic films in order to demonstrate the application of the reusable vapor cell for characterization of anti-relaxation coating quality. Figure 6.21 displays measurements taken with multilayer OTS, showing that the model developed in Section 6.3.1 agrees well with our observations. Measurements taken two days apart exhibit a decrease in the number of bounces from 450 to 125, likely due to condensation of potassium on the slides; the higher number is consistent with observations of coating effectiveness in the cells listed in Table 6.1. For comparison, uncoated samples display much shorter relaxation times that are consistent with the prediction for only one bounce. Data taken at multiple values of buffer gas pressure allow for more accurate determination of the number of bounces allowed by a coating, and all future measurements will be made by varying the pressure. However, most of the observations described below were made at only a single pressure.

Chemical	Substrate	He Pressure (Torr)	T_1 (ms)	Bounces (Approx.)
Uncoated	Glass	5	0.3	<10
		10	0.5	<10
OTS (1)	Glass	5	0.7	30
		10	1.2	40
OTS (2)	Glass	5	1.1	50
		10	1.5	60
OTS (3)	Silicon	5	3.5	450
ODPA	Glass	5	0.3	<10
		10	0.6	<10
Paraffin (1)	Glass	15	1.9	80
Paraffin (2)	Glass	15	1.1	30
BTS*	Silicon	10	1.2	40
DTS*	Silicon	10	2.4	130
OTS*	Silicon	10	0.9	25

Table 6.2: List of coatings studied with the reusable vapor cell, including the approximate number of bounces measured for each. Coatings marked with an asterisk are monolayers.

We summarize our results in Table 6.2. In addition to multilayer OTS, we also consider monolayer OTS as well as monolayers of butyltrichlorosilane (BTS, $\text{CH}_3(\text{CH}_2)_3\text{SiCl}_3$) and dodecyltrichlorosilane (DTS, $\text{CH}_3(\text{CH}_2)_{11}\text{SiCl}_3$), which are similar to OTS but with shorter hydrocarbon chains.¹⁶ All three silane monolayers exhibit anti-relaxation properties at temperatures of at least 95°C, and in particular we confirm that BTS functions at 145°C. As discussed in Section 6.2, the effectiveness of monolayer coatings may be limited by the presence of domain boundaries, but the reproducibility of monolayer surface characteristics makes them ideal for future study.

We observe an interesting dependence of DTS coating efficiency on temperature, as shown in Figure 6.22. At 95°C the measured relaxation time is much less than at 65°C, and

¹⁶ The monolayer silane coatings were provided by Sandrine Rivillon-Amy, and the coating procedure is detailed in Seltzer *et al.* (2008).

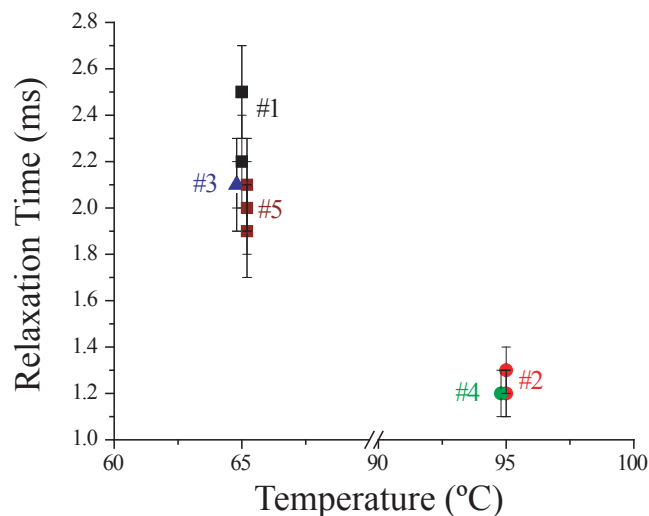


Figure 6.22: Variation with temperature of T_1 measured for a DTS monolayer coating at 10 Torr of helium. The numbers label the order in which the temperature is varied, with multiple measurements at a single temperature taken hours apart.

the difference is too large to be accounted for by the slight changes in the diffusion constant and spin-destruction rates. We switch back and forth between the two temperatures multiple times and find that the change in T_1 is reversible and repeatable. This is the first observation of dependence of anti-relaxation properties on the temperature of a film, other than a film melting or decomposing above some critical temperature. We speculate that the DTS monolayer may experience a phase change at some point between 65–95°C that alters the molecular alignment within the film, potentially exposing a greater fraction of the underlying surface. It is unknown whether this behavior is common to all DTS monolayer films, as well as why similar temperature dependence is not observed for BTS and OTS monolayers. Further investigation of DTS monolayers as anti-relaxation coatings may elucidate this behavior.

We also consider octadecylphosphonic acid (ODPA, $\text{CH}_3(\text{CH}_2)_{17}\text{P}(\text{OH})_2\text{O}$) monolayers, which resemble OTS except with a phosphorous atom attached to the surface rather than a silicon atom. The procedure for coating ODPA on silicon is given by [Hanson *et al.* \(2003\)](#),

and it is repeated multiple times on various samples to ensure high surface coverage. For all ODPA slides we observe $N \sim 1$, indicating that ODPA is ineffective as an anti-relaxation coating. In addition, Cell #3 (see Table 6.1) is coated with ODPA and exhibits a polarization lifetime consistent with only 15 bounces. Finally, we consider samples coated with paraffin that perform substantially worse than the 10,000 bounces observed in some paraffin-coated cells. The poor anti-relaxation quality is probably the result of our inexperience at coating paraffin, and we intend to repeat these measurements using slides that are coated professionally.

As noted in Section 6.3, one of the advantages of using flat slides is that they can be easily characterized before and after exposure to alkali vapor in order to investigate the effect of exposure on coatings. For instance, the AFM images shown in Figure 6.10 demonstrate that coating quality may be related to surface coverage by defects. In addition, IR spectroscopy can be performed on the samples; Figure 6.23 shows absorbance peaks observed in the monolayer OTS coating due to symmetric and asymmetric vibrational modes of the carbon-hydrogen bonds within the OTS molecules. These bonds remain after exposure to potassium vapor, revealing that the film survives intact, and narrowing of the absorption peaks indicates an increase in monolayer order due to thermal annealing (Snyder *et al.*, 1978; Venkataraman and Vasudevan, 2002). On the other hand, IR spectroscopy on ODPA shows no evidence of any carbon-hydrogen bonds after exposure to potassium, indicating that the film is likely damaged or destroyed. ODPA has been observed to be highly reactive with alkali vapor (Stephens *et al.*, 1994), so we speculate that phosphonate head groups are more susceptible to attack by alkali atoms than silane groups, possibly explaining the disparity in effectiveness as anti-relaxation coatings.

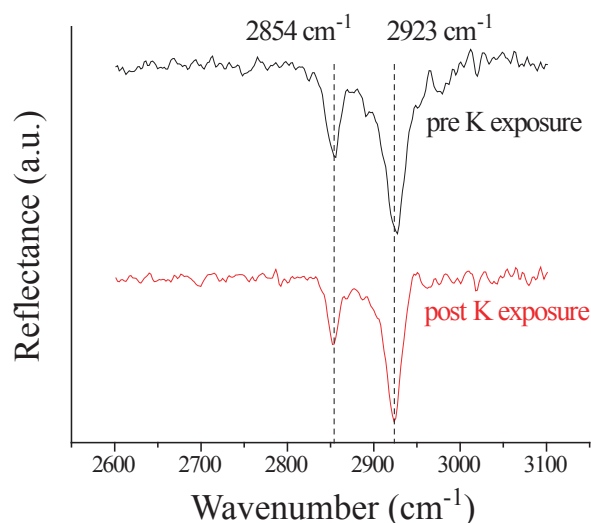


Figure 6.23: Infrared spectra of an OTS monolayer taken before and after exposure to potassium vapor. The absorbance peaks are a signature of the carbon-hydrogen bonds within the coating, and their presence after exposure to potassium indicates that the film remains intact.

6.3.3 Improvements and Future Prospects

We have several improvements planned for the reusable alkali vapor cell in order to increase the efficiency of testing multiple surface coatings. Loading the prototype results in the formation of a thin oxidation layer on the potassium droplets inside the cell, and after each loading the alkali vapor density observed at a particular temperature decreases as a result of contamination. We have designed a modified glovebox that allows us to open the cell in an inert environment, hopefully eliminating the oxidation of alkali metal within the cell. A residual gas analyzer evaluates the environment inside the glovebox to confirm that there is minimal oxygen or water vapor present; the analyzer can also be attached directly to the cell in order to identify the by-products of any reaction that may occur between alkali vapor and the coating materials. Figure 6.24 shows a more recent version of the cell and the glovebox. In addition, we intend to employ a thermoelectric cooler (TEC) to create a cold spot on the back of the cell, set about 10°C colder than the rest of the cell, to prevent

condensation on the samples from reducing the measured value of T_1 and causing us to underestimate coating effectiveness.

These improvements should enable us to efficiently investigate numerous coatings in an effort to discover the relationship between alkali depolarization and surface properties such as stiffness, thickness, degree of cross-linking, chemical composition, and surface coverage. We will have the ability to vary individual characteristics of a film and observe how the changes affect the number of bounces. We may also consider coatings that are completely different from the organic molecules currently used, such as chiral films, to see if they hold any promise as anti-relaxation coatings; once the experiment is fully operational it will be an invaluable tool for testing any and all films that can be grown on glass or silicon surfaces. The understanding that we gain of alkali-surface interactions will hopefully lead to the development of high-quality surface coatings for vapor cells operating at high temperatures.

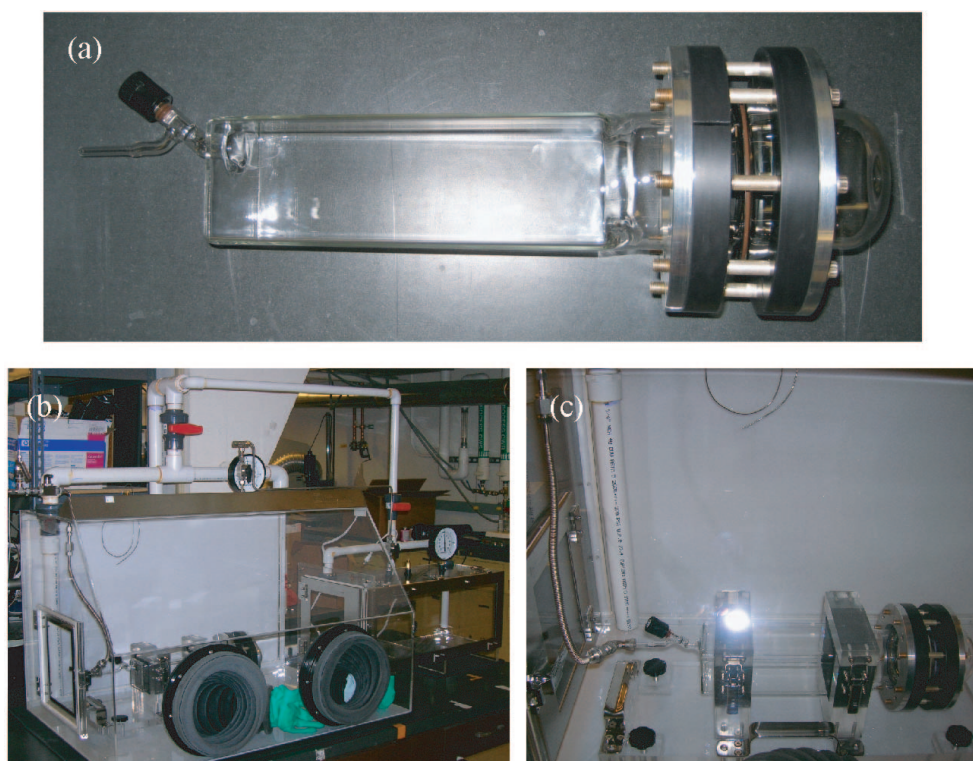


Figure 6.24: The reusable alkali vapor cell and its glovebox. Clearly seen are the o-ring flange and nylon clamp that hold the cell together, as well as the input port for connecting to a helium supply and vacuum pump; not seen is the retort which contains the alkali metal reservoir. The cell can be connected to a residual gas analyzer when sitting inside the glovebox.

Chapter 7

Towards a Cs-Xe Electric Dipole Moment

Experiment

THE SERF MAGNETOMETER can act as a probe of fundamental physics in addition to serving for the practical applications described earlier. We introduce a polarized noble gas species in order to operate as a comagnetometer, rendering the alkali spins insensitive to external magnetic fields but allowing us to detect other interactions, such as coupling to electric fields. We propose to employ a Cs- ^{129}Xe SERF comagnetometer to measure the electric dipole moments (EDMs) of the cesium and xenon spins. Here we discuss several of the challenges involved with this experiment, including the application of electric fields to a vapor cell without systematic effects, attainment of high field sensitivity, and suppression of magnetic fields. Work on this experiment is ongoing, and it has the potential to achieve sensitivity about two orders of magnitude better than previous EDM searches.

7.1 Search for Permanent Electric Dipole Moments

Experimental searches for permanent electric dipole moments of elementary particles and of atoms have been ongoing for decades (Purcell and Ramsey, 1950; Smith *et al.*, 1957), and there are currently numerous experiments underway. Initial interest was due to the fact that EDMs violate parity inversion (P) and time reversal (T) symmetries. In addition, EDMs are predicted by many extensions of the standard model, including supersymmetry theory, so there has been recent interest in using the results of EDM searches to place constraints on the parameters of these theories. On a basic level, the discovery of electric dipole moments of particles such as the electron or neutron would provide new knowledge about these fundamental particles. Here we provide only a cursory overview of the motivation and status of the search for EDMs; a basic introduction is given by Fortson *et al.* (2003), a more detailed review is given by Khriplovich and Lamoreaux (1997), and a summary of current results is given by Commins (2007).

We consider the electric dipole moment $\mathbf{d}$ of a particle, which is a vector with a definite direction. According to the Wigner-Eckart theorem, this vector must be either parallel or anti-parallel to the spin $\mathbf{s}$ of the particle:

$$\mathbf{d} = d \frac{\mathbf{s}}{|\mathbf{s}|}, \quad (7.1)$$

where the moment d can be either positive or negative. Otherwise additional quantum numbers would be necessary to define the alignment of $\mathbf{d}$ with respect to $\mathbf{s}$, and there are apparently no such quantum numbers since chemistry and nuclear physics are well-explained by existing theory. By the same reasoning, the magnetic dipole moment $\boldsymbol{\mu}$ must also be parallel or anti-parallel to the spin, and therefore to the EDM as well.

To show that the existence of a nonzero EDM violates P symmetry, we consider the case that the EDM and spin lie parallel, as shown in Figure 7.1; the argument that follows also holds in the case that they are anti-parallel. Under parity inversion, all polar vectors

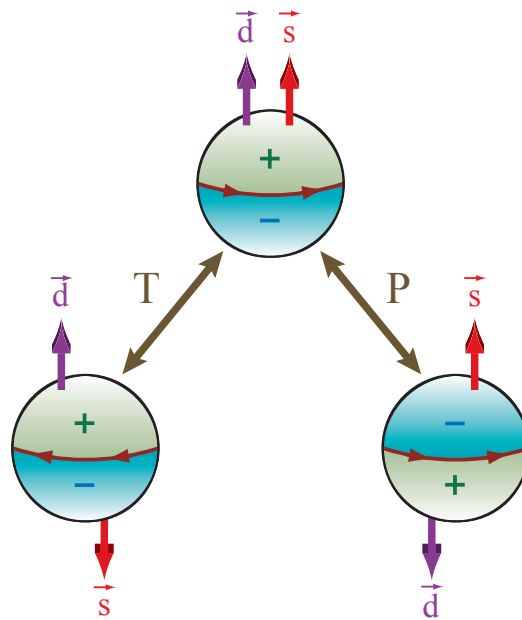


Figure 7.1: Electric dipole moments violate parity and time-reversal symmetries: Under parity inversion the EDM $\vec{d}$ changes sign but the spin $\vec{s}$ does not, while under time reversal the spin changes sign but the EDM does not. Both P and T transformations therefore reverse the relative orientation of the spin and EDM.

change sign, so $\mathbf{d} \rightarrow -\mathbf{d}$. However, the spin is an axial vector and remains unchanged by parity inversion, as we see by noting that angular momenta arise as cross products of vectors, so for instance $\mathbf{j} = \mathbf{r} \times \mathbf{p} \rightarrow (-\mathbf{r}) \times (-\mathbf{p}) = \mathbf{j}$. After a parity transformation the EDM and spin therefore lie antiparallel, and thus P symmetry requires that $d=0$.

Similar reasoning shows that a nonzero EDM also violates T symmetry. Time reversal changes the sign of velocity vectors, so $\mathbf{s} \rightarrow -\mathbf{s}$, while the EDM remains unchanged. A more detailed discussion of time reversal symmetry is given by [Sachs \(1987\)](#). It is presently believed that the combined CPT symmetry is inviolate in the universe, where C is charge conjugation (*i.e.*, reversal of sign of all electric charges); T asymmetry therefore implies CP asymmetry as well, so that the total CPT transformation leaves the relative orientation of the EDM and spin unchanged. While direct violation of T was recently observed in the decay of kaons ([Angelopoulos *et al.*, 1998](#)), the discovery of an EDM would be the first observation of T or CP violation outside of the quark sector.

The Hamiltonian of a particle or atom with nonzero electric and magnetic dipole moments is given by

$$\mathcal{H} = -(\boldsymbol{\mu} \cdot \mathbf{B} + \mathbf{d} \cdot \mathbf{E}). \quad (7.2)$$

An electric field therefore causes the spin of a particle with $d \neq 0$ to precess, just as a magnetic field causes spin precession (see Section 2.6). The basic principle behind EDM experiments is simply to apply an electric field and monitor the spin in order to determine whether its precession frequency changes as a result; the precession frequency serves as a measurement of the electric dipole moment. Alternatively, the direction of the field can be reversed in order to change the sign of the precession frequency. This technique is similar to the principle behind atomic magnetometers, which observe spin precession in order to measure magnetic fields using the magnetic dipole moment as a conversion factor. The coupling to magnetic fields is significantly stronger than the coupling to electric fields, since $\mu \gg d$, so the largest systematic problems with EDM experiments are often caused

by extraneous magnetic fields. Experiments often use comagnetometers to cancel magnetic effects, and we propose to do the same, as described in Section 7.2.

All EDM experiments thus far have measured $d \sim 0$, and the upper bounds on the dipole moments of various particles and atoms have grown progressively smaller as the result of increasingly sensitive searches. There are currently numerous EDM searches being conducted, all of which require precise execution to achieve sensitivity to electric dipole moments smaller than the existing experimental bounds. Our experiment exploits the extremely high sensitivity of a SERF magnetometer to spin precession and is designed to detect the EDMs of the electron and of the ^{129}Xe atom. The projected sensitivity of our experiment to these moments is listed in Table 7.2 along with the current best experimental limits, as well as the present limits for the ^{199}Hg atom (which has the smallest limit of any diamagnetic atom) and the neutron for reference. The electric dipole moment is typically given in units of $e\text{-cm}$, where e is the electron charge. Detection of a nonzero EDM would be evidence of physics beyond the standard model, and even just a reduction of the experimental bounds would provide information about possible extensions of the standard model.

7.2 The SERF Comagnetometer

The SERF magnetometer described in Chapter 5 is extremely sensitive to the spin precession frequency of alkali atoms. Although we usually characterize a magnetometer in terms of its sensitivity to magnetic fields, in fact the magnetometer is specifically sensitive to spin precession, and a magnetic field is only one possible cause of precession. A SERF magnetometer with sensitivity of $\delta B = 1 \text{ fT}/\sqrt{\text{Hz}}$ to magnetic fields has sensitivity of $\delta\omega \sim 2\pi \times 10^{-6} \text{ Hz}/\sqrt{\text{Hz}}$ to the atomic spin precession frequency. As discussed in Section 7.1, if the valence electron of an alkali atom has a nonzero electric dipole moment,

then an electric field also causes spin precession, and there is generally no means of distinguishing between precession caused by electric and magnetic interactions.

The magnetic field experienced by the alkali spins therefore must be minimized so that any observed spin precession can be attributed solely to the applied electric field. We plan to use a comagnetometer with two spin species for suppression of magnetic field effects. Specifically, the SERF comagnetometer employs a buffer gas with nonzero nuclear spin (Kornack and Romalis, 2002), such as ^3He , ^{21}Ne , or ^{129}Xe , that is polarized through spin-exchange collisions with the optically pumped alkali atoms (Walker and Happer, 1997). The noble gas magnetization cancels all external fields, so the alkali spins can respond only to non-magnetic fields, which we refer to as “anomalous” fields; the coupling constant between the spin and an anomalous field must be different from the magnetic moment for the comagnetometer to produce a signal. Previous experiments have used the SERF comagnetometer to search for coupling of alkali and/or noble gas spins to *CPT* and Lorentz violating fields (Kornack *et al.*, 2008) and to long-range spin-dependent forces (Vasilakis *et al.*, 2009), and the comagnetometer can also be employed as a highly sensitive gyroscope (Kornack *et al.*, 2005). Here the anomalous field is the electric field, and its coupling constants with the alkali and noble gas spins are the electric dipole moments.

We only discuss briefly the operation of the SERF comagnetometer, which is described in significantly greater detail by Kornack (2005). We define the pumping direction as $\hat{z}$ and the probing direction as $\hat{x}$; the alkali spins polarize the noble gas nuclei along the pumping direction. As shown in Figure 7.2(a), if we apply an external magnetic field $\mathbf{B}'' = B''\hat{z}$ to cancel the fields produced by the magnetization of the alkali and noble gas spins, then there is nearly zero total field inside the vapor cell. The alkali spins experience only a small longitudinal field proportional to their own magnetization that does not affect suppression of external fields,

$$B_z = \frac{32\pi^2\kappa_0\mu_B P_z n}{3\mu_0}, \quad (7.3)$$

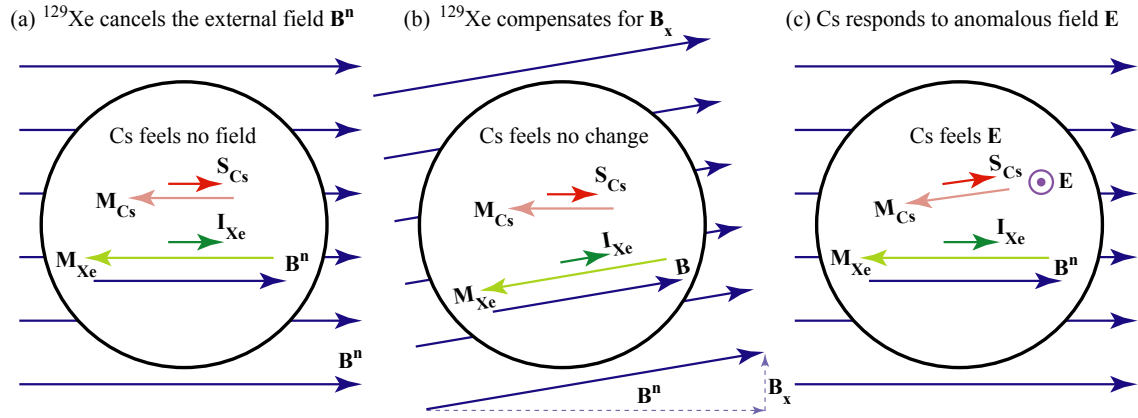


Figure 7.2: Principle of operation of the SERF comagnetometer: polarized noble gas nuclei shield the alkali spins from feeling external magnetic fields, but anomalous non-magnetic fields result in a precession signal. Adapted from Kornack (2005).

where κ_0 is the magnetic field enhancement factor due to the overlap of the alkali atom and noble gas spin wavefunctions.¹ Walker (1989) calculates that $\kappa_0=880$ for cesium and ^{129}Xe . Upon application of a small transverse magnetic field $B_x \ll B^n$, the noble gas spins rotate to cancel out the field and maintain the alkali atoms at zero field, as shown in Figure 7.2(b). The alkali atoms therefore remain in the SERF regime and are insensitive to external magnetic fields and gradients. Operation as a comagnetometer is particularly effective at suppressing low-frequency noise, such as the magnetic noise produced by μ -metal and ferrite shields.

This perfect cancelation of external fields only works if the noble gas magnetization precisely balances the field experienced by the alkali atoms. If the noble gas nuclei couple to an anomalous field, then they precess and no longer compensate for the magnetic field which then interacts with the alkali spins, resulting in a magnetometer signal. Similarly, if the alkali spins couple to an anomalous field not compensated for by the noble gas nuclei as shown in Figure 7.2(c), then the alkali precession again causes a magnetometer signal.

¹ This formula applies specifically to spins in a spherical cell, but we may use it as an approximation for other cell geometries.

The response of the comagnetometer is complicated and includes sensitivity to several effects, including the magnetic field, light shifts, nonorthogonality of the pump and probe beams, and rotation of the system, in addition to the anomalous field. If all of these effects except the last are canceled or accounted for, then the signal given in response to an electric field $E = E_y \hat{y}$ is

$$S_x = \frac{S_0 \gamma^e}{R_{\text{OP}} + R_{\text{rel}}} E_y \left(\frac{d^a}{\frac{1}{2} g_s \mu_B} - \frac{d^n}{I g_I \mu_N} \right). \quad (7.4)$$

Here d^a is the electric dipole moment of the alkali atom, and $\frac{1}{2} g_s \mu_B$ is the magnetic moment of the electron, where $g_s \approx 2$ is the electron g-factor, and μ_B is the Bohr magneton; similarly, d^n is the electric dipole moment of the noble gas atom, and $I g_I \mu_N$ is its magnetic moment, where I is the nuclear spin, g_I is the nuclear g-factor of the atom, and μ_N is the nuclear magneton.

A magnetometer uses a known coupling constant (the magnetic dipole moment μ) to determine the strength of a field, whereas an EDM experiment uses a known field strength to determine the coupling constant. From Equation 7.2, we see that the conversion between the magnetometer sensitivity and the EDM sensitivity is²

$$\delta d = \frac{\mu \times \delta B}{E} = \frac{\hbar \times \delta \omega}{2E}. \quad (7.5)$$

In order to minimize δd it is therefore necessary to achieve the lowest possible noise level at the measurement frequency; while the comagnetometer suppresses the effect of magnetic field noise, technical noise remains a problem at time scales longer than ~ 1 second. For example, air currents can cause slight variation in the direction of the probe beam, so it is important to minimize the path length of the beam. Slight temperature drift of various optical components can also steer the probe beam as well as cause fluctuation of polarization due to birefringence, and frequency drift of the lasers can cause light shifts (in the case of the pump beam) and changes in the optical rotation signal (in the case of the probe beam).

² Note that the energy splitting between adjacent Zeeman levels is given by twice the interaction energy with the electric and magnetic fields, i.e., $\hbar \omega = 2\mu B + 2dE$.

By carefully controlling these effects, [Vasilakis *et al.* \(2009\)](#) demonstrate sensitivity of about $1 \text{ fT}/\sqrt{\text{Hz}}$ at a frequency of 0.1 Hz, although the sensitivity is worse at lower frequencies. We expect to operate in the range of .02-.05 Hz by reversing the electric field direction with a time period of 20-40 seconds, comparing the comagnetometer signal before and after each reversal.

The spatial profile of the electron cloud of an atom distorts in response to an external electric field in order to keep the total field inside the atom at zero, so application of a field should have no effect on an atom regardless of whether it has an electric dipole moment. However, in heavy atoms the valence electrons are relativistic near the nucleus and actually enhance the internal electric field rather than null it ([Sandars, 1965, 1966](#)). This effect is quantified in terms of the enhancement factor K of the electric field, such that the dipole moment of the atom d^a and the moment of the bare electron d^e are related by

$$d^a = Kd^e. \quad (7.6)$$

The enhancement factor is on the order of $\alpha^2 Z^3$, where $\alpha \sim 1/137$ is the fine-structure constant, and Z is the atomic number. We therefore select cesium as our alkali species because it is the heaviest stable alkali metal with an enhancement factor of roughly 114 ([Hartley *et al.*, 1990](#)), rendering the experiment highly sensitive to the EDM of the electron. Similarly, we choose xenon as the buffer gas because it is the heaviest stable noble gas; we use isotopically enriched ^{129}Xe ($I=1/2$) because the more abundant isotope ^{131}Xe ($I=3/2$) has a nuclear quadrupole moment that causes fast spin relaxation ([Warren and Norberg, 1966](#)). All previous demonstrations of the SERF comagnetometer have employed potassium and ^3He .

7.3 Application of Electric Fields to Alkali Vapor Cells

The sensitivity of the experiment to the electric dipole moments of cesium and ^{129}Xe scales linearly with the strength of the applied electric field (see Equation 7.5), so we apply the largest possible field inside the vapor cell. We fill the cell with about 2-3 atm of nitrogen gas to prevent electrical breakdown at fields up to 20 kV/cm (Meek and Craggs, 1953); although higher pressures of nitrogen would permit application of stronger fields, the large spin-destruction cross-section between cesium and nitrogen would limit the magnetic linewidth. The nitrogen also serves as the buffer gas, and we include only about 5-10 Torr of ^{129}Xe in the cell, enough to allow operation as a comagnetometer without greatly affecting its linewidth. We fill the oven with sulfur hexafluoride [SF_6], which is a better insulator and has a higher breakdown voltage than air or dry nitrogen.

We observe that the cesium vapor density slowly decreases upon application of an electric field in an uncoated cell made of aluminosilicate glass.³ The rate at which the density drops depends on the amplitude of the field, and the density does not immediately recover if the field disappears; when we allow the density to drop to zero and then turn off the field, the vapor takes about one day to recover its initial density because new cesium atoms must diffuse from the end of the stem back into the main cell body. We speculate that the electric field induces the bare aluminosilicate glass to absorb cesium atoms, and the atoms remain inside the glass even after the field turns off. Figure 7.3 shows the transmission of a laser tuned to the D1 resonance through a cesium cell with a field of 6 kV/cm, displaying a steady decrease of vapor density with time.

Any effect of the electric field on the vapor density is unacceptable for continuous operation of the comagnetometer to detect electric dipole moments, especially since a field of

³ We use aluminosilicate rather than borosilicate (pyrex) glass because it has a higher bulk resistivity; see Section 7.3.2.

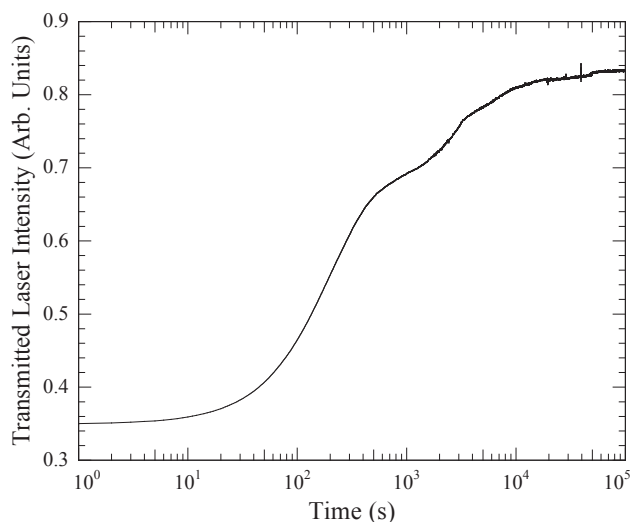


Figure 7.3: Increase in the transmission of a resonantly tuned laser through a cesium cell upon application of a 6 kV/cm electric field, demonstrating that the optical depth of the cell, and thus the vapor density, decreases slowly to zero with time.

20 kV/cm causes the density to drop by a substantial fraction of its value within a few seconds; fields as small as 0.1 kV/cm have a noticeable effect on the cesium density. However, cells coated with OTS (see Section 6.2) exhibit no change in density upon application of an electric field. In particular, we observe a fractional change of density less than 10^{-4} over the course of an hour in a field of 20 kV/cm. We believe that the OTS coating effectively forms a barrier between the glass surface and the cell, preventing atoms from reaching the glass and getting absorbed by it. The coating is unaffected by the operating temperature $\sim 120^\circ\text{C}$ of the experiment, so all comagnetometer cells used for the EDM experiment will have a multilayer OTS coating prepared according the procedure described in Section 6.2.2. While a similar dependence of vapor density on the electric field has not been detected in pyrex cells, [Hunter *et al.* \(1988\)](#) observe that a silane coating improves the field homogeneity in such cells.

7.3.1 Measuring the Stark Shift

We verify the existence and amplitude of the electric field inside the vapor cell by measuring the dc Stark shift of the optical resonance frequency. We use a single linearly polarized laser as a probe beam, and we tune its frequency to approximately one halfwidth away from the D1 resonance, where the slope of the transmission curve given by Equations 2.7 and 2.55 is greatest. We heat the cell to 90°C, so the vapor is optically thick, and small shifts of the resonance frequency thus result in a noticeable change in probe beam transmission. We calibrate the measurement by modulating the laser frequency by several megahertz and observing the change in transmission as the laser frequency moves closer to and farther from resonance; for small modulation amplitudes the fractional change in transmission is linear in the frequency shift. We measure the dc Stark shift by turning the electric field on and off with a period of approximately 250 seconds and comparing the probe beam transmission with the field applied and not applied.

The scalar shift of the optical resonance frequency ν_0 is given by

$$\Delta\nu_0 = k|E|^2, \quad (7.7)$$

where $k=115.219 \text{ kHz}/(\text{kV}/\text{cm})^2$ for the cesium D1 transition (Hunter *et al.*, 1992). The change in probe beam transmission therefore should not depend on the polarity of the electric field, but its sign should depend on whether the probe beam frequency is higher or lower than the resonance frequency. We show our measurements in one particular OTS-coated cell in Figure 7.4(a), and we see that the transmission change behaves as expected, indicating that the signal is indeed due to the electric field. The Stark shift can be used to monitor the strength of the electric field in the cell, and during the EDM experiment it will be measured periodically to check that the appropriate field is applied. We show in Figure 7.4(b) that the observed field is systematically smaller than expected; this discrepancy must be resolved before the experiment is conducted.

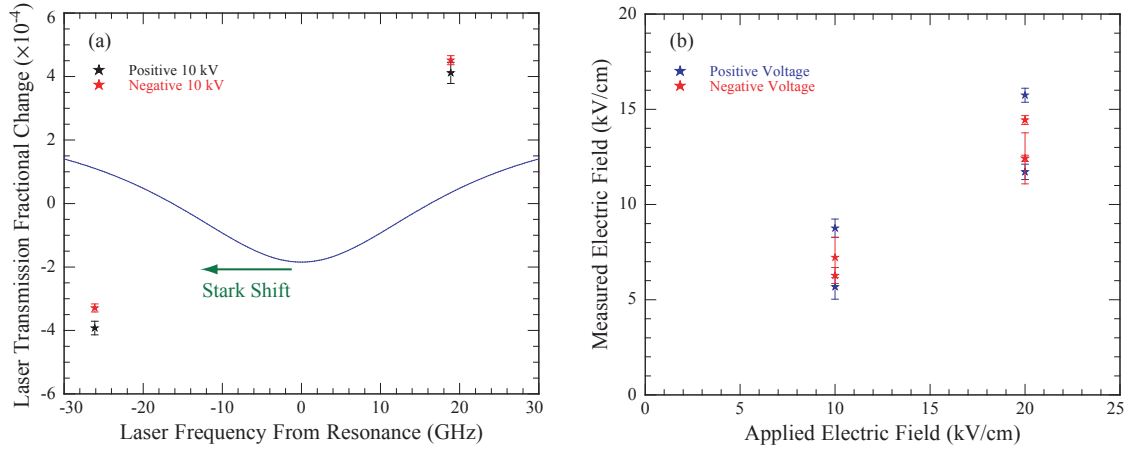


Figure 7.4: (a) Measured change in fractional transmission of a detuned probe beam due to the dc Stark shift of the cesium D1 transition upon application of an electric field. The cesium cell contains 3 amg of nitrogen to prevent electrical breakdown, and we heat it to 90°C. (b) Measured electric field amplitude compared to the applied field amplitude.

7.3.2 Leakage Currents

Perhaps the most important systematic effect that we must control in this experiment is the leakage current across the comagnetometer vapor cell. Although glass is an extremely good electrical insulator, its resistivity is finite, and the application of a high voltage to generate an electric field causes a small current to flow through the glass. This current creates a magnetic field that is experienced by the cesium spins, and their response to the magnetic field mimics the existence of a nonzero EDM: the magnetic field scales with the strength of the applied electric field, and its polarity depends on the sign of the electric field. While the current is likely to flow straight between the electrodes, we consider the worst-case scenario that the leakage current flows in a complete loop around the cell in order to estimate the maximum effect that the current can have on the EDM measurement.

For example, a current of 1 pA in a cell with an outer diameter of 2 inches generates a magnetic field of 25 aT. This field prevents the experiment from detecting an electric dipole moment with amplitude smaller than $\mu B/E$, and a magnetometer operating with

a sensitivity of $1 \text{ fT}/\sqrt{\text{Hz}}$ would be limited by this field after about 1600 seconds of integration. However, [Vasilakis *et al.* \(2009\)](#) demonstrated that a $\text{K-}^3\text{He}$ SERF comagnetometer suppresses magnetic fields by a factor of more than 1000, and a $\text{Cs-}^{129}\text{Xe}$ comagnetometer should be able to perform just as well. A suppression factor of only 100 is sufficient to reduce the magnetic field due to the leakage current below the level of sensitivity after 100 days of integration,⁴ so we consider a leakage current below 1 pA to be acceptable for operation of the EDM experiment, although we have yet to demonstrate field suppression at this level by the $\text{Cs-}^{129}\text{Xe}$ comagnetometer.

We therefore design the vapor cell to minimize the leakage current generated by application of the high voltages necessary for the experiment. One of the prototype cells is shown in [Figure 7.5](#), displaying a characteristic ‘H’-shaped cross-section. This design allows the electrodes to be placed close together (about 5 mm apart), while the leakage current has a long path length to flow across (about 100 mm), thus maximizing the electric field and minimizing the current that result from a given voltage. We use external electrodes, and we cover the glass at the electrode regions with conductive, but nonmagnetic, silver paint. The electric field in the central region is most homogeneous if the electrodes are solid discs without any gaps, so the stem will likely be placed at the bottom of the outer edge of the cells used for the actual experiment. Care must be taken to prevent cesium condensation outside of the stem, since metal deposits on the glass surface reduce the resistance of the leakage current path.

We construct the vapor cell out of aluminosilicate glass because it has the highest bulk resistivity of the glass types that are unreactive with alkali metal. In order to determine the suitability of a particular kind of glass, we measure the leakage current across “dummy cells,” which are 2 cm long pieces of tubing with an outer diameter of 25 mm and wall thickness of 1.5 mm; such pieces are easier to produce and to clean than actual vapor cells,

⁴ Specifically, the magnetic field due to the leakage current after 100 times suppression is equivalent to 189 days of integration at a sensitivity of $1 \text{ fT}/\sqrt{\text{Hz}}$.



Figure 7.5: Picture of a prototype cell designed for the EDM experiment, showing the 'H'-shaped cross-section that maximizes the electric field and minimizes the leakage current produced by a given voltage.

and they allow for efficient testing of multiple samples. We heat the cells in an oven filled with SF_6 , and we measure the current with a high-precision ammeter identical to the one used by Romalis *et al.* (2001) and described in Griffith (2005), giving us a precision of 0.1 pA. As an example we show in Figure 7.6 the leakage current measured in a quartz dummy cell during application of 10 kV both before and after cleaning with chromic acid,⁵ displaying an increase with temperature due to the exponential decrease of the bulk resistivity that is characteristic of electrical insulators, as well as the significant improvement that results from cleaning the surface of the glass. We find that an OTS coating does not adversely affect the measured leakage current and may in fact improve the resistance of the quartz dummy cells.

⁵ Chromic acid is a saturated solution of chromium trioxide [CrO_3] in sulfuric acid. Following Griffith (2005), we first clean the glass with soap and water, and then we sonicate it in acetone. We let the glass sit in chromic acid at 80°C for about 15 minutes, and then we let it sit in two changes of boiling deionized water for 15 minutes each before sonicating it in acetone and then methanol.

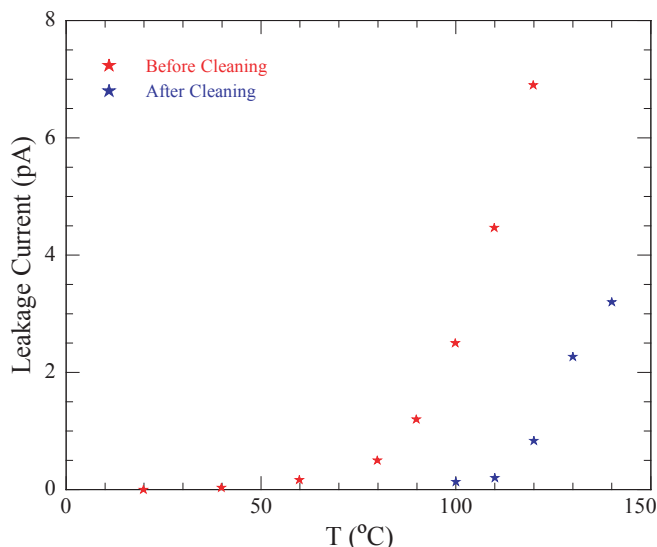


Figure 7.6: Leakage current measured in a quartz dummy cell during application of 10 kV, showing a dramatic decrease after cleaning with chromic acid. The resistivity of the glass decreases exponentially with temperature.

Our initial efforts focused on Corning 1720, the aluminosilicate glass that the prototype EDM cells are made of. We measure currents on the order of 50-100 pA resulting from application of 10 kV at 100°C in the prototype cells even after cleaning with chromic acid, making these cells unacceptable for use in the EDM experiment. Corning 1720 dummy cells exhibit currents of 10-20 pA under the same conditions, and there is evidence that the resistivity may decrease after the glass is heated and melted for glassblowing purposes. We therefore conclude that this glass type is inappropriate for use in the EDM experiment.

Instead we consider alternate compositions of aluminosilicate glass that have higher resistivity than Corning 1720, including particularly Schott 8252 and GE 180. The resistivity measured of dummy cells made of each kind of aluminosilicate glass are presented in Figure 7.7, along with the measured resistivity of quartz for reference. The predicted values are extrapolated from the datasheets provided by the manufacturers. We observe

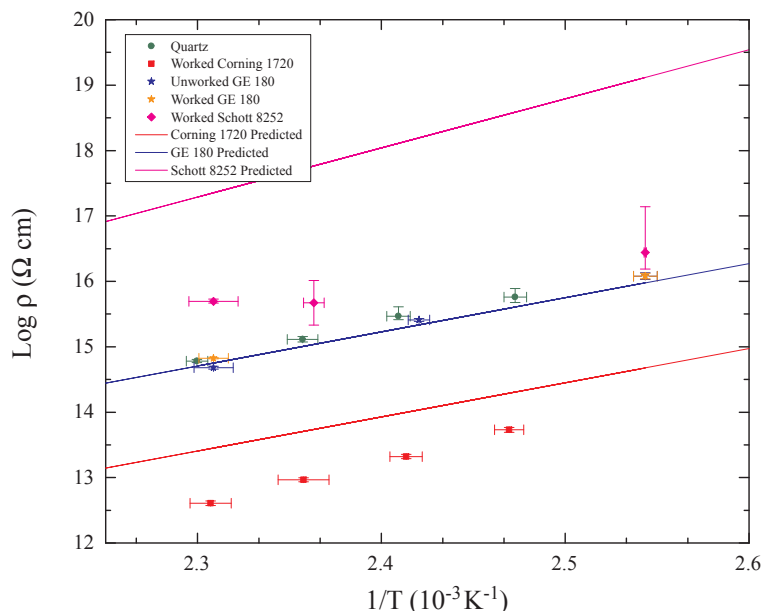


Figure 7.7: Resistivity of several varieties of aluminosilicate glass, as well as quartz, measured using dummy cells in an SF_6 atmosphere. Pieces labeled as “worked” have been resized by a glassblower in order to test if the electrical properties are affected by flame heating.

significantly smaller leakage currents for Schott 8252 and GE 180 than for Corning 1720,⁶ making them potentially suitable for the EDM experiment. Blowing these types of glass does not appear to affect their resistivity, and cleaning them with chromic acid is unnecessary; we find it sufficient to sonicate them sequentially in acetone, trichloroethylene, and methanol. We measure leakage currents of 0.2 ± 0.2 pA at 120°C and 1.0 ± 0.1 pA at 160°C in an unfilled ‘H’-shaped cell made of Schott 8252. The current at 160°C is slightly higher than expected, but the resistance of the cell may improve after cleaning and coating the interior surface and then baking out any residual moisture for several days. The observed properties of Schott 8252 make it an ideal candidate for future testing.

The cell and electrodes act as a capacitor when the high voltage ramps up and down, so we measure a current of several hundred picoamps that depends on the rate of change of

⁶ In fact, the currents detected for the Schott 8252 dummy cells are consistent with zero and are limited by the precision of the measurement. Corning 1720 has a much higher alkali metal content than Schott 8252 or GE 180, which may be the cause of its lower resistivity.

the voltage. Once the voltage reaches its set point, the current requires a finite amount of time to settle to its minimum value. In teflon dummy cells we find that the current drops to zero within a few seconds, but in glass cells the current can take several minutes to reach the value determined by the resistivity. The settling time is likely due to the properties of the glass; for instance, we observe that the current settles more quickly for Schott 8252 than for GE 180. This effect must be studied further in order to determine if there is a way to hasten the decay of the charging current. The frequency at which the EDM experiment operates will likely be determined by the settling time, as it will be necessary to wait until the current drops below some value, such as 1 pA, before recording data. This level is much higher than the minimum current we observe for the Schott 8252 cells, so we will not need to wait for the current to settle completely, and we believe that the experiment may be able to run with a period of 20-40 seconds.

7.3.3 Density Matrix Simulation

We calculate the effect of the applied electric field on the cesium spins in order to check that the field does not disrupt operation as a magnetometer. The evolution of the density matrix is nonlinear in the SERF regime because of rapid spin-exchange collisions (see the second term on the right-hand side of Equation 2.185), so we use a numerical simulation similar to that described in Section 3.4. In Figure 7.8 we compare the calculated magnetic linewidth to that predicted by Equation 5.5 as a function of the magnetic field strength, showing that the density matrix simulation describes accurately the behavior of alkali spins in a magnetic field under conditions of rapid spin-exchange. Here we present the first calculation of spin evolution in the SERF regime under application of an electric field. The applied field causes a quadratic dc Stark shift of the ground-state Zeeman levels $|F, m_F\rangle$ given by

$$\Delta H(F, m_F) = -\frac{1}{2}\alpha(F)\frac{3m_F^2 - F(F+1)}{2I(I+1)}(3\cos^2\theta - 1)|E|^2, \quad (7.8)$$

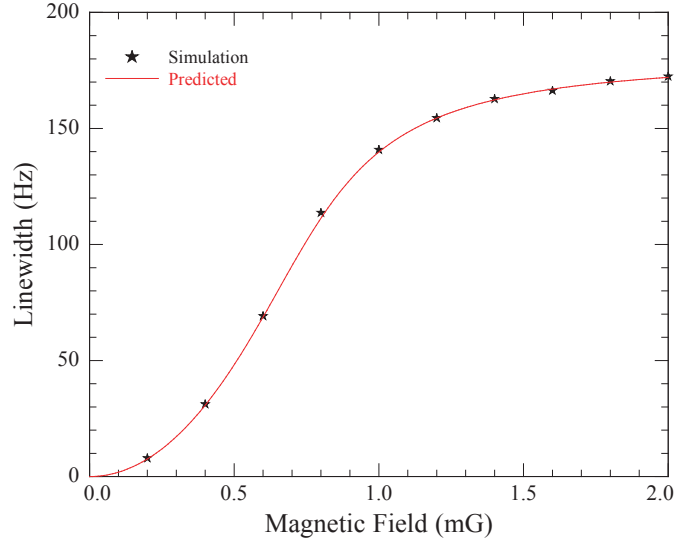


Figure 7.8: Comparison of the spin-exchange broadening calculated by the density matrix simulation to the value predicted by Equation 5.5 for cesium density $8 \times 10^{12} \text{ cm}^{-3}$, corresponding to operation at 90°C .

where θ is the angle between the electric field and the spin quantization axis, and $\alpha(4) \approx -\alpha(3) = -2\pi\hbar \times 0.0372 \text{ Hz}/(\text{kV}/\text{cm})^2$ is the tensor polarizability of the cesium ground state (Ulzega *et al.*, 2006).⁷ We do not include the energy shift common to all levels with the same value of F , and we ignore it in our calculations because it is significantly smaller than the hyperfine splitting. Figure 7.9 shows the ground-state energy splitting due to the electric field, where we describe the Stark shift in terms of the frequency ω_S given by

$$\omega_S = \frac{3\alpha(F)}{\hbar(2I+1)} |E|^2. \quad (7.9)$$

We find that the Hamiltonian describing the interaction between the cesium spins and the electric field is

$$\mathcal{H} = -\frac{4\alpha(F)}{(2I+1)^2} (E_x F_x + E_y F_y - E_z F_z)^2. \quad (7.10)$$

⁷ Here we use H rather than E to refer to the energy of the $|F, m_F\rangle$ state in order to avoid confusion with the electric field E .

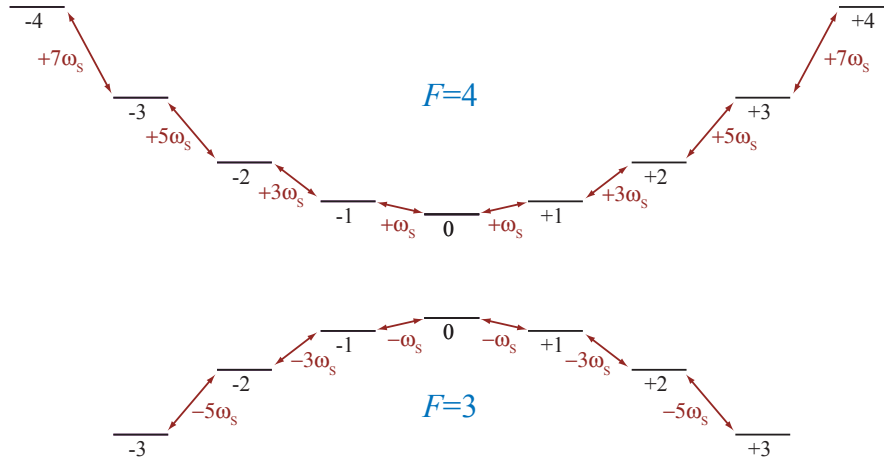


Figure 7.9: Quadratic ground-state energy splitting of cesium spins under application of an electric field due to the dc Stark shift. Energy levels are labeled by m_F .

We combine this expression with Equation 3.17 to form the total ground-state Hamiltonian due to electric and magnetic fields and the hyperfine interaction, and we calculate the evolution of the density matrix according to Equation 2.185. In addition to the electric field, we may also apply the longitudinal magnetic field given by Equation 7.3 for comagnetometer operation, with $B_z=0.3$ mG at 90°C and $B_z=1.7$ mG at 120°C , assuming that $P=0.5$.

The Stark shift is quadratic in Em_F , so an electric field orthogonal to the spin direction does not cause precession, unlike an orthogonal magnetic field. While the directions of individual spins change in response to the electric field, they do so symmetrically about the polarization axis to create an aligned state rather than an oriented one, so no net transverse polarization of the ensemble develops. However, the amplitude of the ensemble polarization vector does decrease as individual spins tilt away from the pumping axis. We show the reduction of longitudinal polarization with increasing electric field amplitude in Figure 7.10; at higher vapor density, the spins maintain larger polarization at a given electric field strength because more rapid spin-exchange collisions preserve a quasi-spin-temperature distribution. The calculated polarization does not depend greatly on the optical pumping or spin-destruction rates. We intend to operate the EDM experiment using

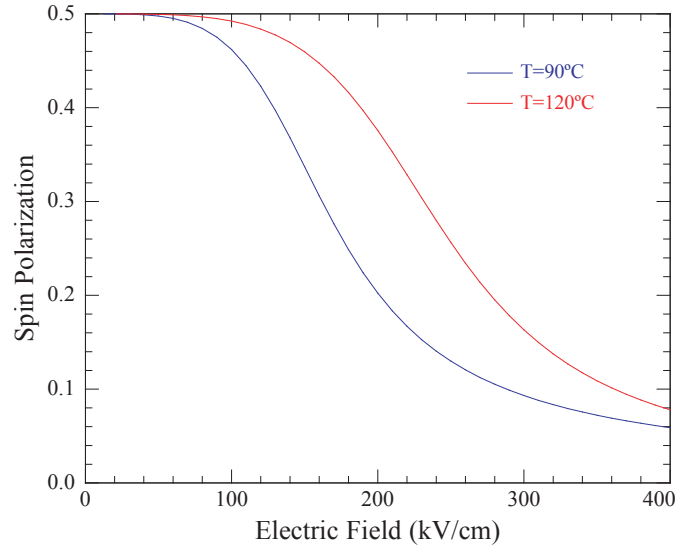


Figure 7.10: Longitudinal spin polarization $2S_z$ of an ensemble of cesium atoms in an electric field, calculated from a density matrix simulation. To achieve nominal polarization of 50%, we set the optical pumping rate equal to the spin-destruction rate caused by 2.5 amg of nitrogen and 5 Torr of xenon, with $R_{SD}=3554 \text{ s}^{-1}$ at 90°C , and $R_{SD}=3985 \text{ s}^{-1}$ at 120°C ; the increase is due to more rapid cesium-cesium spin-destruction collisions at higher vapor density. The atoms experience the longitudinal magnetic field given by Equation 7.3, which has only a small effect on the longitudinal polarization.

a field of about 20 kV/cm, so this effect should not prevent us from achieving the spin polarization $P=0.5$ necessary for sensitive operation of a SERF magnetometer.

However, if the electric field is not orthogonal to the polarization axis, then the symmetry of the system is broken and the alkali spins develop nonzero transverse components. Figure 7.11(a-c) shows the three spin components in a 20 kV/cm electric field as a function of the angle θ between the polarization axis and the electric field, where we set the magnetic field equal to zero in order to remove the effect of spin precession. We find that for

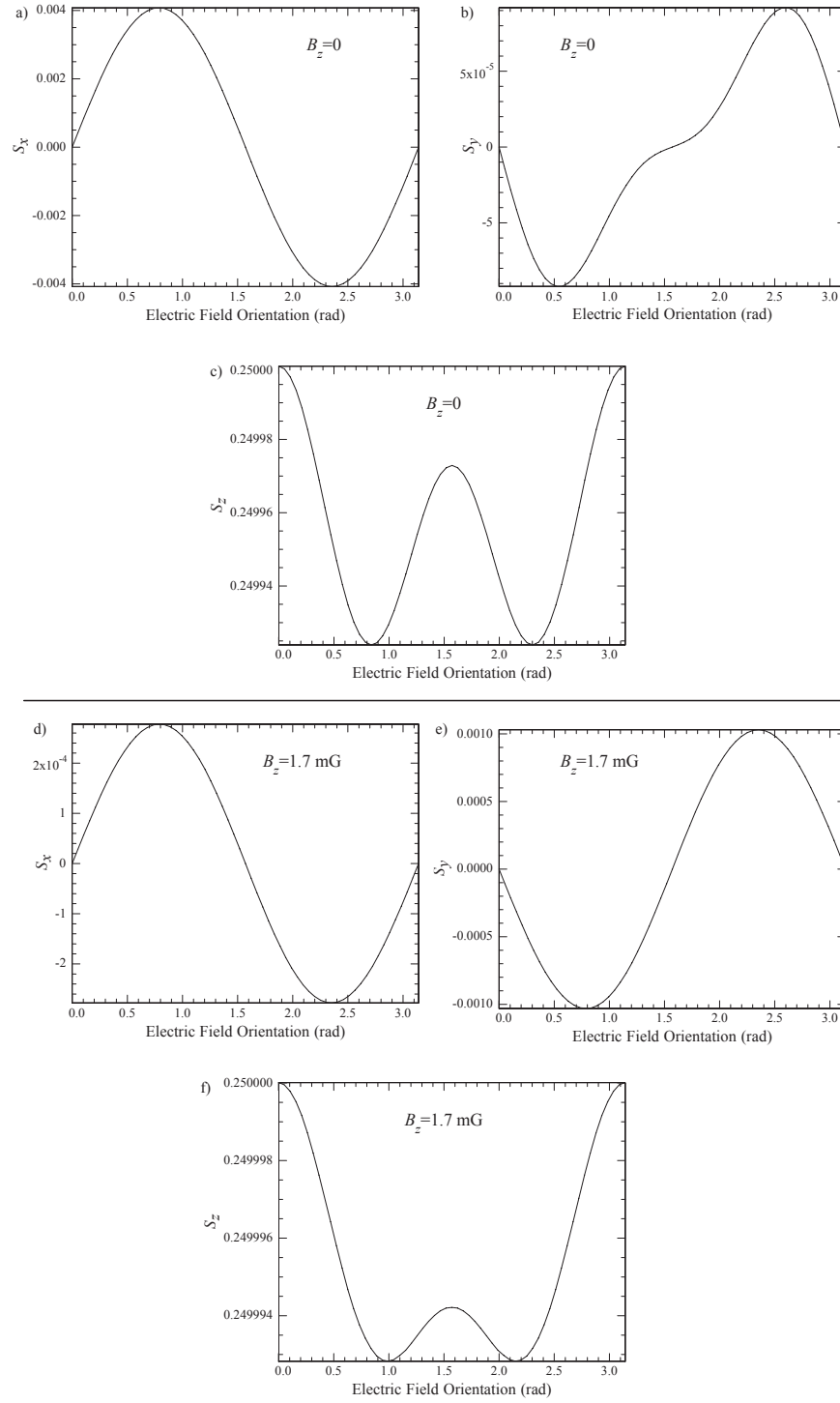


Figure 7.11: Components of the cesium spin polarization as a function of the relative angle between the polarization direction and a 20 kV/cm electric field, as calculated with $R_{OP}=R_{SD}=4000 \text{ s}^{-1}$. We consider both zero magnetic field and a longitudinal magnetic field $B_z=1.7 \text{ mG}$ given by Equation 7.3 for operation at 120°C with $P=0.5$.

$P \approx 0.5$, these components may be written as

$$S_x = \frac{\alpha |E|^2}{R_{OP} + R_{SD}} (a_x \sin \theta \cos \theta) \quad (7.11)$$

$$S_y = \pm \frac{\alpha^2 |E|^4}{(R_{OP} + R_{SD})^2} (a_y \sin 2\theta + b_y \sin 4\theta) \quad (7.12)$$

$$S_z = \pm \left[0.25 - \frac{\alpha^2 |E|^4}{(R_{OP} + R_{SD})^2} (a_z \cos 2\theta + b_z \cos 4\theta + c_z) \right], \quad (7.13)$$

where the coefficients are listed in Table 7.1 and are functions of the optical pumping, spin-relaxation, and spin-exchange rates. Figure 7.12 compares the analytic forms given in the table to the values calculated by the density matrix simulation for a_y and b_y as examples. S_x is even under reversal of the spin direction, as determined by the pump beam polarization, while S_y and S_z are odd. Following Murthy *et al.* (1989), who considered several effects of the dc Stark shift on ground-state cesium atoms at low magnetic field, we write the change in magnetometer signal ΔS in terms of the angular momentum of the pump beam photons $\mathbf{s}$, the electric field $\mathbf{E}$, and the angular momentum of the analyzing photons $\mathbf{J}$:⁸

$$\begin{aligned} \Delta S = & \frac{\alpha}{R_{OP} + R_{SD}} [a_x (\mathbf{s} \cdot \mathbf{E})(\mathbf{s} \times \mathbf{E}) \cdot \mathbf{J}] \\ & + \frac{\alpha^2}{(R_{OP} + R_{SD})^2} (\mathbf{s} \cdot \mathbf{E})(\mathbf{J} \cdot \mathbf{E}) [a_y |E|^2 + b_y ((\mathbf{s} \cdot \mathbf{E})^2 - (\mathbf{J} \cdot \mathbf{E})^2)] \\ & + (\mathbf{s} \cdot \mathbf{J}) \left\{ .25 - \frac{\alpha^2}{(R_{OP} + R_{SD})^2} [a_z ((\mathbf{s} \cdot \mathbf{E})^2 - (\mathbf{J} \cdot \mathbf{E})^2) |E|^2 \right. \\ & \quad \left. + b_z ((\mathbf{s} \cdot \mathbf{E})^4 - 6(\mathbf{s} \cdot \mathbf{E})^2(\mathbf{J} \cdot \mathbf{E})^2 + (\mathbf{J} \cdot \mathbf{E})^4) - c_z |E|^4] \right\}. \end{aligned} \quad (7.14)$$

The dc Stark shift therefore results in a probe beam signal proportional to S_x , which is linear near the nominal operating condition of $\theta = \pi/2$; the slope $\partial S_x / \partial \theta$ is approximately the same regardless of density but depends strongly on the magnetic field. We may consider the nonzero spin component S_x that develops at $B_z = 0$ as being due to precession of S_z in an effective magnetic field B'_y , which is given by combining Equations 5.19 and 7.11:

$$\gamma^e B'_y = 3\alpha |E|^2 \sin \theta \cos \theta, \quad (7.15)$$

⁸ Here we refer to photons measuring the particular spin component under consideration. For example, if we consider S_y , then the analyzing photons have angular momentum $\mathbf{J} = J\hat{y}$.

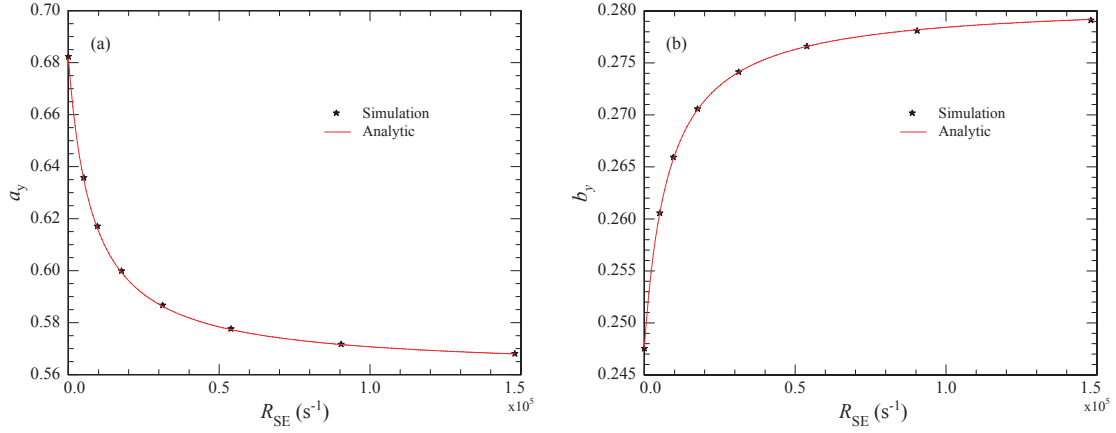


Figure 7.12: Comparison of the analytic expressions for a_y and b_y given in Table 7.1 to the values calculated by the density matrix simulation for $R_{OP}=R_{SD}=4000 \text{ s}^{-1}$.

Coefficient	Value
a_x	$\frac{3}{4}$
a_y	$\frac{15}{22} - \frac{3R_{SE}}{25(R_{SE}+R_{OP}+R_{SD})}$
b_y	$\frac{26}{105} + \frac{R_{SE}}{30(R_{SE}+R_{OP}+R_{SD})}$
a_z	$\frac{3}{16} - \frac{3R_{SE}}{16(R_{SE}+R_{OP}+R_{SD})}$
b_z	$\frac{26}{105} + \frac{R_{SE}}{30(R_{SE}+R_{OP}+R_{SD})}$
c_z	$\frac{4}{9} - \frac{8R_{SE}}{49(R_{SE}+R_{OP}+R_{SD})}$

Table 7.1: Coefficients for Equations 7.11-7.14; the numerical values may not be exact but are all accurate to within 1%.

where γ^e is the gyromagnetic ratio of the bare electron. The value of S_x for an arbitrary longitudinal magnetic field is then given by Equation 5.19 as

$$S_x \approx S_z \frac{\beta'_y}{1 + \beta_z^2}, \quad (7.16)$$

where $\beta = \gamma^e \mathbf{B} / (R_{\text{OP}} + R_{\text{SD}})$, and we assume that $\beta'_y \ll 1$.⁹ For example, at zero magnetic field with $|E|=20$ kV/cm and $R_{\text{OP}}=R_{\text{SD}}=4000$ s⁻¹, the slope is .008 rad⁻¹, while at $B_z=50$ mG the slope is 6×10^{-7} rad⁻¹. During comagnetometer operation, the cesium spins experience the longitudinal magnetic field given by Equation 7.3, which is proportional to the vapor density. As a result, the slope of the comagnetometer signal is 6×10^{-3} rad⁻¹ for operation at 90°C, but it drops to 6×10^{-4} rad⁻¹ for operation at 120°C; compare Figures 7.11(a) ($B_z=0$) and 7.11(d) ($B_z=1.7$ mG at 120°C).

The electric field must therefore remain nearly orthogonal to the pump beam direction so as not to cause a false magnetometer signal larger than the EDM that we attempt to measure. We can zero the nonorthogonality by periodically turning the electric field off, comparing the magnetometer signal to that with the field turned on and using the error signal to adjust the pump beam direction. From Equation 5.22, we find that magnetic field sensitivity of 1 fT/ $\sqrt{\text{Hz}}$ is equivalent to sensitivity to the transverse polarization of $\delta S_x = 1 \times 10^{-9}$ for a SERF magnetometer with the anticipated operating conditions of $R_{\text{OP}}+R_{\text{SD}} \sim 7000$ s⁻¹ and $P=0.5$. With the slope $\partial S_x / \partial \theta = 6 \times 10^{-4}$ rad⁻¹ for a comagnetometer at 120°C, this corresponds to $\delta \theta = 2 \mu\text{rad} / \sqrt{\text{Hz}}$; for example, measuring for 4 seconds enables us to ensure orthogonality within 1 μrad . This procedure can be repeated as frequently as necessitated by drift in the beam direction. For comparison, we estimate that Murthy *et al.* (1989) required orthogonality within about 300 μrad in their experiment, which measured $\delta d^e < 10^{-25}$ e-cm using cesium atoms near zero magnetic field with $|E|=4$ kV/cm.

⁹ Similarly, we may describe the nonzero S_y that develops as being due to an effective magnetic field B'_x , but $S_y \ll S_x$ for $B_z=0$, so we ignore it in this discussion.

In addition, the change in S_x due to the Stark shift is even under reversal of either E or s , whereas the signal due to an EDM is odd in both, so we may subtract the magnetometer signal before and after reversal of each parameter to cancel the effect of the Stark shift while doubling the EDM signal.¹⁰ The error signal is quadratic in both E and s , so if we achieve perfect reversal of the amplitudes of both to within 1%, then we suppress the effect of nonorthogonality by a factor of 2500. If the nonorthogonality is maintained such that its instantaneous value is within $\pm 1 \mu\text{rad}$, then over long periods of integration its average value should be approximately zero. By applying feedback to null the nonorthogonality, we can therefore reduce the systematic uncertainty due to the Stark shift below the level of the statistical uncertainty of the experiment.

The reversal of E can be imperfect in direction as well as amplitude. For example, a buildup of electrical charge somewhere on the surface of the cell could create a small electric field $E_{\text{stat}}\hat{z}$ along the pumping direction that remains constant when the applied field reverses direction. This static field causes the total electric field to tilt at an angle $\Delta\theta \approx E_{\text{stat}}/E$ from its nominal orientation, with $\Delta\theta$ changing sign upon field reversal. The resulting shift in S_x also reverses sign, mimicking an electric dipole moment. The static field must be less than 0.02 V/cm to ensure orthogonality with 1 μrad . Similarly, we must ensure that the applied field itself reverses direction perfectly so as not to produce a false EDM signal.

The longitudinal magnetic field affects S_y and S_z as well as S_x , as shown in Figure 7.11(e-f), although both remain odd under reversal of the spin direction. In particular, the longitudinal field B_z causes mixing of S_x and S_y due to spin precession about the polarization axis. Equations 5.19-5.20 give S_y in terms of S_x ,

$$S_y = \beta_z S_x = \frac{\gamma^e B_z}{R_{\text{OP}} + R_{\text{SD}}} S_x. \quad (7.17)$$

¹⁰ Reversal of s also requires reversal of the ^{129}Xe spin direction, which can be achieved either by adiabatic fast passage or by application of a π pulse.

In fact, we see that a nonorthogonal electric field results in a greater value of S_y than of S_x for comagnetometer operation at 120°C, with $\beta_z=3.7$ for $R_{OP}=R_{SD}=4000 \text{ s}^{-1}$. If the electric field is nonorthogonal to the pumping direction to $2 \mu\text{rad}$, then $S_y=4 \times 10^{-9}$ while $S_x=1 \times 10^{-9}$. This can create a problem if the electric field is also not orthogonal to the probing direction, with the relative angle $\varphi \neq \pi/2$, effectively causing the magnetometer to measure a combination of S_x and S_y .¹¹ However, the slope $\partial S_x / \partial \varphi = 3 \times 10^{-9} \text{ rad}^{-1}$ near $\varphi = \pi/2$, so this effect is extremely small; φ must change by 0.25 rad in order to double the false magnetometer signal. It should therefore not be necessary to apply a feedback routine for canceling the nonorthogonality of the electric field and probe beam. However, note that we can distinguish between tilt of the electric field into the pumping and probing directions because they cause error signals that are even and odd under reversal of the spin direction, respectively. If necessary we can therefore cancel both effects using the procedure described above, comparing the error signals with the electric field on and off but also including reversals of the spin direction.

Nonorthogonality of the pump and probe beams can also affect the magnetometer signal. Previous implementations of the SERF comagnetometer have used a feedback routine for zeroing this nonorthogonality to the order of $1 \mu\text{rad}$ (Kornack, 2005), and the EDM experiment will do the same, so changes in S_z due to the electric field should have minimal effect on the magnetometer signal. A deviation from normal of the angle θ between the electric field and pumping direction of $2 \mu\text{rad}$ results in a decrease of S_z of only 4×10^{-11} , or about $1/(7 \times 10^9)$ of its nominal value. The effect of the dc Stark shift on S_z should therefore not be a concern.

¹¹ Conceptually, the electric field tilting into the probe direction is equivalent to the electric field remaining along $\hat{y}$ and the probe beam tilting into that direction.

7.4 Prospects for the Cs-Xe EDM Experiment

We project the sensitivity of this experiment to the electric dipole moments of the electron and the ^{129}Xe atom assuming optimal conditions, including magnetometer sensitivity of $1 \text{ fT}/\sqrt{\text{Hz}}$ and an applied field strength of 20 kV/cm . We find from Equation 7.5 that $\delta d^e = 9 \times 10^{-30} \text{ e-cm}$ and $\delta d^{Xe} = 4 \times 10^{-31} \text{ e-cm}$ after 100 days of integration, using an electric field enhancement factor of 114 for the cesium atom (Hartley *et al.*, 1990) and a magnetic moment of $\mu = -0.778 \mu_N$ for ^{129}Xe (Stone, 2005), where $\mu_N = 5.05 \times 10^{-27} \text{ J/T}$ is the nuclear magneton. The SERF comagnetometer can operate indefinitely with minimal maintenance, so a run time of this length is reasonable, although the duty cycle is likely to be about 50% because of the various feedback routines required by the comagnetometer (Kornack, 2005). The anticipated sensitivities are listed in Table 7.2, along with the current experimental limits. Under ideal conditions this experiment will improve the bound on the electron EDM by more than one order of magnitude in one day of operation and two orders of magnitude after 100 days. The experiment will offer a similar improvement over the best diamagnetic atomic EDM limit, currently set with ^{199}Hg , and four orders of magnitude improvement over the existing limit on the EDM of the ^{129}Xe atom. While these numbers may be optimistic, they nevertheless show that the Cs-Xe EDM experiment can still offer a substantial improvement over previous experiments even if it does not achieve optimal sensitivity.

There are several issues that we must resolve before the experiment reaches this level of performance. For example, the vapor cell must contain nitrogen and xenon gas, both of which have large spin-destruction cross-sections with cesium (see Table A.2), leading to a broader magnetic linewidth than is typical for a SERF magnetometer. The linewidth observed in the prototype EDM cells is about 50 Hz at $90\text{--}125^\circ\text{C}$ with $R_{\text{OP}} \ll R_{\text{SD}}$, as shown in Figure 5.4(a), due to contributions from the spin-destruction rates of cesium with 3 amg of nitrogen gas and 5 Torr of xenon gas. Such a broad linewidth could make it difficult to

	This Experiment (Projected)	Present Experimental Limit Bound	Reference
Electron	9×10^{-30}	1.6×10^{-27}	(Regan <i>et al.</i> , 2002)
^{129}Xe	4×10^{-31}	4.1×10^{-27}	(Rosenberry and Chupp, 2001)
^{199}Hg		2.1×10^{-28}	(Romalis <i>et al.</i> , 2001)
Neutron		2.9×10^{-26}	(Baker <i>et al.</i> , 2006)

Table 7.2: Projected sensitivity δd of the Cs-Xe EDM experiment after 100 days of integration, assuming sensitivity of $1 \text{ fT}/\sqrt{\text{Hz}}$ and an applied field of 20 kV/cm . For comparison we also list the current experimental limits on the electric dipole moments of the electron, ^{129}Xe , ^{199}Hg (which has the lowest bound of any diamagnetic atom), and the neutron. All electric dipole moments are in units of $e\text{-cm}$.

realize high magnetic field sensitivity. We attain sensitivity of about $50 \text{ fT}/\sqrt{\text{Hz}}$ in a SERF magnetometer with a cesium cell containing 650 Torr of helium and 90 Torr of nitrogen with a much narrower linewidth of 4 Hz, as displayed in Figure 7.13, and Ledbetter *et al.* (2008) demonstrate sensitivity of $40 \text{ fT}/\sqrt{\text{Hz}}$ in a cesium SERF magnetometer with similar linewidth. Improving the sensitivity of a cesium SERF magnetometer to the level of $1 \text{ fT}/\sqrt{\text{Hz}}$ will require great effort, especially in the EDM cells with a minimum magnetic linewidth of about 100 Hz necessary for 50% polarization.

In addition to spin destruction caused by collisions with the xenon and nitrogen gases, we must also consider the effect of spin-exchange collisions. During comagnetometer operation, the cesium atoms experience a nonzero longitudinal field given by Equation 7.3, so spin-exchange relaxation is not completely eliminated.¹² Combining Equations 5.5 and 7.3, we find that the contribution of spin-exchange collisions to the magnetic linewidth increases linearly with density, as shown in Figure 7.14(a). At 90°C the linewidth broadening is 16 Hz, and at 120°C it is 89 Hz. This effect is significantly larger than the broadening due to cesium-cesium spin-destruction collisions, and it places a limit on the density at

¹² The longitudinal field is much smaller for a $\text{K-}^3\text{He}$ SERF comagnetometer because $\kappa_0=5.9$ (Ben-Amar Baranga *et al.*, 1998), compared to $\kappa_0=880$ for a $\text{Cs-}^{129}\text{Xe}$ comagnetometer (Walker, 1989). Previous implementations of the SERF comagnetometer therefore remained nearly unaffected by spin-exchange relaxation.

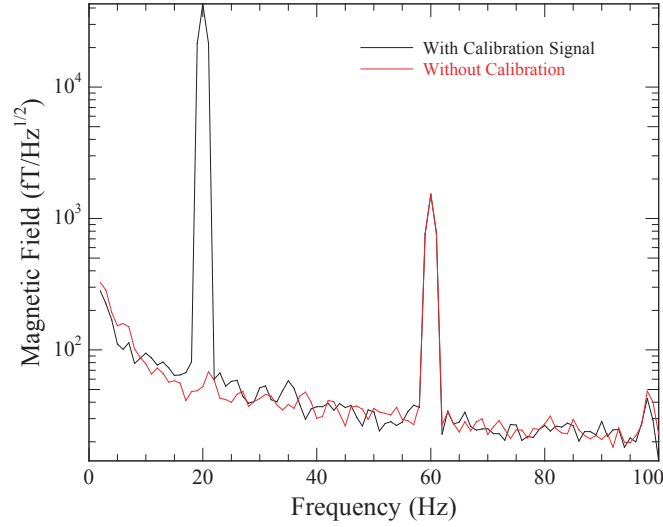


Figure 7.13: Noise spectrum recorded with a cesium SERF magnetometer at 122°C, displaying sensitivity of about $50 \text{ fT}/\sqrt{\text{Hz}}$ with a linewidth of 4 Hz.

which we may operate the comagnetometer without greatly affecting the magnetic linewidth. We show in Figure 7.14(b) the spin-exchange broadening as a function of operating temperature. Ideally the contribution of cesium-cesium collisions to the linewidth should be comparable to the contribution from collisions with other gas species, so the EDM experiment should operate in the range of 110-120°C.

We see from Equation 5.22 that the spin component S_x measured by the probe beam is inversely proportional to the magnetic linewidth, so a broader linewidth results in a smaller magnetometer signal. For a cesium SERF magnetometer with 50% polarization and linewidth $\Delta\omega = 2\pi \times 100 \text{ Hz}$,¹³ a magnetic field $B_y = 1 \text{ fT}$ gives transverse spin $S_x = 9 \times 10^{-10}$. Such small polarization leads to optical rotation angles of about 50 nrad for a probe beam tuned close to the D1 resonance near the anticipated operating density, as shown in Figure 7.15(a). However, we see that the rotation angle is larger in a cell containing less nitrogen gas because of the reduced optical and magnetic linewidths. We

¹³ Note that the contribution from spin-destruction collisions is $\Delta\omega = (R_{\text{OP}} + R_{\text{SD}})/q$, where the slowing-down factor $q=13$ for cesium with $P=0.5$. The contribution from spin-exchange collisions should be scaled appropriately so that it may be used in Equation 5.22.

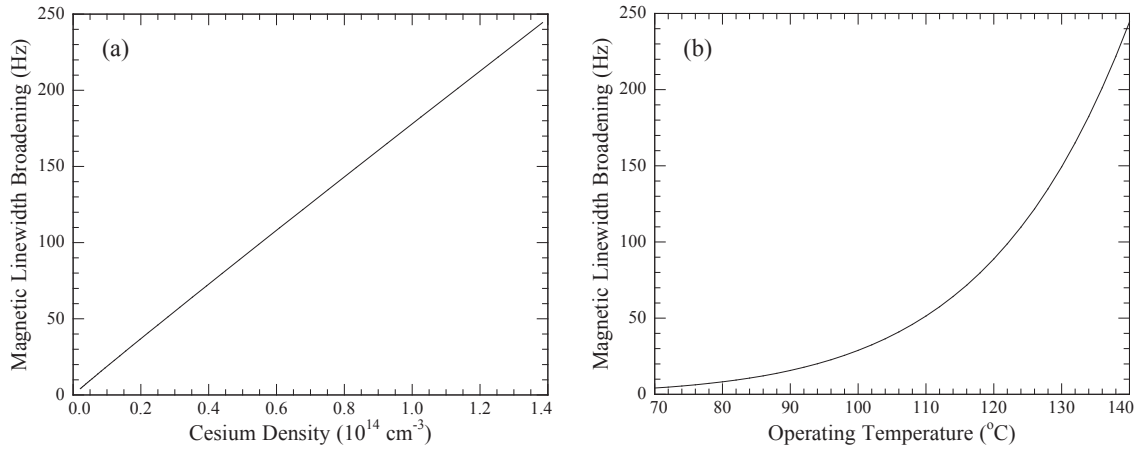


Figure 7.14: (a) Spin-exchange broadening of the magnetic linewidth calculated from Equation 5.5 as a function of vapor density in a Cs- ^{129}Xe SERF comagnetometer, assuming 50% polarization and a longitudinal magnetic field proportional to the cesium magnetization given by Equation 7.3. (b) Spin-exchange broadening as a function of temperature, assuming that the operating density is given by the saturated vapor pressure (Equation A.1).

must therefore select the nitrogen pressure that best balances the competing requirements of narrow linewidths and high breakdown voltage, likely closer to 2 amg than to 3 amg. The xenon pressure should be chosen to have roughly the same contribution to the spin-relaxation rate as the cesium vapor and nitrogen gas, in order to achieve large ^{129}Xe spin magnetization for comagnetometer operation while maintaining a reasonable magnetic linewidth.

Unfortunately, we see from Figure 7.15(b) that the probe beam is completely absorbed when tuned close to resonance at such high cesium vapor density. In order to obtain a useable optical rotation signal, we must instead tune the probe beam 100-150 GHz away from resonance, giving a rotation angle in the range of 2-6 nrad.¹⁴ While shot-noise-limited angular sensitivity of less than $5 \text{ nrad}/\sqrt{\text{Hz}}$ can be easily achieved in the radio-frequency regime (Savukov *et al.*, 2005), it is significantly more challenging at frequencies below 1 Hz due to convection effects and mechanical vibrations. In an effort to reduce long-term laser

¹⁴ This probe beam detuning is ideal because the frequency that gives 50% transmission through the cell falls within this range for operation at 125°C with 2-3 amg of nitrogen.

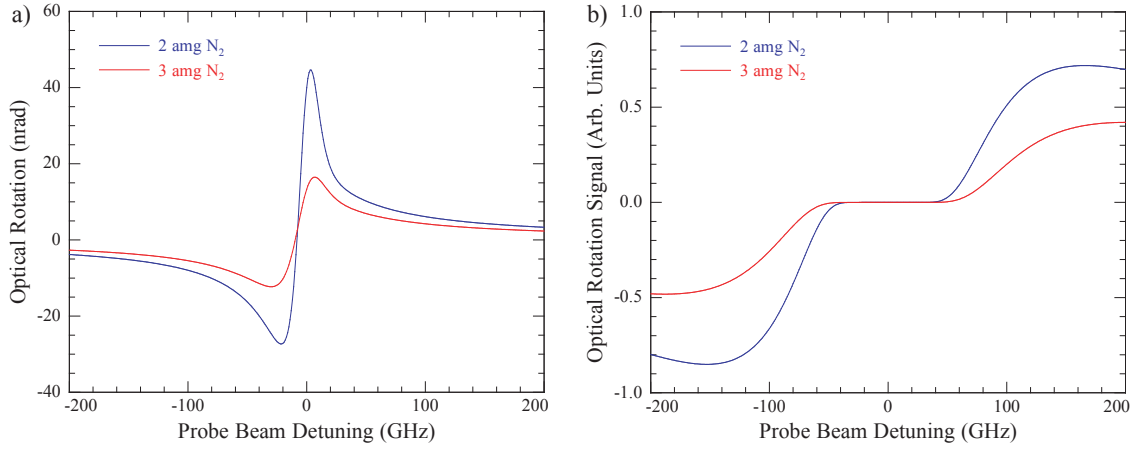


Figure 7.15: (a) Optical rotation angles calculated from Equations 2.99 and 5.22 due to a magnetic field $B_y=1$ fT for a cesium vapor cell with density $n=6\times 10^{13}$ cm⁻³, corresponding to operation at 125°C. We set $R_{OP}=R_{SD}$ to each be twice the cesium-nitrogen spin-destruction rate in order to account for collisions with xenon atoms and to set the polarization $P=0.5$. (b) Optical rotation signal accounting for absorption of the probe beam near resonance in a cell with length 5 cm.

noise, we plan to enclose the optics to reduce air flow, to use optical fibers to minimize the beam path length through air, and to evacuate the beam path close to the cell. We also consider the possibility of reflecting the probe beam so that it passes twice through the cell to amplify the optical rotation angle, and we calculate that the useable optical rotation signals increase by slightly less than a factor of 2 when accounting for the increased absorption of the beam; this method has the added benefit of canceling birefringence due to the cell and optical components.

The experiment has several parameters that must be optimized, so we perform a numerical analysis of the magnetometer signal and fundamental noise limit, accounting for the effects of pump and probe beam attenuation and the resulting inhomogeneity of spin polarization throughout the cell. We find that absorption of the pump beam is the larger concern, as the magnetometer exhibits better fundamental sensitivity if the cell is shorter in the pumping direction and longer in the probing direction. We may attempt to construct oval-shaped cells to take advantage of this, although round cells (like the prototype

displayed in Figure 7.5) are probably much easier to produce. Use of counter-propagating pump beams (see Section 4.1.4) may also assist in achieving more uniform polarization within the cell, improving magnetometer performance, but we include only one pump beam in our analysis. Depending on the size and shape of the cell, the power of both the pump and probe beams should be in the range of 200-600 mW because of the high optical density and broad magnetic linewidth. We calculate from Equation 5.16 that the noise limit of the magnetometer is in the range of 0.5-1.0 fT/ $\sqrt{\text{Hz}}$ if all parameters are optimized, so the anticipated sensitivity of 1 fT/ $\sqrt{\text{Hz}}$ requires us to achieve near-ideal performance of the magnetometer.

All previous implementations of the SERF comagnetometer have used potassium and ^3He , so we must demonstrate operation as a comagnetometer using cesium and ^{129}Xe . The suppression factor of external magnetic fields is determined by the magnetization of the xenon spin ensemble; we are unable to increase the xenon pressure in the cell above 5-10 Torr because of the large spin-destruction cross-section between cesium and xenon, so we must instead achieve high polarization of the xenon spins. The time scale of the xenon spin response for cancelation of external magnetic fields is set by the xenon resonance frequency (Kornack and Romalis, 2002), which must be much larger than the operating frequency of the EDM experiment. Like the cesium atoms, during comagnetometer operation the xenon nuclei experience a magnetic field that is proportional to their own magnetization, which is given by replacing the electron magnetic moment in Equation 7.3 with the magnetic moment of ^{129}Xe :

$$B_{z,\text{Xe}} = \frac{32\pi^2\kappa_0(-0.778\mu_N)P_{\text{Xe}}n_{\text{Xe}}}{3\mu_0}, \quad (7.18)$$

where P_{Xe} and n_{Xe} are the polarization and density of the xenon spins, respectively. Multiplying the magnetic field by the ^{129}Xe gyromagnetic ratio $\gamma^{\text{Xe}}=2\pi\times 1177 \text{ Hz/G}$, we find a resonance frequency of 3.0 Hz for 5 Torr of ^{129}Xe with 50% polarization (or for 10 Torr

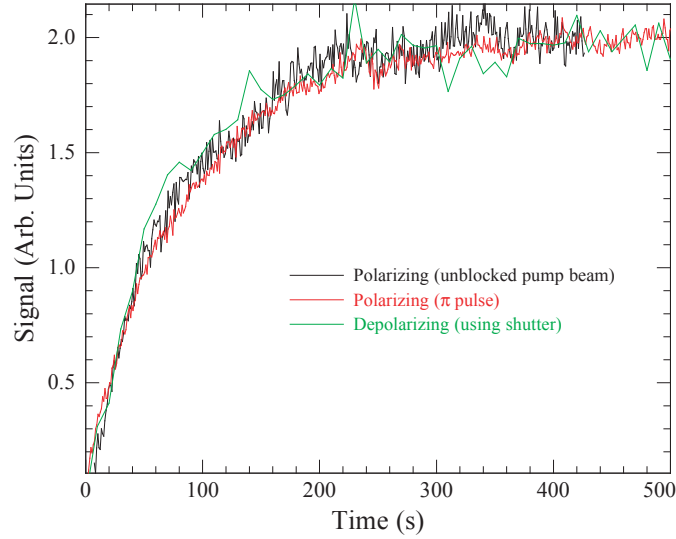


Figure 7.16: Measured polarization lifetime $T_1 \sim 60$ s of ^{129}Xe spins in an OTS-coated cesium cell at 95°C . We measure the lifetime using three different methods, including observation of both polarization and depolarization of the spins, and we scale all three curves to appear together.

with 25% polarization). This frequency should be sufficiently high for operation of the experiment.

We measure the spin polarization lifetime $T_{1,\text{Xe}}$ of the xenon spins in one of the prototype EDM cells to be about 60 seconds at 90°C , as shown in Figure 7.16, with the lifetime decreasing to about 30 seconds at 125°C . The xenon polarization lifetime is given by the sum of the spin-exchange rate with cesium atoms $R_{\text{SE,Xe}}^{\text{Cs}}$ and the rate of depolarizing collisions with the OTS-coated cell wall R_{wall} :

$$\frac{1}{T_{1,\text{Xe}}} = R_{\text{SE,Xe}}^{\text{Cs}} + R_{\text{wall}} \quad (7.19)$$

$$R_{\text{SE,Xe}}^{\text{Cs}} = n_{\text{Cs}} \sigma_{\text{SE}} \bar{v}, \quad (7.20)$$

where n_{Cs} is the density of the cesium vapor. Using the cesium-xenon spin-exchange cross section $\sigma_{\text{SE}} = 2.7 \times 10^{-20} \text{ cm}^2$ calculated by Walker (1989), we estimate that $1/R_{\text{wall}} \sim 120$ seconds. In another prototype cell we measure $T_{1,\text{Xe}}$ of only 10 seconds, indicating a much larger rate of depolarization due to collisions with the wall. The effectiveness of the OTS

coating at maintaining ^{129}Xe polarization does not degrade after application of electric fields within the cell. The polarization of the ^{129}Xe spins is given by an expression similar to that for alkali atoms, with the spin-exchange rate taking the place of the optical pumping rate:

$$P_{\text{Xe}} = P_{\text{Cs}} \frac{R_{\text{SE,Xe}}^{\text{Cs}}}{R_{\text{SE,Xe}}^{\text{Cs}} + R_{\text{wall}}} , \quad (7.21)$$

where P_{Cs} is the cesium polarization. A depolarization rate of $1/R_{\text{wall}} \sim 100\text{-}200$ seconds should therefore be sufficient to achieve high xenon spin polarization at an operating temperature near 120°C . ^{129}Xe spin lifetimes on the order of 1000 seconds have been observed using similar silane coatings (Zeng *et al.*, 1983; Driehuys *et al.*, 1995), so further study of the interaction of polarized ^{129}Xe with OTS may lead to improvement.

Before the EDM experiment can begin taking data, we must be able to implement the Cs-Xe SERF comagnetometer and simultaneously achieve each of the tasks described above: application of a 20 kV/cm electric field without affecting the cesium vapor density, measurement of the dc Stark shift of the optical resonance in agreement with the field strength, leakage current below the level of 1 pA with a short settling time, magnetic field sensitivity of $1 \text{ fT}/\sqrt{\text{Hz}}$ at a frequency of .02-.05 Hz, near-perfect orthogonality of the electric field to the pump and probe beam directions, and sufficient ^{129}Xe spin polarization to enable suppression of external magnetic fields. This is a challenging task, but all of these goals are reasonable and work is underway on each of them. There are no insurmountable problems that we are aware of, so we anticipate that this experiment will reach the level of performance needed to attain sensitivity that is two orders of magnitude better than previous electric dipole moment searches. This experiment will therefore be a valuable tool for probing the symmetries of nature and for investigating the standard model and its extensions.

Chapter 8

Summary and Conclusions

SUMMARIZING THE RESULTS of this thesis, we have presented several recent developments in alkali-metal magnetometry for both practical applications and fundamental physics experiments. We demonstrate synchronous optical pumping of quantum revival beats in the ground state of potassium atoms by double modulation of the pumping rate, in order to excite the atomic resonances at both the Larmor and revival frequencies. This technique increases the polarization of alkali spins in the Earth's magnetic field and thus improves the sensitivity of a scalar magnetometer; in particular, we observe an increase in polarization of ^{39}K spins by a factor of 3.9 at 0.5 G over the traditional single modulation technique. We also show through numerical simulation that double modulation can highly suppress the orientation-dependent heading error that limits the accuracy of alkali-metal scalar magnetometers, and this should be demonstrated in field operation. The method of synchronous pumping of revivals should also be attempted in other multi-level quantum systems in order to determine its general applicability for enhancing alignment.

We demonstrate a radio-frequency magnetometer that achieves sensitivity of $0.2 \text{ fT}/\sqrt{\text{Hz}}$ to oscillating magnetic fields, with potential for improvement by more than an order of

magnitude. The use of counter-propagating pump beams results in more uniform polarization throughout the cell compared to just one beam, and in practice it can assist with achieving high magnetic field sensitivity in one or multiple detection channels. The rf magnetometer can be useful for low-field magnetic resonance applications, and we demonstrate detection of NMR from polarized water at 62 kHz. Better magnetic shielding and gradiometric measurement should lead to substantial improvement, enabling the magnetometer to outperform surface pick-up coils at frequencies below about 50 MHz for NMR and MRI. We also observe nuclear quadrupole resonance signals from ammonium nitrate at 423 kHz with much better sensitivity than a pick-up coil of comparable volume. Although a fieldable version of the rf magnetometer for detection of NQR signals can only have partial shielding and would require subfemtotesla sensitivity, it has the potential to be an extremely useful tool for discovering hidden explosives and narcotics with minimal measurement times.

We show that the spin-exchange relaxation-free magnetometer can serve to measure all three vector components of the magnetic field simultaneously. Active feedback locks the alkali atoms at zero magnetic field, and the compensation currents act as a measurement of the field components. It should be possible to extend this technique to measure all five gradient components of the magnetic field as well. We use three-axis feedback to operate the SERF magnetometer in an unshielded environment, and we demonstrate sensitivity on the order of $1 \text{ pT}/\sqrt{\text{Hz}}$, limited by ambient field noise and gradients; we expect that operation in a remote location would provide significant improvement. As a vector sensor, the SERF magnetometer can provide more information about the magnetic field than a scalar detector, and it may find use in portable applications requiring recognition of slight field variations, such as magnetic anomaly detection.

We also study the application of anti-relaxation surface coatings for high-temperature magnetometry. We find that a multilayer of octadecyltrichlorosilane allows alkali atoms

to collide with the cell wall up to 2100 times without depolarizing, although most OTS-coated cells exhibit a level of several hundred bounces. OTS functions at temperatures up to about 170°C in the presence of potassium and rubidium vapor, making it the first known anti-relaxation coating for use at temperatures above the melting point of paraffin, and we demonstrate its use in a SERF magnetometer. In addition to magnetometry, it should be tested for other applications of alkali vapor, including atomic frequency references. Our discovery of the effectiveness of OTS raises the possibility that even better coatings may be found, including potentially a paraffin-level coating that allows up to 10,000 bounces. To this end we have developed a reusable alkali vapor cell for comparing the polarization lifetimes allowed by multiple varieties of surface coatings, and the use of flat coated slides allows us to perform additional surface characterization tests. Hopefully we will gain a better understanding of the interaction between alkali atoms and coated surfaces, enabling us to design superior high-temperature anti-relaxation coatings.

Finally, we explore the challenges of employing a SERF comagnetometer for a measurement of the permanent electric dipole moments of the electron and the ^{129}Xe atom. We find that the cell walls must be coated with OTS (or perhaps a similar molecule) in order to prevent absorption of cesium atoms into the surface upon application of an electric field. Preliminary measurements indicate that leakage currents may not affect the experiment if the cell is constructed from Schott 8252 or GE 180 aluminosilicate glass. We intend to reverse the direction of a 20 kV/cm electric field with a period of 20-40 seconds, and we anticipate achieving near-ideal magnetic field sensitivity of $1 \text{ fT}/\sqrt{\text{Hz}}$ at frequencies below 0.1 Hz, requiring polarimeter sensitivity to optical rotation at the level of several $\text{mrad}/\sqrt{\text{Hz}}$. We must demonstrate effective comagnetometer operation, with a high degree of ^{129}Xe polarization compensating for external magnetic fields by a factor of at least 100. We also must investigate experimentally whether our density matrix simulation accurately predicts the effect of the dc Stark shift on the cesium spins, and we must show that the

nonorthogonality of the electric field to the pump and probe directions can be controlled well enough so as not to cause an unacceptable false magnetometer signal. Work on these problems will proceed over the next several years, and once the experiment is online it has the potential to lower the existing experimental bounds by several orders of magnitude; we project the sensitivity to EDMs after 100 days of integration to be $\delta d^e = 9 \times 10^{-30}$ e-cm and $\delta d^{Xe} = 4 \times 10^{-31}$ e-cm.

Appendix A

Properties of the Alkali Metals

Table A.1 is a compilation of properties of the alkali metal isotopes that are most commonly used for magnetometry applications: ^{39}K , ^{41}K , ^{85}Rb , ^{87}Rb , and ^{133}Cs . The gyromagnetic ratio is $\gamma = 2\pi \times \nu_{\text{L}}/B$, where ν_{L} is the Larmor frequency as defined in Equation 3.28. The quantum beats revival and super-revival frequencies are given by ν_{rev} and ν_{suprev} , respectively, and are defined in Equations 3.29 and 3.30.

Table A.2 is a compilation of interaction properties for potassium, rubidium, and cesium. Interaction cross-sections are given for spin exchange σ_{SE} , spin destruction σ_{SD} , and quenching σ_{Q} . Diffusion constants D_0 are given at 1 amg and 273 K. Pressure broadening widths Γ (fullwidth at half-maximum, FWHM) and pressure shifts $\Delta\nu_0$ are given at 273 K. The references listed in both tables may themselves include compilations of earlier data, in which case they give appropriate citations for each measurement.

Alkali Isotopes:	³⁹ K	⁴¹ K	⁸⁵ Rb	⁸⁷ Rb	¹³³ Cs	References
Natural Abundance (%)	93.3	6.7	72.2	27.8	100	
Nuclear Spin I	3/2	3/2	5/2	3/2	7/2	
Nuclear g factor g_I	0.261	0.143	0.539	1.827	0.732	(Arimondo <i>et al.</i> , 1977)
D1 Transition in vacuum (nm)	770.1	770.1	795.0	795.0	894.6	(Kratz, 1949)
D2 Transition in vacuum (nm)	766.7	766.7	780.2	780.2	852.3	(Kratz, 1949)
Oscillator Strength, f_{D1}	0.324	0.324	0.332	0.332	0.347	(Migdalek and Kim, 1998)
f_{D2}	0.652	0.652	0.668	0.668	0.721	
Hyperfine Splitting, $^2S_{1/2}$ (MHz)	461.7	254.0	3036	6835	9193	(Arimondo <i>et al.</i> , 1977;
$^2P_{1/2}$ (MHz)	57.7	30.4	362.2	812.4	1168	Bendali <i>et al.</i> , 1981)
Natural Lifetime, $^2P_{1/2}$ (ns)	26.8	26.8	27.7	27.7	34.8	(Volz and
$^2P_{3/2}$ (ns)	26.5	26.5	26.2	26.2	30.4	Schmoranzner, 1996)
ν_L/B , $F = I + 1/2$ (kHz/G)	700.4	700.4	466.7	699.5	349.8	
$F = I - 1/2$ (kHz/G)	-700.7	-700.6	-467.5	-702.2	-350.9	
ν_{rev}/B^2 (Hz/G ²)	1063	1932	71.86	71.87	13.35	
ν_{suprev}/B^3 (Hz/G ³)	3.23	10.66	0.022	0.015	0.001	

Table A.1: Properties of the alkali metal isotopes that are most commonly used in magnetometry applications.

Alkali Metals:	Potassium	Rubidium	Cesium	References
σ_{SE}^{Self}	1.8×10^{-14}	1.9×10^{-14}	2.1×10^{-14}	(Ressler <i>et al.</i> , 1969; Aleksandrov <i>et al.</i> , 1999)
σ_{SD}^{Self}	1.0×10^{-18}	1.6×10^{-17}	2.0×10^{-16}	(Walker and Happer, 1997; Kadlecek <i>et al.</i> , 1998)
σ_{SD}^{He}	8.0×10^{-25}	9×10^{-24}	2.8×10^{-23}	(Allred <i>et al.</i> , 2002)
σ_{SD}^{Xe}	1.8×10^{-19}	2.0×10^{-19}	2.3×10^{-19}	(Walker, 1989)
$\sigma_{SD}^{N_2}$	7.9×10^{-23}	1×10^{-22}	5.5×10^{-22}	(Kadlecek <i>et al.</i> , 1998; Allred <i>et al.</i> , 2002)
$\sigma_Q^{N_2}, ^2P_{1/2}$	3.5×10^{-15}	5.8×10^{-15}	5.5×10^{-15}	(McGillis and Krause, 1967, 1968;
$^2P_{3/2}$	3.9×10^{-15}	4.3×10^{-15}	6.4×10^{-15}	Hryciyshyn and Krause, 1970)
D_0^{He}	0.35	0.50	0.29	(Franz and Sooriamoorthi, 1974;
$D_0^{N_2}$	0.20	0.19	0.10	Franz and Volk, 1976, 1982; Silver, 1984)
$\Gamma^{He}, D1$	13.3	18.0	19.92	All K data from Lwin and McCartan (1978); Rb data from Romalis <i>et al.</i> (1997); Cs data from Andalkar and Warrington (2002)
D2	17.7	15.8	17.86	
$\Delta\nu_0^{He}, D1$	3.9	3.2	3.39	
D2	2.1	0.30	0.57	
$\Gamma^{N_2}, D1$	21.0	17.8	14.83	
D2	21.0	18.1	17.24	
$\Delta\nu_0^{N_2}, D1$	-15.7	-8.25	-6.25	
D2	-11.3	-5.9	-5.11	

Table A.2: Interaction properties for potassium, rubidium, and cesium atoms. Cross-sections are in units of cm^2 , diffusion constants are in units of cm^2/s and are given at 1 amg and 273 K, pressure broadening widths are in units of GHz/amg and are fullwidth at half-maximum (FWHM) at 273 K, and pressure shifts are in units of GHz/amg at 273 K.

	Melting Point (°C)	Solid		Liquid	
		A	B	A	B
Potassium	63.5	4.961	4646	4.402	4453
Rubidium	39.3	4.857	4215	4.312	4040
Cesium	28.5	4.711	3999	4.165	3830

Table A.3: Parameters for Equation A.1 to give the alkali vapor density at a given temperature. Note that the parameters are different depending on whether the metal is above or below the melting point. Density parameters are taken from [Alcock *et al.* \(1984\)](#).

A.1 Alkali Vapor Density

The saturated density of the alkali vapor in units of cm^{-3} at a temperature T in Kelvin is given by the equation

$$n = \frac{1}{T} 10^{21.866 + A - B/T}. \quad (\text{A.1})$$

Appropriate values for the parameters A and B are given in Table A.3; note the difference between the solid and liquid phases. The saturated vapor density is plotted in Figure A.1 for potassium, rubidium, and cesium over the normal temperature range of magnetometer operation. In general the observed density in a vapor cell is less than predicted, often by a factor of 2 or more, especially in cells with anti-relaxation coatings.

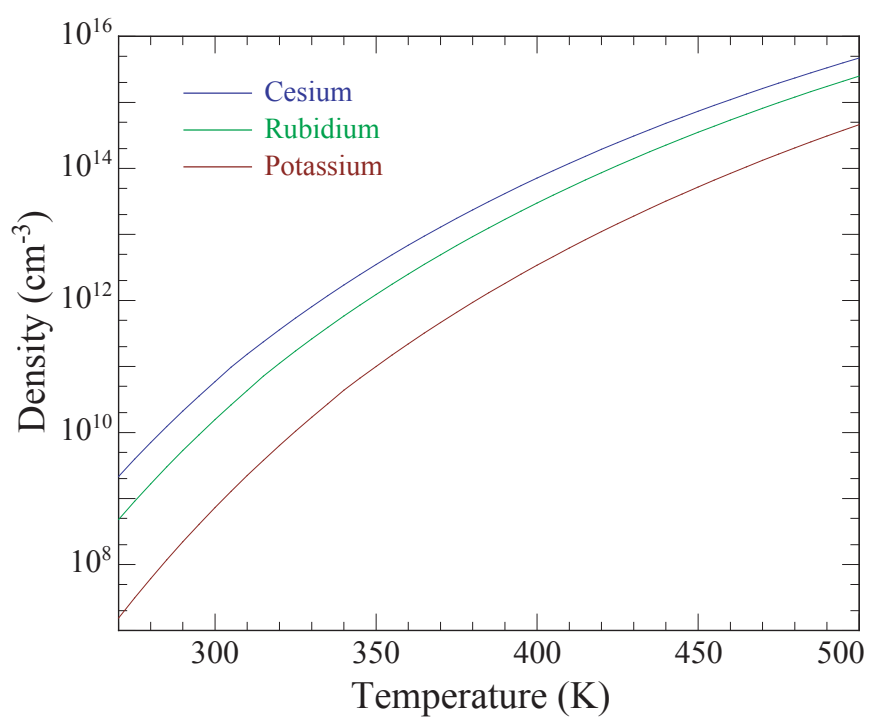


Figure A.1: Saturated density of potassium, rubidium, and cesium vapor as a function of temperature, as determined by Equation A.1.

Appendix B

Calculation of the Physical Eigenstates of the Alkali Atoms

As discussed in Section 2.9, the Zeeman interaction between a magnetic field and the electron and nuclear spins of an alkali atom causes mixing of the $|I + 1/2, m_F\rangle$ and $|I - 1/2, m_F\rangle$ states, so that the atom may no longer be properly described using the pure $|F, m_F\rangle$ basis. The ground-state atomic Hamiltonian (with $J = S$) in a magnetic field $\mathbf{B}$ is given by

$$\mathcal{H} = A_J \mathbf{I} \cdot \mathbf{J} + g_s \mu_B \mathbf{S} \cdot \mathbf{B} - g_I \mu_N \mathbf{I} \cdot \mathbf{B}, \quad (\text{B.1})$$

where $A_J = 2\hbar\omega_{\text{hf}}/(2I + 1)$ is the hyperfine coupling coefficient, g_I is the nuclear g-factor of the atom, and μ_N is the nuclear magneton. If the magnetic field is oriented along the $\hat{\mathbf{z}}$ -axis (*i.e.*, $\mathbf{B} = B\hat{\mathbf{z}}$), then the Hamiltonian simplifies:

$$\mathcal{H} = A_J \mathbf{I} \cdot \mathbf{J} + g_s \mu_B B S_z - g_I \mu_N B I_z. \quad (\text{B.2})$$

The physical basis that describes the atomic energy states is given by the eigenvectors of the Hamiltonian. If we wish to write the physical eigenstates in terms of the pure eigenstates, then we need to determine the action of the operators $\mathbf{I} \cdot \mathbf{J}$, S_z , and I_z on the state $|F, m_F\rangle$.

The angular momentum of an atom with nuclear spin I and electron spin $J = S = 1/2$ can be described using either the $|F, m_F\rangle$ basis or the $|m_I, m_s\rangle$ basis. A state in one basis can be decomposed in terms of states in the other basis using Clebsch-Gordan coefficients:

$$|F, m_F\rangle = C_{I, m_F+1/2, 1/2, -1/2}^{F, m_F} |m_F + 1/2, -1/2\rangle + C_{I, m_F-1/2, 1/2, +1/2}^{F, m_F} |m_F - 1/2, +1/2\rangle \quad (\text{B.3})$$

$$|m_I, m_s\rangle = C_{I, m_I, 1/2, m_s}^{I+1/2, m_I+m_s} |I + 1/2, m_I + m_s\rangle + C_{I, m_I, 1/2, m_s}^{I-1/2, m_I+m_s} |I - 1/2, m_I + m_s\rangle. \quad (\text{B.4})$$

See [Varshalovich *et al.* \(1988\)](#) for a detailed description of the properties of the Clebsch-Gordan coefficients. In particular,

$$C_{I, m_F-1/2, 1/2, +1/2}^{I+1/2, m_F} = \sqrt{\frac{I + 1/2 + m_F}{2I + 1}} \quad (\text{B.5})$$

$$C_{I, m_F+1/2, 1/2, -1/2}^{I+1/2, m_F} = \sqrt{\frac{I + 1/2 - m_F}{2I + 1}} \quad (\text{B.6})$$

$$C_{I, m_F-1/2, 1/2, +1/2}^{I-1/2, m_F} = \sqrt{\frac{I + 1/2 - m_F}{2I + 1}} \quad (\text{B.7})$$

$$C_{I, m_F+1/2, 1/2, -1/2}^{I-1/2, m_F} = -\sqrt{\frac{I + 1/2 + m_F}{2I + 1}}. \quad (\text{B.8})$$

Using Equations [B.3](#) and [B.4](#), we can calculate the action of an electron or nuclear spin operator on the state $|F, m_F\rangle$ by decomposing into the $|m_I, m_s\rangle$ basis, applying the operator, and then converting back to the $|F, m_F\rangle$ basis. Using the identities [B.5-B.8](#), this procedure

gives

$$S_z |I + 1/2, m_F\rangle = \frac{1}{2I + 1} (m_F |I + 1/2, m_F\rangle + \sqrt{(2I + 1)^2 - m_F^2} |I - 1/2, m_F\rangle) \quad (\text{B.9})$$

$$S_z |I - 1/2, m_F\rangle = \frac{1}{2I + 1} (\sqrt{(2I + 1)^2 - m_F^2} |I + 1/2, m_F\rangle - m_F |I - 1/2, m_F\rangle) \quad (\text{B.10})$$

$$I_z |I + 1/2, m_F\rangle = \frac{1}{2I + 1} (2I m_F |I + 1/2, m_F\rangle - \sqrt{(2I + 1)^2 - m_F^2} |I - 1/2, m_F\rangle) \quad (\text{B.11})$$

$$I_z |I - 1/2, m_F\rangle = \frac{1}{2I + 1} (-\sqrt{(2I + 1)^2 - m_F^2} |I + 1/2, m_F\rangle + (2I + 2)m_F |I - 1/2, m_F\rangle). \quad (\text{B.12})$$

Note that the operators S_z and I_z do not change the value of $m_F = m_I + m_s$. If we wish to determine the transverse spin components S_x and S_y , for example to calculate the optical rotation signal, then we can define them in terms of the raising and lowering operators

$$S_x = \frac{S_+ + S_-}{2} \quad (\text{B.13})$$

$$S_y = \frac{S_+ - S_-}{2i}, \quad (\text{B.14})$$

which operate on a state with spin S according to

$$S_+ |m_I, m_s\rangle = \sqrt{S(S + 1) - m_s(m_s + 1)} |m_I, m_s + 1\rangle \quad (\text{B.15})$$

$$S_- |m_I, m_s\rangle = \sqrt{S(S + 1) - m_s(m_s - 1)} |m_I, m_s - 1\rangle. \quad (\text{B.16})$$

Using the same procedure as before, we get that

$$S_+|I + 1/2, m_F\rangle = \frac{1}{2I+1}(\sqrt{(I + 1/2 - m_F)(I + 3/2 + m_F)} |I + 1/2, m_F + 1\rangle + \sqrt{(I + 1/2 - m_F)(I - 1/2 - m_F)} |I - 1/2, m_F + 1\rangle) \quad (\text{B.17})$$

$$S_+|I - 1/2, m_F\rangle = \frac{-1}{2I+1}(\sqrt{(I + 1/2 + m_F)(I + 3/2 + m_F)} |I + 1/2, m_F + 1\rangle + \sqrt{(I + 1/2 + m_F)(I - 1/2 - m_F)} |I - 1/2, m_F + 1\rangle) \quad (\text{B.18})$$

$$S_-|I + 1/2, m_F\rangle = \frac{1}{2I+1}(\sqrt{(I + 1/2 + m_F)(I + 3/2 - m_F)} |I + 1/2, m_F - 1\rangle - \sqrt{(I + 1/2 + m_F)(I - 1/2 + m_F)} |I - 1/2, m_F - 1\rangle) \quad (\text{B.19})$$

$$S_-|I - 1/2, m_F\rangle = \frac{1}{2I+1}(\sqrt{(I + 1/2 - m_F)(I + 3/2 - m_F)} |I + 1/2, m_F - 1\rangle - \sqrt{(I + 1/2 - m_F)(I - 1/2 + m_F)} |I - 1/2, m_F - 1\rangle), \quad (\text{B.20})$$

where the effect of the operator $S_{\pm}$ is to change m_F by ± 1 . Finally, we note that $\mathbf{I} \cdot \mathbf{J}$ commutes with the quantum numbers F and m_F , so that

$$\mathbf{I} \cdot \mathbf{J}|F, m_F\rangle = \frac{F(F+1) - I(I+1) - J(J+1)}{2} |F, m_F\rangle. \quad (\text{B.21})$$

The physical eigenstates are given by the eigenvectors of the Hamiltonian (Equation B.2) as

$$|\bar{F}, m_{\bar{F}}\rangle = A_J \mathbf{I} \cdot \mathbf{J}|F, m_F\rangle + g_s \mu_B B S_z |F, m_F\rangle - g_I \mu_N B I_z |F, m_F\rangle. \quad (\text{B.22})$$

These are the states that should be used as the basis for the atomic density matrix and used in all calculations. As an example, we present the physical eigenstates for the case of

$I = 3/2$:

$$\begin{aligned}
\overline{|2, -2\rangle} &= \left[\frac{3}{4}A_J + B\left(-\frac{1}{2}g_s\mu_B + \frac{3}{2}g_I\mu_N\right) \right] |2, -2\rangle \\
\overline{|2, -1\rangle} &= \left[\frac{3}{4}A_J + B\left(-\frac{1}{4}g_s\mu_B + \frac{3}{4}g_I\mu_N\right) \right] |2, -1\rangle \\
&\quad + \frac{\sqrt{3}}{4}B(g_s\mu_B + g_I\mu_N) |1, -1\rangle \\
\overline{|2, 0\rangle} &= \frac{3}{4}A_J |2, 0\rangle \\
&\quad + \frac{1}{2}B(g_s\mu_B + g_I\mu_N) |1, 0\rangle \\
\overline{|2, +1\rangle} &= \left[\frac{3}{4}A_J + B\left(\frac{1}{4}g_s\mu_B - \frac{3}{4}g_I\mu_N\right) \right] |2, +1\rangle \\
&\quad + \frac{\sqrt{3}}{4}B(g_s\mu_B + g_I\mu_N) |1, +1\rangle \\
\overline{|2, +2\rangle} &= \left[\frac{3}{4}A_J + B\left(\frac{1}{2}g_s\mu_B - \frac{3}{2}g_I\mu_N\right) \right] |2, +2\rangle \\
\overline{|1, -1\rangle} &= \frac{\sqrt{3}}{4}B(g_s\mu_B + g_I\mu_N) |2, -1\rangle \\
&\quad - \left[\frac{5}{4}A_J - B\left(\frac{1}{4}g_s\mu_B + \frac{5}{4}g_I\mu_N\right) \right] |1, -1\rangle \\
\overline{|1, 0\rangle} &= \frac{1}{2}B(g_s\mu_B + g_I\mu_N) |2, 0\rangle \\
&\quad - \frac{5}{4}A_J |1, 0\rangle \\
\overline{|1, +1\rangle} &= \frac{\sqrt{3}}{4}B(g_s\mu_B + g_I\mu_N) |2, +1\rangle \\
&\quad - \left[\frac{5}{4}A_J + B\left(\frac{1}{4}g_s\mu_B + \frac{5}{4}g_I\mu_N\right) \right] |1, +1\rangle.
\end{aligned}$$

Similarly, the energies of the atomic physical eigenstates are given by the eigenvalues of the diagonalized Hamiltonian, as shown in Section 3.2.

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